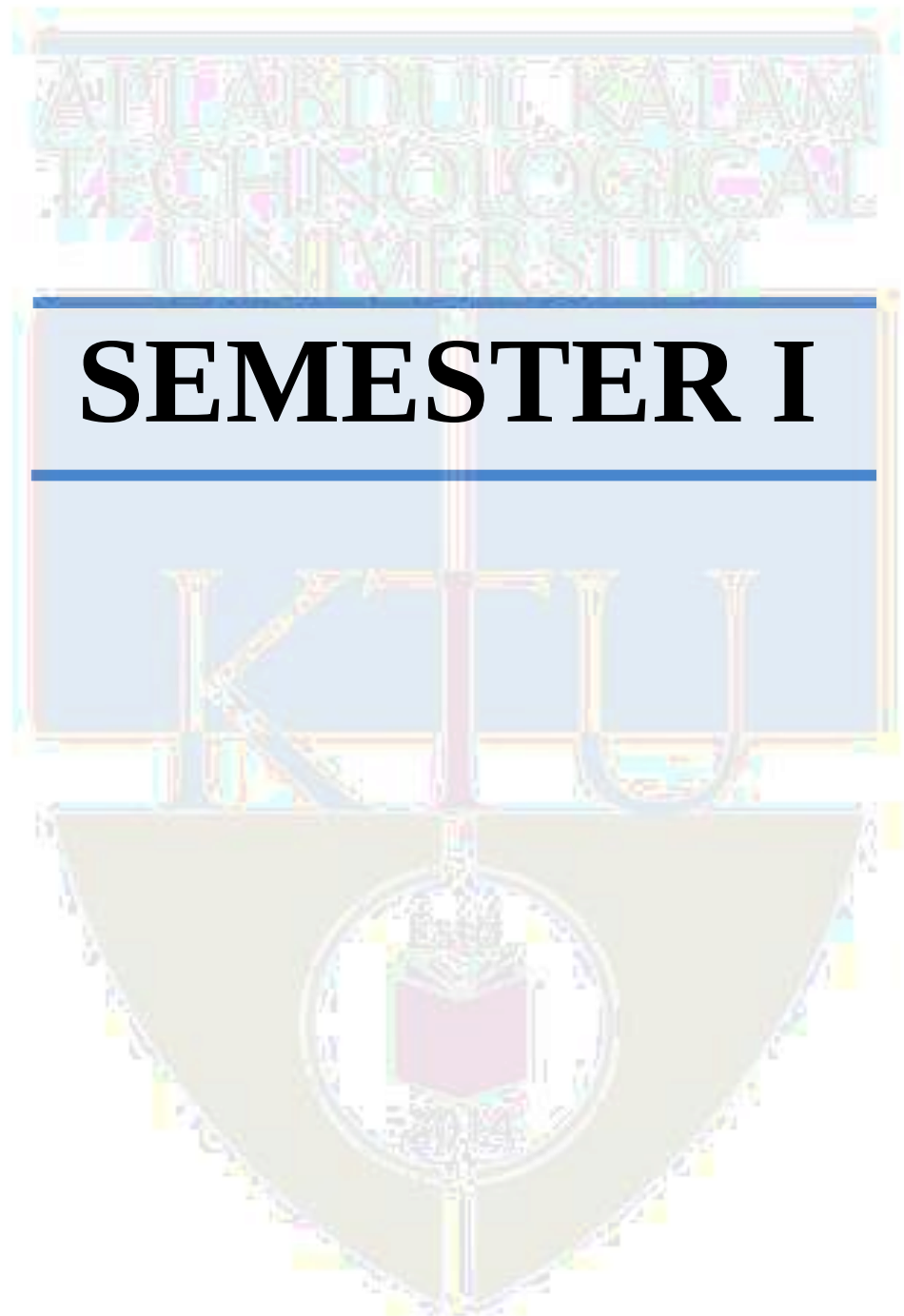




Discipline: INFORMATION TECHNOLOGY
Stream : IT1

INFORMATION TECHNOLOGY-IT1

CODE	COURSE NAME	CATEGORY	L	T	P	CREDIT
221TIT100	MATHEMATICAL FOUNDATION FOR NETWORKS AND SECURITY	DISCIPLINE CORE	3	0	0	3

Preamble: The students are expected to learn fundamentals of mathematical concepts used to design and analyse network and security protocols. These concepts will help them to develop network and security models and analyse them before being used in various applications.

Course Outcomes: After the completion of the course the student will be able to

- CO 1** Apply the fundamental mathematical concepts in Symmetric-Key cryptography (Level : Apply)
- CO 2** Apply the fundamental mathematical concepts in Asymmetric- Key cryptography (Level : Apply)
- CO 3** Apply the fundamental concepts in graph theory (Level : Apply)
- CO 4** Identify an appropriate probability distribution and solve problems related to a given discrete or continuous random variable and use its properties to calculate probabilities. (Level : Apply)
- CO 5** Solve statistics such as the mean and variance of common probability distributions (Level : Apply)
- CO 6** Solve queueing theory problems along with the stochastic processes techniques useful for modelling queueing systems and queueing networks (Level : Apply)

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	√	√	√	√	√	√	
CO 2	√	√	√	√	√	√	
CO 3	√	√		√	√	√	
CO 4	√	√		√	√	√	
CO 5	√	√	√	√	√	√	
CO 6	√	√	√	√	√	√	

Assessment Pattern

Bloom's Category	End Semester Examination
Apply	100%
Analyse	0%
Evaluate	0%
Create	0%

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation (CIE) Pattern:

Micro project/Course based project: 20 Marks

Course-based task/Seminar/Quiz: 10 Marks

Test paper (one only): 10 Marks

The project shall be done individually. Group projects are not permitted. The test paper shall include a minimum of 80% of the syllabus.

End Semester Examination Pattern: The End-Semester Examination shall be conducted by the University. There shall be two parts; Part A and Part B. Part A shall contain five (numerical) questions (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation, and understanding of the students), with one question from each module, having five marks for each question. Students shall answer all questions. Part B shall contain seven questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem-solving, and quantitative evaluation), with at least one question from each module, of which, the student shall answer any five. Each question shall carry seven marks

MODEL QUESTION PAPER FOR END-SEMESTER EXAMINATION

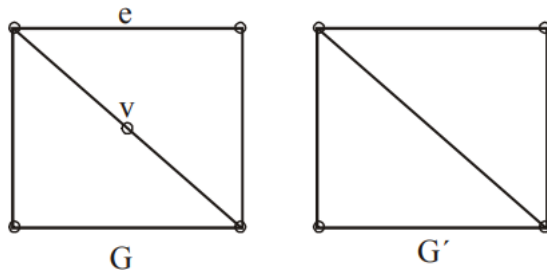
Time: 2½ Hours, Maximum marks: 60

PART A (Answer all questions. Each question carries 5 marks)

1. a. Find the multiplicative inverse of 11 in Z_{26} ? (2 Marks)
 b. In $GF(2^4)$, find the inverse of (x^2+1) modulo (x^4+x+1) (3 Marks)
2. Does the number 561 pass the Miller-Rabin test? (5 Marks)
3. In a given connected weighted graph G , suppose there exists an edge e_s whose weight is smaller than that of any other in G . Prove that every shortest spanning tree in G must contain e_s . (5 Marks)
4. Let X and Y be independent random variables, each having an exponential distribution with parameter λ .
 a. Find the density function of $Z = X + Y$. (5 marks)
5. Consider a small post office in a village where on the average 70 customers arrive according to a Poisson process during a day. Let us assume that the service times are exponentially distributed with rate 10 clients per hour and the office operates 10 hours daily. Find the mean queue length, and the probability that the number of waiting customer is greater than 2. What is the mean waiting time and the probability that the waiting time is greater than 20 minutes? (5 Marks)

PART B (Answer any five questions. Each question carries 7 marks)

6. a. Find the particular and general solutions to the equation $21x + 14y = 35$ (4 Marks)
 b. Using the Euclidean algorithm, find the Greatest Common Divisor of the following pairs of integers:
 a. 88 and 220
 b. 401 and 700 (3 Marks)
7. a. Use the trial division algorithm to find the factors of 1233 (3 Marks)
 b. Find x in $4 \equiv 3^x \pmod{7}$ (4 Marks)
8. Calculate $17^{22} \pmod{21}$ using square-and-multiply algorithm (7 marks)
9. a. In a graph G let p_1 and p_2 be two different paths between two given vertices. Prove that $p_1 \oplus p_2$ is a circuit or a set of circuits in G . (5 Marks)
 b. Is the graph G' a subgraph of G ?



(2 Marks)

10. Let X be the continuous random variable with density function f defined by

$$f(x) = \begin{cases} 0 & \text{for } x < 1 \\ 1 & \text{for } x = 1 \\ \frac{1}{x^2} & \text{for } x > 1 \end{cases}$$

Analyse the existence of higher order moments.

(7 Marks)

11. Let X and Y be independent random variables, each having an exponential distribution with parameter

a. Find the density function.

(7 Marks)

12. 14. Traffic to a message switching centre for one of the outgoing communication lines arrives in a random pattern at an average rate of 240 messages per minute. The line has a transmission rate of 800 characters per second. The message length distribution (including control characters) is approximately exponential with an average length of 176 characters. Calculate the principal statistical measures of system performance assuming that a very large number of message buffers are provided. What is the probability that 10 or more messages are waiting to be transmitted? (7 marks)

INFORMATION TECHNOLOGY-IT1 SYLLABUS

Module I: Mathematics of Cryptography I (6 Hours)

Greatest common divisor- GCD, Euclidean algorithm Fundamental theorem of arithmetic, Linear Diophantine Equations, Integer arithmetic, Modular arithmetic, Matrices, Linear congruences, Algebraic structures, $GF(2^n)$ Fields

Module II: Mathematics of Cryptography II (10 Hours)

Prime numbers, Fermat's and Euler's Theorem, Generating Primes, Primality Testing: Deterministic algorithms, Probabilistic algorithms, Quadratic Congruences- Quadratic Residues, The Legendre Symbol, Quadratic Reciprocity, The Jacobi Symbol, Factorization: Factorization methods- Trial division method, Fermat method, Pollard $p-1$ method, Pollard Rho method, Exponentiation and Logarithm: Square-and-multiply algorithm, Discrete Logarithm, Solution to modular logarithm using Discrete Logs

Module III: Graph Theory and Applications (8 Hours)

What is a Graph? Application of Graphs, Finite and Infinite Graphs, Incidence and Degree, Isolated Vertex, Pendant Vertex, and Null Graph. PATHS AND CIRCUITS – Isomorphism, Subgraphs, Walks, Paths, and Circuits, Connected Graphs, Disconnected Graphs, and Components, Euler Graphs, Operations on Graphs, More on Euler Graphs, Hamiltonian Paths and Circuits, The Travelling Salesman Problem, Trees And Fundamental Circuits – Trees, Properties of Trees, Pendant Vertices in a Tree, Distance and Centers in a Tree, Rooted and Binary Trees, On Counting Trees, Spanning Trees, Fundamental Circuits, Finding All Spanning Trees of a Graph, Spanning Trees in a Weighted Graph.

Module IV: Random Variables and Distributions (8 Hours)

The Concept of a Random Variable – Introduction, Distribution and Density Functions, Specific Random Variables, Conditional Distributions, Asymptotic Approximations for Binomial Random Variable, Functions of One Random Variable - The Random Variable $g(x)$, The Distribution of $g(x)$, Mean and Variance, Moments, Characteristic Functions, Two Random Variables - Bivariate Distributions, One Function of Two Random Variables, Two Functions of Two Random Variables, Joint Moments, Joint Characteristic Functions, Conditional Distributions, Conditional Expected Values.

Module V: Queuing Models (8 Hours)

General concepts, Arrival pattern, service pattern, Queue Disciplines, FIFO Queuing systems, $M/M/1$, $M/M/c$, $M/M/\infty$, $M/G/1$, $M/M/m/m$ Markov models, non-Markov models, Network queues, Burke's theorem, Jackson's theorem.

Reference Books

1. Behrouz A. Fourouzan, Debdeep Mukhopadhyay “ Cryptography and Network Security”, Second Edition, Tata McGrawHill Education
2. Thomas Koshy, “Elementary Number Theory with Applications” Second Edition, Elsevier Science, 2007
3. Judith L. Gersting, “Mathematical Structures for Computer Science - Discrete Mathematics and Its Applications”, 7e, W. H. Freeman and Company, 2014
4. Ralph P. Grimaldi, “Discrete and Combinatorial Mathematics”, 5th edition, Pearson, 2018
5. Arnold O. Allen, “Probability, Statistics, and Queuing Theory with Computer Science Applications”, Elsevier Inc, 1990
6. eth H. Rosen, “Elementary Number Theory and its Applications” Addison-Wesley Publishing Company, 1986

INFORMATION TECHNOLOGY-IT1

7. Nitish Mukhopadhyay, "Probability and Statistical Inference", Library of Congress Cataloging-in-Publication Data, Marcel Dekker, Inc. 2000
8. Gary Haggard, John Schlipf, Sue Whitesides, "Discrete Mathematics for Computer Science", Thomson Brooks/Cole, 2006.
9. K.S. Trivedi, "Probability and Statistics with Reliability, Queueing and Computer Science Applications", John Wiley and Sons, 2nd edition, 2002.
10. J.F. Shortle, J.M. Thompson, D. Gross and C.M. Harris, Fundamentals of Queueing Theory, 5th Edition, Wiley, 2018

COURSE PLAN

No	Topic	No. of Lectures
1	Mathematics of Cryptography I	
1.1	Greatest common divisor- GCD, Euclidean algorithm Fundamental theorem of arithmetic, Linear Diophantine Equations	1 Hour
1.2	Linear Diophantine Equations	1 Hour
1.3	Integer arithmetic, Modular arithmetic, Matrices, Linear congruences	1 Hour
1.4	Matrices, Linear congruences	1 Hour
1.5	Algebraic structures	1 Hour
1.6	GF(2 ⁿ) Fields	1 Hour
2	Mathematics of Cryptography II	
2.1	Prime numbers, Fermat's Theorem	1 Hour
2.2	Euler's Theorem, Generating Primes	1 Hour
2.3	Primality Testing: Deterministic algorithms	1 Hour
2.4	Probabilistic algorithms	1 Hour
2.5	Quadratic Congruences- Quadratic Residues, The Legendre Symbol	1 Hour
2.6	Quadratic Reciprocity, The Jacobi Symbol	1 Hour
2.7	Factorization: Factorization methods- Trial division method	1 Hour
2.8	Fermat method, Pollard p-1 method, Pollard Rho method	1 Hour
2.9	Exponentiation and Logarithm: Square-and-multiply algorithm	1 Hour
2.10	Discrete Logarithm, Solution to modular logarithm using Discrete Logs	1 Hour
3	Graph Theory and Applications	
3.1	Introduction - What is a Graph? Application of Graphs, Finite and Infinite Graphs, Incidence and Degree, Isolated Vertex, Pendant Vertex, and Null Graph.	1 Hour
3.2	Paths and Circuits – Isomorphism, Subgraphs, A Puzzle With Multicolored Cubes, Walks, Paths, and Circuits	1 Hour
3.3	Connected Graphs, Disconnected Graphs, and Components, Euler Graphs, Operations On Graphs, More on Euler Graphs	1 Hour
3.4	Hamiltonian Paths and Circuits, The Traveling Salesman Problem	1 Hour
3.5	Trees And Fundamental Circuits – Trees, Properties of Trees, Pendant Vertices in a Tree	1 Hour

3.6	Distance and Centers in a Tree, Rooted and Binary Trees, On Counting Trees	1 Hour
3.7	Spanning Trees, Fundamental Circuits, Finding All Spanning Trees of a Graph, Spanning Trees in a Weighted Graph	2 Hours
4	Random Variables and Distributions	
4.1	The Concept of a Random Variable – Introduction, Distribution and Density Functions, Specific Random Variables	1 Hour
4.2	Conditional Distributions, Asymptotic Approximations for Binomial Random Variable	1 Hour
4.3	Functions of One Random Variable - The Random Variable $g(x)$	1 Hour
4.4	The Distribution of $g(x)$	1 Hour
4.5	Mean and Variance, Moments, Characteristic Functions	1 Hour
4.6	Two Random Variables - Bivariate Distributions	1 Hour
4.7	One Function of Two Random Variables, Two Functions of Two Random Variables	1 Hour
4.8	Joint Moments, Joint Characteristic Functions, Conditional Distributions, Conditional Expected Values	1 Hour
5	Queuing Models	
5.1	General concepts, Arrival pattern, service pattern	1 Hour
5.2	Queue Disciplines, FIFO Queuing systems	1 Hour
5.3	M/M/1, M/M/c	1 Hour
5.4	M/M/ ∞ , M/G/1	1 Hour
5.5	M/M/m/m and other Markov models,	1 Hour
5.6	Non-Markov models	1 Hour
5.7	Network queues.	1 Hour
5.8	Burke's theorem, Jackson's theorem	1 Hour

CODE	COURSE NAME	CATEGORY	L	T	P	CREDIT
221TIT001	ADVANCED COMPUTER NETWORKS	INFORMATION TECHNOLOGY-IT1 PROGRAM CORE 1	3	0	0	3

Preamble: The students are expected to understand and analyze the protocols used in various layers of the network stack.

Course Outcomes: After the completion of the course the student will be able to

CO 1	Explain the working of application layer protocols.(Level: Understand)
CO 2	Analyze the transport protocols used on the internet. (Level: Analyze)
CO 3	Describe IPv6, multicast, and flow-based routing protocols. (Level: Understand)
CO 4	Analyze the protocols used in the link layer.(Level: Analyze)
CO 5	Apply error detection and error correcting codes.(Level: Apply)
CO 6	Summarize network management and transport and network services for multimedia applications.(Level: Understand)

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

PO1: An ability to independently carry out research/investigation and development work in engineering and allied streams

PO2: An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

PO3: An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor's program

PO4: An ability to apply stream knowledge to design or develop solutions for real-world problems by following the standards

PO5: An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tool to model, analyse and solve practical engineering problems.

PO6: An ability to engage in life-long learning for the design and development related to the stream-related problems taking into consideration sustainability, societal, ethical and environmental aspects

PO7: An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1		☑	☑	☑	☑		
CO 2	☑	☑	☑	☑	☑	☑	☑
CO 3		☑	☑	☑	☑		
CO 4	☑	☑	☑	☑	☑	☑	☑
CO 5	☑	☑	☑	☑	☑	☑	☑
CO 6		☑	☑	☑	☑		

Assessment Pattern

Bloom's Category	End Semester Examination
Apply	80
Analyse	20
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation Pattern:

Micro project/Course based project: 20 Marks

Course-based task/Seminar/Quiz: 10 Marks

Test paper (one only): 10 Marks

The project shall be done individually. Group projects are not permitted. The test paper shall include a minimum of 80% of the syllabus.

End Semester Examination Pattern:

There shall be two parts; Part A and Part B. Part A shall contain five (numerical) questions (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation, and understanding of the students), with one question from each module, having five marks for each question. Students shall answer all questions. Part B shall contain seven questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer

questions relating to theoretical/practical knowledge, derivations, problem-solving, and quantitative evaluation), with at least one question from each module, of which, the student shall answer any five. Each question shall carry seven marks.

Course Level Assessment Questions

Course Outcome 1 (CO1):

1. What are the components in a content distribution network?
2. When an HTTP server initiates a close() at its end of a connection, it must then wait in TCP state FIN WAIT 2 for the client to close the other end. What mechanism within the TCP protocol could help an HTTP server deal with non-cooperative or poorly implemented clients that don't close from their end?
3. Explain the message structure of SOAP

Course Outcome 2 (CO2)

1. Assuming that loss occurs during the first slow-start and that the file transfer is long enough, obtain the duration of the second slow-start and the number of packets transmitted during it.
2. Derive the packet throughput if the TCP receiver returns a cumulative ACK for every b^{th} packet.
3. Explain the features of datagram congestion control protocol

Course Outcome 3 (CO3):

1. What is MPLS?
2. Explain the format of route table entry in a RIPng message
3. How do routers that use reverse path multicasting learn about the location of group members?

Course Outcome 4 (CO4):

1. Derive the expression to compute the throughput of unslotted ALOHA
2. For a time-slotted network, where the slots are small compared with the packet lengths, consider the following variation of slotted Aloha. When a node has a packet to transmit, it begins transmission at the beginning of a slot. If there was no collision in the first slot, then it has captured the next $X - 1$ slots, where X is the packet transmission time and all other stations will defer. If there was a collision in the first slot, then the node makes a randomized retransmission attempt (as in slotted Aloha) and continues to do so until it succeeds. All nodes will know of the end of transmission of this packet when they sense the channel idle again. If slotted Aloha with an adaptive protocol were to yield a throughput of η when the packet length is equal to the slot length, what would be the throughput of this network? Note the similarities with the CSMA/CD protocol.

3. Show that for unslotted CSMA/CD, the maximum interval of time over which a transmitting node can hear a collision is 2β , where β is the propagation delay.

Course Outcome 5 (CO5):

1. Show that the final parity check in a horizontal and vertical parity check code, if taken as the modulo 2 sum of all the data bits, is equal to the modulo 2 sum of the horizontal parity checks and also equal to the modulo 2 sum of the vertical parity checks.

2. Let $g(D) = D^4 + D^2 + D + 1$, and let $s(D) = D^3 + D + 1$. Find the remainder when $D^4 s(D)$ is divided by $g(D)$, using modulo 2 arithmetic.

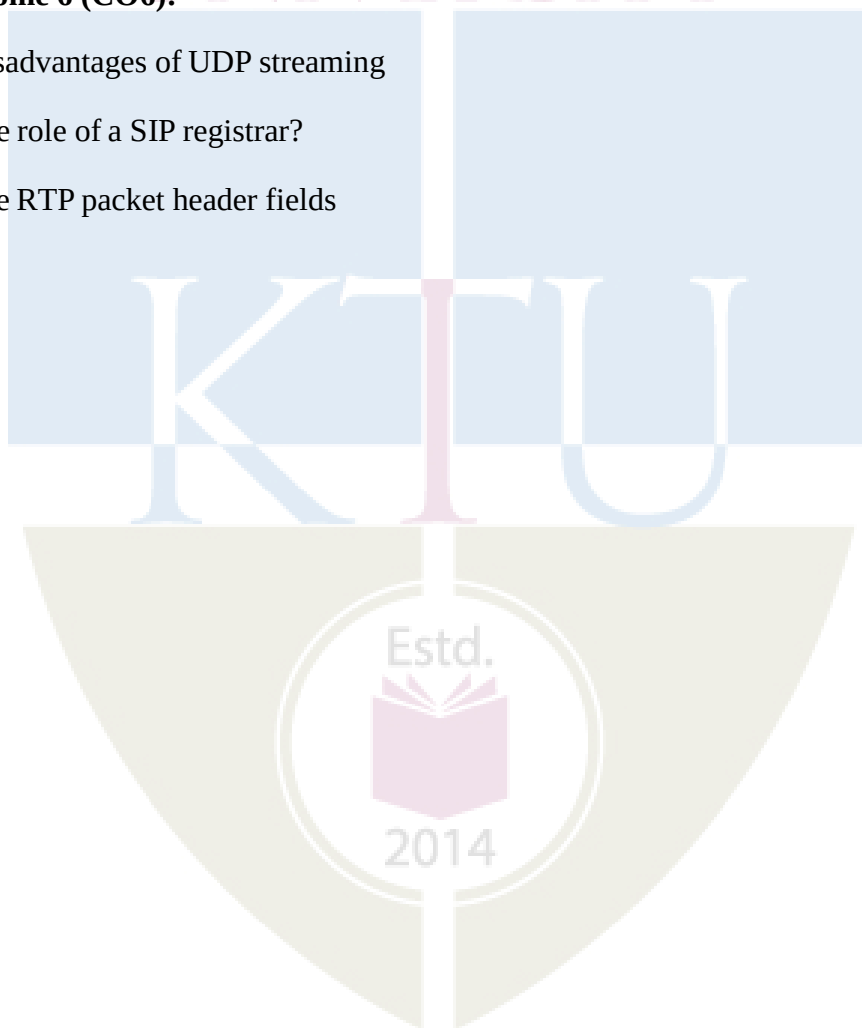
3. Explain how error correction is done by hamming codes

Course Outcome 6 (CO6):

1. List the disadvantages of UDP streaming

2. What is the role of a SIP registrar?

3. Explain the RTP packet header fields



Model Question Paper

INFORMATION TECHNOLOGY-IT1

A P J ABDUL KALAM TECHNOLOGICAL UNIVERSITY

M.TECH DEGREE EXAMINATION

.....SEMESTER

Branch:

Time: 2½ Hours, Maximum marks: 60

PART A (Answer all questions. Each question carries 5 marks)

1. What is the difference between persistent HTTP with pipelining and persistent HTTP without pipelining? Which of the two is used by HTTP/1.1?
2. How does TCP prevent a fast transmitter from flooding a slow receiver that has limited storage?
3. Explain the features of multicast routing
4. Compare pure ALOHA and slotted ALOHA
5. List the audio and video file formats supported by RTP

PART B (Answer any five questions. Each question carries 7 marks)

6. Explain the working of DNS
7. Explain DCCP in detail
8. What are the measures taken by TCP to control congestion?
9. Prove that the maximum throughput for pure ALOHA is $\frac{1}{2e}$
10. Explain the encoding and decoding of low-density parity codes
11. What is MPLS? What are the advantages of MPLS routing over conventional routing?
12. Explain the standard SNMP model in detail.

Syllabus

INFORMATION TECHNOLOGY-IT1

Module I: Introduction (7 Hours)

Computer Networks and Internet – Network edge, network core, Delay, loss, and throughput in packet-switched networks, protocol layers, and their service model. Application Layer – Principles of network applications – network application architectures, process communication, Web and HTTP, FTP, Email – SMTP, MIME, POP3, IMAP, DNS, Custom application protocols (WSDL, SOAP), Generic application protocol (REST), Overlay networks – routing overlays, Peer-to-peer applications(P2P file distribution, BitTorrent, distributed hash tables), content distribution networks

Module II: Transport Layer (8 Hours)

Internet's adaptive window protocol TCP: slow start and congestion avoidance, receiver buffer and cumulative ACKs, packet loss recovery and congestion control, single connection: analysis with buffer overflow (analysis of slow start phase, short transfer throughput, slow start: evolution after buffer overflow, analysis of the congestion avoidance phase, congestion avoidance: evolution after buffer overflow, congestion avoidance: timeout based recovery), Stochastic model for a wide area TCP connection, TCP with Explicit Feedback (RED and ECN), Stream control transmission protocol, datagram congestion control

Module III: Network Layer (10 Hours)

Introduction, datagram networks and virtual circuits, Internet Protocol: Forwarding, IPv4 and IPv6 addressing, Routing algorithms, Routing in the Internet – RIPng, Changes in OSPF to support IPv6, BGP multiprotocol extensions for IPv6, Multicast forwarding and routing information, basic multicast forwarding paradigms, Consequences of TRPF, Multicast trees, reverse path multicasting, Multicast routing protocols: distance vector multicast routing protocol and tunneling, Core based trees, protocol independent multicast, Traffic Engineering – ECMP, Modifying path costs, routing IP flows, service-based routing, dynamic traffic engineering, routing extensions for traffic engineering, MPLS

Module IV: Link Layer (8 Hours)

Introduction to the link layer, Error detection –single parity checks, horizontal and vertical parity checks, parity check codes, cyclic redundancy checks, Error correction techniques – Hamming codes, binary convolutional codes, Reed Solomon codes, Low-density parity-check codes, Multi-access communication: Stability, maximum throughput, and delay analysis using Markov chain - slotted multi-access and the ALOHA system, unslotted ALOHA, CSMA slotted ALOHA, CSMA unslotted ALOHA. Switched LANs, Link virtualization, data center networking

Module V: Multimedia Networking & Network Management (7 Hours)

Multimedia networking applications – properties of video, properties of audio, types of multimedia network applications, Streaming stored video – UDP streaming, HTTP streaming, prefetching video, client application buffer and TCP buffers, analysis of video streaming,

early termination and repositioning the video, Adaptive streaming, and DASH, Case studies: Netflix, YouTube, Voice over IP, Protocols for real-time conversational applications: RTP, SIP, Network support for multimedia applications, Infrastructure for network management, SNMP

Reference Books

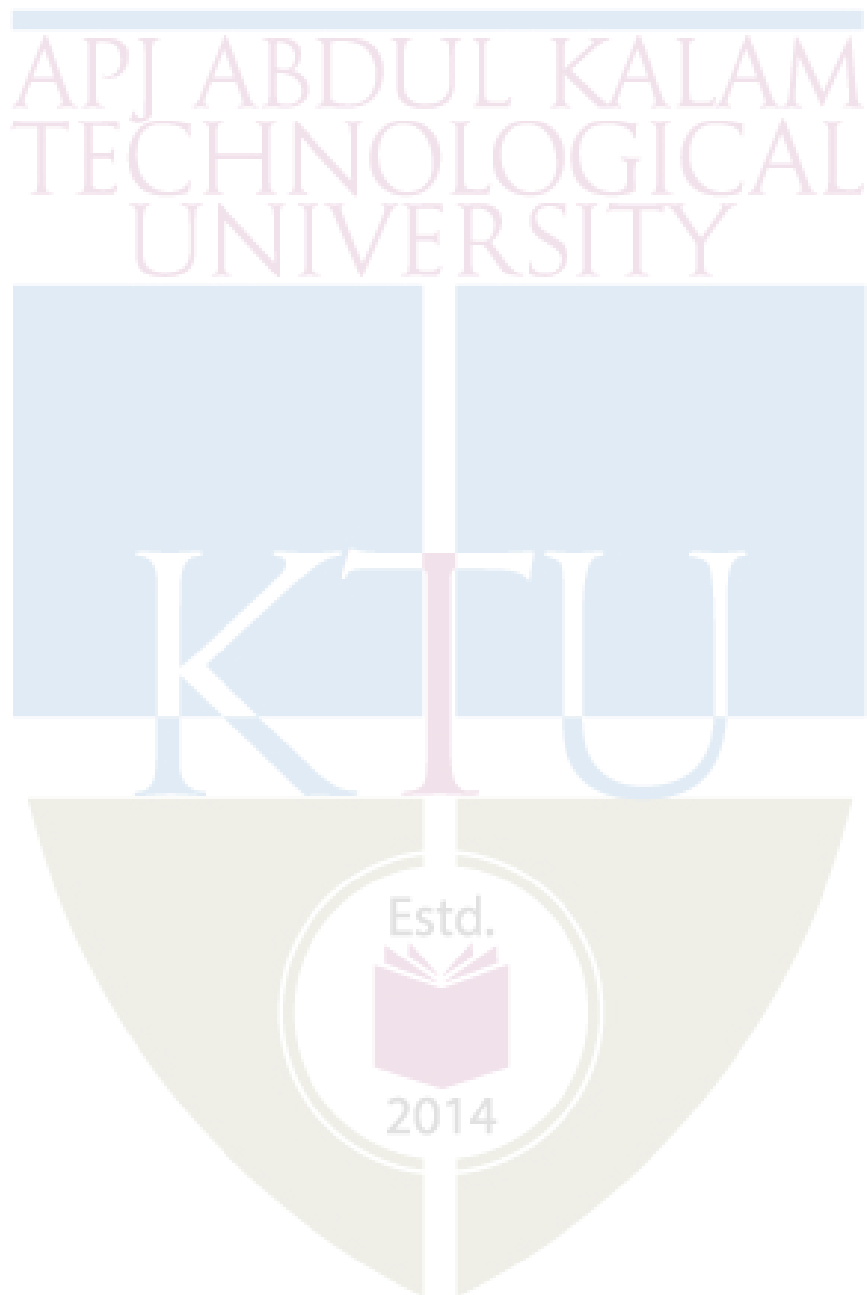
1. James F. Kurose, Keith W. Ross, “Computer networking: a top-down approach”, 6th ed., Pearson.
2. Larry Peterson and Bruce S Davis “Computer Networks :A System Approach” 5 th edition , Elsevier -2014
3. Dimitri P. Bertsekas, Robert G. Gallager, “Data Networks”, 2nd Edition, Prentice Hall
4. Anurag Kumar, D. Manjunath, Joy Kuri, “Communication Networking : An Analytical Approach”, Morgan Kaufman Publishers
5. Oliver C Ibe, “Fundamentals of data communication networks”, 1st edition, Wiley
6. Douglas E Comer, “Internetworking with TCP/IP, Principles, Protocols and Architecture” 6th Edition, PHI – 2014
7. Ranjan Bose, “Information theory, coding, and cryptography”, 3rd Edition, Mc Graw Hill
8. W. R. Stevens, TCP/IP Illustrated, Volume 1: The Protocols, Addison Wesley, 1994.

Course Plan

No	Topic	No. of Lectures (40) Hours
1	Introduction 7 Hours	
1.1	Computer Networks and Internet – network edge, network core, delay, loss, and throughput in packet switched networks, protocol layers and their service model.	1
1.2	Application Layer – principles of network applications – network application architectures, process communication, web and HTTP, FTP	2
1.3	Email – SMTP, MIME, POP3, IMAP, DNS, custom application protocols (WSDL, SOAP), generic application protocol (REST)	2
1.4	Overlay networks – routing overlays, peer-to-peer applications (P2P file distribution, BitTorrent, distributed hash tables),	1
1.5	Content distribution networks	1
2	Transport Layer 8 Hours	
2.1	Internet’s adaptive window protocol TCP: slow start and congestion avoidance, receiver buffer and cumulative ACKs, packet loss recovery and congestion control	2
2.2	Single connection: analysis with buffer overflow (analysis of	2

	slow start phase, short transfer throughput, slow start: evolution after buffer overflow, analysis of the congestion avoidance phase, congestion avoidance: evolution after buffer overflow, congestion avoidance: timeout-based recovery)	INFORMATION TECHNOLOGY-IT
2.3	Stochastic model for a wide area TCP connection	1
2.4	TCP with Explicit Feedback (RED and ECN)	1
2.5	Stream control transmission protocol, datagram congestion control	2
3	Network Layer 10 Hours	
3.1	Introduction, datagram networks and virtual circuits	1
3.2	Internet Protocol: Forwarding, IPv4 and IPv6 addressing	2
3.3	Routing algorithms, Routing in the Internet – RIPng, Changes in OSPF to support IPv6, BGP multiprotocol extensions for IPv6	2
3.4	Multicast forwarding and routing information, basic multicast forwarding paradigms, Consequences of TRPF, Multicast trees, reverse path multicasting	1
3.5	Multicast routing protocols: distance vector multicast routing protocol and tunnelling, Core based trees, protocol-independent multicast	1
3.6	Traffic Engineering – ECMP, Modifying path costs, routing IP flows, service-based routing, dynamic traffic engineering, routing extensions for traffic engineering	2
3.7	MPLS	1
4	Link Layer 8 Hours	
4.1	Introduction to the link layer, Error detection –single parity checks, horizontal and vertical parity checks, parity check codes, cyclic redundancy checks	1
4.2	Error correction techniques – Hamming codes, binary convolutional codes, Reed Solomon codes, Low density parity check codes	2
4.3	Multi-access communication: Stability, maximum throughput, and delay analysis using Markov chain - slotted multi-access and the ALOHA system, unslotted ALOHA, CSMA slotted ALOHA, CSMA unslotted ALOHA.	2
4.4	Switched LANs	1
4.5	Link virtualization, data center networking	2
5	Multimedia Networking & Network Management 7 Hours	
5.1	Multimedia networking applications – properties of video, properties of audio, types of multimedia network applications, Streaming stored video – UDP streaming	1
5.2	HTTP streaming, prefetching video, client application buffer and TCP buffers, analysis of video streaming, early termination and repositioning the video, Adaptive streaming and DASH	2
5.3	Case studies: Netflix, You Tube	1

5.4	Voice over IP, Protocols for real time conversational applications : RTP, SIP	INFORMATION TECHNOLOGY-IT
5.5	Network support for multimedia applications	1
5.6	Infrastructure for network management, SNMP	1



CODE:	COURSE NAME:	CATEGORY	L	T	P	CREDIT
221TIT002	TOPICS IN SECURITY	PROGRAM CORE 2	3	0	0	3

Preamble: This course is intended for students to have a deeper knowledge of selected topics in network security, system security, and cryptography. The first two modules of the course cover the Symmetric-key and Asymmetric-key cryptosystems in depth. The third module deals with Cryptographic schemes that provide Integrity and Authentication. The fourth module discusses the practical applications of security schemes in computer networks and network applications. The last module deals with Operating system security and database system security.

Course Outcomes:

After the completion of the course, the student will be able to:

CO 1	Examine the mathematical principles behind modern symmetric-key cryptosystems and compare various symmetric-key distribution methods. (Level: Analyze)
CO 2	Examine the mathematical principles behind asymmetric-key cryptosystems and cryptographic commitment schemes, and compare various asymmetric-key distribution methods. (Level: Analyze)
CO 3	Contrast modern hash functions, MACs, and Digital Signature Schemes and their security strength, and apply them in appropriate contexts. (Level: Apply)
CO 4	Describe the security schemes employable in modern networks and network applications and apply them in appropriate contexts. (Level: Apply)
CO 5	Examine the security requirements of, and the security threats to Operating Systems and Database Systems, and identify appropriate access control policies. (Level: Analyze)

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	√		√	√	√		

CO 2	√		√	√	√		
CO 3	√		√	√	√		
CO 4	√	√	√	√	√		√
CO 5	√	√	√	√	√		√

Assessment Pattern

Bloom's Category	End Semester Examination
Apply	80%
Analyze	20%
Evaluate	0%
Create	0%

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

The evaluation shall only be by employing application-based, analysis-based, or design-based questions(for both internal and end-semester examinations)

Continuous Internal Evaluation (CIE) Pattern:

Micro project/Course based project: 20 Marks

Course-based task/Seminar/Quiz: 10 Marks

Test paper (one only): 10 Marks

The project shall be done individually. Group projects are not permitted. The test paper shall include a minimum of 80% of the syllabus.

End Semester Examination (ESE) Pattern:

The End-Semester Examination shall be conducted by the University. There shall be two parts; Part A and Part B. Part A shall contain five(numerical) questions (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation, and understanding of the students), with one question from each module, having five marks for each question. Students shall answer all questions. Part B shall contain seven questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem-solving, and quantitative evaluation), with at least one question from each module, of which, the student shall answer any five. Each question shall carry seven marks.

Model Question Paper

A P J ABDUL KALAM TECHNOLOGICAL UNIVERSITY

M. TECH DEGREE EXAMINATION

.....SEMESTER

Branch:

MODEL QUESTION PAPER FOR END-SEMESTER EXAMINATION

Time: 2½ Hours, Maximum marks: 60

PART A (Answer all questions. Five marks each)

1. Suppose that two parties Alice and Bob wish to set up a common secret key between themselves using the Diffie-Hellman key exchange technique. They agree on 7 as the modulus and 3 as the primitive root. Alice chooses 2 and Bob chooses 5 as their respective secrets. Find the secret key that will be established between Alice and Bob.
2. Prove that Pedersen Commitment Scheme satisfies *binding* property and *hiding* property.
3. Show how the AES block cipher can be used to build a 128-bit hash function. Comment on its security

4. Describe any five of the major cloud-specific security threats, and countermeasures for them.
5. Compare the security provided by Fence registers and Base/Bound registers in Operating Systems.

PART B

(Answer any five questions. Seven marks each)

6. Let the following state matrix be input to the *MixColumn* transformation of AES (Values are in hexadecimal form). Find the resultant state matrix.

63	EB	9F	A0
2F	93	92	C0
AF	C7	AB	30
A2	20	CB	2B

7. Suppose Alice and Bob use El-Gamal Cryptosystem with a common prime $q = 71$ and a primitive root $\alpha = 7$.
 - (a) If Bob has public key $Y_B = 3$ and Alice chose the random integer $k = 2$, what is the ciphertext corresponding to the plaintext $M = 30$?
 - (b) If Alice now chooses a different value of k so that the encoding of $M = 30$ is $C = (59, C_2)$, what is the integer C_2 ?
8. Describe the requirements of an ideal hash function. Consider the following hash function. Messages are in the form of a sequence of numbers in $\mathbf{Z}_n$ ($M = (a_1, a_2, \dots, a_t)$). The hash value h is calculated as $h(M) = \sum_{i=1}^t a_i$ for some predefined value n . What requirements of an ideal hash function are satisfied by h ? Justify your answer.
9. Consider the following threats to Web security and describe how each one is countered by a particular feature of TLS.
 - (a) Brute-Force Cryptanalysis attack: An exhaustive search of the key space for a conventional encryption algorithm.
 - (b) Replay attack: TLS handshake messages replayed
 - (c) Man-in-the-Middle attack: An attacker interposes during the key exchange, acting as the client to the server and as the server to the client.
 - (d) Password sniffing: Passwords in HTTP or other application traffic are eavesdropped.
 - (e) IP spoofing: Uses forged IP addresses to fool a host into accepting bogus data.
 - (f) IP hijacking: An active, authenticated connection between two hosts is disrupted and the attacker takes the place of one of the hosts.
 - (g) SYN flooding: An attacker sends TCP SYN messages to request a connection but does not respond to the final message to establish the connection fully. The attacked TCP module typically leaves the “half-open connection” around for a few minutes. Repeated TCP SYN can clog the TCP module.

10. Consider two Operating Systems OS1 and OS2. OS1 has all its security features implemented as part of the kernel. OS2 has all its security features implemented in a 'Security Kernel' (which is isolated from the rest of the kernel). Compare and contrast the two Operating System implementations with respect to security.
11. Explain the steps in the signing process and verification process for the El-Gamal digital signature scheme. For implementing the El-Gamal digital signature scheme between them, Alice and Bob select a common prime $q = 19$ and a primitive root $\alpha = 10$. Compute Alice's key pairs. Suppose Alice wants to sign a message with the hash value $m = 14$. Explain how the signing and verification steps are done in this case at Alice's side and Bob's side respectively.
12. Alice and Bob use an Elliptic Curve Cryptosystem with parameters $E_{11}(1, 6)$ and $G = (2, 7)$. Bob's private key is $n_B = 7$. Find Bob's public key.

Syllabus

Module I: Introduction, Symmetric-Key Cryptography: (9 Hours)

Security Goals and Attacks threatening them, Symmetric-key Cryptography v/s Asymmetric-key (Public-key) Cryptography, Block cipher v/s Stream Cipher, Cryptanalysis, AES Mathematical Background, AES, XTS-AES mode of operation, Format Preserving Encryption, Symmetric-key Management and Distribution - KDC, Distribution using Asymmetric-key cryptosystems, Diffie-Hellman Key Exchange Protocol – Anonymous and Ephemeral (with Mathematical Background).

Module II: Public-Key Cryptography: (8 Hours)

Factorization Problem, Discrete Logarithm Problem, RSA Cryptosystem, El-Gamal Cryptosystem, Elliptic Curve Arithmetic, Elliptic Curve Cryptosystem, Cryptographic Commitment - Pedersen Commitment, Distribution of Public Keys - Public-key Certificates, PKI.

Module III: Integrity and Authentication: (8 Hours)

Cryptographic hash functions, Secure Hash Function - SHA3, Message Authentication Codes – HMAC, CMAC, Digital Signatures - El-Gamal Digital Signature, Schnorr Digital Signature, DSA, RSA-PSS.

Module IV: Network Security and Internet Security: (8 Hours)

IP Security – ESP, Security Association, TLS overview, HTTPS, SSH, Network Access Control – Elements, Enforcement methods, EAP, Cloud Security – Elements, Risks, and Countermeasures, Data protection, Cloud Security as a Service.

Module V: Operating System Security and Database System Security: (7 Hours)

Security features in OS, Security in the design of OS, Rootkit, Access Control - Role-based and Procedure oriented, Security requirements of Databases, Reliability and Integrity, Database Disclosure.

Reference Books

1. Cryptography and Network Security, Principles and Practice, William Stallings, 7th Edition
2. Cryptography and Network Security, Forouzan and Mukhopadhyay, 3rd Edition
3. Security in Computing, Pfleeger, Pfleeger and Margulies, 5th Edition
4. Introduction to Modern Cryptography, Katz and Lindell, 3rd Edition

Course Plan

No	Topic	No. of Lectures (40 Hours)
1	Introduction, Symmetric-Key Cryptography(9 Hours)	
1.1	Security Goals and Attacks threatening them, Symmetric-key Cryptography v/s Asymmetric-key (Public-key) Cryptography, Block cipher v/s Stream Cipher, Cryptanalysis	2
1.2	AES Mathematical Background	1
1.3	AES	2
1.4	XTS-AES mode of operation	1
1.5	Format Preserving Encryption	1
1.6	Symmetric-key Management and Distribution, KDC, Distribution using Asymmetric-key cryptosystems.	1

1.7	Diffie-Hellman Key Exchange Protocol – Anonymous and Ephemeral (with Mathematical Background).	1
2	Public-Key Cryptography(8 Hours)	
2.1	Factorization Problem, Discrete Logarithm Problem.	1
2.2	RSA Cryptosystem	1
2.3	El-Gamal Cryptosystem	1
2.4	Elliptic Curve Arithmetic	1
2.5	Elliptic Curve Cryptosystem	2
2.3	Cryptographic Commitment, Pedersen Commitment.	1
2.4	Distribution of Public Keys, Public-key Certificates, PKI.	1
3	Integrity and Authentication(8 Hours)	
3.1	Cryptographic hash functions	1
3.2	Secure Hash Function - SHA3	2
3.3	Message Authentication Codes – HMAC, CMAC	1
3.4	Digital Signatures, El-Gamal Digital Signature	1
3.5	Schnorr Digital Signature	1
3.6	DSA	1
3.7	RSA-PSS	1
4	Network Security and Internet Security(8 Hours)	
4.1	IP Security – ESP, Security Association.	2
4.2	TLS overview, HTTPS, SSH.	2
4.3	Network Access Control – Elements, Enforcement methods, EAP	2
4.4	Cloud Security – Elements, Risks and Countermeasures, Data protection, Cloud Security as a Service	2

5	Operating System Security and Database System Security (7 Hours)	
5.1	Security features in OS, Security in the design of OS, Rootkit.	3
5.2	Access Control - Role-based and Procedure oriented.	1
5.3	Security requirements of Databases, Reliability and Integrity, Database Disclosure.	3



CODE	COURSE NAME:	CATEGORY	L	T	P	CREDIT
221LIT100	ADVANCED NETWORKING LAB	LABORATORY 1	0	0	2	1

Preamble: This course will cover different experiments on advanced network programming and will provide the students with an environment to work on network simulation tools. This helps students to develop and implement protocols and evaluate their performance.

Course Outcomes: After the completion of the course, the student will be able to

CO 1	Implement various network protocols for LAN and Internet (Level: Apply)
CO 2	Implement and analyze the performance of various network protocols(Level: Analyze)
CO 3	Implement and evaluate different network routing protocols. (Level: Apply)
CO 4	Simulate and compare the performance of various routing protocols, congestion control mechanisms, and wireless networks.(Level: Analyze)

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	✓	✓		✓	✓		
CO 2	✓	✓	✓	✓	✓		
CO 3	✓	✓	✓	✓	✓		
CO 4	✓	✓	✓	✓	✓		

Continuous Internal Evaluation Pattern:

CIE: 100%

Attendance : 10%

Preparation and Completion of Experiment by the Student : 60%

Final Assessment : 30%

Syllabus and Course Plan

No	Topic	No. of Hours
1	Experiments on Network Programming using C/Java	
1.1	Implementation of ARP and improvisation of protocol by resolving various security problems involved in it.	2 Hrs
1.2	Implementation of Remote Command Execution with the various scenarios such as Remote File copy, Remote Distribution etc.	2 Hrs
1.3	Program to integrate Link State and Distance Vector Routing Protocols.	2 Hrs
1.4	Implement a protocol which ensures reliable QoS to transfer a file across a network and measure its performance in comparison with TCP.	2 Hrs
1.5	Implementation of network protocol used on the Internet or local area networks to provide a bidirectional interactive communications facility.	2 Hrs
1.6	Implement a protocol for Authenticated Routing in LAN networks.	2 Hrs
2	Network Simulation and Performance Evaluation Using Simulator	
2.1	Simulation and Performance Comparison of various Routing	2 Hrs

	Protocols.	
2.2	Simulation of Wireless Networks (Eg: Wifi, Adhocetc).	2 Hrs
2.3	Simulation and Performance Comparison of different congestion control mechanisms for TCP/IP protocols.	2 Hrs

Reference Books

1. W. Richard Stevens, "UNIX Network Programming", PHI , Eastern Economy Edition
2. J.F. Kurose and K.W. Ross, Computer Networking: A Top-Down Approach Featuring Internet,3/e, Pearson Education, 2005.
3. Using Java2 Platform – Joseph Weber (AWL)
4. Douglas E.Comer, Hands on Networking with Internet Technologies, Pearson Education.
5. Network and System Security, edited by John R. Vacca



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SEMESTER I

PROGRAM ELECTIVE I



CODE	COURSE NAME	CATEGORY	L	T	P	CREDIT
221EIT100	NETWORK ROUTING PROTOCOLS	PROGRAM ELECTIVE I	3	0	0	3

Preamble: The students are expected to understand the architectures and routing protocols used in different types of networks

Course Outcomes: After the completion of the course the student will be able to

CO No.	Course Outcome (CO)	Bloom's Category Level
CO 1	Apply shortest path and widest path routing algorithms in communication network routing.	Level 3: Apply
CO 2	Design single and multi-commodity network flow problems	Level 4 : Analyze
CO 3	Describe IP routing protocols & the routing protocols used in PSTN	Level 2 : Understand
CO 4	Explain the routing protocols used in high-speed and mobile networks	Level 2 : Understand
CO 5	Apply the routing protocols for ad hoc networks	Level 3: Apply

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

PO1: An ability to independently carry out research/investigation and development work in engineering and allied streams

PO2: An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

PO3: An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor's program

PO4: An ability to apply stream knowledge to design or develop solutions for real world problems by following the standards

PO5: An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tool to model, analyse and solve practical engineering problems.

PO6: An ability to engage in life-long learning for the design and development related to the stream related problems taking into consideration information technology and IT1 environmental aspects

PO7: An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1			✓	✓	✓		
CO 2	✓	✓	✓	✓	✓	✓	
CO 3			✓	✓	✓		
CO 4	✓	✓	✓	✓	✓	✓	
CO 5	✓	✓	✓	✓	✓	✓	

Assessment Pattern

Bloom's Category	End Semester Examination
Apply	80
Analyse	20
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

The evaluation shall only be by employing application-based, analysis-based, or design-based questions (for both internal and end-semester examinations)

Continuous Internal Evaluation Pattern:

Evaluation shall only be based on application, analysis or design based questions (for both internal and end semester examinations).

Continuous Internal Evaluation: 40 marks**INFORMATION TECHNOLOGY-IT1**

- i. Preparing a review article based on peer reviewed original publications (minimum 10 publications shall be referred) : 15 marks
- ii. Course based task / Seminar/ Data collection and interpretation : 15 marks
- iii. Test paper (1 number) : 10 marks

Test paper shall include minimum 80% of the syllabus.

End Semester Examination Pattern:

The end semester examination will be conducted by the respective College.

There will be two parts; Part A and Part B.

Part A will contain 5 numerical/short answer questions with 1 question from each module, having 5 marks for each question. Students should answer all questions. Part B will contain 7 questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem solving and quantitative evaluation), with minimum one question from each module of which student should answer any five. Each question can carry 7 marks

Total duration of the examination will be 150 minutes.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly.

For example if the average end semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is $40+20 = 60\%$.

Course Level Assessment Questions**Course Outcome 1 (CO1):**

1. In what ways, are the Bellman-Ford algorithm and the distance vector algorithm different?
2. Develop a specialized k-shortest paths algorithm, given that a path cannot consist of more than two links.
3. What is the difference between minimum hop routing and shortest path routing?

Course Outcome 2 (CO2)

1. Consider a four-node ring network where nodes are connected as follows: 1-2-3-4-1. Assume that demand volume between 1 and 3 is 25, between 2 and 4 is 30, and between 2 and 3 is 10. Capacity on each link is 50.

- (a) Formulate an optimization problem in which the goal is to maximize free capacity availability. Determine the optimal flow for this objective.
- (b) Formulate an optimization problem in which the goal is to load balance the network. Determine the optimal flow for this objective.

2. Consider a four-node network in which nodes are numbered 1, 2, 3, 4. All nodes are directly connected to each other except that there is no link between nodes 1 and 2. Link capacities are given as follows: 30 on link 1-3, 5 on link 1-4, 15 on link 2-3, 10 on link 2-4, and 10 on link 3-4. Demand volumes are given to be 15 for pair 1:2, 20 for pair 1:3, and 10 for pair 2:3. Formulate the load balancing optimization problem, and determine the optimal solution.

Course Outcome 3(CO3):

1. Why do some routing protocols message identification at above the transport layer while some other do so directly over the IP layer?
2. For a given IP address, how would you find out its home AS number?
3. Explain how hierarchical routing works

Course Outcome 4 (CO4):

1. Describe deflection routing? Explain how buffers can reduce deflection probability
2. With a neat diagram explain the mobile network architecture
3. Explain DARPA packet radio network

Course Outcome 5 (CO5):

1. What are the characteristics of MANET?
2. Compare and contrast DSDV and AODV
3. Explain DSR with an example



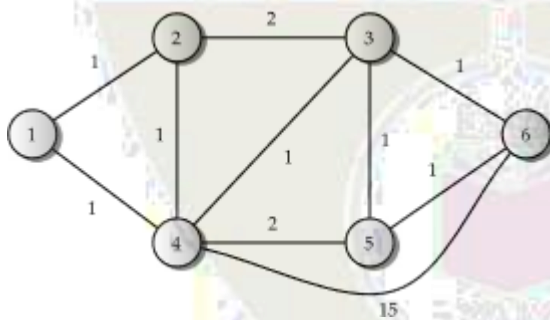
Time: 2½ Hours, Maximum marks: 60

PART A (Answer all questions. Each question carries 5 marks)

1. What are the main differences between shortest path routing and widest path routing?
2. Explain the basic principle behind a path vector protocol through a simple illustration
3. What is crankback?
4. What are the routing modes in packet level routing of PLANET?
5. TORA builds and maintains a DAG rooted at a destination. Why?

PART B (Answer any five questions. Each question carries 7 marks)

6. Consider the network topology given below: Assume that the links have the following bandwidth: 1-2: 1, 1-4: 1, 2-3: 2, 2-4: 2, 3-4: 1, 3-5: 1, 3-6: 4-5: 2; 4-6: 3; 5-6: 2. Determine the widest paths from node 6 to all the other nodes.



7. Explain the packet header of EIGRP
8. Explain how real time network routing works
9. Discuss about the different implementation strategy of PNNI routing in ATM
10. Explain about Cluster head Gateway Switch Routing
11. Describe the routing algorithms used for large sized packet radio networks

12. What are the different optimization techniques used in dynamic source routing protocol?

INFORMATION TECHNOLOGY-IT1

Syllabus

Module I (7 Hours)

Router architecture, Network topology architecture, Network management architecture, Public switched telephone network (PSTN), Routing algorithms: Shortest path and widest path – Bellman ford algorithm and distance vector approach (centralized view, distributed view), Dijkstra's algorithm – centralized and distributed approach, Comparison of the Bellman-Ford Algorithm and Dijkstra's Algorithm, Shortest Path Computation with Candidate Path Caching, Widest Path Computation with Candidate Path Caching, Widest path algorithm, k-shortest path algorithm.

Module II (8 Hours)

Path vector routing protocol: Basic principle, link failure case, path vector with path caching, node failure case, Network flow modeling: Single commodity network flow, Multi commodity network flow: Three node example, general formulation, IP routing: IGRP – packet formats, computing composite metric, EIGRP, Route redistribution, IS-IS – Key features, comparison with OSPF.

Module III (7 Hours)

Internet routing architecture – current architectural view of the Internet, Allocation of IP prefixes and AS number, Policy-based routing, Implications of traffic engineering, Internet routing instabilities, Routing in the PSTN – Hierarchical routing, Limitations of hierarchical routing, Dynamic routing, Real-time network routing, Dynamic routing and its relation to other routing – dynamic routing and link state protocol, Path Selection in Dynamic Routing in Telephone Networks and IP Routing

Module IV (8 Hours)

High Speed Networks -Routing in optical networks-The optical layer, Node Designs, Network design and operation, Optical layer cost tradeoffs, Routing in ATM networks-ATM address structure, ATM Routing, PNNI protocol, Routing in the PLANET network and Deflection Routing. MOBILE NETWORKS - Routing in Cellular Mobile Radio communication networks, Mobile Network Architecture. Mobility and Routing in Cellular Digital Packet Data (CDPD) network, Packet Radio Routing- DARPA packet radio network, Routing algorithms for small, medium and large sized packet radio networks.

Module V (10 Hours)

Internet-based mobile ad-hoc networking, communication strategies, routing algorithms, Table-driven routing –Destination Sequenced Distance vector (DSDV), Source initiated on-demand routing –Dynamic Source Routing (DSR). Ad-hoc On-demand Distance Vector (AODV), Hierarchical based routing, Cluster head Gateway Switch Routing (CGSR) and temporally-Ordered Routing Algorithm (TORA), Quality of Service.

Reference Books

INFORMATION TECHNOLOGY-IT1

1. Deepankar Medhi, Karthikeyan Ramasamy, "Network Routing Algorithms, Protocols, and Architecture, ", The Morgan Kaufmann Series in Networking.
2. M. Steen strub, "Routing in Communication networks", PH International, NY, 1995.
3. William Stallings, "ISDN & Broadband ISDN with Frame Relay & ATM", PHI, ND, 2004.
4. Behrouz A Forouzan, "Data Communications and Networking", 3rd Edition, TMH, 2004
5. William Stallings, "High Speed Networks TCP/IP and ATM Design Principles", Prentice Hall International, 1998.
6. Mohammad Ilyas, "The Handbook of Ad hoc Wireless Networks", CRC Press, 2002
7. Vijay K.Garg, "Wireless Network Evolution: 2G to 3G", Pearson Education, ND, 2003.
8. Rajiv Ramaswami and Kumar N.Sivarajan, "Optical Networks", Morgan Kaufmann Publishers, 1998.
9. Sumit Kasera and Pankaj sethi, "ATM Networks", TMH Publishing Co. Ltd, ND, 2001.

Course Plan

No	Topic	No. of Lectures (40 Hours)
1	Network Routing : Basics and Foundations	7 Hours
1.1	Router architecture, Network topology architecture, Network management architecture, Public switched telephone network (PSTN)	1
1.2	Routing algorithms : Shortest path and widest path – Bellman ford algorithm and distance vector approach (centralised view, distributed view)	1
1.3	Dijkstra's algorithm – centralised and distributed approach	1
1.4	Comparison of the Bellman–Ford Algorithm and Dijkstra's Algorithm	1
1.5	Shortest Path Computation with Candidate Path Caching	1
1.6	Widest Path Computation with Candidate Path Caching	1
1.7	Widest path algorithm, k-shortest path algorithm	1
2	Network flow modeling & IP routing	8 Hours
2.1	Path vector routing protocol : Basic principle, link failure case, path vector with path caching, node failure case	1
2.2	Network flow modeling: Single commodity network flow	1
2.3	Multi commodity network flow: Three node example, general formulation	2

2.4	IP routing : IGRP – packet formats, computing composite metric, EIGRP, Route redistribution	2
2.5	IP routing : IGRP – packet formats, computing composite metric, EIGRP, Route redistribution	1
2.6	IS-IS – Key features, comparison with OSPF	1
3	Internet routing architecture & Routing in the PSTN	7 Hours
3.1	Internet routing architecture – current architectural view of the Internet, Allocation of IP prefixes and AS number	1
3.2	Policy based routing, Implications of traffic engineering, Internet routing instabilities	2
3.2	Routing in the PSTN – Hierarchical routing, limitations of hierarchical routing, Dynamic routing	2
3.3	Real time network routing, Dynamic routing and its relation to other routing - dynamic routing and link state protocol, Path Selection in Dynamic Routing in Telephone Networks and IP Routing	2
4	High speed and Mobile Network Routing	9 Hours
4.1	HIGH SPEED NETWORKS -Routing in optical networks-The optical layer, Node Designs, Network design and operation, Optical layer cost tradeoffs	1
4.2	Routing in ATM networks-ATM address structure, ATM Routing, PNNI protocol	2
4.3	Routing in the PLANET network and Deflection Routing.	1
4.4	MOBILE NETWORKS - Routing in Cellular Mobile Radio Communication networks, Mobile Network Architecture.	1
4.5	Mobility and Routing in Cellular Digital Packet Data (CDPD) network	1
4.6	Packet Radio Routing- DARPA packet radio network	1
4.7	Routing algorithms for small, medium and large sized packet radio networks.	2
5	Ad hoc Networks	9 Hours
5.1	Internet-based mobile ad-hoc networking, communication strategies, routing algorithms	1
5.2	Table-driven routing –Destination Sequenced Distance vector (DSDV)	1
5.3	Source initiated on-demand routing - Dynamic Source Routing (DSR).	2
5.4	Ad-hoc On-demand Distance Vector (AODV), Hierarchical based routing	2
5.5	Cluster head Gateway Switch Routing (CGSR) and temporally-Ordered Routing Algorithm (TORA)	2
5.6	Quality of Service.	1

INFORMATION TECHNOLOGY-IT1						
CODE	COURSE NAME	CATEGORY	L	T	P	CREDIT
221EIT001	WIRELESS AND AD HOC NETWORKS	PROGRAM ELECTIVE I	3	0	0	3

Preamble: The students are expected to understand the basic concepts of wireless communication, ad hoc networks, and sensor networks

Course Outcomes: After the completion of the course the student will be able to

CO No.	Course Outcomes (CO)	Bloom's Category Level
CO 1	Identify the characteristics of wireless communication systems and cellular systems	Level2: Understand
CO 2	Explain the operation of wireless internet	Level 2: Understand
CO 3	Analyze the MAC and routing protocols used in ad hoc networks	Level4: Analyse
CO 4	Explain the transport and security protocols for ad hoc networks	Level 2: Understand
CO 5	Explain the fundamental concepts of wireless sensor networks	Level 2: Understand

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

PO1: An ability to independently carry out research/investigation and development work in engineering and allied streams

PO2: An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

PO3: An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program

PO4: An ability to apply stream knowledge to design or develop solutions for real world problems by following the standards

PO5: An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tool to model, analyse and solve practical engineering problems.

PO6: An ability to engage in life-long learning for the design and development related to the stream related problems taking into consideration sustainability, societal, ethical and environmental aspects

PO7: An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1			✓	✓	✓	✓	
CO 2	✓		✓	✓	✓	✓	
CO 3	✓		✓	✓	✓	✓	
CO 4	✓	✓	✓	✓	✓		
CO 5			✓	✓	✓		

Assessment Pattern

Bloom's Category	End Semester Examination
Apply	80
Analyse	20
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

The evaluation shall only be by employing application-based, analysis-based, or design-based questions (for both internal and end-semester examinations)

Continuous Internal Evaluation: 40 marks

- Preparing a review article based on peer reviewed original publications (minimum 10 publications shall be referred) : 15 marks
- Course based task / Seminar/ Data collection and interpretation : 15 marks
- Test paper (1 number) : 10 marks

Test paper shall include minimum 80% of the syllabus.

End Semester Examination Pattern:

The end semester examination will be conducted by the respective College.

There will be two parts; Part A and Part B.

Part A will contain 5 numerical/short answer questions with 1 question from each module, having 5 marks for each question. Students should answer all questions. Part B will contain 7 questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem solving and quantitative evaluation), with minimum one question from each module of which student should answer any five. Each question can carry 7 marks

Total duration of the examination will be 150 minutes.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly.

For example if the average end semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is $40+20 = 60\%$.

Course Level Assessment Questions

Course Outcome 1 (CO1):

1. Illustrate with necessary expressions how frequency reuse helps in increasing cellular capacity
2. Examine the features of scattering. Why does it occur?
3. What are the factors that influence small scale fading?

Course Outcome 2 (CO2)

1. Explain wireless application protocol stack
2. What is co-located care of address?
3. Briefly discuss about agent discovery in Mobile IP

Course Outcome 3(CO3):

1. Explain the features of MACAW protocol

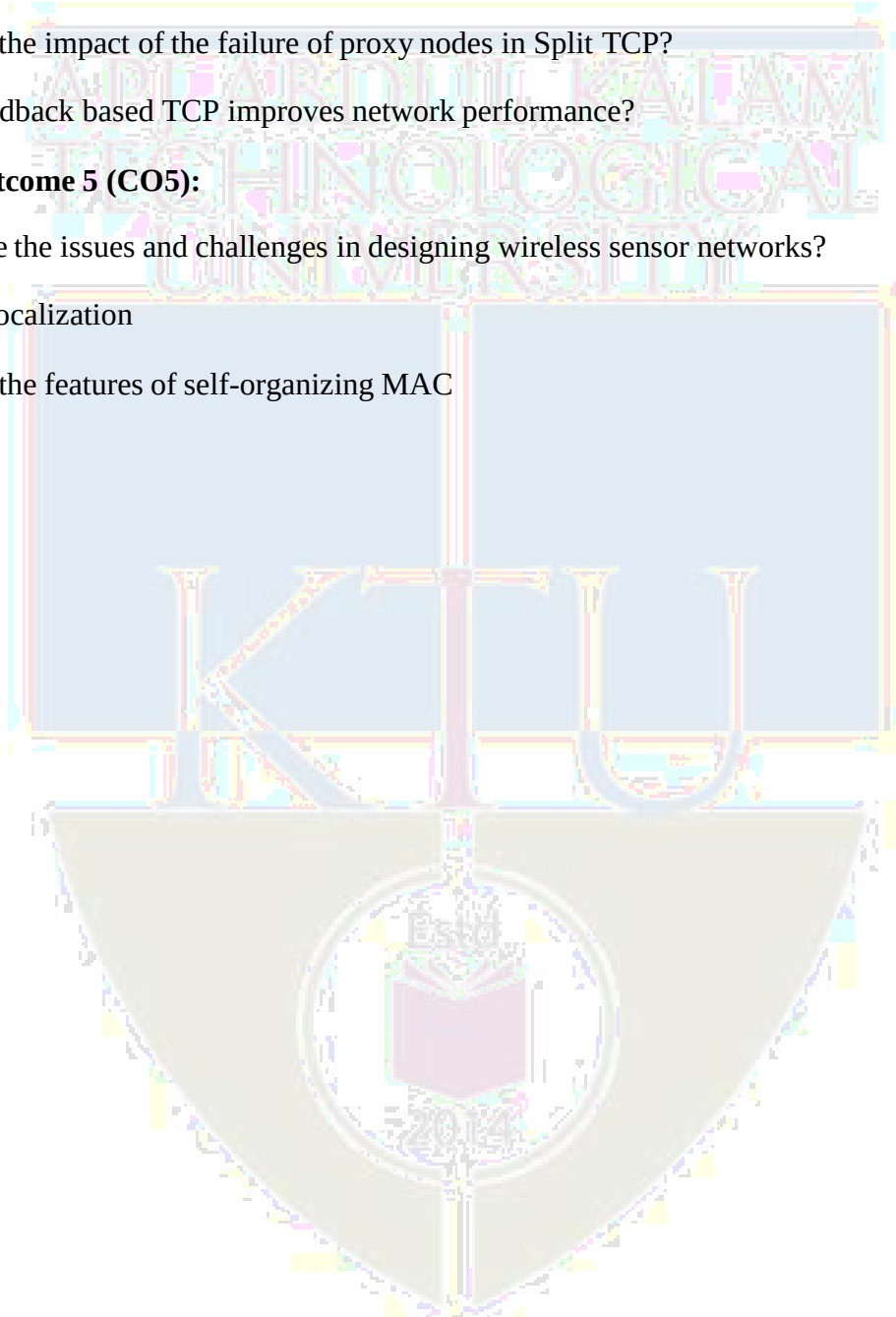
2. Compare DSR protocol and AODV protocol
3. Analyze DSDV routing protocol in detail

Course Outcome 4 (CO4):

1. What are the network security requirements for mobile ad hoc networks?
2. What is the impact of the failure of proxy nodes in Split TCP?
3. How feedback based TCP improves network performance?

Course Outcome 5 (CO5):

1. What are the issues and challenges in designing wireless sensor networks?
2. Define localization
3. Explain the features of self-organizing MAC



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M.TECH DEGREE EXAMINATION
.....SEMESTER
Branch:

Time: 2½ Hours, Maximum marks: 60

PART A (Answer all questions. Each question carries 5 marks)

1. A spectrum of 30 MHz is allocated to a wireless FDD cellular system which uses two 25 KHz simplex channels to provide full duplex voice and control channels, compute the number of channels available per cell. Compute the number of channels available per cell if it uses 4-cell reuse.
2. What are the advantages of I-TCP?
3. Explain the issues in designing routing protocols for ad hoc networks
4. How does TCP bus sender probe the congestion in network?
5. Write the properties and the main design challenges of data dissemination protocols in a wireless sensor network.

PART B (Answer any five questions. Each question carries 7 marks)

6. “In practice fast fading only occurs for very low data rate (communications)”. Justify.
7. Explain multichannel CSMA MAC protocol
8. Consider the network given below. Here ‘S’ is source node and ‘D’ is target node. Illustrate the process of route discovery, route reply, data delivery and route caching using DSR.



9. Explain ad hoc TCP in detail

10. Explain CSMA based MAC protocols for wireless sensor networks
11. What is triangular routing?
12. Compare the data gathering methods used in wireless sensor networks

Syllabus

Module I: Introduction (8 Hours)

Introduction to wireless communication systems: Evolution of mobile radio communications, Mobile radio systems around the world, Example of wireless communication systems, Modern wireless communication systems: 2G, 3G, 4G and 5G, Wireless local loop, Wireless local area networks, Bluetooth and personal area networks, The Cellular Concept: Frequency reuse, channel assignment and handoff strategies, Interference and system capacity, trunking and grade of service, Improving coverage and capacity in cellular systems, Signal Propagation – path loss of radio signals, additional signal propagation effects, multipath propagation. Spread spectrum – Direct sequence spread spectrum, Frequency-hopping spread spectrum, Medium access control – motivation for a specialized MAC, SDMA, FDMA, TDMA

Module II: Wireless Internet and Adhoc Wireless Networks (8 Hours)

Wireless Internet – introduction, Mobile IP, TCP in wireless domain, Wireless Application Protocol – model, protocol stack. Ad hoc wireless networks – introduction, issues in ad hoc wireless networks, MAC protocols for ad hoc wireless networks – introduction, issues in designing a MAC protocol for ad hoc wireless networks, classification of MAC protocols. MAC Protocols – MACAW, Distributed packet reservation multiple access protocol, distributed priority scheduling and medium access in ad hoc networks, multi-channel CSMA MAC protocol

Module III: Routing Protocols for Ad hoc wireless networks (8 Hours)

Introduction, issues in designing a routing protocol for ad hoc wireless networks, classification of routing protocols, Table driven routing protocols – destination sequenced distance vector routing protocol, wireless routing protocol, On-demand routing protocols – dynamic source routing protocol, AODV, Hybrid routing protocols: core extraction distributed ad hoc routing protocol, zone routing protocols, Power-aware routing protocols

Module IV: Transport and Security protocols for ad hoc wireless networks (8 Hours)

Transport layer – design issues, classification of transport layer solutions, TCP over ad hoc wireless networks – Feedback based TCP, TCP with explicit link failure notification, TCP Bus, Ad hoc TCP, Split TCP, Application controlled transport protocol, ad hoc transport

protocol, Security in Ad hoc wireless networks, network security requirements, issues and challenges in security provisioning, network security attacks

Module V: Wireless Sensor Networks (8 Hours)

Introduction, applications of sensor networks, comparison with ad hoc wireless networks, issues and challenges in designing a sensor network, Sensor network architecture – layered architecture, clustered architecture, Data dissemination – flooding, gossiping, rumor routing, sequential assignment routing, directed diffusion, sensor protocols for information via negotiation, geographic hash table, Data gathering – direct transmission, power-efficient gathering for sensor information systems, binary scheme, chain based three-level scheme, MAC protocols for sensor networks – Self-organizing MAC, Hybrid TDMA/FDMA, CSMA based MAC protocols, Location discovery – indoor localization, sensor network localization

Reference Books

1. Theodore S Rappaport, “Wireless Communications: Principles and Practice”, 2nd edition, Pearson
2. C. Siva Ram Murthy, B.S. Manoj, “Ad hoc Wireless Networks: Architectures and Protocols”, Pearson Education
3. Jochen Schiller, “Mobile Communications”, 2nd Edition, Pearson

Course Contents and Lecture Schedule

No	Topic	No. of Lectures (40 Hours)
1	Introduction 8 Hours	
1.1	Introduction to wireless communication systems: Evolution of mobile radio communications, Mobile radio systems around the world. Example of wireless communication systems. Modern wireless communication systems: 2G, 3G, 4G and 5G.	1
1.2	Wireless local loop, Wireless local area networks, Bluetooth and personal area networks.	2
1.3	The Cellular Concept: Frequency reuse, channel assignment and handoff strategies	1
1.4	Interference and system capacity, trunking and grade of service. Improving coverage and capacity in cellular systems.	1
1.5	Signal Propagation – path loss of radio signals, additional signal propagation effects, multi path propagation	1
1.6	Spread spectrum – Direct sequence spread spectrum, Frequency-hopping spread spectrum	1
1.7	Medium access control – motivation for a specialized MAC, SDMA, FDMA, TDMA	1

2	Wireless Internet and Adhoc Wireless Networks	8 Hours
2.1	Wireless Internet – introduction, Mobile IP	1
2.2	TCP in wireless domain	1
2.3	Wireless Application Protocol – model, protocol stack	1
2.4	Ad hoc wireless networks – introduction, issues in ad hoc wireless networks	2
2.5	MAC protocols for ad hoc wireless networks – introduction, issues in designing a MAC protocol for ad hoc wireless networks, classification of MAC protocols	2
2.6	MAC Protocols – MACAW, Distributed packet reservation multiple access protocol, distributed priority scheduling and medium access in ad hoc networks, multi-channel CSMA MAC protocol	1
3	Routing Protocols for Ad hoc wireless networks	8 Hours
3.1	Introduction, issues in designing a routing protocol for ad hoc wireless networks, classification of routing protocols	1
3.2	Table driven routing protocols – destination sequenced distance vector routing protocol, wireless routing protocol	2
3.3	On-demand routing protocols – dynamic source routing protocol, AODV	2
3.4	Hybrid routing protocols: core extraction distributed ad hoc routing protocol, zone routing protocols	2
3.5	Power-aware routing protocols	1
4	Transport and Security protocols for ad hoc wireless networks	8 Hours
4.1	Transport layer – design issues, classification of transport layer solutions	2
4.2	TCP over ad hoc wireless networks – Feedback-based TCP, TCP with explicit link failure notification, TCP Bus	2
4.3	Ad hoc TCP, Split TCP, Application controlled transport protocol, ad hoc transport protocol	2
4.4	Security in Ad hoc wireless networks, network security requirements, issues and challenges in security provisioning, network security attacks	2
5	Wireless Sensor Networks	8 Hours
5.1	Introduction, applications of sensor networks, comparison with ad hoc wireless networks, issues and challenges in designing a sensor network	1
5.2	Sensor network architecture – layered architecture, clustered architecture	2
5.3	Data dissemination – flooding, gossiping, rumor routing, sequential assignment routing, directed diffusion, sensor protocols for information via negotiation, geographic hash table	2
5.4	Data gathering – direct transmission, power-efficient gathering	1

	for sensor information systems, binary scheme, chain-based three-level scheme	
5.5	MAC protocols for sensor networks – Self-organizing MAC, Hybrid TDMA/FDMA, CSMA-based MAC protocols	1
5.6	Location discovery – indoor localization, sensor network localization	1

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CODE:	COURSE NAME:	CATEGORY	L	T	P	CREDIT
221EIT002	CLOUD COMPUTING AND SECURITY	PROGRAM ELECTIVE I	3	0	0	3

Preamble: The syllabus is designed with the view of preparing the students capable of understanding the security aspects of cloud computing. The students will get to know about the cloud architecture, vulnerabilities and threats occurring in various cloud systems and methods to secure the same.

Course Outcomes: Upon completion, the student will be able to

CO 1	Apply the basics of virtualization to implement virtual machines(Cognitive Level: Apply)
CO 2	Apply the fundamentals of cloud computing and its architecture for implementing cloud services.(Cognitive level: Apply)
CO 3	Deploy an application in cloud.(Cognitive level: Apply)
CO 4	Analyse the security issues and challenges faced in a given cloud environment and implement appropriate security measures(Cognitive level: Apply)

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

- PO1:** An ability to independently carry out research/investigation and development work in engineering and allied streams
- PO2:** An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.
- PO3:** An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program

PO4: An ability to apply stream knowledge to design or develop solutions for real world problems by following the standards

PO5: An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tool to model, analyse and solve practical engineering problems.

PO6: An ability to engage in life-long learning for the design and development related to the stream related problems taking into consideration sustainability, societal, ethical and environmental aspects

PO7: An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1							
CO 2							
CO 3							
CO 4							

Assessment Pattern	
Bloom's Category	End Semester Examination
Apply	100%
Analyse	
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation: 40 marks

- i. Preparing a review article based on peer reviewed original publications (minimum 10 publications shall be referred) : 15 marks
- ii. Course based task / Seminar/ Data collection and interpretation : 15 marks
- iii. Test paper (1 number) : 10 marks

Test paper shall include minimum 80% of the syllabus.

End Semester Examination Pattern:

The end semester examination will be conducted by the respective College.
There will be two parts; Part A and Part B.

Part A will contain 5 numerical/short answer questions with 1 question from each module, having 5 marks for each question. Students should answer all questions. Part B will contain 7 questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem solving and quantitative evaluation), with minimum one question from each module of which student should answer any five. Each question can carry 7 marks

Total duration of the examination will be 150 minutes.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly.

For example if the average end semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is $40+20 = 60\%$.

Course Level Assessment Questions

Course Outcome 1 (CO1):

1. Demonstrate how you will a Configure a VM instance using virtual box in your local machine and allocate CPU, memory and storage space as per a specified requirement.
2. Illustrate the steps to perform VM migration.
3. A degree of caution is advised when launching a server virtualization. Clarify this statement by stating the drawbacks of the same.

Course Outcome 2 (CO2):

1. With the help of examples, differentiate between SaaS, PaaS and IaaS?
2. To meet the requirements of a single organization, which one out of the private, public and hybrid cloud providers, is recommended? Why?
3. How will you classify massive systems? Explain.

Course Outcome 3 (CO3):

1. “OpenNebula and OpenStack serve different needs”. Justify this statement.
2. Demonstrate the challenges in designing and developing a cloud application, with the help of an example.
3. How is OpenStack used in a cloud environment?

Course Outcome 4 (CO4):

1. Given an e-healthcare application deployed in cloud. Identify the challenges faced and the security objectives to be achieved.
2. How is security provided to data at various stages in the context of the cloud?
3. How virtualization supports disaster recovery?

Model Question Paper:

QP CODE:

Reg No:

Name:

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

FIRST SEMESTER M.TECH DEGREE EXAMINATION, MONTH & YEAR

221EIT002

CLOUD COMPUTING AND SECURITY

Max. Marks : 60

Duration: 2.5 hrs

PART A

Answer All Questions. Each Question Carries 5 Marks

1. Give the steps to install a virtual machine. (5)
 2. In which scenario, users can opt IaaS, PaaS and SaaS? Find suitable application for each. (5)
 3. How inter-cloud resource management is performed? (5)
 4. Among openNebula and openStack, which is suited best for enterprise cloud computing? (5)
 5. Why is it harder to establish security in the cloud? (5)
- (5x5=25)**

Part B

(Answer any five questions. Each question carries 7 marks)

6. How is memory virtualization implemented? Provide necessary examples and diagrams wherever necessary. (7)
7. Illustrate the steps to perform live VM migration. (7)

8. Depict the cloud architecture. Explain the same with the help of an example application. (7)
9. Explain PaaS model for cloud computing. With a neat diagram, write about Google App Engine for PaaS applications. (7)
10. Consider a restaurant application on the cloud that has serverless apps connecting to a backend NoSQL database. There are two types of users in the application - manager (who manages the catalog) and customer (who views the catalog). There are separate serverless functions for the users. During development, you need to create a set of roles that can define what operations are permitted on the NoSQL database by the serverless functions. In addition, the entire application and the data should be secured. Propose solutions for the following. (7)
 - a) What permissions are required for the role assigned to manager function?
 - b) What permissions are required for the role assigned to customer function?
 - c) How can you ensure application security?
 - d) How can you ensure data security?
 - e) How can you create an auditable trail of modifications to the database?
11. Explain Security Architecture Design in cloud. (7)
12. Explain the life cycle of Secure software Development (SecSDLC). (7)

Syllabus

Module I:

Introduction to component virtualization: Basics of Virtualization - Types of Virtualization - Implementation Levels of Virtualization -Virtualization of CPU, Memory, I/O Devices – Virtual clusters and resource management- Desktop Virtualization – Server Virtualization- Storage Virtualization – Network Virtualization.

Module II: Introduction to Cloud Computing: System Models for Distributed and Cloud Computing - NIST Cloud Computing Reference Architecture - Features of Cloud Computing – Cloud Services – IaaS, PaaS, SaaS – Cloud service Providers – Public, Private and Hybrid Clouds.

Module III: Architectural Design of Compute and Storage Clouds – Layered Cloud Architecture Development – Design Challenges - Inter Cloud Resource Management – Resource Provisioning and Platform Deployment – Global Exchange of Cloud Resources.

Module IV: Parallel and Distributed Programming Paradigms – Map Reduce, Hadoop Library from Apache – Mapping Applications - Programming Support – EC2, Google App Engine, Amazon AWS - Cloud Software Environments - Eucalyptus, Open Nebula, OpenStack.

Security in the cloud: Overview – Cloud Security Challenges- Software-as-a-Service Security – Security Governance – Risk Management.

Module V:SecSDLC-Security Monitoring – Security Architecture Design – Data Security – Application Security. VM security. Cloud Security Architecture: Trusted cloud, Identity and access management, Access control in cloud, Autonomic security, protection, self-healing. Firewalls-Web application firewall.

Reference Books

1. Kai Hwang, Geoffrey C Fox, Jack G Dongarra, “Distributed and Cloud Computing, From Parallel Processing to the Internet of Things”, 1sted. ,Morgan Kaufmann Publishers.
2. John W.Rittinghouse and James F.Ransome, “Cloud Computing: Implementation, Management, and Security”, CRC Press, 2010.
3. Tim Mather, S. Kumaraswamy and S. Latif, “Cloud Security and Privacy: An EnterprisePerspective on Risks and Compliance”, O’Reilly Media, 2009.
4. Ronald L. Krutz Russell Dean Vines “Cloud Security: A Comprehensive Guide to SecureCloud Computing”, Wiley ,2010.

5. Toby Velte, Anthony Velte, Robert Elsenpeter, “Cloud Computing, A Practical Approach”, TMH, 2009.

6. George Reese, “Cloud Application Architectures: Building Applications and Infrastructure in the Cloud” O'Reilly, 2009.

7. James E. Smith, Ravi Nair, “Virtual Machines: Versatile Platforms for Systems and Processes”, Elsevier/Morgan Kaufmann, 2005.

8. Katarina Stanoevska-Slabeva, Thomas Wozniak, Santi Ristol, “Grid and Cloud Computing – A Business Perspective on Technology and Applications”, Springer, 2010.

Course Contents and Lecture Schedule

No	Topic	No. of Lectures (40)
1	Introduction to component virtualization:	8 Hours
1.1	Basics of Virtualization – Types of Virtualization	1 hour
1.2	Implementation Levels of Virtualization	1 hour
1.3	Virtualization of CPU, Memory, I/O Devices	1 hour
1.4	Virtual clusters and resource management	1 hour
1.5	Live VM migration steps	1 hour
1.6	Migration of memory, files and network resources	1 hour
1.7	Desktop Virtualization –Server Virtualization Storage Virtualization	1 hour
1.8	Network Virtualization	1 hour
2	Introduction to Cloud Computing	6 Hours

INFORMATION TECHNOLOGY-IT1

2.1	System Models for Distributed and Cloud Computing	1 hour
2.2	NIST Cloud Computing Reference Architecture	1 hour
2.3	Features of Cloud Computing	1 hour
2.4	Cloud Services – IaaS, PaaS, SaaS	1 hour
2.5	Cloud service Providers – Public	1 hour
2.6	Cloud service Providers – Private and Hybrid Clouds.	1 hour
3	Architectural Design of Compute and Storage Clouds	6 Hours
3.1	Architectural Design of Compute and Storage Clouds	1 hour
3.2	Layered Cloud Architecture Development	1 hour
3.3	Design Challenges	1 hour
3.4	Inter Cloud Resource Management	1 hour
3.5	Resource Provisioning and Platform Deployment	1 hour
3.6	Global Exchange of Cloud Resources.	1 hour
4	Paradigms and Environments	10 Hours
4.1	Parallel and Distributed Programming Paradigms – Map Reduce	1 hour
4.2	Twister and Iterative MapReduce	1 hour
4.3	Hadoop Library from Apache – Mapping Applications – Programming Support	1 hour
4.4	Google App Engine	1 hour

4.5	Amazon AWS	1 hour
4.6	Cloud Software Environments – Eucalyptus	1 hour
4.7	Open Nebula, OpenStack.	1 hour
4.8	Security in the cloud-overview	1 hour
4.9	Cloud security challenges	1 hour
4.10	Software-as-a-Service Security – Security Governance – Risk Management	1 hour
5	Cloud security	10 Hours
5.1	Secure Software Development Lifecycle	1 hour
5.2	Security Monitoring	1 hour
5.3	Security Architecture Design	1 hour
5.4	Data Security – Application Security. VM security	1 hour
5.5	Cloud Security Architecture: Trusted cloud,	1 hour
5.6	Identity and access management,	1 hour
5.7	Access control in cloud, RBAC	1 hour
5.8	Autonomic security, protection, self-healing.	1 hour
5.9	Firewalls	1 hour
5.10	Web application firewall.	1 hour

CODE	COURSE NAME	INFORMATION	IT	CH	NO	CREDITS
221EIT003	ETHICAL HACKING	PROGRAM ELECTIVE I	3	0	0	3

Preamble: The students are expected to understand the basic concepts of Ethical Hacking and Penetration testing.

Course Outcomes: After the completion of the course the student will be able to

CO 1	Perform system security vulnerability testing
CO 2	Learn various hacking methods
CO 3	Perform system vulnerability exploit attacks
CO 4	Produce a security assessment report
CO 5	Recognize tools and methods to carry out penetration testing.

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

PO1:An ability to independently carry out research/investigation and development work in engineering and allied streams

PO2:An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

PO3:An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor's program

PO4:An ability to apply stream knowledge to design or develop solutions for real-world problems by following the standards

PO5:An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tools to model, analyze and solve practical engineering problems.

PO6:An ability to engage in lifelong learning for the design and development related to the stream-related problems taking into consideration sustainability, societal, ethical, and environmental aspects

PO7:An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4 INFC	PO 5 INFORMATION	PO 6 TECHNOLOGY	PO 7-IT1
CO 1	✓		✓	✓	✓	✓	
CO 2	✓		✓	✓	✓	✓	
CO 3	✓		✓	✓	✓	✓	
CO 4	✓	✓	✓	✓	✓	✓	
CO 5	✓	✓	✓	✓	✓	✓	

Assessment Pattern

Bloom's Category	End Semester Examination
Apply	80
Analyse	20
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

The evaluation shall only be by employing application-based, analysis-based, or design-based questions (for both internal and end-semester examinations)

Continuous Internal Evaluation Pattern: Estd.

The evaluation shall only be based on application, analysis or design-based questions (for both internal and end-semester examinations).

Continuous Internal Evaluation: 40 marks

- Preparing a review article based on peer-reviewed original publications (minimum 10 publications shall be referred) : 15 marks
- Course based task / Seminar/ Data collection and interpretation : 15 marks
- Test paper (1 number) : 10 marks

Test paper shall include a minimum of 80% of the syllabus.

Course-based task/test paper questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation, and understanding of the students.

End Semester Examination Pattern:

The end semester examination will be conducted by the respective College.

There will be two parts; Part A and Part B.

Part A will contain 5 numerical/short answer questions with 1 question from each module, having 5 marks for each question. Students should answer all questions. Part B will contain 7 questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem-solving and quantitative evaluation), with a minimum of one question from each module of which student should answer any five. Each question can carry 7 marks

The total duration of the examination will be 150 minutes.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly.

For example, if the average end-semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is $40+20 = 60\%$.



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M.TECH DEGREE EXAMINATION
.....SEMESTER**

Branch:

Time: 2½ Hours, Maximum marks: 60

PART A (Answer all questions. Each question carries 5 marks)

1. Explain the terms Threat, Vulnerability, and Risk.
2. What are the different methods for managing insider threats?
3. What do you mean by 'Vulnerability Scanner'? Draw the scanner and briefly explain
4. Explain in detail computer fraud classification according to the data processing model?
5. Is there any method to demonstrate whether network infrastructure is adequate or not? Justify your answer.

PART B (Answer any five questions. Each question carries 7 marks)

6. Being an expert in hacking, you are asked to analyze the free, open-source software being used to run FTP services on a server. You noticed that there is an excessive number of `gets()` and `get()` on the source code. These C++ functions do not check bounds. What kind of attack is this program susceptible to? Comment on your recommendation to prevent this attack.
7. Explain the type of attack faced by the stack data structure. Also, explain the uses of two pointers used in a stack.
8. Illustrate how SQL injection attacks are done by attackers.
9. Discuss some architecture strategies for fraud prevention with respect to computer network-enabled system
10. Explain IT infrastructure.
11. An attacker uses a distributed group of computers to shut down a single machine or network making it inaccessible to its intended users? Explain the attack with the help of a neat diagram. What are the ways to prevent this attack?
12. Illustrate the process involved in understanding the inherent risk profile of a business with relevant tables and assumptions

Module I: (8 Hours)

Introduction: CIA Triads, Hacking Impact, The Hacker Framework: Planning the test, Sound operations, Reconnaissance, Enumeration, Vulnerability Analysis, Exploitation, Final Analysis, and Deliverable. Integration Information Security Models: Computer Security, Network Security, Service Security, Application Security. Security Architecture Information Security Program: The Process of Information Security, Component Parts of Information Security Program, Risk Analysis, and Ethical Hacking.

Module II: (8 Hours)

The Business Perspective: Business Objectives, Security Policy, Previous Test Result. Business Challenges Planning for a Controlled Attack: Inherent Limitations Imposed Limitations, Timing Everything, Attack Type, Source Point, Required Knowledge, Multi-Phased Attack, Teaming and Attack Structure, Engagement Planner, The Right Security Consultant, The Tester, Logistics, Intermediates, Law Enforcement.

Module III: (8 Hours)

Preparing for a Hack: Technical Preparation, Managing the Engagement, Reconnaissance: Social Engineering, Physical Security, and Internet Reconnaissance.

Module IV: (8 Hours)

Enumeration: Enumeration Techniques, Soft Objective, Looking Around or Attack, Elements of Enumeration. Preparing for the Next Phase Exploitation: Intuitive Testing, Evasion, Threads and Groups, Operating Systems, Password Crackers, Rootkits, Applications, Wardialing, Network, Services, and Areas of Concern

Module V: (8 Hours)

Deliverable: The Deliverable, The Document, Overall Structure, Aligning Findings. Presentation Integration: Integrating the Results, Integration Summary, Mitigation, Defense Planning, Incident Management, Security Policy

Reference Books

1. Hacking: The Art of Exploitation Jon Erikson 2nd edition
2. The Basics of Hacking and Penetration Testing Patrick Engebretson 2nd edition
3. The Hacker Playbook 2 Peter Kim 1st edition
4. Penetration Testing Georgia Weldman 1st edition.

5. The Web Application Hacker's Handbook Dafydd Stuttard 2nd edition.
6. Stuart McClure, Joel Scambray and George Invernizzi, "Information Technology Security Secrets & Solutions", Tata McGraw-Hill Publishers, 2010 (6th Edition)
7. Ben Smith and Brian Komer, "Microsoft Windows Security Resource Kit", Prentice Hall of India, 2010

Syllabus and Course Plan (For 3 credit courses, the content can be for 40 hrs, and for 2 credit courses, the content can be for 26 hrs. The audit course in the third semester can have content for 30 hours).

No	Topic	No. of Lectures (40 Hours)
1	Module 1	8 Hours
1.1	Introduction: CIA Triads, Hacking Impact.	2
1.2	The Hacker Framework: Planning the test, Sound Operations, Reconnaissance, Enumeration, Vulnerability Analysis, Exploitation, Final Analysis, and Deliverable.	2
1.3	Integration Information Security Models: Computer Security, Network Security, Service Security, Application Security	2
1.4	Security Architecture Information Security Program: The Process of Information Security, Component Parts of Information Security Program, Risk Analysis, and Ethical Hacking	2
2	Module 2	9 Hours
2.1	The Business Perspective: Business Objectives, Security Policy, Previous Test Result	2
2.2	Business Challenges Planning for a Controlled Attack: Inherent Limitations Imposed Limitations,	2
2.3	Timing Everything, Attack Type, Source Point, Required Knowledge, Multi-Phased Attack	1
2.4	Teaming and Attack Structure, Engagement Planner,	2
2.5	The Right Security Consultant, The Tester, Logistics, Intermediates, Law Enforcement.	2
3	Module 3	7 Hours
3.1	Preparing for a Hack: Technical Preparation,	1
3.2	Managing the Engagement	2
3.3	Reconnaissance: Social Engineering	2
3.4	Physical Security, and Internet Reconnaissance.	2
4	Module 4	8 Hours
4.1	Enumeration: Enumeration Techniques, Soft Objective,	1

4.2	LookingAround or Attack, Elements of Enumeration.	2
4.3	Preparing for the Next Phase Exploitation: Intuitive Testing, Evasion, Threads and Groups, Operating Systems, INFORMATION TECHNOLOGY	2
4.4	Password Crackers, Rootkits,Applications	2
4.5	Wardialing, Network, Services and Areas of Concern.	1
5	Module 5	8 Hours
5.1	Deliverable: The Deliverable, The Document, Overall Structure, Aligning Findings	2
5.2	Presentation Integration: Integrating the Results, Integration Summary	2
5.3	Mitigation, Defense Planning	2
5.4	Incident Management, Security Policy	2



INFORMATION TECHNOLOGY-IT1						
CODE	COURSE NAME	CATEGORY	L	T	P	CREDIT
221EIT004	SOFTWARE DEFINED NETWORKS	PROGRAM ELECTIVE 1	3	0	0	3

Preamble: The modern computer network has evolved into a complex beast that is challenging to manage and that struggles to scale to the requirements of some of today's environments. SDN represents a new approach that attempts to address these weaknesses of the current paradigm. This course introduces a highly scalable and centralized network control SDN architecture introduced that is better suited to the extremely large networks prevalent in today's mega scale data centers.

Course Outcomes: After the completion of the course the student will be able to

CO 1	Explain the evolution of networking technology (Level: Understand)
CO 2	Analyse the basic components of a Software Defined Networking system, their roles, and how they interact with one another (Level: Analyze)
CO 3	Identify alternative SDN technologies that propose SDN solutions (Level: Understand)
CO 4	Analyse the specific shortcomings that exist in data center networks today and identify the technologies that can be employed in the data centers with the advent of SDN (Level: Analyze)
CO 5	Examine the applicability of SDN in various domains (Level: Apply)

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

PO1: An ability to independently carry out research/investigation and development work in engineering and allied streams

PO2: An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

PO3: An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program

PO4: An ability to apply stream knowledge to design or develop solutions for real world problems by following the standards

PO5: An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tool to model, analyse and solve practical engineering problems.

INFORMATION TECHNOLOGY-IT1

PO6: An ability to engage in life-long learning for the design and development related to the stream related problems taking into consideration sustainability, societal, ethical and environmental aspects

PO7: An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1		✓	✓				
CO 2	✓				✓		
CO 3	✓		✓	✓	✓	✓	
CO 4			✓	✓	✓		
CO 5	✓			✓	✓		

Assessment Pattern

Bloom's Category	End Semester Examination
Apply	70%-80%
Analyse	20%-30%
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation: 40 marks

- i) Preparing a review article based on peer reviewed original publications (minimum 10 publications shall be referred) : 15 marks
- ii) Course based task / Seminar/ Data collection and interpretation : 15 marks
- iii) Test paper (1 number) : 10 marks

Test paper shall include minimum 80% of the syllabus.

End Semester Examination Pattern:

The end semester examination will be conducted by the respective College.

There will be two parts: Part A and Part B.

Part A will contain 5 numerical/short answer questions with 1 question from each module, having 5 marks for each question. Students should answer all questions. Part B will contain 7 questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem solving and quantitative evaluation), with minimum one question from each module of which student should answer any five. Each question can carry 7 marks.

Total duration of the examination will be 150 minutes.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly.

For example, if the average end semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is $40+20 = 60\%$.

Course Level Assessment Questions**Course Outcome 1 (CO1):**

1. Explain the traditional switching architecture.
2. What are the new requirements emerging from the technological advances taking place now in data center environments. How it is changing the traditional switching architecture?

Course Outcome 2 (CO2)

1. What are the functionalities of a SDN controller? Demonstrate the working of SDN Controller.
2. Explain how an SDN devices communicate. Illustrate the working using a software switch. How it is different from a hardware switch?

Course Outcome 3(CO3):

1. “Open SDN is often depicted as a single controller responsible for overseeing the operation of the entire network”, What is its main disadvantage? What are the alternative solutions available to over this disadvantage?
2. Explain how legacy APIs can be used to implement SDN?

Course Outcome 4 (CO4):

1. What are the demands of Data centers on the capacity of network components? Suggest two the methods to overcome these demands.
2. Explain the stateless transport tunnelling. How is it used for network virtualization?

Course Outcome 5 (CO5):

1. Simulate SDN applied on a WAN.
2. What are the characteristics of a campus network? Demonstrate how SDN can be applied to on a campus network.

Model Question Paper

QP CODE:

Reg No: _____

Name: _____

PAGES : 2

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

FIRST SEMESTER M.TECH DEGREE EXAMINATION, MONTH & YEAR

Course Code: 221EIT004

Course Name: SOFTWARE DEFINED NETWORKS

Max. Marks : 60

Duration: 2.5 Hours

PART A

Answer All Questions. Each Question Carries 5 Marks(5x5=25 Marks)

1. Explain traditional switch architecture. Analyse how it affects the modern networking applications.
2. Demonstrate how moving the control plane off the switching device and onto a centralized controller improves forwarding of incoming packet?
3. What are the limitations of Open SDN implementations? Identify alternative solutions available.
4. Inspect the SDN implementation of through SDN via Opening Up the Device.
5. Distinguish the between two general styles of SDN applications, based on communication between the switch and the controller.

Part B**(Answer any five questions. Each question carries 7 marks)**

6. Illustrate the evolution of switches and control planes that has culminated in a fertile playing field for SDN. (7)
7. Explain the fundamental characteristics of SDN. (7)
8. Demonstrate the communication between the controller and the device using OpenFlow protocol. (7)
9. Using a Suitable example, explain SDN implemented via hypervisor-based overlay technology. (7)
10. What is multipath routing? What are the technologies are available in order to achieve multipath routing? (7)
11. From a data center's perspective, how is multitenancy achieved in SDN via Overlays, Open SDN and SDN via APIs? (7)
12. How is SDN applied to service providers and carrier networks? Give an example. (7)

Syllabus**Module 1: Why SDN?**

Introduction - Basic Packet-Switching Terminology, Historical background, The Modern Data Center, traditional Switch Architecture, Autonomous and Dynamic Forwarding Tables, Can We Increase the Packet- Forwarding IQ, Open Source and Technological Shifts, Evolution of Switches and Control Planes, Cost, SDN Implications for Research and Innovation, DataCenter Innovation, Data Center Needs, Genesis of SDN – The Evolution of Networking Technology, Forerunners of SDN, Software Defined Networking is Born Sustaining SDN Interoperability ,Open Source Contributions, Legacy Mechanisms Evolve Toward SDN, Network Virtualization

Module 2: How SDN Works?

Fundamental Characteristics of SDN, SDN Operation, SDN Devices, SDN Controller, SDN Applications, Alternate SDN Methods, Open Flow Overview, Basics, Additions, Limitations.

Module 3: Alternative Definitions of SDN

Potential Drawbacks of Open SDN, SDN via APIs, SDN via Hypervisor-Based Overlays, SDN via Opening Up the Device, Network Functions Virtualization, Alternatives Overlap and Ranking.

Module 4: SDN in Data Centers

Data Center Definition, Data Center Demands, Tunneling Technologies for the Data Center, Path Technologies in the Data Center, Ethernet Fabrics in the Data Center, SDN Use Cases in

the Data Center, Comparison of OpenSDN, Overlays and APIs in the Data Center, Real-World Data Center Implementations.

Module 5: Application of SDN

Implementations of SDN in Other Environments - Wide Area Networks, Service Provider and Carrier Networks, Campus Networks, Hospitality Networks, Mobile Networks, In-Line, Network Functions, Optical Networks, SDN vs. P2P/Overlay Networks Players in the SDN Eco-system, SDN Applications - Application Types, Using Floodlight for Training Purposes, A Simple Reactive Java Application, Controller Considerations, Network Device Considerations, Offloading Flows in the Data Center, Access Control for the Campus

Reference Books

1. Paul Goransson and Chuck Black, —Software Defined Networks: A Comprehensive Approach, Second Edition, Morgan Kaufmann, 2014.
2. Thomas D. Nadeau, Ken Gray, —SDN: Software Defined Networks, O'Reilly Media, 2013.
3. Siamak Azodolmolky, —Software Defined Networking with Open Flow, Packet Publishing, 2013.
4. Cheng Sheng, Jie Bai, Qi Sun, “Software-Defined Wide Area Network Architectures and Technologies”, CRC Press, 2021
5. Dijiang Huang, Ankur Chowdhary, Sandeep Pisharody, “Software-Defined Networking and Security from Theory to Practice”, CRC Press 2019.

Course Plan

No	Topic	No. of Lectures 40 Hours
1	Why SDN? (9 Hours)	
1.1	Introduction - Basic Packet-Switching Terminology, Historical background, The Modern Data Center, traditional Switch Architecture	2 Hours
1.2	Autonomous and Dynamic Forwarding Tables, Can We Increase the Packet- Forwarding IQ, Open Source and Technological Shifts	2 Hours
1.3	Evolution of Switches and Control Planes, Cost, SDN Implications for Research and Innovation, Data Center Innovation, Data Center Needs	2 Hours
1.4	Genesis of SDN – The Evolution of Networking Technology, Forerunners of SDN	1 Hour
1.5	Software Defined Networking is Born Sustaining SDN Interoperability	1 Hour
1.6	Open-Source Contributions, Legacy Mechanisms Evolve Toward SDN, Network Virtualization	1 Hour
2	How SDN Works?(7 Hours)	
2.1	Fundamental Characteristics of SDN, SDN Operation, SDN Devices	2 hours

2.2	SDNController	2 Hours
2.3	SDN Applications, Alternate SDN Methods	1 Hour
2.4	OpenFlow Overview, Basics, Additions, Limitations.	2 Hours
3	Alternative Definitions of SDN(8 Hours)	
3.1	Potential Drawbacks of Open SDN, SDN via APIs	2 Hours
3.2	SDN via Hypervisor-Based Overlays	2 Hours
3.3	SDN via Opening Up the Device	2 Hours
3.4	Network Functions Virtualization, Alternatives Overlap and Ranking	2 Hours
4	SDN in Data Centers(8 Hours)	
4.1	Data Center Definition, Data Center Demands, Tunneling Technologies for the Data Center	2 Hours
4.2	Path Technologies in the Data Center	1 Hour
4.3	Ethernet Fabrics in the Data Center	2 Hours
4.4	SDN Use Cases in the Data Center	1 Hour
4.5	Comparison of OpenSDN, Overlays and APIs in the Data Center, Real-World Data Center Implementations.	2 Hours
5	Application of SDN(8 Hours)	
5.1	Implementations SDN in Other Environments - Wide Area Networks, Service Provider and Carrier Networks	2 Hours
5.2	Campus Networks, Hospitality Networks, Mobile Networks,	2 Hours
5.3	In-Line, Network Functions, Optical Networks, SDN vs. P2P/Overlay Networks Players in the SDN Eco-system	2 Hours
5.4	SDN Applications - Application Types, Using Floodlight for Training Purposes, A Simple Reactive Java Application, Controller Considerations, Network Device Considerations, Offloading Flows in the Data Center, Access Control for the Campus	2 Hours

Estd.



2014

CODE	COURSE NAME	CATEGORY	L	T	P	CREDIT
221EIT005	INTRUSION DETECTION AND PREVENTION SYSTEMS	INFORMATION TECHNOLOGY-III PROGRAM ELECTIVE I	3	0	0	3

Preamble: The students are expected to understand the intrusion detection tools and techniques in order to provide security in a network.

Course Outcomes: After the completion of the course, the student will be able to

CO 1	Explain the possible attacks in computer networks (Level : Understand)
CO 2	Describe the principles and techniques in intrusion detection (Level: Understand)
CO 3	Analyse data collection for host-based and network-based IDSs (Level: Analyse)
CO 4	Use various protocol analyzers and Network Intrusion Detection Systems as security tools to detect network attacks and troubleshoot network problems. (Level : Apply)
CO 5	Understand intrusion infrastructure (Level: Understand)

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

PO1: An ability to independently carry out research/investigation and development work in engineering and allied streams

PO2: An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

PO3: An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program

PO4: An ability to apply stream knowledge to design or develop solutions for real world problems by following the standards

PO5: An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tool to model, analyse and solve practical engineering problems.

PO6: An ability to engage in life-long learning for the design and development related to the stream related problems taking into consideration sustainability, societal, ethical and environmental aspects

PO7: An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	☑		☑			☑	
CO 2	☑		☑			☑	
CO 3	☑		☑			☑	
CO 4	☑		☑	☑	☑		☑
CO 5	☑	☑	☑	☑	☑		☑

Assessment Pattern

Bloom's Category	End Semester Examination
Understand	50 %
Apply	30 %
Analyse	20 %
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation: 40 marks

- i. Preparing a review article based on peer reviewed original publications (minimum 10 publications shall be referred) : 15 marks
- ii. Course based task / Seminar/ Data collection and interpretation : 15 marks
- iii. Test paper (1 number) : 10 marks

Test paper shall include minimum 80% of the syllabus.

End Semester Examination Pattern:

The end semester examination will be conducted by the respective College.

There will be two parts; Part A and Part B.

Part A will contain 5 numerical/short answer questions with 1 question from each module, having 5 marks for each question. Students should answer all questions. Part B will contain 7 questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem solving and quantitative evaluation), with minimum one question from each module of which student should answer any five. Each question can carry 7 marks

Total duration of the examination will be 150 minutes.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly.

For example if the average end semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is $40+20 = 60\%$.

Course Level Assessment Questions**Course Outcome 1 (CO1):**

1. What type of attack is privilege escalation?
2. What kind of attack methods are used in a network?
3. How to prevent Mitnick attack?

Course Outcome 2 (CO2)

1. What are the methods of anomaly detection?
2. What is hybrid-based intrusion detection system?
3. Do routers play a role in intrusion detection? Explain

Course Outcome 3(CO3):

1. What is the significance of TCP dump filters?
2. What type of data is collected by intrusion detection systems?
3. What does a host-based IDS monitor?

Course Outcome 4 (CO4):

1. How protocol analyzer provide security to the network?
2. Write the format of snort options

Course Outcome 5 (CO5):

1. Compare centralized and distributed intrusion detection systems
2. How can you prevent Mitnick attack?
3. How can you differentiate automated and manual response?

Model Question Paper

QP CODE:

Reg No: _____

Name: _____

PAGES : 4

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

FIRST SEMESTER M.TECH DEGREE EXAMINATION, MONTH & YEAR

Course Code:

Course Name: INTRUSION DETECTION AND PREVENTION SYSTEMS

Max. Marks : 60

Duration: 2.5

Hours

PART A

Answer All Questions. Each Question Carries 5 Marks

1. Do routers play a role in intrusion detection? Explain.
2. Write the function of packet-filtering devices.
3. What is meant by Host-based IDS?
4. What is TCPdump filter?
5. How to prevent Mitnick attack?

(5x5=25)

Part B

(Answer any five questions. Each question carries 7 marks)

6. (a) (4)
Differentiate Denial of Service and Distributed denial of Service attacks.
- (b) Explain Buffer Overflow Attacks (3)
7. (a) Explain smurf attack . (5)
- (b) What is DNS resolution Process? (2)
8. Compare centralized and distributed intrusion detection systems (7)

9. Discuss about Snort Rules. (7)
- 10 Write notes on Architectural issues. (7)
- 11 (a) What is quantitative risk assessment? (3)
- (b) Explain threat assessment and analysis. (4)
- 12 Discuss about Automated responses. (7)

Syllabus

Module 1: Introduction

Network Attacks: Attack taxonomies, Probes, Privilege Escalation Attacks, Denial of Service and distributed Denial of service attacks, Worms attack, Routing attacks Detection approaches – Anomaly detection -Misuse detection - specification based detection – hybrid detection

Module 2: TCP/IP

IP concepts, Introduction to TCPdump and TCP Fragmentation, ICMP Stimulus and Response, DNS Packet Dissection using TCPdump, Examining IP header fields Examining embedded protocol Header fields, Real world Analysis

Module 3: Data Collection

Data Collection for Host-Based IDSs- Audit Logs, System Call Sequences Data Collection for Network-Based IDSs- SNMP, Packets, Limitations of Network -Based IDSs, Data Collection for Application- Based IDSs, Data Collection for Application -Integrated IDSs, Hybrid Data Collection

Module 4: Filters/Rules for Network Monitoring

Writing TCPdump filters, Introduction to Snort Rules, Snort Rules -Part II, Detection of Intelligent Gathering

Module 5: Intrusion Infrastructure

Mitnick attack, Architectural issues, Organizational issues, Automated and manual response, Business case for Intrusion Detection, Future directions

Reference Books

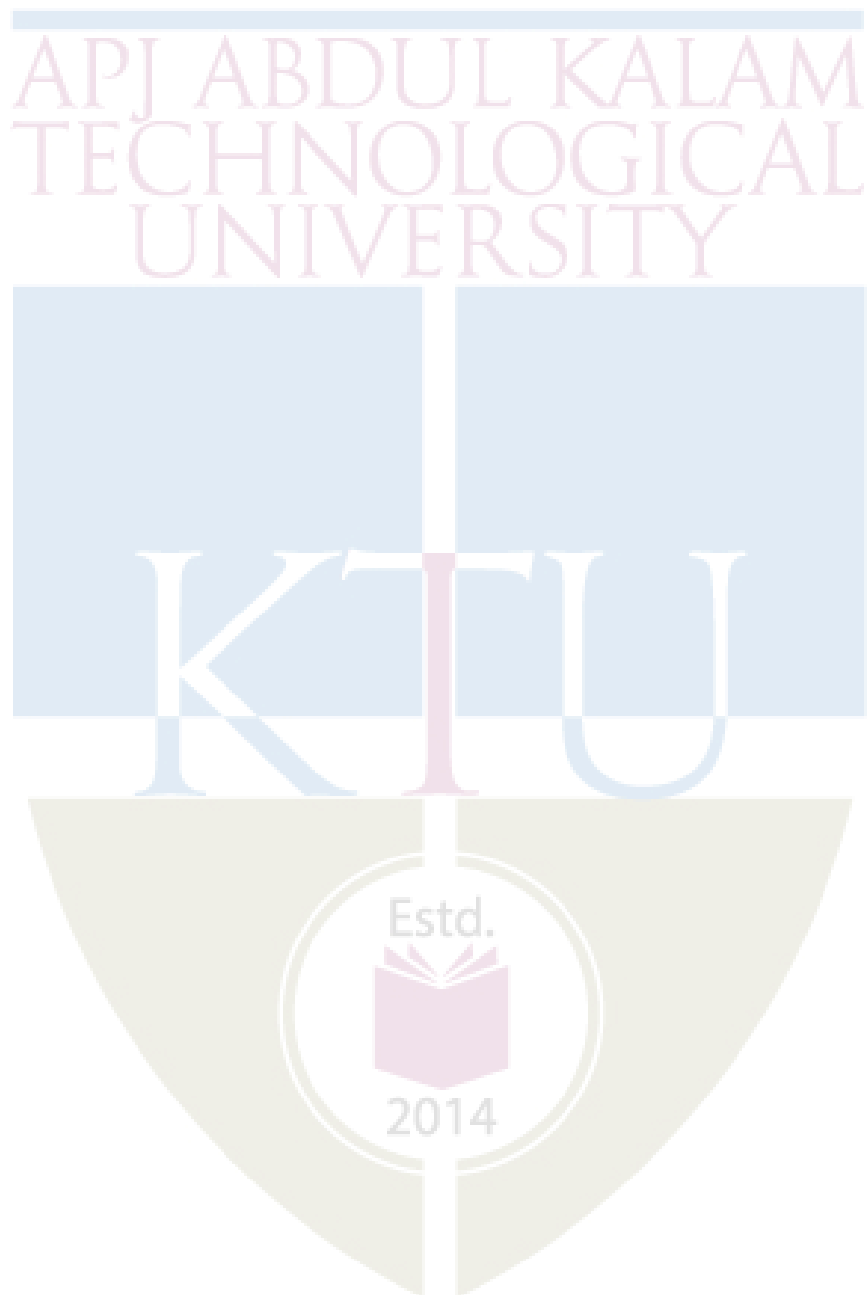
1. Ali A. Ghorbani, Wei Lu, “Network Intrusion Detection and Prevention: Concepts and Techniques”, Springer, 2010
2. Stephen Northcutt, Judy Novak, “Network Intrusion Detection and Prevention”, New Riders Publishing , 2002

3. Carl Enrolf, Eugene Schultz, Jim Mellander, "Intrusion detection and Prevention", McGraw Hill, 2004
3. Paul E. Proctor, "The Practical Intrusion Detection Handbook ",Prentice Hall , 2001.
4. Ankit Fadia and Mnu Zacharia, "Intrusion Alert", Vikas Publishing house Pvt., Ltd, 2007.
5. Earl Carter, Jonathan Hogue, "Intrusion Prevention Fundamentals", Pearson Education, 2006.

Course Plan

No	Topic	No. of Lectures (40)
1	Module 1: Introduction	6 Hours
1.1	Network Attacks: Attack taxonomies, Probes, Privilege Escalation Attacks,	2 Hour
1.2	Denial of Service and distributed Denial of service attacks, Worms attack, Routing attacks	2 Hour
1.3	Detection approaches – Anomaly detection -Misuse detection - specification based detection – hybrid detection	2 Hours
2	Module 2: TCP/IP	10 Hours
2.1	IP concepts, Introduction to TCPdump and TCP	2 Hour
2.2	Fragmentation, ICMP	2 Hours
2.3	Stimulus and Response, DNS	2 Hours
2.4	Packet Dissection using TCPdump, Examining IP header fields	2 Hours
2.5	Examining embedded protocol Header fields, Real world Analysis	2 Hours
3	Module 3: Data Collection	8 Hours
3.1	Data Collection for Host-Based IDSs- Audit Logs, System Call Sequences	2 Hour
3.2	Data Collection for Network-Based IDSs- SNMP, Packets, Limitations of Network -Based IDSs	2 Hour
3.3	Data Collection for Application- Based IDSs	2 Hour
3.4	Data Collection for Application -Integrated IDSs, Hybrid Data Collection	2 Hours
4	Module 4: Filters/Rules for Network	8 Hours
4.1	Writing TCPdump filters	2 Hours
4.2	Introduction to Snort Rules	2 Hours
4.3	Snort Rules -Part II	2 Hours
4.4	Detection of Intelligent Gathering	2 Hours
5	Module 5: Intrusion Infrastructure	8 Hours
5.1	Mitnick attack, Architectural issues	2 Hours
5.2	Organizational issues	2 Hours

5.3	Automated and manual response	INFORMATION TECHNOLOGY-IT1	2 Hours
5.4	Business case for Intrusion Detection, Future directions		2 Hours



APJ ABDUL KALAM
TECHNOLOGICAL
UNIVERSITY

SEMESTER I

PROGRAM ELECTIVE II



INFORMATION TECHNOLOGY-IT1						
CODE	COURSE NAME	CATEGORY	L	T	P	CREDIT
221EIT006	WEB TECHNOLOGIES AND WEB SECURITIES	PROGRAM ELECTIVE 2	3	0	0	3

Preamble: This course helps the learner to understand the fundamental ideas of Web Application Development using HTML5, CSS3, JavaScript and detailed knowledge about server-side programming using PHP and Frameworks.

Course Outcomes: After the completion of the course the student will be able to:

CO 1	Demonstrate an understanding in web application development using HTML and CSS3 (Cognitive Knowledge Level: Apply)
CO 2	Implement web applications using client-side scripting (Cognitive Knowledge Level: Apply)
CO 3	Implement web applications using server-side scripting (Cognitive Knowledge Level: Apply)
CO 4	Implement session management and object-oriented programming using PHP. (Cognitive Knowledge Level: Apply)
CO 5	Understand the principles of web security and encoding schemes (Cognitive Knowledge Level: Understand)

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	✓	✓	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓	✓	✓
CO 4	✓	✓	✓	✓	✓	✓	✓
CO 5	✓	✓	✓	✓	✓	✓	✓

Assessment Pattern

Bloom's Category	End Semester Examination
Understand	20%
Apply	80%
Analyse	
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation: 40 marks

Preparing a review article based on peer reviewed

Original publications (minimum 10 15 marks

Publications shall be referred):

Course based task/Seminar/Data 15 marks

Collection and interpretation:

Test paper, 1 no.: 10 marks

Test paper shall include minimum 80% of the syllabus.

End Semester Examination Pattern: 60 marks

The end semester examination will be conducted by the respective College. There will be two parts; Part A and Part B. Part A will contain 5 numerical/short answer questions with 1 question from each module, having 5 marks for each question (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students). Students should answer all questions. Part B will contain 7 questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem solving and quantitative evaluation), with minimum one question from each module of which student should answer any five. Each question can carry 7 marks.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly. For example if the average end semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is $40 + 20 = 60$ %.

Syllabus**Module – 1 (8 hours)**

Introduction: Web essentials – Clients – Servers – Communications – Mark-up Languages – HTML 5 – XHTML – Style sheets – CSS3.

Module – 2 (8 hours)

Client-side Scripting: Client-side Scripting using JavaScript – Syntax – Control Structures – Arrays and Objects – In-built objects-Host Objects: Browsers and the DOM

Module: 3 (7 hours)

Server-side Scripting: Introduction to PHP – Variables – Control Structures – Syntax – Arrays – Built-in Arrays – Form Processing-Object Oriented PHP.

Module: 4 (8 hours)

Database Applications: PHP with MySQL – mysql interface – MySQLi interface – Object Oriented Version of MySQLi

Module: 5 (8 hours)

Introduction to security testing Fundamentals - HTTP – Client-side scripting - Server-side scripting, Basic observation - observing live request headers, observing live post data, highlighting and detecting JavaScript events. Web oriented data encoding – working with base-36, base-64, URL encoded and HTML entity data.



Model Question Paper

**A P J ABDUL KALAM TECHNOLOGICAL UNIVERSITY
M.TECH DEGREE EXAMINATION
.....SEMESTER**

Branch:

Time: 2½ Hours, Maximum marks: 60

PART A (Answer all questions. Five marks each)

1. List all of the generic Internet top-level domains
2. Differentiate between MySQL and MySQLi
3. List out the features of Security Testing
4. Explain the usage of switch case statement in PHP with example
5. Write JavaScript code that will create an Object with a property named color having a String value of red

PART B(Answer any five questions. Seven marks each)

6. Write a style sheet that will cause the li elements within any ol element to be numbered in an outline style: the top-level li elements should use uppercase Roman numerals, the next level uppercase letters, the next level lowercase Roman numerals, then lowercase letters, and finally decimal numerals at the fifth level
7. What is the purpose of the HTML title attribute (not element)? Describe a scenario in which this attribute might be particularly helpful to a Web site user
8. Assume that you are adding JavaScript code to an HTML document that contains a form with a submit button. You did not modify any of the HTML or add an event listener for the click event on this button, and yet the button stopped working after you added your code to the document. Give at least two possible explanations.
9. Contrast the IE6 and DOM approaches to modifying an intrinsic event attribute, giving at least one advantage of each approach over the other.
10. Write in detail about PHP sessions
11. Explain aggregate functions in MySQL with example
12. How Live Request Headers are observed with Firebug
13. Describe in a detailed way different types of Web oriented data encoding techniques

Syllabus and Course Plan

No	Topic	No. of Lectures
Module – 1 (8 hours)		
1.1	Introduction: Web essentials	2
1.2	Clients – Servers – Communications	2
1.3	Mark-up Languages – HTML 5 – XHTML	2
1.4	Style sheets – CSS3	2
Module – 2 (8 hours)		
2.1	Client Side Scripting using JavaScript – Syntax	2
2.2	Control Structures – Arrays and Objects	2
2.3	In-built objects	2
2.4	Host Objects: Browsers and the DOM	2
Module: 3 (8 hrs)		
3.1	Server-side Scripting: Introduction to PHP	2
3.2	Variables – Control Structures – Syntax	2
3.3	Arrays – Built-in Arrays – Form Processing	2
3.4	Object Oriented PHP	2
Module: 4 (8 hrs)		
4.1	Database Applications: PHP with MySQL	3
4.2	mysql interface	2
4.3	MySQLi interface	2
4.4	Object Oriented Version of MySQLi	1
Module: 5 (8 hrs)		
5.1	Introduction to security testing Fundamentals - HTML - HTTP	2
5.2	Client-side scripting - Server-side scripting	2
5.3	Basic observation observing live request headers, observing live post data, highlighting and detecting JavaScript events.	2
5.4	Web oriented data encoding – working with base36, base 64, URL encoded and HTML entity data.	2

Reference Books

1. Jeffrey C Jackson, —Web Technology – A Computer Science Perspective, Pearson Education, 2007.
2. Brian P Hogan, —HTML5 and CSS3: Develop with Tomorrow's Standards Today, Pragmatic Bookshelf, 2010.
3. W Jason Gilmore, —Beginning PHP and MySQL: From Novice to Professional, Apress, 2010.
4. Paco Hope, Ben Walther, —Web Security Testing Cookbook: Systematic Techniques to Find Problems Fast, O'REILLY media, 2009.



CODE	COURSE NAME	CATEGORY	L	T	P	CREDIT
221EIT007	IOT ARCHITECTURE AND PROTOCOLS	PROGRAM ELECTIVE 2	3	0	0	3

Preamble: The syllabus is designed with the view of preparing the students capable of understanding different architectures, protocols, and security aspects of Internet of Things.

The students will get to know about the IoT Architectures, protocols and security systems, and methods to secure the same.

Course Outcomes: Upon completion, the student will be able to

CO 1	Apply the basic design and architecture of IoT to real-world problems. (Cognitive Level: Apply)
CO 2	Identify Data link layer, network layer and transport layer protocols for IoT. (Cognitive level: Apply)
CO 3	Identify session layer and application layer protocols for IoT. (Cognitive level: Apply)
CO 4	Implement novel Internet of Things applications. (Cognitive level: Apply)
CO5	Analyze the security issues and challenges in IoT, choose Security and Privacy algorithms, and implement appropriate security measures (Cognitive level: Apply)

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

PO1: An ability to independently carry out research/investigation and development work in engineering and allied streams

PO2: An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

PO3: An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program

PO4: An ability to apply stream knowledge to design or develop solutions for real world problems by following the standards

PO5: An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tool to model, analyze and solve practical engineering problems.

PO6: An ability to engage in life-long learning for the design and development related to the stream related problems taking into consideration sustainability, societal, ethical and environmental aspects

PO7: An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1							
CO 2							
CO 3							
CO 4							
CO5							

Bloom's Category	End Semester Examination
Apply	100%
Analyse	
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation Pattern:

Evaluation shall only be based on application, analysis or design based questions (for both internal and end semester examinations).

Continuous Internal Evaluation: 40 marks

- i. Preparing a review article based on peer reviewed original publications (minimum 10 publications shall be referred) : 15 marks
- ii. Course based task / Seminar/ Data collection and interpretation : 15 marks
- iii. Test paper (1 number) : 10 marks

Test paper shall include minimum 80% of the syllabus.

Course based task/test paper questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students.

End Semester Examination Pattern:

The end semester examination will be conducted by the respective College.

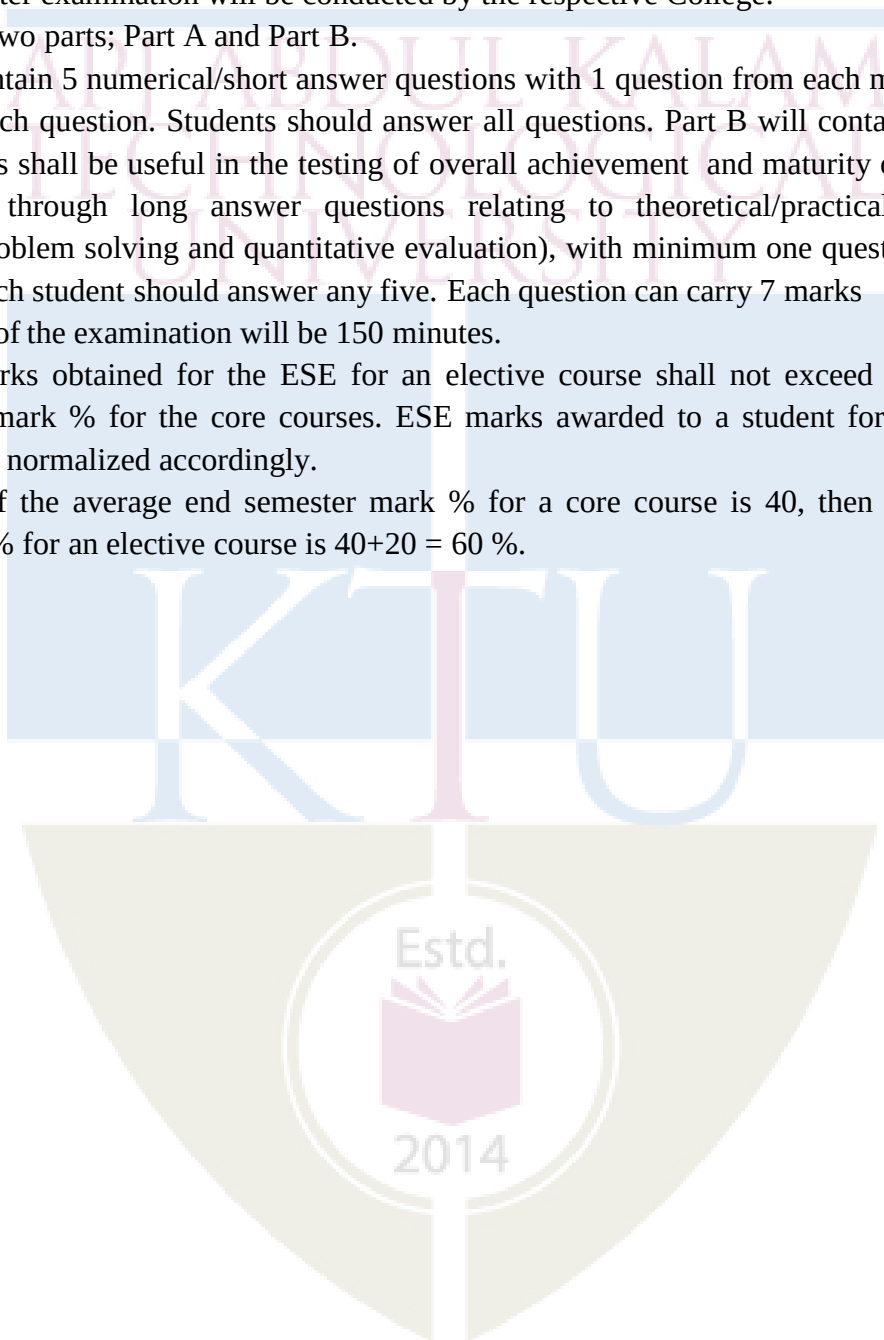
There will be two parts; Part A and Part B.

Part A will contain 5 numerical/short answer questions with 1 question from each module, having 5 marks for each question. Students should answer all questions. Part B will contain 7 questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem solving and quantitative evaluation), with minimum one question from each module of which student should answer any five. Each question can carry 7 marks

Total duration of the examination will be 150 minutes.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly.

For example if the average end semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is $40+20 = 60\%$.



Model Question Paper:**QP CODE:**

Reg No:**Name:****APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY****FIRST SEMESTER M.TECH DEGREE EXAMINATION, MONTH & YEAR****221EIT007****IoT Architectures and Protocols****Max. Marks : 60****Duration: 2.5 hrs****PART A****Answer All Questions. Each Question Carries 5 Marks**

1.	Explain any two challenges and its solution to implement smart charging of vehicles.	(5)
2.	In which scenario, users can opt 6LoWPAN and RPL? Find suitable application.	(5)
3.	Why HTTP is not suitable for IoT applications?	(5)
4.	What is the basic difference between the IoT network and Wireless Sensor Network?	(5)
5.	Why Light weight cryptographic algorithms are used to establish security in IoT?	(5) (5x5=25)
Part B (Answer any five questions. Each question carries 7 marks)		
6.	Develop a physical design.for smart parking in IoT systems	(7)

7.	Depict the IoT reference architecture. Explain the same with the help of neat diagram.	(7)
8.	Illustrate the role of data protocol MQTT and CoAP in IoT.	(7)
9.	In which scenario users can opt WiFi, Bluetooth and Zigbee? Find suitable application.	(7)
10.	How IOT deployment is done in Raspberry Pi /Arduino/Equivalent platform? Provide necessary examples and diagrams wherever necessary	(7)
11.	Describe the technologies that enable IoT .How to create a new mote using cooja simulator?	(7)
12.	Explain the Homomorphic Encryption Schemes and Key Agreement in IoT security.	(7)

Syllabus

Module I:Architecture for IoT

Architectural approach towards the future Internet of Things- Definitions and Functional Requirements, Possible Architecture for the Future Internet of Things.Characteristics of IoT - Challenges and Issues - Physical Design of IoT, Logical Design of IoT , Software Design- Control Units,IoT Functional Blocks.IETF Architecture for IoT, IoT Reference Architecture. Examples of IoT infrastructure,

Module II:IOT protocols –DDL,NWL,TL

IoT Data Link Layer & Network Layer Protocols PHY/MAC Layer (3GPP MTC, IEEE 802.11, IEEE 802.15), Wireless HART, Z Wave, WiFi, Bluetooth Low Energy, Zigbee Smart Energy, Network Layer-IPv4,IPv6, 6LoWPAN, DHCP, ICMP, RPL, CORPL, CARP, IoT Transport Layer Protocols: -TCP, MPTCP, UDP, DCCP, TLS, DTLS

Module III:

IoT protocols-Session Layer,Application Layer andService LayerIoT Session Layer Protocols andApplication Layer Protocols-: CoAP, XMPP, AMQP, MQTT,DDS, IoT Service Layer

Protocols & Security Protocols: -M2M, OMA, BBF, IPsec VPN, secure DNS, Secure HTTP, HTTPS

Module IV:

Technologies behind IoT, Four pillars of IoT paradigm, - RFID, Wireless Sensor Networks, SCADA (Supervisory Control and Data Acquisition), M2M - IoT Enabling Technologies - BigData Analytics, Cloud Computing, Embedded Systems.

Programming the microcontroller for IoT:-Working principles of sensors – IoT deployment for Raspberry Pi /Arduino/Equivalent platform – Reading from Sensors, Communication: Connecting microcontroller with mobile devices –communication through Bluetooth, wifi, and USB - Contiki OS- Cooja Simulator.

Module V:

Security and Privacy in the IoT, Security Issues in the IoT, Security Mechanisms -Traditional vs Lightweight security- Symmetric-key LWC Algorithms- Public-key (Asymmetric) LWC Algorithms- Lightweight Cryptographic Hash Functions. Homomorphic Encryption Schemes, Key Agreement, Distribution, and Security Bootstrapping. Case studies: IoT for smart cities, health care, agriculture, smart meters. M2M, Web of things, Cellular IoT, Industrial IoT, Industry 4.0, IoT standards.

Course Plan(40 Hours)

No	Topic	No. of Lectures (40)
1	Architecture for IoT(8 hours)	
1.1	Architectural approach towards the future Internet of Things- Definitions and Functional Requirements, Possible Architecture for the Future Internet of Things.	1 hour
1.2	Characteristics of IoT - Challenges and Issues	1 hour
1.3	Physical Design of IoT, Logical Design of IoT	1 hour
1.4	Software Design-Control Units,	1 hour
1.5	IoT Functional Blocks	1 hour

1.6	IETF Architecture for IoT	1 hour
1.7	IoT Reference Architecture.	1 hour
1.8	Examples of IoT infrastructure	1 hour
2	IoT protocols –DDL,NWL,TL(8 hours)	
2.1	IoT Data Link Layer -PHY/MAC Layer (3GPP MTC, IEEE 802.11, IEEE 802.15)	2hour
2.2	Wireless HART, Z Wave	1 hour
2.3	WiFi, Bluetooth Low Energy	1 hour
2.4	Zigbee Smart Energy	1 hour
2.5	Network Layer-IPv4,IPv6, 6LoWPAN, DHCP, ICMP, RPL, CORPL, CARP	2hour
2.6	IoT Transport Layer Protocols : -TCP, MPTCP, UDP, DCCP, TLS, DTLS	1 hour
3	IoT protocols- Session Layer, Application Layer and Service Layer(8 hours)	
3.1	IoT Session Layer Protocols and Application Layer Protocols-: CoAP, XMPP	2hour
3.2	AMQP	1 hour
3.3	MQTT,DDS	1 hour
3.4	IoT Service Layer Protocols: -M2M, OMA, BBF	2hour
3.5	Security Protocols-IPSec VPN	1 hour
3.6	secure DNS, Secure HTTP,HTTPS	1 hour

4	Technologies behind IoT (8 hours)	
4.1	Four pillars of IOT paradigm, - RFID, Wireless Sensor Networks, SCADA,M2M	1 hour
4.2	IOT Enabling Technologies - BigData Analytics, Cloud Computing, Embedded Systems.	1 hour
4.3	Programming the microcontroller for IoT :-Working principles of sensors	1 hour
4.4	IOT deployment for Raspberry Pi /Arduino/Equivalent platform	1 hour
4.4	Reading from Sensors, Communication: Connecting microcontroller with mobile devices	1 hour
4.5	communication through Bluetooth, wifi and USB	1 hour
4.6	Contiki OS- Cooja Simulator.	2 hour
5	Security and Privacy in the IoT(8 hours)	
5.1	Security Issues in the IoT , Security Mechanisms - Traditional vs Lightweight security	1 hour
5.2	Symmetric-key LWC Algorithms- Public-key (Asymmetric) LWC Algorithms	1 hour
5.3	Lightweight Cryptographic Hash Functions	1 hour
5.4	Homomorphic Encryption Schemes	1 hour
5.5	Key Agreement, Distribution, and Security Bootstrapping.	1 hour
5.6	Case studies: IoT for smart cities, health care, agriculture, Smart meters.	1 hour

5.7	M2M, Web of things	1 hour
5.8	Cellular IoT, Industrial IoT, Industry 4.0, IoT standards.	1 hour

Reference Books

1. Simone Cirani, Gianluigi Ferrari, Marco Picone, Luca Veltri. Internet of Things: Architectures, Protocols and Standards, 1st edition, Wiley Publications, 2019.
2. Bahga, Arshdeep, and Vijay Madisetti. Internet of Things: A hands-on approach, 1st edition, University press, 2014
3. Dr. Ovidiu Vermesan, Dr Peter Friess, Internet of Things: Converging Technologies for Smart Environments and Integrated Ecosystems, River Publishers, 2013
4. Vermesan, Ovidiu, and Peter Friess, eds. Internet of things-from research and innovation to market deployment, 1st edition, Aalborg: River publishers, 2014.
5. Tsiatsis, Vlasios, Tsiatsis, Vlasios, Stamatis Karnouskos, Jan Holler, David Boyle, and Catherine Mulligan, Internet of Things: technologies and applications for a new age of intelligence, 2nd edition, Academic Press, 2018.



CODE:	COURSE NAME:	CATEGORY	L	T	P	CREDIT
221EIT008	STORAGE MANAGEMENT AND SECURITY	PROGRAM ELECTIVE 2	3	0	0	3

Preamble: This course aims at introducing the students to the various storage architectures, their management, and security aspects.

Course Outcomes: After the completion of the course, the student will be able to

CO No.	Course Outcome (CO)	Bloom's Category Level
CO 1	Choose from various storage technologies the one that suits the required application	Apply
CO 2	Explain the storage system principles including structure, host I/O processing, and core algorithms	Understand
CO 3	Explain key characteristics, services, deployment models, and infrastructure components for a cloud computing	Understand
CO 4	Design business continuity plan	Apply
CO 5	Analyze networked storage capabilities	Apply

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1							
CO 2							
CO 3							

CO 4							
CO 5							

Assessment Pattern

Bloom's Category Levels	End Semester Examination
BL 2: Understand	60 %
BL 3: Apply	40 %
BL 4: Analyze	
BL 5: Evaluate	
BL 6: Create	

Mark distribution

Total Marks	Continuous Internal Evaluation (CIE)	Estd. End Semester Examination (ESE)	ESE Duration
100	40	60	2.5 hours

The evaluation shall only be by employing application-based, analysis-based, or design-based questions (for both internal and end-semester examinations)

Continuous Internal Evaluation (CIE) Pattern:

Preparing a review article based on peer reviewed original publications (minimum 10 publications shall be referred): 15 marks

Course-based task/Seminar/Data collection and interpretation: 15 Marks

Test paper (one only): 10 Marks

Each student shall prepare the review article based on peer reviewed publications (refer minimum 10 original publications). The test paper shall include a minimum of 80% of the syllabus.

End Semester Examination (ESE) Pattern:

There shall be two parts; Part A and Part B. Part A shall contain five (numerical) questions (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation, and understanding of the students), with one question from each module, having five marks for each question. Students shall answer all questions. Part B shall contain seven questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem-solving, and quantitative evaluation), with at least one question from each module, of which, the student shall answer any five. Each question shall carry seven marks.



Model Question Paper

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY FIRST SEMESTER M.TECH

DEGREE EXAMINATION, MONTH & YEAR

Course Code: 221EIT008

Course Name: Storage Management and Security

Max.Marks: 60

Duration: 2½ Hours,

PART A (Answer all questions. Five marks each)

1. What is a distributed storage system and why is it important?
2. Describe the concept of RAID and different RAID levels in detail?
3. Write short notes on Storage Management activities.
4. Write short notes on Backup Granularity.
5. What are different file sharing protocols and what are the I/O operations of a file?

PART B (Answer any five questions. Seven marks each)

6. Explain Information Storage.
7. Explain Explain the SCSI command model.
8. What are the components of NAS? Explain NAS Implementations.
9. Explain various forms of virtualization & Storage virtualization challenges.
10. Explain the concept of Storage security domains,
11. Explain Risk Triad.
12. Write a brief note on cloud computing infrastructure.

Syllabus

Module 1: Introduction to Storage and Management:	
Introduction, History: computing, networking, storage, Intelligent Storage Array. Need for storage networking - SAN, NAS, SAN/NAS, Distributed Storage Systems, SCSI command Model and fiber channel connectivity,	8 Hours
Module 2: Storage System Environment	
Storage Components; Data organization: File vs. Block, Object, Data store; File sharing protocols, I/O operations, Error Management: Disk error Management – RAID Error Management, Distributed Systems Error Management	8 Hours
Module 3: Large Storage Systems	
Cloud/Web-based systems, Storage Management: - Device Management – NAS Management, Storage Virtualization -Forms of Virtualization, Types of Storage Virtualization, Block level virtualization, File level virtualization.	8 Hours
Module 4: Introduction to Business connectivity	
Information Availability, BC terminology, Life cycles, Backup Purpose - Granularity, Backup and restore operations	8 Hours
Module 5: Storage Security Domains	
Overview of storage security, Storage Security Framework, Risk Triad – Storage Security Domains, Storage management challenges and ideal solutions	8 Hours

Reference Books

1. EMC Education Services” Information Storage and Management: Storing, Managing and Protecting Digital Information”, John Wiley & Sons, 2010
2. David Alexander, Amanda French, Dave Sutton “Information Security Management Principles” BCS, The Chartered Institute 2008
3. John Chirillo, Scott Baul “Storage Security: Protecting SANs, NAS and DAS”, Wiley, 2003
4. Meeta Gupta, Storage Area Network Fundamentals, Pearson Education Ltd 2002 Network Management: Principles and Practice, Mani Subramanian, 2/e, Pearson, 2010
5. Foster Stockwell, “A history of information storage and retrieval” McFarland, 2001

Course Plan

Sl. No.	Topic	No. of Lectures
1	Introduction to Storage and Management 8 Hours	
1.1	Introduction, History: computing, networking, storage, Intelligent Storage Array	1
1.2	Need for storage networking - SAN, NAS, SAN/NAS	3
1.3	Distributed Storage Systems	2
1.4	SCSI command Model and fiber channel connectivity	2
2	Storage System Environment 8 Hours	
2.1	Storage Components; Data organization: File vs. Block, Object, Data store;	3
2.2	File sharing protocols, I/O operations	2
2.3	Error Management: Disk error Management – RAID Error Management	2

2.4	Distributed Systems Error Management	1
3	Large Storage Systems 8 Hours	
3.1	Cloud/Web-based systems	2
3.2	Storage Management: - Device Management – NASManagement	3
3.3	Storage Virtualization -Forms of Virtualization	2
3.4	Types of Storage Virtualization, Block level virtualization, File level virtualization	1
4	Introduction to Business connectivity 8 Hours	
4.1	Information Availability	2
4.2	BC terminology, Life cycles	2
4.3	Backup Purpose -Granularity	2
4.4	Backup and restore operations	2
5	Storage security Domains 8 Hours	
5.1	Overview of storage security	2
5.2	Storage Security Framework	2
5.3	Risk Triad – Storage Security Domains	2
5.4	Storage management challenges and ideal solutions	2

CODE:	COURSE NAME:	CATEGORY	L	T	P	CREDIT
221EIT009	NETWORK PROGRAMMING AND MANAGEMENT	PROGRAM ELECTIVE 2	3	0	0	3

Preamble: This course is intended for students to develop the necessary skills for creating robust and scalable network applications and to impart basic knowledge for managing networks.

Course Outcomes: After the completion of the course, the student will be able to

CO No.	Course Outcome (CO)	Bloom's Category Level
CO 1	Compare the features of elementary TCP/IP protocols and sockets	BL 2: Understand
CO 2	Develop applications using TCP sockets and use various socket options	BL 3: Apply
CO 3	Illustrate the functions of UDP and SCTP sockets	BL 2: Understand
CO 4	Develop high-performance scalable applications using the concept of multithreading	BL 3: Apply
CO 5	Summarize simple network management protocols and their practical issues	BL 2: Understand

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1							
CO 2							
CO 3							

CO 4							
CO 5							

Assessment Pattern

Bloom's Category Levels	End Semester Examination
BL 2: Understand	60 %
BL 3: Apply	40 %
BL 4: Analyze	
BL 5: Evaluate	
BL 6: Create	

Mark distribution

Total Marks	Continuous Internal Evaluation (CIE)	End Semester Examination (ESE)	ESE Duration
100	40	60	2.5 hours

The evaluation shall only be by employing application-based, analysis-based, or design-based questions (for both internal and end-semester examinations)

Continuous Internal Evaluation (CIE) Pattern:

Review article preparation/Mini project: 15 Marks

Course-based task/Seminar/Data collection and interpretation: 15 Marks

Test paper (one only): 10 Marks

Each student shall prepare the review article based on peer reviewed publications (refer minimum 10 original publications). The test paper shall include a minimum of 80% of the syllabus.

End Semester Examination (ESE) Pattern:

There shall be two parts; Part A and Part B. Part A shall contain five (numerical) questions (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation, and understanding of the students), with one question from each module, having five marks for each question. Students shall answer all questions. Part B shall contain seven questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem-solving, and quantitative evaluation), with at least one question from each module, of which, the student shall answer any five. Each question shall carry seven marks.



Model Question Paper

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY FIRST SEMESTER M.TECH

DEGREE EXAMINATION, MONTH & YEAR

Course Code: 221EIT009

Course Name: Network Programming and Management

Max.Marks: 60

Duration: 2½ Hours,

PART A (Answer all questions. Five marks each)

1. Explain the fields present in Sockaddr_In structure.
2. What are the boundary conditions for crashing of TCP server host?
3. Develop a simple UDP application server and client program. The server should echo back the message sent by the client.
4. Why is a mutex always associated with a condition available?
5. What are the limitations of using SNMP?

PART B (Answer any five questions. Seven marks each)

6. Explain any five functions meant for byte order transformation.
7. Explain the general format for getsockopt and setsockopt for specifying TCP socket options.
8. Develop a UDP socket program to enable a client to send a float value to the server. The server process should increment the value of the number it receives by 2. The server should print both the value it receives and the value that it sends. The client should print the value it receives from the server.
9. Write a simple TCP/IP client-server application program that accepts client connections and echoes back whatever text is sent to the server. The server should be multithreaded, i.e., it can serve multiple concurrent clients.
10. Explain the key elements for TCP/IP network management.
11. Write a program that uses TCP sockets to accept a binary IP address and return the host name corresponding to that address from the host address structure.
12. Compare and contrast the key features of SNMPv3 with SNMPV2.

Syllabus

Module 1: Introduction to TCP/IP and Socket Programming	
Introduction: Overview of Transport Layer protocols - TCP, UDP, and SCTP; Socket Programming – Introduction to Sockets – Socket address Structures – Byte ordering and manipulation functions – address conversion functions – Elementary TCP Sockets – socket, connect, bind, listen, accept, read, write, close functions – Iterative Server – Concurrent Server.	8 Hours
Module 2: TCP/IP Application Development and Socket Options	
TCP Echo Server – TCP Echo Client – POSIX Signal handling – Server with multiple clients – boundary conditions: Server process Crashes, Server host Crashes, Server Crashes and reboots, Server Shutdown – I/O multiplexing – I/O Models – select function – shutdown function – TCP echo Server (with multiplexing) – poll function – TCP echo Client (with Multiplexing). Socket options – getsockopt and setsockopt functions – generic socket options – IP socket options – ICMP socket options – TCP socket options.	9 Hours
Module 3: Elementary UDP and SCTP Sockets	
Elementary UDP sockets – UDP echo Server – UDP echo Client – Multiplexing TCP and UDP sockets – Domain name system – gethostbyname function – Ipv6 support in DNS – gethostbyaddr function – getservbyname and getservbyport functions, Elementary SCTP sockets, interface models, functions for bind, connect, address mapping and shutdown.	8 Hours
Module 4: Advanced Socket Programming	
Ipv4 and Ipv6 interoperability – threaded servers – thread creation and termination – TCP echo server using threads – Mutex – condition variables – raw sockets – raw socket creation – raw socket output – raw socket input – ping program – traceroute program.	7 Hours
Module 5: Network Management	
SNMP network management concepts – SNMP management information – standard MIBs – SNMPv1 protocol and Practical issues, RMON – Basic concepts, statistics collection, alarms and filters, Practical issues, RMON2 – Overview, Protocol Directory and Protocol Distribution Groups, RMON2 Host and Matrix Groups, Practical Issues, SNMPv2 – Protocol Operations, MIBs, SNMPv3 – Architecture and Applications, Cryptographic algorithms.	8 Hours

Reference Books

1. UNIX Network Programming, Volume 1: Networking APIs - Sockets and XTI, W Richard Stevens, Prentice Hall, 1997
2. SNMP, SNMPv2, SNMPv3, and RMON 1 and 2, William Stallings, 3/e, Addison Wesley, 1998
3. Internetworking with TCP/IP Vol. II: Design, Implementation, and Internals, Douglas E Comer, David L Stevens, 3/e, Pearson, 2015
4. Network Management: Principles and Practice, Mani Subramanian, 2/e, Pearson, 2010

Course Plan

Sl. No.	Topic	No. of Lectures
1	Introduction to TCP/IP and Socket Programming 8 Hours	
1.1	Introduction: Overview of Transport Layer protocols - TCP, UDP, and SCTP	2
1.2	Socket Programming – Introduction to Sockets – Socket address Structures – Byte ordering and manipulation functions – address conversion functions	2
1.4	Elementary TCP Sockets – socket, connect, bind, listen, accept, read, write, close functions	2
1.5	Iterative Server – Concurrent Server	2
2	TCP/IP Application Development and Socket Options 9 Hours	
2.1	TCP Echo Server – TCP Echo Client – POSIX Signal handling – Server with multiple clients	2
2.2	Boundary conditions: Server process Crashes, Server host Crashes, Server Crashes and reboots	2
2.3	Server Shutdown – I/O multiplexing – I/O Models – select function	2

	– shutdown function	
2.4	TCP echo Server (with multiplexing) – poll function – TCP echo Client (with Multiplexing)	1
2.5	Socket options – getsocket and setsocket functions – generic socket options – IP socket options – ICMP socket options – TCP socket options	2
3	Elementary UDP and SCTP Sockets8 Hours	
3.1	Elementary UDP sockets – UDP echo Server – UDP echo Client	2
3.2	Multiplexing TCP and UDP sockets – Domain name system – gethostbyname function	2
3.3	Ipv6 support in DNS – gethostbyaddr function – getservbyname and getservbyport functions	2
3.4	Elementary SCTP sockets, interface models, functions for bind, connect, address mapping, and shutdown	2
4	Advanced Socket Programming7 Hours	
4.1	Ipv4 and Ipv6 interoperability – threaded servers – thread creation and termination	2
4.2	TCP echo server using threads – Mutex – condition variables	2
4.3	raw sockets – raw socket creation – raw socket output – raw socket input	2
4.4	ping program – trace route program	1
5	Network Management8 Hours	

5.1	SNMP network management concepts – SNMP management information – standard MIBs	1
5.2	SNMPv1 protocol and Practical issues	1
5.3	RMON – Basic concepts, statistics collection, alarms and filters, Practical issues	2
5.4	RMON2 – Overview, Protocol Directory and Protocol Distribution Groups, RMON2 Host and Matrix Groups, Practical Issues	2
5.5	SNMPv2 – Protocol Operations, MIBs, SNMPv3 – Architecture and Applications, Cryptographic algorithms	2



CODE:	COURSE NAME:	CATEGORY	L	T	P	CREDIT
221EIT010	NETWORK FORENSICS	PROGRAM ELECTIVE 2	3	0	0	3

Preamble: This course is intended for students to have sound technical knowledge in basic and advanced topics in network forensics including the application of traffic acquisition software, packet and flow traffic analysis, statistical traffic protocols network tunneling strategies, and malware evolution trends.

Course Outcomes: After the completion of the course, the student will be able to

CO No.	Course Outcome (CO)	Bloom's Category Level
CO 1	Familiarize various traffic acquisition software used for network forensics	BL 2: Understand
CO 2	Analyze network traffic in various levels – protocol, packet and flow analysis	BL 4: Analyze
CO 3	Analyze various statistical wireless traffic protocols	BL 4: Analyze
CO 4	Identify the different types of network devices and logs	BL 2: Understand
CO 5	Characterize various network tunneling strategies and malware evolution trends	BL 2: Understand

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 7
CO 1								
CO 2								

CO 3								
CO 4								
CO 5								

Assessment Pattern

Bloom's Category Levels	End Semester Examination
BL 2: Understand	60 %
BL 3: Apply	20 %
BL 4: Analyze	20 %
BL 5: Evaluate	
BL 6: Create	

Mark distribution

Total Marks	Continuous Internal Evaluation (CIE)	End Semester Examination (ESE)	ESE Duration
100	40	60	2.5 hours

The evaluation shall only be by employing application-based, analysis-based, or design-based questions (for both internal and end-semester examinations)

Continuous Internal Evaluation (CIE) Pattern:

Preparing a review article based on peer reviewed original publications (minimum 10 publications shall be referred): 15 marks

Course-based task/Seminar/Data collection and interpretation: 15 Marks

Test paper (one only): 10 Marks

Each student shall prepare the review article based on peer reviewed publications (refer minimum 10 original publications). The test paper shall include a minimum of 80% of the syllabus.

End Semester Examination (ESE) Pattern:

There shall be two parts; Part A and Part B. Part A shall contain 5 (numerical) questions (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation, and understanding of the students), with one question from each module, having 5 marks for each question. Students shall **answer all questions**. Part B shall contain 7 questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem-solving, and quantitative evaluation), with at least one question from each module, of which, the student shall answer **any five**. Each question shall carry 7 marks.



Model Question Paper

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY FIRST SEMESTER M.TECH

DEGREE EXAMINATION, MONTH & YEAR

Course Code: 221EIT010

Course Name: Network Forensics

Max.Marks: 60

Duration: 2½ Hours,

PART A (Answer all questions. Five marks each)

1. Give the syntax of the Berkeley packet filter interface.
2. Compare flow analysis versus packet analysis.
3. What are the types of sensors based on their deployment?
4. List and briefly explain any four attacks in wireless networks.
5. What happens in TLS protocol implementation when a client contacts a server for communication?

PART B (Answer any five questions. Seven marks each)

6. List and explain the different types of digital evidence.
7. What are protocol analyzers? Explain the options in Wireshark for packet filtering.
8. What is network flow monitoring? Compare the features of Netflow, sFlow and IPFIX for monitoring network traffic.
9. Explain the roles played by switches, routers and firewalls in a network.
10. Explain any three strategies for covert tunneling.
11. Explain protocol awareness and behavioural analysis in network intrusion detection.
12. Explain the features of libpcap and WinPcap for acquiring network evidence.

Syllabus

Module 1: Technical Fundamentals of Network Forensics	
Introduction to Network Forensics: Concepts in digital evidence - Challenges related to network evidence - Network forensic investigative methodology; Technical Fundamentals: Sources of network-based evidence - On the wire, In the air, Switches, Routers, DHCP servers, Name servers, Authentication servers, Central log servers; Evidence Acquisition - Physical interception of cable, Radio frequency, Hubs, Switches; Traffic acquisition software - Libpcap and WinPcap, The Berkeley packet filter (BPF) language, Tcpdump, Wireshark, Tshark, Dumpcap; Active acquisition - Common interfaces, Inspection without access strategy.	9 Hours
Module 2: Traffic Analysis	
Traffic Analysis: Protocol Analysis – Protocol analysis tool, Protocol analysis technique; Packet analysis - Packet analysis tool, Packet analysis technique; Flow analysis - Flow analysis tool, Flow analysis technique; Higher-layer traffic analysis - Common higher-layer protocols, Higher-layer analysis tool, Higher layer analysis techniques.	7 Hours
Module 3: Statistical Flow Analysis	
Statistical flow analysis - Processor overview – Sensors - sensors types, Sensor software, Sensor placement; Flow record export protocols – Netflow, IPFIX, sFlow - Collection and aggregation; Wireless access point - Type of WAP, WAP evidence; Wireless traffic capture and analysis: Spectrum analysis, Wireless passive evidence acquisition; Common attacks: Sniffing, Rogue wireless access points, Evil twin, WEP cracking, Locating wireless devices.	8 Hours
Module 4: Network Devices	
Network Devices: Intrusion detection and analysis – Typical NIDS/NIPS functionality – sniffing - Higher layer protocol awareness - Alerting on suspicious bits modes of detection - Modes of Detection –Signature-based analysis - Protocol awareness - Behavioural analysis - Type of NIDS/NIPS, NIDS/NIPS evidence acquisition - Comprehensive packet logging - Event log aggregation - Correlation and analysis - Source of logs, Operating system logs, Application logs, Physical device logs, Network equipment logs; Network log architecture - Three type of logging architecture; Remote logging - Common pitfall and strategy, Log aggregation and analysis tool; Collecting and Analysing Evidence, Switches: Content-addressable memory table - ARP, Switch evidence; Routers - Type of	8 Hours

routers, Router evidence; Firewalls - Type of firewalls, Firewall evidence.	
Module 5:Network Tunneling and Malware Forensics	
Advanced Topics - Web Proxies - Web proxy functionality: Caching, URI Filtering, Content Filtering; Distributed cache – Squid – Squid configuration, Squid access logfile, Squid Cache; Encrypted web traffic - Transport Layer Security – Gaining access to encrypted content – Commercial TLS/SSL Interception Tools; Network Tunneling - Tunneling for functionality – Inter-switch link (ISL), Generic routing encapsulation (GRE), Tunneling for confidentiality; Internet protocol security (IPSec) - Transport layer security (TLS) and Secure socket layer (SSL) – Covert Tunneling: Covert Tunneling strategy, TCP sequence number; DNS Tunneling - ICMP tunnels – Malware forensics - Trends in malware evolution, Network behaviour of malware - Propagation, Command and control Communications, Payload behaviour.	8 Hours

Reference Books

1. Network Forensics: Tracking Hackers through Cyberspace, Sherri Davidoff, Jonathan Ham, Pearson, 2012.
2. Computer Forensics: Computer Crime Scene Investigation, John R Vacca, Jones & Bartlett Learning, LLC, 3/e, 2010
3. Network Forensics, Ric Messier, Wiley Publishers, 2017
4. Computer Evidence: Collection & Preservation, Christopher L T Brown, Charles River Media, 2005
5. Investigating Network Intrusions and Cybercrime (CHFI), Computer Forensic Series by EC-Council, Cengage Learning, 2016

Course Plan and Lecture Schedule

Sl. No.	Topic	No. of Lectures
1	Technical Fundamentals of Network Forensics (9 hours)	
1.1	Introduction to Network Forensics: Concepts in digital evidence - Challenges related to network evidence - Network forensic investigative methodology	2
1.2	Technical Fundamentals: Sources of network-based evidence - On the wire, In the air, Switches, Routers, DHCP servers, Name servers, Authentication servers, Central log servers	2
1.3	Evidence Acquisition - Physical interception of cable, Radio frequency, Hubs, Switches	1
1.4	Traffic acquisition software - Libpcap and WinPcap, The Berkeley packet filter (BPF) language, Tcpdump, Wireshark, Tshark, Dumpcap	2
1.5	Active acquisition - Common interfaces, Inspection without access strategy	2
2	Traffic Analysis(7 hours)	
2.1	Traffic Analysis: Protocol Analysis – Protocol analysis tool, Protocol analysis technique	2
2.2	Packet analysis - Packet analysis tool, Packet analysis technique	2
2.3	Flow analysis - Flow analysis tool, Flow analysis technique	1
2.4	Higher-layer traffic analysis - Common higher-layer protocols, Higher-layer analysis tool, Higher layer analysis techniques	2
3	Statistical Flow Analysis (8 hours)	
3.1	Statistical flow analysis - Processor overview – Sensors - sensors	2

	types, Sensor software, Sensor placement	
3.2	Flow record export protocols – Netflow, IPFIX, sFlow - Collection and aggregation	2
3.3	Wireless access point - Type of WAP, WAP evidence; Wireless traffic capture and analysis: Spectrum analysis, Wireless passive evidence acquisition	2
3.4	Common attacks: Sniffing, Rogue wireless access points, Evil twin, WEP cracking, Locating wireless devices	2
4	Network Devices(8 hours)	
4.1	Network Devices: Intrusion detection and analysis – Typical NIDS/NIPS functionality – sniffing	2
4.2	Higher layer protocol awareness - Alerting on suspicious bits modes of detection - Modes of Detection –Signature-based analysis - Protocol awareness - Behavioural analysis - Type of NIDS/NIPS, NIDS/NIPS evidence acquisition	2
4.3	Comprehensive packet logging - Event log aggregation - Correlation and analysis - Source of logs, Operating system logs, Application logs, Physical device logs, Network equipment logs	2
4.4	Network log architecture - Three types of logging architecture; Remote logging - Common pitfall and strategy, Log aggregation and analysis tool; Collecting and Analysing Evidence, Switches: Content-addressable memory table - ARP, Switch evidence; Routers - Type of routers, Router evidence; Firewalls - Type of firewalls, Firewall evidence	2
5	Network Tunneling and Malware Forensics(8 hours)	
5.1	Advanced Topics - Web Proxies - Web proxy functionality: Caching, URI Filtering, Content Filtering; Distributed cache –	2

	Squid – Squid configuration, Squid access logfile, Squid Cache	
5.2	Encrypted web traffic - Transport Layer Security – Gaining access to encrypted content – Commercial TLS/SSL Interception Tools	2
5.3	Network Tunneling - Tunneling for functionality – Inter-switch link (ISL), Generic routing encapsulation (GRE), Tunneling for confidentiality; Internet protocol security (IPSec) - Transport layer security (TLS) and Secure socket layer (SSL), Covert Tunneling: Covert Tunneling strategy, TCP sequence number; DNS Tunneling - ICMP tunnels	2
5.4	Malware forensics - Trends in malware evolution, Network behaviour of malware - Propagation, Command and control Communications, Payload behaviour.	2



CODE:	COURSE NAME:	CATEGORY	L	T	P	CREDIT
221EIT011	MATHEMATICAL MODEL FOR INTERNET	PROGRAM ELECTIVE 2	3	0	0	3

Preamble: The course aims to impart advanced concepts on complex networks and game theory. Using these mathematical concepts, the course aims to enable students to model research problems related to the Internet.

Course Outcomes: After the completion of the course the student will be able to

CONo.	Course Outcomes (CO)	Bloom's Category Level
CO 1	Explain the evolution of random networks based on graph theoretical concepts.	Level2: Understand
CO 2	Interpret the nature of scale-free networks based on Barabasi-Albert model.	Level 2: Understand
CO 3	Analyse the robustness of the network using complexity theory	Level4: Analyse
CO 4	Calculate Nash Equilibrium of Normal form and Extensive form games	Level 3: Apply
CO 5	Model problems related to internet pricing, congestion control, and cyber security using game theoretical concepts.	Level4: Analyse

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

PO1:An ability to independently carry out research/investigation and development work in engineering and allied streams

PO2:An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

PO3:An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program

PO4:An ability to apply stream knowledge to design or develop solutions for real world problems by following the standards

PO5:An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tool to model, analyse and solve practical engineering problems.

PO6:An ability to engage in life-long learning for the design and development related to the stream related problems taking into consideration sustainability, societal, ethical and environmental aspects

PO7:An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	-	-	✓	-	-	-	-
CO 2	-	-	✓	-	-	-	-
CO 3	✓	✓	✓	✓	-	-	✓
CO 4	-	-	✓	-	-	-	-
CO 5	✓	✓	✓	✓	✓	✓	✓

Assessment Pattern

Bloom's Category	End Semester Examination
Understand	40%
Apply	40%
Analyse	20%
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation: 40 marks

- i. Preparing a review article based on peer reviewed original publications (minimum 10 publications shall be referred) : 15 marks
- ii. Course based task / Seminar/ Data collection and interpretation : 15 marks
- iii. Test paper (1 number) : 10 marks

Test paper shall include minimum 80% of the syllabus.

End Semester Examination Pattern:

The end semester examination will be conducted by the respective College. There will be two parts; Part A and Part B. Part A will contain 5 numerical/short answer questions with 1 question from each module, having 5 marks for each question (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students). Students should answer all questions. Part B will contain 7 questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem solving and quantitative evaluation), with minimum one question from each module of which student should answer any five. Each question can carry 7 marks.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly. For example, if the average end semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is $40+20 = 60$ %.

Course Level Assessment Questions

Course Outcome 1 (CO1):

1. Explain the evolution of random network and the emergence of giant component.
2. Can structure of the Internet be explained using random graphs? Discuss.

Course Outcome 2 (CO2)

1. Explain briefly the Scale Free phenomenon and its significance.
2. Explain Barabasi-Albert model of growth of complex network.

Course Outcome 3(CO3):

1. Does the phenomenon of inverse percolation help in assessing robustness of a network? Investigate.
2. “Mitigation of Cascaded attacks can be done using Lazarus effect” Do you agree with this statement? Substantiate your answer with example analysis.

Course Outcome 4 (CO4):

1. Calculate Nash equilibrium for Cournot duopoly game.
2. Comment on the uniqueness of Nash equilibrium with the example of Stag Hunt Game.

Course Outcome 5 (CO5):

1. Model congestion control problem using game theory. Investigate on the existence and uniqueness of Nash Equilibrium in the model.
2. Using a game theoretical model of DDoS attack in a computer network, Investigate on the effect of having firewall as the countermeasure.



QP CODE:

Reg No: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

FIRST SEMESTER M.TECH DEGREE EXAMINATION, MONTH & YEAR

IT12022-PE2 (F)

Mathematical Models for Internet

Max. Marks : 60

Duration: 2.5 hrs

PART A

(Answer all questions. Each Question Carries 5 Marks)

1. Compare the structure of random network and real network based on cluster size distribution.
2. Use degree distribution to mathematically explain the structure of Internet.
3. Discuss small world phenomena in scale free networks with examples.
4. Give an example of Zero-Sum game and explain its importance in game theoretical analysis.
5. Analyze the Nash equilibrium of advertising game from the perspectives of normal form game and extensive form game.

PART B

(Answer any FIVE Questions. Each Question Carries 7 Marks)

6. Define Average Path length, Clustering Coefficient and Degree Distribution. Compare the behavior of these in the context of random networks and real networks.
7. Discuss on the emergent behavior of random networks
- 8 Use degree distribution to mathematically explain the structure of Internet.
9. Compare and contrast coordination game with coalitional game with examples.

10. Explain extensive form game and Backward Induction.

11. Analyze the Nash equilibrium of advertising game from the perspectives of normal form game and extensive form game..

12. Explain the modelling of DDoS attack using game theory. Explore the possibility of solving the problem as a non-zero-sum game.

Syllabus

Module 1: Graph models (8 Hours)
Relevant Concepts From Graph Theory - Undirected, directed, signed, weighted and spatial networks, Paths, connected components, Directed Acyclic Graphs, Bipartite graphs, Max-flow/min-cut, Clustering coefficients. The Random Network Model - Introduction, Number of Links, Degree Distribution in random network, The Evolution of a Random Network, Small Worlds, Clustering Coefficient in random network, Watts-Strogatz model.
Module 2: Scale-Free Networks and The Barabasi-Albert model (7 Hours)
Power Laws and Scale-Free Networks, Discrete Formalism, Continuum Formalism, Hubs, The Meaning of Scale-Free, Universality, Ultra-Small Property, The Role of the Degree Exponent. The Barabasi-Albert model-Growth and Preferential Attachment, The Barabási-Albert Model, Degree Dynamics, Degree Distribution in BA Model, Measuring Preferential Attachment, Non-linear Preferential Attachment, The Origins of Preferential Attachment, Diameter and Clustering Coefficient.
Module 3: Modelling Network Robustness(8 Hours)
Network Robustness- Percolation Theory, Inverse Percolation Transition and Robustness, Robustness of Scale-free Networks, Molloy-Reed Criterion, Critical Threshold, Attack Tolerance, Cascading Failures, Modelling Cascading Failures, Failure Propagation Model, Branching Model, Building Robustness, Designing Robust Networks.
Module 4: Game theory-Introduction (8 Hours)
Games in Normal Form, Dominant Strategies, Nash Equilibrium, Randomization and Mixed Strategies. Sequentiality, Extensive Form Games, and Backward Induction
Module 5: Game Theoretical Modelling (9 Hours)
Example games and analysis, application of game theory in internet pricing, congestion control and cyber security

Reference Books

1. A-L. Barabási , Network Science,Cambridge University Press 2016, Available online:<http://barabasi.com/networksciencebook/>
2. M.E.J. Newman, Networks - An introduction, Oxford Univ Press, 2010.
3. D. Easley and J. Kleinberg, Networks, Crowds and Markets, Cambridge Univ Press.
4. R. Cohen and S. Havlin, Complex Networks - Structure, Robustness and Function , Cambridge Univ Press, 2010.
5. Frank Kelly, Mathematical Modeling of the Internet, Proc. Fourth International Congress on Industrial and Applied mathematics, Edinburgh, 1999.
6. Frank Kelly, Models for a Self Managed Internet, Philosophical Transactions of the Royal Society A358, pp. 2335-2348, 2000
7. A Brief Introduction to the Basics of Game Theory-Matthew O. Jackson, Stanford University
8. Algorithms, Games, and the Internet, Christos H. Papadimitriou,University of California, Berkeley,christos@cs.berkeley.edu
9. A Game-Theoretic Framework for Congestion Control in General Topology Networks, Tansu Alpcan, Tamer Basar (alpcan, tbasar)@control.csl.uiuc.edu
10. Wu, Q., Shiva, S., Roy, S., Ellis, C. and Datla, V., 2010, April. On modeling and simulation of game theory-based defense mechanisms against DoS and DDoS attacks. In *Proceedings of the 2010 spring simulation multiconference* (pp. 1-8).

Course Contents and Lecture Schedule

No	Topic	No. of Lectures 40 Hours
1	Graph models	8 Hours
1.1	Relevant Concepts From Graph Theory - Undirected, directed, signed, weighted and spatial networks ,Paths, connected components.	2 Hours
1.2	Directed Acyclic Graphs, Bipartite graphs Max-flow/min-cut, Clustering coefficients.	1 Hour
1.3	The Random Network Model - Introduction, Number of Links, Degree Distribution in random network.	2 Hours
1.4	The Evolution of a Random Network.	1 Hour
1.5	Small Worlds, Clustering Coefficient in random network	1 Hour
1.6	Watts-Strogatz model.	1 Hour
2	Scale Free Property and The Barabasi-Albert model	7 Hours
2.1	Scale Free Property - Introduction, Power Laws and Scale-Free Networks, Discrete Formalism, Continuum Formalism, Hubs.	2 Hours
2.2	The Meaning of Scale-Free, Universality	1 Hour
2.3	Ultra-Small Property, The Role of the Degree Exponent.	1 Hour
2.4	Growth and Preferential Attachment, The Barabási-Albert	2 Hours

	Model, Degree Dynamics, Degree Distribution in BA Model, Measuring Preferential Attachment, Non-linear Preferential Attachment.	
2.5	The Origins of Preferential Attachment, Diameter and Clustering Coefficient.	1 Hour
3	Degree correlations and Network Robustness	8 Hours
3.1	Network Robustness - Introduction, Percolation Theory	1 Hour
3.2	Inverse Percolation Transition and Robustness, Robustness of Scale-free Networks.	2 Hours
3.3	Molloy-Reed Criterion, Critical Threshold, Attack Tolerance.	2 Hours
3.4	Modelling Cascading Failures, Failure Propagation Model, Branching Model	1 Hour
3.5	Building Robustness, Designing Robust Networks.	2 Hours
4	Game theory-Introduction	8 Hours
4.1	Games in Normal Form, Dominant Strategies	2 Hours
4.2	Nash Equilibrium	1 Hour
4.3	Randomization and Mixed Strategies.	2 Hours
4.4	Sequentiality, Extensive Form Games	1 Hour
4,5	Backward Induction	2Hours
5	Game Theoretical Modelling	9 Hours
5.1	Study of Example games	2 Hours
5.2	Analysis of Nash equilibrium of example games	2 Hours
5.3	Application of game theory in internet pricing	2 Hours
5.4	Game theoretical modelling of congestion control	1 Hour
5.5	Game theoretical modelling in cyber security	2 Hours

CODE	COURSE NAME	CATEGORY	L	T	P	CREDIT
221RGE100	RESEARCH METHODOLOGY & IPR	General Course	2	0	0	2

Preamble:

This course introduces the strategies and methods related to scientific research. The students are also trained in the oral presentation with visual aids and writing technical thesis/reports/research papers. The salient aspects of publication and patenting along with the crucial role of ethics in research is discussed.

Course Outcomes

After the completion of the course the student will be able to

CO 1	Approach research projects with enthusiasm and creativity.
CO 2	Conduct literature survey and define research problem
CO 3	Adopt suitable methodologies for solution of the problem
CO 4	Deliver well-structured technical presentations and write technical reports.
CO 5	Publish/Patent research outcome.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	☑	☑				☑	
CO 2	☑	☑				☑	
CO 3	☑	☑				☑	
CO 4	☑	☑				☑	
CO 5	☑	☑				☑	
CO 6	☑	☑				☑	

Assessment Pattern

Bloom's Category	End Semester Examination
Apply	70 %
Analyse	30 %
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation Pattern:

Course based task: 15 marks

Some sample course based tasks that can be performed by the student given below.

- *Conduct a group discussion based on the good practices in research.*
- *Conduct literature survey on a suitable research topic and prepare a report based on this.*

Seminar: 15 marks

Test paper: 10 marks

End Semester Examination Pattern:

Total Marks: 60

The examination will be conducted by the respective college with the question provided by the University. The examination will be for 150 minutes and contain two parts; Part A and Part B. Part A will contain 6 short answer questions with 1 question each from modules 1 to 4, and 2 questions from module 5. Each question carries 5 marks. Part B will contain only 1 question based on a research article from the respective discipline and carries 30 marks. The students are to answer the questions based on that research article.

Model Question paper

QP Code:		Total Pages:
Reg No.: _____		Name: _____
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY FIRST SEMESTER M. TECH DEGREE EXAMINATION, Month & Year		
Course Code: 221RGE100		
Course Name: RESEARCH METHODOLOGY & IPR		
Max. Marks: 60		Duration: 2.5 Hours
PART A		
	Answer all questions. Each question carries 5 marks	Marks
1	Discuss the salient recommendations for great research recommended by Richard Hamming in his famous talk "You and Your Research"	30
2	What are the characteristics of a good research question? Discuss with an example.	
3	Explain the difference between continuum, meso-scale and micro scale approaches for numerical simulation.	
4	Discuss any four rules of scientific writing.	
5	What are the requirements for patentability?	
6	What are the differences between copyright and trademark protection?	
	Read the given research paper and write a report that addresses the following issues (The paper given can be specific to the discipline concerned)	
7	What is the main research problem addressed?	3
8	Identify the type of research	3
9	Discuss the short comings in literature review if any?	6
10	Discuss appropriateness of the methodology used for the study	6
11	Discuss the significance of the study and summarize the important results and contributions by the authors	6
12	Identify limitations of the article if any.	6

Syllabus and Course Plan

No	Topic	No. of Lectures
1	Introduction	
1.1	Meaning and significance of research, Skills, habits and attitudes for research, Types of research,	1
1.2	Characteristics of good research, Research process	1
1.3	Motivation for research: Motivational talks on research: "You and Your Research"- Richard Hamming	1
1.4	Thinking skills: Levels and styles of thinking, common-sense and scientific thinking, examples, logical thinking, division into sub-problems, verbalization and awareness of scale.	1
1.5	Creativity: Some definitions, illustrations from day to day life, intelligence versus creativity, creative process, requirements for creativity	1
2	Literature survey and Problem definition	
2.1	Information gathering – reading, searching and documentation, types of literature.	1
2.2	Integration of research literature and identification of research gaps	1
2.3	Attributes and sources of research problems, problem formulation, Research question, multiple approaches to a problem	1
2.4	Problem solving strategies – reformulation or rephrasing, techniques of representation, Importance of graphical representation, examples.	1
2.5	Analytical and analogical reasoning, examples, Creative problem solving using Triz, Prescriptions for developing creativity and problem solving.	1
3	Experimental and modelling skills	
3.1	Scientific method, role of hypothesis in experiment, units and dimensions, dependent and independent variables, control in experiment	1
3.2	precision and accuracy, need for precision, definition, detection, estimation and reduction of random errors, statistical treatment of data, definition, detection and elimination of systematic errors,	1
3.3	Design of experiments, experimental logic, documentation	1

3.4	Types of models, stages in modelling, curve fitting, the role of approximations, problem representation, logical reasoning, mathematical skills.	1
3.5	Continuum/meso/micro scale approaches for numerical simulation, Two case studies illustrating experimental and modelling skills.	1
4	Effective communication - oral and written	
4.1	Examples illustrating the importance of effective communication, stages and dimensions of a communication process.	1
4.2	Oral communication –verbal and non-verbal, casual, formal and informal communication, interactive communication, listening, form, content and delivery, various contexts for speaking- conference, seminar etc.	1
4.3	Guidelines for preparation of good presentation slides.	1
4.4	Written communication – Rules of scientific writing, form, content and language, layout, typography and illustrations, nomenclature, reference and citation styles, contexts for writing – paper, thesis, reports etc. Tools for document preparation-LaTeX.	1
4.5	Common errors in typing and documentation	1
5	Publication and Patents	
5.1	Relative importance of various forms of publication, Choice of journal and reviewing process, Stages in the realization of a paper.	1
5.2	Research metrics-Journal level, Article level and Author level, Plagiarism and research ethics	1
5.3	Introduction to IPR, Concepts of IPR, Types of IPR	1
5.4	Common rules of IPR practices, Types and Features of IPR Agreement, Trademark	1
5.5	Patents- Concept, Objectives and benefits, features, Patent process – steps and procedures	2

Reference Books

1. E. M. Phillips and D. S. Pugh, "How to get a PhD - a handbook for PhD students and their supervisors", Viva books Pvt Ltd.
2. G. L. Squires, "Practical physics", Cambridge University Press
3. Antony Wilson, Jane Gregory, Steve Miller, Shirley Earl, Handbook of Science Communication, Overseas Press India Pvt Ltd, New Delhi, 1st edition 2005
4. C. R. Kothari, Research Methodology, New Age International, 2004
5. Panneerselvam, Research Methodology, Prentice Hall of India, New Delhi, 2012.

6. Leedy P. D., Practical Research: Planning and Design, McMillan Publishing Co.
7. Day R. A., How to Write and Publish a Scientific Paper, Cambridge University Press, 1989.
8. William Strunk Jr., Elements of Style, Fingerprint Publishing, 2020
9. Peter Medawar, 'Advice to Young Scientist', Alfred P. Sloan Foundation Series, 1979.
10. E. O. Wilson, Letters to a Young Scientist, Liveright, 2014.
11. R. Hamming, You and Your Research, 1986 Talk at Bell Labs.



API ABDUL KALAM
TECHNOLOGICAL
UNIVERSITY

Semester II

Discipline: INFORMATION TECHNOLOGY

Stream: IT1 (NETWORK ENGINEERING,
NETWORKS AND SECURITY)

222TIT101	ADVANCED MACHINE LEARNING	CATEGORY	L	T	P	CREDIT
		DISCIPLINE CORE	3	0	0	3

Preamble: This course deals with various algorithms to enable computers to learn data without being explicitly programmed. It covers an insight into various types of machine learning algorithms, optimization techniques and strategies for model generation and evaluation. The fundamental and advanced machine learning algorithms helps the learner to meet the required research and industry standards.

Course Outcomes:

After the completion of the course the student will be able to

CO 1	Explain the Machine Learning algorithms, basic parameter estimation and optimization methods. (Cognitive Knowledge Level: Understand)
CO 2	Illustrate the concepts of regression and classification techniques (Cognitive Knowledge Level: Apply)
CO 3	Use unsupervised learning methods and dimensionality reduction techniques in real world problems. (Cognitive Knowledge Level: Apply)
CO 4	Apply reinforcement learning for AI applications. (Cognitive Knowledge Level: Apply)
CO 5	Choose suitable model parameters for different machine learning techniques and to evaluate a model performance. (Cognitive Knowledge Level: Apply)

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

PO1: An ability to independently carry out research/investigation and development work in engineering and allied streams

PO2: An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

INFORMATION TECHNOLOGY

P03: An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program

P04: An ability to apply stream knowledge to design or develop solutions for real world problems by following the standards

P05: An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tool to model, analyse and solve practical engineering problems.

P06: An ability to engage in life-long learning for the design and development related to the stream related problems taking into consideration sustainability, societal, ethical and environmental aspects

P07: An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	P07
CO 1	√		√	√	√	√	√
CO 2	√		√	√	√	√	
CO 3	√		√	√	√	√	
CO 4	√		√	√	√	√	
CO 5	√		√	√	√	√	

Assessment Pattern

Bloom's Category	End Semester Examination
Understand	20%
Apply	80%
Analyse	
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation (CIE) Pattern:

Evaluation shall only be based on application, analysis or design-based questions (for both internal and end semester examinations).

Continuous Internal Evaluation: 40 marks

Mini project/ Assignment: 20 Marks

Course-based task/Seminar/Quiz: 10 Marks

Test paper (one only): 10 Marks

Course based task/test paper questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students.

End Semester Examination Pattern:

The End-Semester Examination shall be conducted by the University. There shall be two parts:

Part A and Part B. Part A shall contain five (numerical) questions (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation, and understanding of the students), with one question from each module, having five marks for each question. Students shall answer all questions.

Part B shall contain seven questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem-solving, and quantitative evaluation), with at least one question from each module, of which, the student shall answer any five. Each question shall carry seven marks

Course Level Assessment Questions**Course Outcome 1 (CO1):**

1. Distinguish between supervised learning and Reinforcement learning. Illustrate with an example.
2. Differentiate between overfitting and underfitting with suitable diagram
3. Suppose $X_1, X_2, \dots, X_n$ are i.i.d. random variables with density function $f(x|\sigma) = 1/2\sigma \exp^{-3}(-|x|/\sigma)$. Find the maximum likelihood estimate of σ
4. Use the method of moment to estimate the parameters μ and σ for the normal density

$$f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$$

based on a random sample $X_1, \dots, X_n$.

5. Suppose that $X_1, \dots, X_n$ form a random sample from uniform distribution on the interval $(0, \theta)$, where the parameter $\theta > 0$ but is unknown. Find MLE of θ .
6. Let $X_1, \dots, X_n$ be an i.i.d. sample from a Poisson distribution with parameter λ , i.e.,

$$P(X = x|\lambda) = \frac{\lambda^x e^{-\lambda}}{x!}.$$

find the MLE of the parameter λ .

7. Suppose that $X_1, \dots, X_n$ form a random sample from distribution for which the pdf $f(x|\theta)$ is as follows:

$$f(x|\theta) = \begin{cases} \theta x^{\theta-1}, & \text{for } 0 < x < 1 \\ 0, & \text{for } x \leq 0 \end{cases}$$

Also suppose that the value of θ is unknown ($\theta > 0$). Find the MLE of θ .

8. What is the difference between Batch gradient descent and Stochastic gradient descent?

Course Outcome 2 (C02)

1. Given the set of values $X = (3, 9, 11, 5, 2)^T$ and $Y = (1, 8, 11, 4, 3)^T$. Evaluate the regression coefficients.
2. What is meant by kernel trick in context of support vector machines how is it used to find a SVM classifier?
3. Suppose you are using RBF kernel in SVM with high Gamma value. What does this signify?
4. What are some ways to choose data for a regression without running into a multi-collinearity problem?
5. What is the effect of collinearity on Lasso vs Ridge regression? Which is better in the case of collinearity?
6. Describe various types of regression models based on type of functions.
7. What are the benefits of decision tree pruning? Also mention the different types of pruning methods.
8. Why doesn't ridge regression force some of the coefficients to be exactly zero for sufficiently large lambda whereas the LASSO does?
9. Why is it that the lasso, unlike ridge regression, results in coefficient estimates that are exactly equal to zero?
10. What is the significance of optimal separating hyperplanes in SVM?
11. What is meant by kernel trick in context of support vector machines? How is it used to find a SVM classifier?
12. Describe the features of soft margin hyperplane and explain how it is computed.
13. Describe the significance of Kernel functions in SVM. List any two kernel functions.
14. Explain any two model combination scheme to improve the accuracy of a classifier.
15. How do we learn the conditional probability tables (CPT) in Bayesian networks if information about some variables is missing? How are these variables called?
16. Consider the training data in the following table where Play is a class attribute. In the table, the Humidity attribute has values "L" (for low) or "H" (for high), Sunny has values "Y" (for yes) or "N" (for no), Wind has values "S" (for strong) or "W" (for weak), and Play has values "Yes" or "No". What is class label for the following day (Humidity=L, Sunny=N, Wind=W), according to naïve Bayesian classification?

Humidity	Sunny	Wind	Play
L	N	S	No
H	N	W	Yes
H	Y	S	Yes
H	N	W	Yes
L	Y	S	No

Course Outcome 3(CO):

1. Cluster the following eight points (with (x, y) representing locations) into three clusters:

A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9)

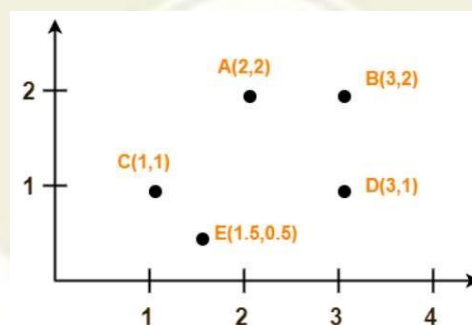
Initial cluster centers are: A1(2, 10), A4(5, 8) and A7(1, 2).

The distance function between two points $a = (x_1, y_1)$ and $b = (x_2, y_2)$ is defined as-

$$P(a, b) = |x_2 - x_1| + |y_2 - y_1|$$

Use K-Means Algorithm to find the three cluster centers after the second iteration.

2. Use K-Means Algorithm to create two clusters-



3. Explain Gaussian Mixture Model model.
4. Explain DBSCAN algorithm for density based clustering. List out its advantages compared to K-means
5. How linear discriminant analysis is used for dimensionality reduction?
6. Compare Feature Extraction and Feature Selection techniques. Explain how dimensionality can be reduced using subset selection procedure

7. What are 3 ways of reducing dimensionality?
8. How factor analysis is used for dimensionality reduction?
9. Write any four commonly used equations for measures of distance between data points of numeric data in Hierarchical Clustering
10. Write the algorithm for DIANA (DlvisiveANAlysis) of hierarchical clustering technique.
11. Is principal component analysis a supervised learning problem? Why? Describe how principal component analysis is carried out to reduce dimensionality of data sets.
12. What is difference between factor analysis and PCA?
13. Consider two data points in two-dimensional A (5,8) and B (8,9). Calculate (a) Manhattan distance (b) Squared Euclidean distance between A & B.
14. Describe the forward selection and backward selection algorithm for implementing the subset selection procedure for dimensionality reduction
15. Distinguish between K-means Clustering and Hierarchical Clustering technique.
16. Consider the two dimensional patterns (2, 1), (3, 5), (4, 3), (5, 6), (6,7), (7, 8). Compute the principal component using PCA Algorithm.

Course Outcome 4 (CO4):

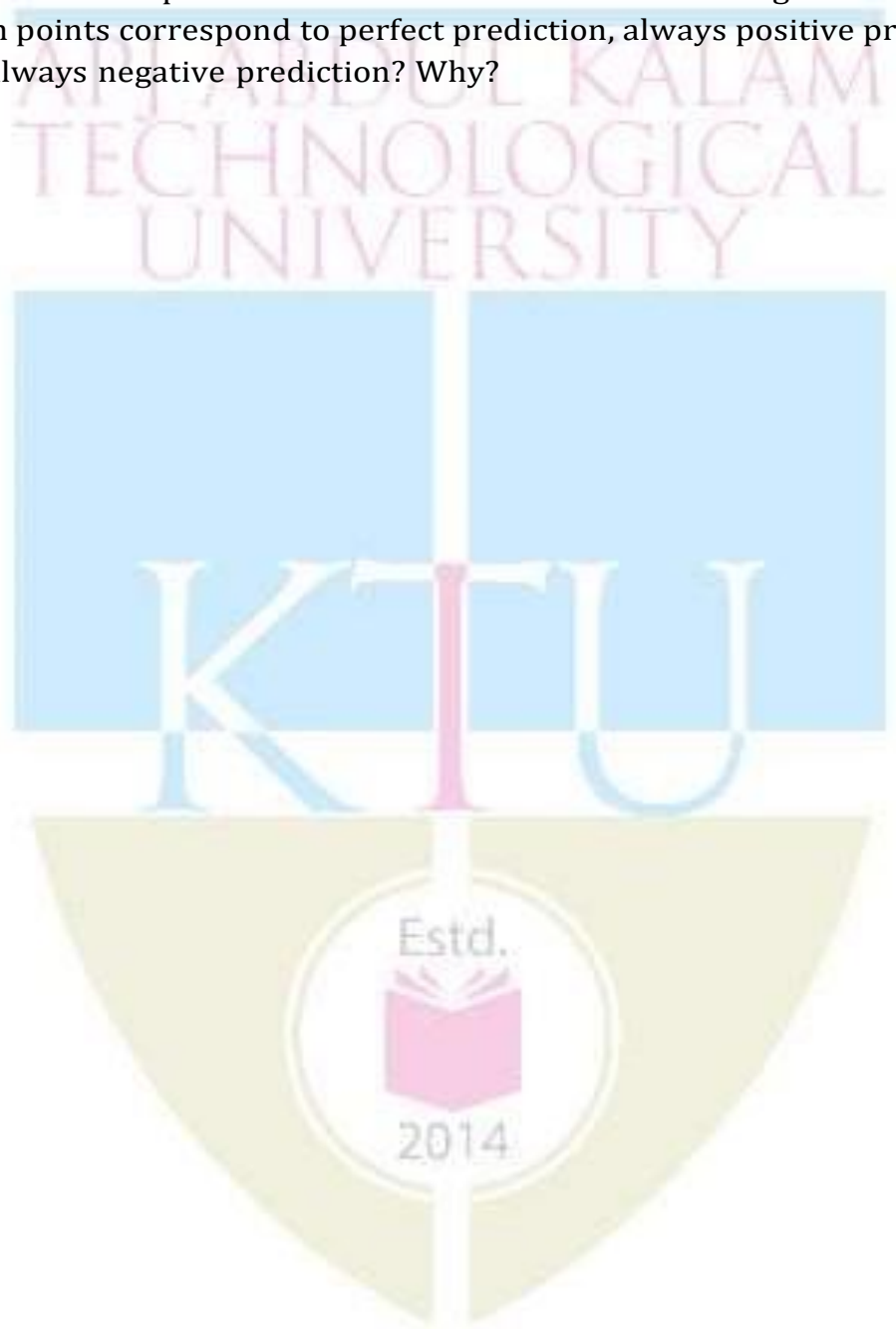
1. Which type of problems can be solved by reinforcement learning?
2. What are the problem solving methods for reinforcement learning?
3. What problems does reinforcement learning solve elaborate with any real time example?
4. Why is the Bellman Equation useful? What are the advantages of Bellman equation to MDP?
5. What is MDP? What are the different ways of classifying MDP? Provide some application of each class.

Course Outcome 5 (CO5):

1. Consider a Machine Learning Model which can detect and hence filter out all the spams. Let's take a sample set of 1000 emails. Out of 1000, 100 are spam and rest 900 are relevant emails. Our Machine Learning Model predicts 30 as spam and 970 as relevant. Compute precision and recall of the model. Also draw the confusion matrix.
2. Express the confusion matrix table for two class datasets.
3. Suppose 10000 patients get tested for flu; out of them, 9000 are actually healthy and 1000 are actually sick. For the sick people, a test was positive for 620 and negative for 380. For the healthy people, the same test was

positive for 180 and negative for 8820. Construct a confusion matrix for the data and compute the precision and recall for the data.

4. With an example classification problem, explain the following terms: a) Hyper parameters b) Training set c) Validation sets d) Bias e) Variance.
5. What are ROC space and ROC curve in machine learning? In ROC space, which points correspond to perfect prediction, always positive prediction and always negative prediction? Why?



Model Question Paper

QP CODE:

Reg No: _____

Name: _____

PAGES: 4

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

FIRST SEMESTER M. TECH DEGREE EXAMINATION, MONTH & YEAR

Course Code: 221TCS100

Course Name: ADVANCED MACHINE LEARNING

Max. Marks : 60

Duration: 2.5 Hours

PART A

Answer All Questions. Each Question Carries 5 Marks

1. Explain the principle of the Batch and Stochastic gradient descent algorithm.
2. Explain the general MLE method for estimating the parameters of a probability distribution
3. State the mathematical formulation of the SVM problem. Give an outline of the method for solving the problem
4. What are differences between value iteration and policy iteration ?
5. What does it mean for a classifier to have a high precision but low recall?

Part B

(Answer any five questions. Each question carries 7 marks)

6. Compute the Maximum Likelihood estimate for the parameter λ in the Poisson distribution whose probability function is

$$f(x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad x = 0, 1, 2, \dots, n$$

7. Consider a naive Bayes classifier with 3 boolean input variables, X_1 , X_2 and X_3 , and one boolean output, Y . How many parameters must be estimated to train such a naive Bayes classifier? How many parameters would have to be estimated to learn the above classifier if we do not make the naive Bayes conditional independence assumption

8. Suppose that $X_1, \dots, X_n$ form a random sample from a uniform distribution on the interval (θ_1, θ_2) , with the pdf as follows:

9.
$$f(x|\theta) = \begin{cases} \frac{1}{\theta_2 - \theta_1}, & \text{for } \theta_1 \leq x \leq \theta_2 \\ 0, & \text{otherwise} \end{cases}$$

- a. Also suppose that the values of θ_1 and θ_2 are unknown $(-\infty < \theta_1 < \theta_2 < \infty)$. Find the MLE's of θ_1 and θ_2 .
10. Given the data in Table, reduce the dimension from 2 to 1 using the Principal Component Analysis (PCA) algorithm.

Feature	Example 1	Example 2	Example 3	Example 4
X_1	4	8	13	7
X_2	11	4	5	14

11. Write and explain the Bellman equations for finite horizon MDP. Explain the optimal policy.
12. Suppose there are three classifiers A, B and C. The (FPR, TPR) measures of the three classifiers are as follows – A (0,), B (1, 1), C (1, 0.5). Which can be considered as a perfect classifier? Justify your answer

SYLLABUS

Module I: Machine Learning Fundamentals I (8 Hours)

Supervised, Semi-supervised, Unsupervised learning and Reinforcement learning. Parameter estimation: Maximum likelihood estimation, Maximum a Posteriori Estimation (MAP), ordinary least squares, weighted least squares. Optimization: Exhaustive search, Gradient Descent Algorithm: Batch, Stochastic and Mini batch, Genetic Algorithm.

Module II: Predictive Supervised Methods II (8 Hours)

Regression Algorithms: Linear, Multivariate, Logistic, Polynomial and Bayesian Linear Regression. Overfitting, Regularization techniques: Ridge regression, lasso (L1 Norm), Dropout.

Classification: Perceptron, Naive Bayes, Decision trees and Support vector machine

Module III: Unsupervised Methods (8 Hours)

Clustering- K-means, DBSCAN, BIRCH, Hierarchical based clustering, Outlier Analysis. Expectation-maximization (EM) of Gaussian mixture model. **Dimensionality Reduction-** Factor analysis, Canonical Correlation Analysis, Linear discriminant Analysis, Non-negative matrix factorization, Probabilistic PCA, Independent components analysis (ICA) randomized optimization.

Module IV: Reinforcement Learning (8 Hours)

The Reinforcement Learning problem: evaluative feedback, non-associative learning, Rewards and returns, Markov Decision Processes, Value functions, optimality, Bellman equation and approximation. Dynamic programming: value iteration, policy iteration, Gambler's Problem, asynchronous DP, generalized policy iteration.

Module V: Evaluation Metrics (8 Hours)

Classification: Accuracy, confusion matrix, log-loss, Precision, Recall, and F1 score. and AUC-ROC. Regression: Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), R-Squared (R^2), Adjusted R-Squared (Adjusted R^2). Clustering: Purity, Jaccard index, Normalized Mutual Information, Clustering Accuracy, Silhouette coefficient and Dunn's Index. Reinforcement learning evaluation.

COURSE PLAN

No	Topics	No. of Lectures (40)
1	Module-I (Machine Learning Fundamentals)	8 hours
1.1	Supervised, Semi-supervised, Unsupervised learning and Reinforcement learning	1
1.2	Parameter estimation: maximum likelihood estimation	1
1.3	Maximum a Posteriori Estimation (MAP)	1
1.4	Ordinary least squares, Weighted least squares	1
1.5	Optimization: Exhaustive search	1
1.6	Gradient Descent Algorithm	1
1.7	Batch, Stochastic and Mini batch	1
1.8	Genetic Algorithm	1
2	Module-II (Predictive Supervised Methods)	8 hours
2.1	Regression Algorithms: Linear, Multivariate	1
2.2	Logistic Regression, Polynomial regression	1
2.3	Bayesian Linear Regression	1
2.4	Overfitting, Regularization techniques: Ridge regression	1
2.5	lasso (L1 Norm), Dropout	1
2.6	Classification: Perceptron, Naive Bayes	1
2.7	Decision trees	1
2.8	Support vector machine	1
3	Module-III (Unsupervised Methods)	8 hours
3.1	Clustering- K-means, DBSCAN	1

INFORMATION TECHNOLOGY

3.2	BIRCH, Hierarchical based clustering	1
3.3	Outlier Analysis	1
3.4	Expectation-maximization (EM) of Gaussian mixture model	1
3.5	Dimensionality Reduction - Factor analysis	1
3.6	Linear discriminant Analysis, Non-negative matrix factorization	1
3.7	Probabilistic PCA	1
3.8	Independent components analysis (ICA) randomized optimization.	1
4	Module-IV (Reinforcement Learning)	8 hours
4.1	The Reinforcement Learning problem: evaluative feedback	1
4.2	Agent-Environment interface, Rewards and returns	1
4.3	Markov Decision Processes (MDP)	1
4.4	Optimal value functions	1
4.5	Bellman equation, Need of approximation	1
4.6	Dynamic Programming : value iteration, policy iteration,	1
4.7	Gambler's Problem	1
4.8	Asynchronous DP, generalized policy iteration	1
5	Module-V (Evaluation Metrics)	8 hours
5.1	Classification: Accuracy, confusion matrix, log-loss, Precision	1
5.2	Recall, and F1 score. and AUC-ROC	1
5.3	Regression: Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), Mean Squared Error (MSE)	1

5.4	Root Mean Squared Error (RMSE), R-Squared (R^2), Adjusted R-Squared (Adjusted R^2)	1
5.6	Clustering: Purity, Jaccard index, Normalized Mutual Information	1
5.7	Clustering Accuracy, Silhouette coefficient and Dunn's Index.	1
5.8	Reinforcement learning evaluation.	1

Reference Books

1. Bishop, C. M. Pattern Recognition and Machine Learning. Springer-Verlag New York. 2006.
2. Kevin P. Murphy. Machine Learning: A Probabilistic Perspective, MIT Press 2012.
3. Trevor Hastie, Robert Tibshirani, Jerome Friedman, The Elements of Statistical Learning,
4. Second edition Springer 2007.
5. Ethem Alpaydin, Introduction to Machine Learning, 2nd edition, MIT Press 2010.
6. Tom Mitchell, Machine Learning, McGraw-Hill, 1997.
7. R. S. Sutton and A. G. Barto. Reinforcement Learning - An Introduction. MIT Press. 1998.



222TIT100	PATH AND FLOW PROBLEMS FOR NETWORKS	CATEGORY	L	T	P	CREDIT
		PROGRAMME CORE	3	0	0	3

Preamble: This course covers Path and Flow Problems which are graph-theoretical problems that find application in the design of computer networks and protocols and their solutions. The syllabus includes core topics such as shortest path problems, maximum flow problems, and minimum cost flow problems, including novel polynomial-time algorithms for solving these problems. The last module throws light on how linear programming can be applied to flow problems.

Course Outcomes: After the completion of the course, the student will be able to:

CO 1	Explain the significance of flow problems in computer networks, and apply them in appropriate contexts. (Level: Understand)
CO 2	Compare and contrast different classes of algorithms to solve the Shortest Path Problem in computer networks. (Level: Analyze)
CO 3	Devise various algorithms to solve the Maximum Flow Problem and analyze them. (Level: Apply)
CO 4	Evaluate various algorithms for computing minimum cost flows. (Level: Analyze)
CO 5	Correlate linear programming with network flows and use linear programming to solve network flow problems. (Level: Apply)

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	√					√	
CO 2	√		√	√	√		

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CO 3	√	√	√	√	√		
CO 4	√	√	√	√	√		
CO 5	√		√	√		√	

Assessment Pattern

Bloom's Category	End Semester Examination
Understand	20%
Apply	80%
Analyze	
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation Pattern:

Micro project/Course-based project/Course-based task: 20 Marks

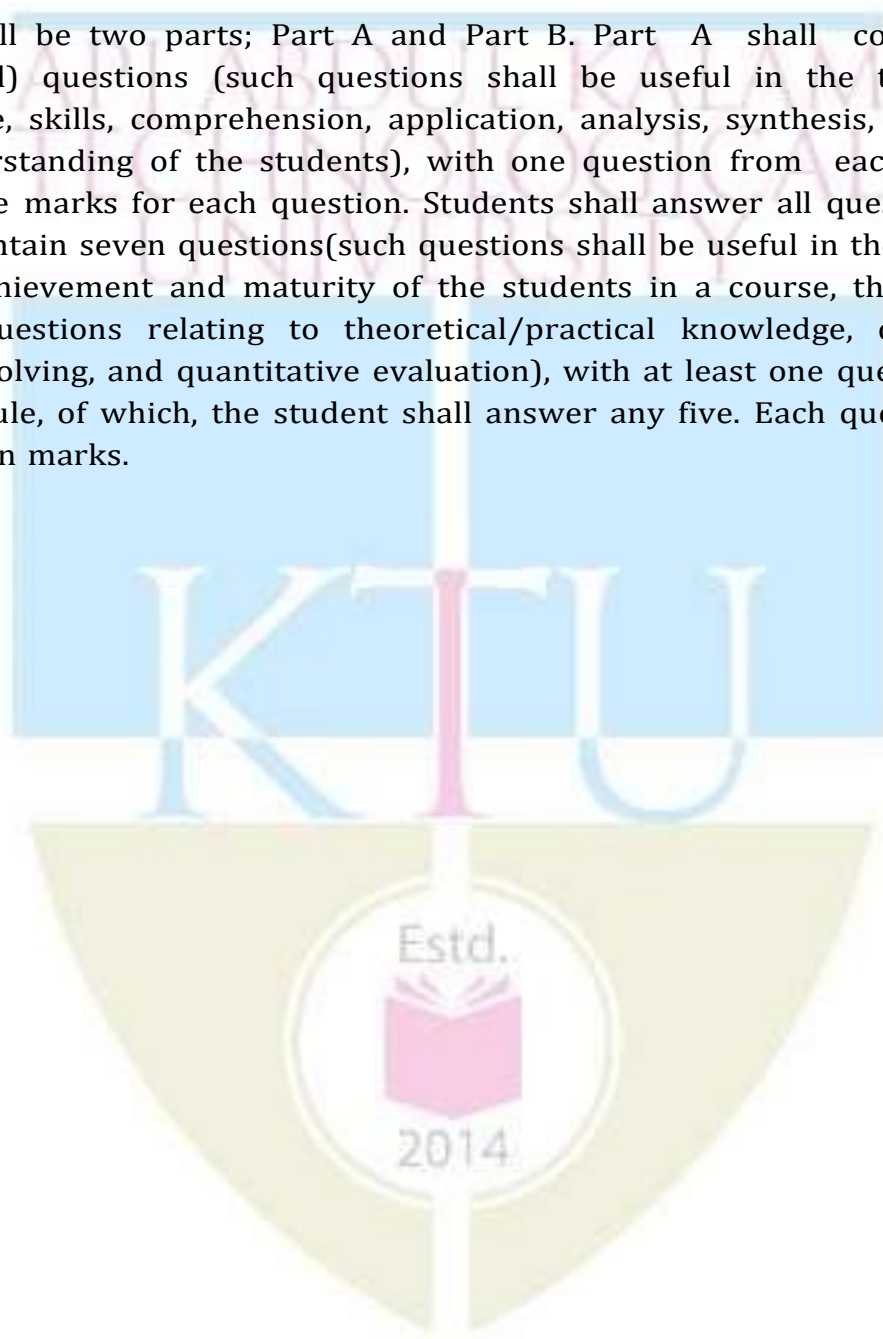
Seminar/Quiz: 10 Marks

Test paper (one only): 10 Marks

The project shall be done individually. Group projects are not permitted. The test paper shall include a minimum of 80% of the syllabus.

End Semester Examination Pattern:

There shall be two parts; Part A and Part B. Part A shall contain five (numerical) questions (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation, and understanding of the students), with one question from each module, having five marks for each question. Students shall answer all questions. Part B shall contain seven questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem-solving, and quantitative evaluation), with at least one question from each module, of which, the student shall answer any five. Each question shall carry seven marks.



MODEL QUESTION PAPER FOR END-SEMESTER EXAMINATION

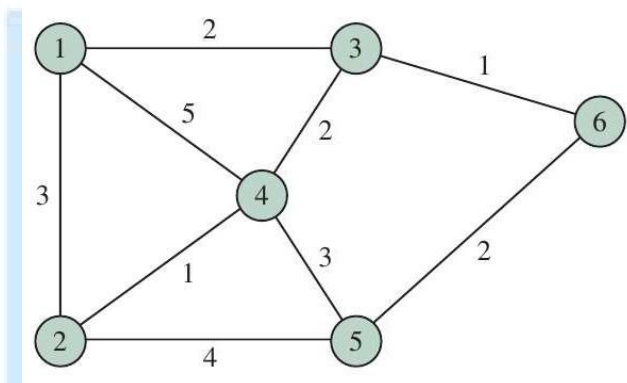
Time: 2½ Hours

Maximum

marks: 60

PART A (Answer all questions. Five marks each)

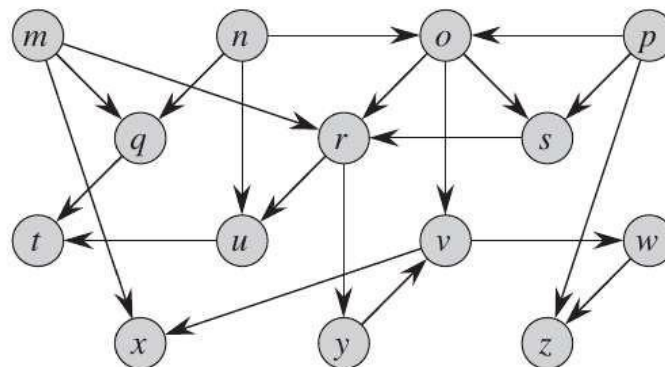
1. State and prove the flow decomposition theorem.
2. Taking node 1 as the source, run the Dijkstra's algorithm on the network shown below:



3. How can the shortest path algorithms be used to detect negative cycles in a graph?
4. Give an overview of the primal-dual algorithm.
5. State the LPP formulation of the weighted bipartite matching problem. How can it be solved?

PART B (Answer any five questions. Seven marks each)

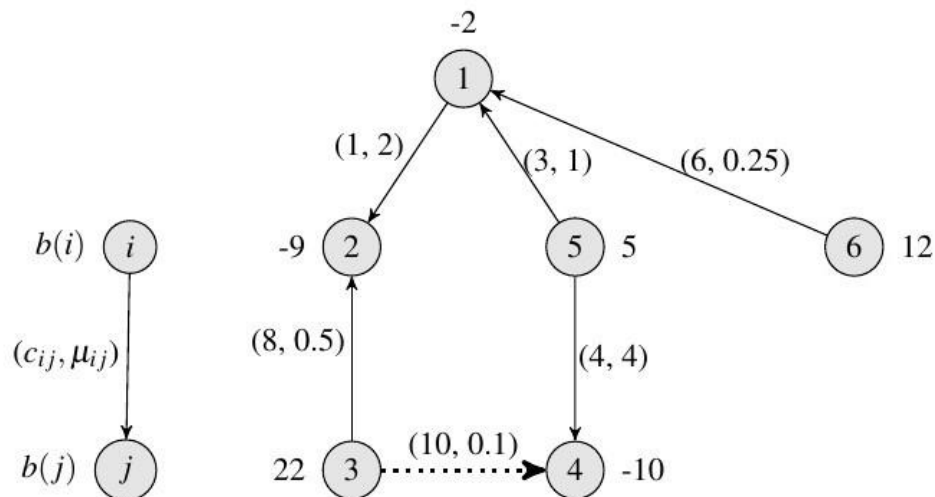
6. Run the topological sort algorithm on the following graph:



7. With the necessary illustrations, explain the radix heap implementation of Dijkstra's algorithm.

8. State and prove the max-flow min-cut theorem.

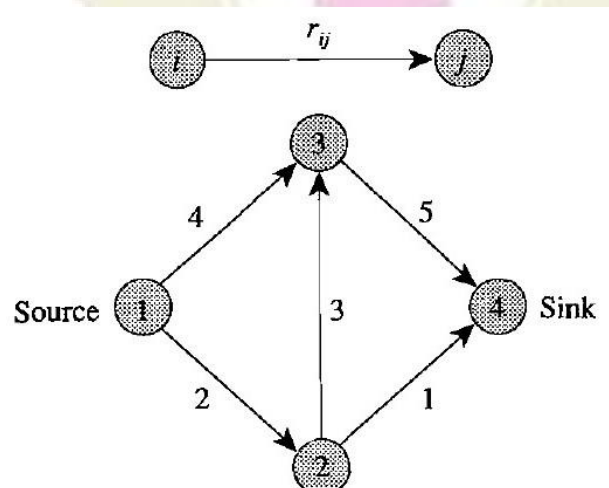
9. Compute node potentials for the graph shown below



10. How are the leaving and entering arcs chosen in the network simplex algorithm?

11. Explain the successive shortest path algorithm with an example.

12. Illustrate the generic augmenting path algorithm on the network shown below:



SYLLABUS

Module I: Network Flow Problems: (6 Hours)

Introduction, Notations, and definitions, Network representations, Search Algorithm, Topological Ordering, Flow Decomposition Algorithms.

Module II: Shortest Path Problem: (10 Hours)

Introduction, Label-Setting, and Label-Correcting Algorithms, Label-Setting algorithms - Dijkstra's Algorithm, Dial's implementation, Radix heap implementation, Label-Correcting Algorithms - Generic Label Correcting Algorithm, Special Implementations of the Modified Label Correcting Algorithm, Detecting Negative Cycles, All Pairs Shortest Path Problem.

Module III: Maximum Flow Problem: (8 Hours)

Generic Augmenting Path Algorithm, Labeling Algorithm and Max-Flow Min-Cut Theorem, Capacity Scaling Algorithm, Generic Pre-Flow Push Algorithm, FIFO Pre-Flow Push Algorithm, Flows in Unit Capacity Networks, Flows in Bipartite Networks, Flows in Planar Undirected Networks.

Module IV: Minimum Cost Flows and Generalized Flows: (9 Hours)

Optimality Conditions, Cycle Canceling Algorithm and the Integrity Property, Successive Shortest Path Algorithm, Primal-Dual Algorithm, Generalized flows – introduction, Augmented Forest Structures, Determining Potentials and Flows for an Augmented Forest Structure.

Module V: Linear programming and network flows: (7 Hours)

Linear programming – modeling, The geometric solution, The Simplex Method for Network Flow Problems, the Representation of a Non-basic Vector in Terms of the Basic Vectors, LP formulation of matching.

COURSE PLAN

No	Topic	No. of Lectures
1	Network Flow Problems (6 hours)	
1.1	Introduction, Notations, and definitions	1
1.2	Network representations	2
1.3	Search Algorithm, Topological Ordering	1
1.4	Flow Decomposition Algorithms	2
2	Shortest Path Problem (10 hours)	
2.1	Introduction, Label-Setting, and Label-Correcting Algorithms	1
2.2	Label-Setting algorithms - Dijkstra's Algorithm	1
2.3	Dial's implementation, Radix heap implementation	2
2.4	Label-Correcting Algorithms - Generic Label Correcting Algorithm,	2
2.5	Special Implementations of the Modified Label Correcting Algorithm	2
2.6	Detecting Negative Cycles, All Pairs Shortest Path Problem	2
3	Maximum Flow Problem (8 hours)	
3.1	Generic Augmenting Path Algorithm	1
3.2	Labeling Algorithm and Max-Flow Min-Cut Theorem	2
3.3	Capacity Scaling Algorithm	1
3.4	Generic Pre-Flow Push Algorithm, FIFO Pre-Flow Push Algorithm	1

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3.5	Flows in Unit Capacity Networks	1
3.6	Flows in Bipartite Networks, Flows in Planar Undirected Networks	2
4	Minimum Cost Flows and Generalized Flows (9 hours)	
4.1	Optimality Conditions	1
4.2	Cycle Canceling Algorithm and the Integrity Property,	1
4.3	Successive Shortest Path Algorithm,	1
4.4	Primal-Dual Algorithm	2
4.5	Generalized flows – introduction, Augmented Forest Structures	2
4.6	Determining Potentials and Flows for an Augmented Forest Structure	2
5	Linear programming and network flows (7 hours)	
5.1	Linear programming - modeling	2
5.2	The geometric solution	1
5.3	The Simplex Method for Network Flow Problems	2
5.4	Representation of a Non-basic Vector in Terms of the Basic Vectors	1
5.5	LP formulation of matching	1

Reference Books

1. Network Flows: Theory, Algorithms, and Applications, Ravindra K. Ahuja, Thomas L. Magnanti, James B. Orlin. Prentice Hall, 1993
2. Higher Engineering Mathematics, B. S. Grewal, Khanna publishers, 2012
3. Operations Research, Prem Kumar Gupta and Dr. D.S. Hira, S Chand Publishers, 2007
4. Linear programming and network flows, Mokhtar S. Bazaraa, John J. Jarvis, Hanif D. Sherali. John Wiley & Sons, 2009



The logo of API Abdul Kalam Technological University is a shield-shaped emblem. The top half is light blue with the university's name in pink. The bottom half is light green and contains a circular seal with a pink book icon, the text 'Estd. 2014', and the acronym 'KJ Somaiya' in the background.

PROGRAM ELECTIVE III

222EIT100	COMPLEX NETWORKS AND APPLICATIONS	CATEGORY	L	T	P	CREDIT
		PROGRAMME ELECTIVE 3	3	0	0	3

Preamble: The course aims to conduct an elaborate study on the topics of the evolution of complex networks, community detection in complex networks, and modelling of spreading phenomena in complex networks. This course also intends to introduce students to the current topics of research in the field of Complex Networks.

Prerequisite: IT12022-PE2 (F) Mathematical Models for Internet

CO No.	Course Outcomes (CO)	Bloom's Category Level
CO 1	Make use of complex network concepts in analysing and modelling real-world phenomena.	Level3: Apply
CO 2	Identify the role of fitness in the formation of complex networks.	Level4: Apply
CO 3	Interpret the evolving network model.	Level4: Apply
CO 4	Identify the presence of communities in complex networks.	Level 4: Apply
CO 5	Analyse patterns of virus spreading in biological and cyber worlds.	Level4: Analyse

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

P01: An ability to independently carry out research/investigation and development work in engineering and allied streams

P02: AN ability to communicate effectively, and write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

P03: An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor's program

P04: An ability to apply stream knowledge to design or develop solutions for real-world problems by following the standards

P05: An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tool to model, analyse and solve practical engineering problems.

PO6: An ability to engage in life-long learning for the design and development related to the stream related problems taking into consideration sustainability, societal, ethical and environmental aspects

PO7: An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	-	-	✓	✓	✓	-	-
CO 2	-	-	✓	✓	✓	-	-
CO 3	-	-	✓	✓	✓	-	-
CO 4	✓	✓	✓	✓	✓	-	-
CO 5	✓	✓	✓	✓	✓	✓	-

Assessment Pattern

Bloom's Category	End Semester Examination
Apply	100%
Analyse	
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation Pattern:

Preparing a review article based on peer reviewed original publications (minimum 10 publications shall be referred) **15 marks**

Course based task/Seminar/Data collection and interpretation **15 marks**

Test paper, 1 no. **10 marks**
(Test paper shall include minimum 80% of the syllabus)

End Semester Examination Pattern:

The end semester examination will be conducted by the respective College. There will be two parts; Part A and Part B. Part A will contain 5 numerical/short answer questions with 1 question from each module, having 5 marks for each question (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students). Students should answer all questions. Part B will contain 7 questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem solving and quantitative evaluation), with minimum one question from each module of which student should answer any five. Each question can carry 7 marks.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly. For example, if the average end-semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is $40 + 20 = 60\%$.

Course Level Assessment Questions

Course Outcome 1 (C01):

1. Based on the studies on congestion in Internet traffic, Compare the congestion behavior of the Internet compared to random networks.
2. "A page with many hyperlinks pointing to it does not necessarily have high PageRank." Do you agree with the statement. Explain your reasons citing relevant research.

Course Outcome 2 (C02)

1. Investigate on the growth of Facebook based on the insight provided by the Bianconi-Barabási Model.
2. Discuss the significance of Bose-Einstein Condensation in explaining scale-free nature of a network

Course Outcome 3(CO3):

1. Explain the phase transition induced by Node removal.
2. Calculate the degree exponent of the directed Barabási-Albert model with accelerated growth, assuming that the degree of the newly arriving nodes increases in time as $m(t) = t^\theta$.

Course Outcome 4 (CO4):

1. Consider a one-dimensional lattice with N nodes that form a circle, where each node connects to its two neighbours. Partition the line into n_c consecutive clusters of size $N_c = N/n_c$. According to the Maximum Modularity Hypothesis, the maximum of M_c corresponds to the best partition. Obtain the community size n_c corresponding to the best partition.
2. Identify the possibility of defining communities in terms of cliques based on the connectedness and density hypothesis.

Course Outcome 5 (CO5):

1. Explain the Time evolution of the fraction of infected individuals, as predicted by Susceptible-Infected (SI) Model.
2. Using the growth of the fraction of infected individuals, compare SI, SIS and SIR models.

Model Question Paper

QP CODE:

Reg No:

Name: _____

**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
SECOND SEMESTER M.TECH DEGREE EXAMINATION,
MONTH & YEAR**

222EIT100 Complex Networks and Applications

Max. Marks: 60

2.5 hrs

Duration:

PART A

(Answer all questions. Each Question Carries 5 Marks)

1. Comment on Milgram's experiment and six degrees of separation in the social network.
2. Explain the Bianconi-Barabási Model for modeling the fitness of the node.
3. Discuss the phenomena of 'accelerated growth' seen in complex networks.
4. Explain the random hypothesis in relation to modularity.
5. Explain Susceptible-Infected-Susceptible (SIS) Model.

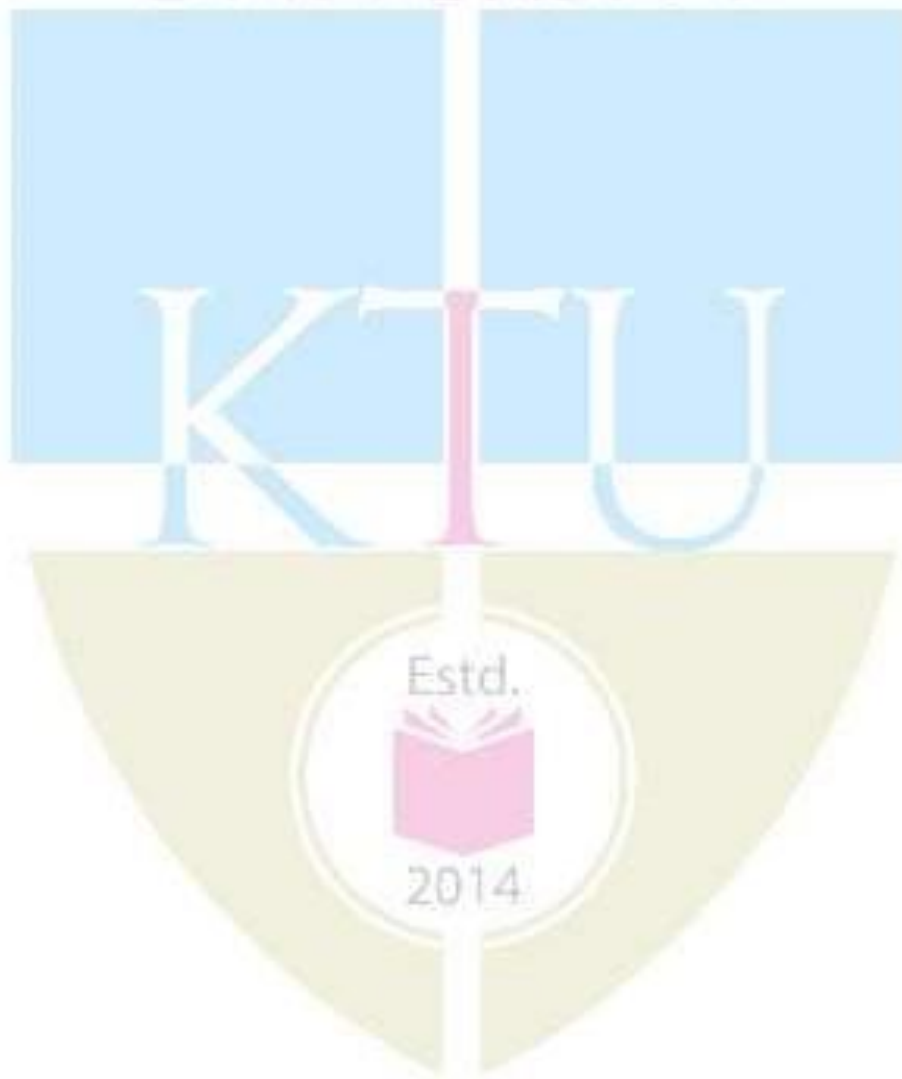
PART B

(Answer any FIVE Questions. Each Question Carries 7 Marks)

6. "viruses spread more quickly on scale-free networks than on small world and random network" Do you agree with the statement? Cite relevant research to justify your argument.
7. Discuss competition in complex network based on Bianconi-

Barabási Model.

8. Explain how to map networks to a Bose Gas Network.
9. Identify the impact of aging on the structure and the dynamics of a growing network.
10. Explain the Ravasz Algorithm for community identification.
11. State Maximal Modularity Hypothesis and explain the modularity maximization algorithm, proposed by Newman.
12. Compare the impact of Random Immunization and Selective Immunization in the case of Scale-free Networks.



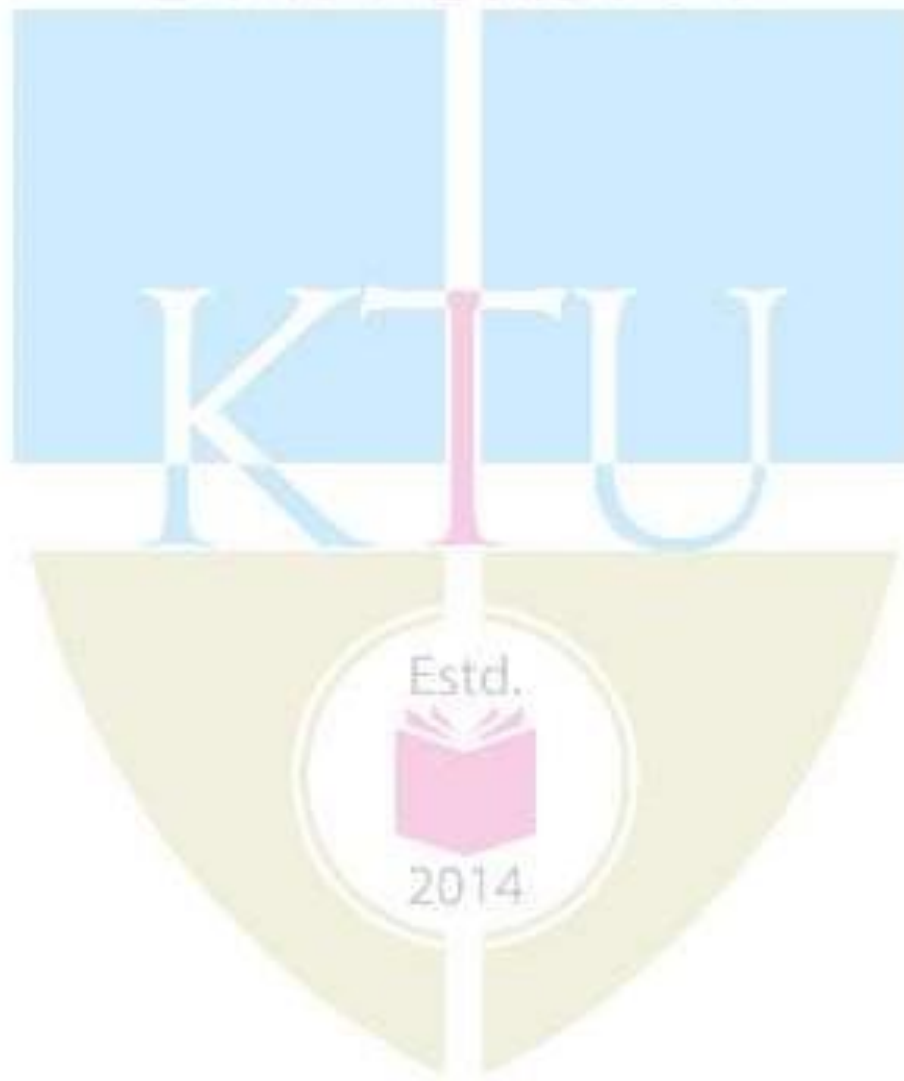
SYLLABUS

Module 1: Analyzing and modeling real-world phenomena with complex networks- survey of selected applications -10 Hours
Studies of the Internet and World Wide Web based on Complex Networks, Social networks- Personal relations- Acquaintances, Sexual relations, Trust, Movie actors, Sports, Scientific collaboration, Citations, Music, Terrorism networks, Communication Networks- Electronic mail, Telephone Computer science- Software architecture, Data sharing, Circuits
Module 2: fitness (6 Hours)
Bianconi-Barabási Model, Degree Dynamics, Degree Distribution, Measuring Fitness, behavioural basis for attachment proportional to fitness, The Fitness of a Web Document, The Fitness of a Scientific Publication, Bose-Einstein Condensation in Networks
Module 3: Evolving Networks (6 Hours)
Initial Attractiveness, Internal Links , Phase Transitions Induced by Node Removal, Accelerated Growth, Aging, The Impact of Aging, Topological Diversity, Modeling Diversity.
Module 4:(9Hours)- Communities
Basics of Communities, Hierarchical Clustering, Agglomerative Procedures: the Ravasz Algorithm, Divisive Procedures: the Girvan-Newman Algorithm, Hierarchy in Real Networks, Modularity, Overlapping Communities, Testing Communities, Characterizing Communities.
Module 5: Spreading Phenomena (9 Hours)
Spreading Phenomena- Epidemic Modelling, Susceptible-Infected (SI) Model, Susceptible-Infected-Susceptible (SIS) Model, Susceptible-Infected-Recovered (SIR) Model, Network Epidemics, Susceptible-Infected (SI) Model on a Network, SIS Model and the Vanishing Epidemic Threshold, Contact Networks, Digital Viruses, Beyond the Degree Distribution, Temporal Networks, Bursty Contact Patterns, Complex Contagion, Immunization, Random Immunization, Vaccination Strategies in Scale-Free Networks, Epidemic Prediction, 'What if' Analysis, Effective Distance.

Reference Books

1. A-L. Barabási , Network Science, Cambridge University Press 2016, Available online: <http://barabasi.com/networksciencebook/>

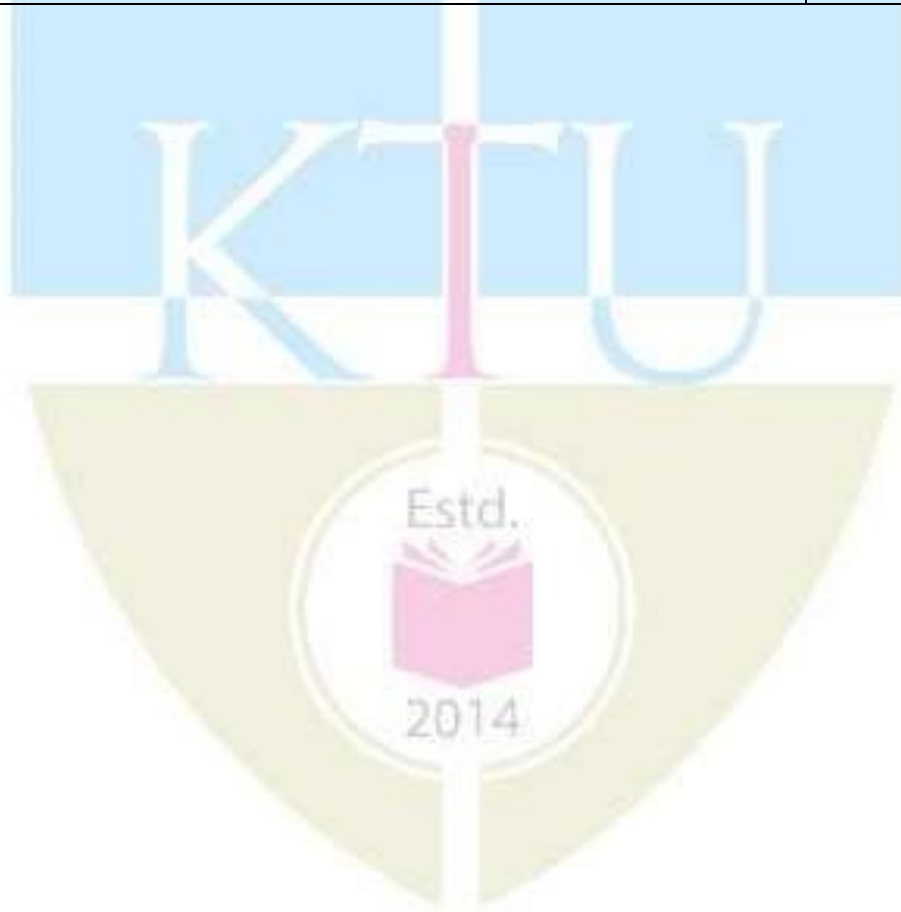
2. Costa, L.D.F., Oliveira Jr, O.N., Travieso, G., Rodrigues, F.A., Villas Boas, P.R., Antiqueira, L., Viana, M.P. and Correa Rocha, L.E., 2011. Analyzing and modeling real-world phenomena with complex networks: a survey of applications. *Advances in Physics*, 60(3), pp.329-412.
3. M.E.J. Newman, *Networks - An introduction*, Oxford University Press, 2010.
4. D. Easley and J. Kleinberg, *Networks, Crowds and Markets*, Cambridge University Press.
5. R. Cohen and S. Havlin, *Complex Networks - Structure, Robustness and Function*, Cambridge University Press, 2010.



COURSE PLAN

No	Topic	No. of Lectures 40 Hours
1	Analyzing and modelling real-world phenomena with complex networks- survey of selected applications	10 Hours
1.1	Studies of Internet and World Wide Web based on Complex Networks	2 Hours
1.2	Social networks- Personal relations- Acquaintances, Sexual relations, Trust, Movie actors, Sports, Scientific collaboration	2Hours
1.3	Citations, Music, Terrorism networks	2 Hours
1.4	Communication Networks- Electronic mail, Telephone	2 Hours
1.5	Computer science- Software architecture, Data sharing, Circuits	2Hours
2	Fitness	6 Hours
2.1	Bianconi-Barabási Model	1Hour
2.2	Degree Dynamics, Degree Distribution	1 Hour
2.3	Measuring Fitness, behavioural basis for attachment proportional to fitness	1 Hour
2.4	The Fitness of a Web Document, The Fitness of a Scientific Publication,	1Hour
2.5	Bose-Einstein Condensation in Networks	2 Hours
3	Evolving Networks	6 Hours
3.1	Initial Attractiveness, Internal Links	1 Hour
3.2	Phase Transitions Induced by Node Removal	2 Hours
3.3	Accelerated Growth, Aging, The Impact of Aging	2 Hours
3.4	Topological Diversity, Modelling Diversity.	1 Hour
4	Communities	9 Hours
4.1	Basics of Communities, Hierarchical Clustering	2 Hours
4.2	Agglomerative Procedures: The Ravasz Algorithm	1 Hour
4.3	Divisive Procedures: the Girvan-Newman Algorithm	1 Hours
4.4	Hierarchy in Real Networks, Modularity	2 Hours
4.5	Overlapping Communities	1Hour

4.6	Testing Communities	1 Hour
4.7	Characterizing Communities.	1 Hour
5	Spreading Phenomena	9 Hours
5.1	Epidemic Modelling, Susceptible-Infected (SI) Model	1 Hour
5.2	Susceptible-Infected-Susceptible (SIS) Model, Susceptible-Infected-Recovered (SIR)	2 Hours
5.3	Network Epidemics, Susceptible-Infected (SI) Model on a Network, SIS Model and the Vanishing Epidemic Threshold	2 Hours
5.4	Contact Networks, Digital Viruses	1 Hour
5.5	Beyond the Degree Distribution, Temporal Networks, Bursty Contact Patterns, Complex Contagion	1 Hour
5.6	Immunization, Random Immunization, Vaccination Strategies in Scale-Free Networks	1 Hour
5.7	Epidemic Prediction, 'What if' Analysis, Effective Distance.	1 Hour



222EIT001	BIG DATA ANALYTICS	CATEGORY	L	T	P	CREDIT
		PROGRAMME ELECTIVE 3	3	0	0	3

Preamble: This course brings together several key big data technologies used for storage, analysis and manipulation of data.

Course Outcomes: After the completion of the course the student will be able to:

CO 1	Explain the need of Big Data, challenges and different analytical architectures(Cognitive Knowledge Level: Understand)
CO 2	Use Hadoop Framework and ecosystems for implementing programs for big data analysis(Cognitive Knowledge Level: Apply)
CO 3	Use Map-reduce programming in Hadoop to analyze big data (Cognitive Knowledge Level: Apply)
CO 4	Develop data analysis programs in Pig, Hive and HBase (Cognitive Knowledge Level: Apply)

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

PO1: An ability to independently carry out research/investigation and development work in engineering and allied streams

PO2: An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

PO3: An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program

INFORMATION TECHNOLOGY

PO4: An ability to apply stream knowledge to design or develop solutions for real world problems by following the standards

PO5: An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tool to model, analyse and solve practical engineering problems.

PO6: An ability to engage in life-long learning for the design and development related to the stream related problems taking into consideration sustainability, societal, ethical and environmental aspects

PO7: An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	P 2	PO 3	PO 4	PO 5	PO	PO 7
CO 1							
CO 2							
CO 3							
CO 4							

Assessment Pattern

Bloom's Category	End Semester Examination
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INFORMATION TECHNOLOGY

Understand	20%
Apply	80%
Analyse	
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation Pattern:

Evaluation shall only be based on application, analysis or design-based questions (for both internal and end semester examinations).

Continuous Internal Evaluation Pattern:

Preparing a review article based on peer reviewed original publications (minimum 10 publications shall be referred) **15 marks**

Course based task/Seminar/Data collection and interpretation **15 marks**

Test paper, 1 no. **10 marks**
(Test paper shall include minimum 80% of the syllabus)

End Semester Examination Pattern:

INFORMATION TECHNOLOGY

The end semester examination will be conducted by the respective College. There will be two parts; Part A and Part B. Part A will contain 5 numerical/short answer questions with 1 question from each module, having 5 marks for each question (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students). Students should answer all questions. Part B will contain 7 questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem solving and quantitative evaluation), with minimum one question from each module of which student should answer any five. Each question can carry 7 marks.

Total duration of the examination will be 150 minutes.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly.

For example if the average end semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is $40 + 20 = 60\%$.

Course Level Assessment Questions

Course Outcome 1 (CO1):

1. Explain the life cycle of Data Analytics.
2. Explain the characteristics of big data.
3. Explain big data architecture.

Course Outcome 2 (CO2):

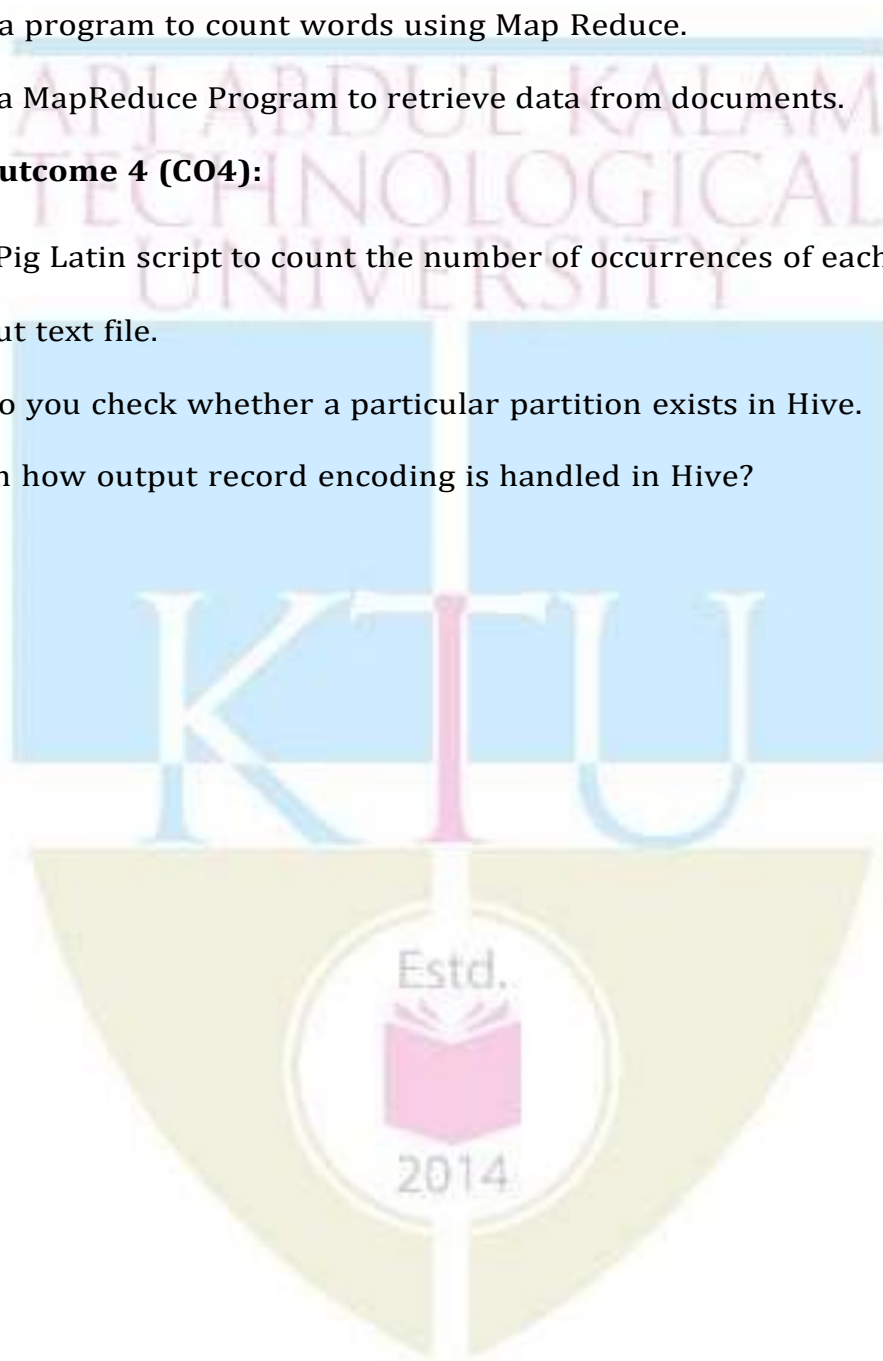
1. Write commands to implement basic file management tasks in Hadoop.
2. Explore the various shell commands in Hadoop.
3. How is HDFS fault tolerant?

Course Outcome 3 (CO3):

1. Implement matrix multiplication with Hadoop Map Reduce
2. Write a program to count words using Map Reduce.
3. Write a MapReduce Program to retrieve data from documents.

Course Outcome 4 (CO4):

1. Write Pig Latin script to count the number of occurrences of each word in an input text file.
2. How do you check whether a particular partition exists in Hive.
3. Explain how output record encoding is handled in Hive?



MODEL QUESTION PAPER

QP CODE:

Reg No: _____

Name: _____

PAGES :

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

SECOND SEMESTER M.TECH DEGREE EXAMINATION, MONTH & YEAR

222EIT001

BIG DATA ANALYTICS

Max. Marks : 60

Duration: 2.5 Hours

PART A

Answer All Questions. Each Question Carries 5 Marks(5x5=25)

- | | | |
|----------|--|----------|
| 1 | How do volume, variety, velocity, and veracity help us decide whether a given data source contributes to big data? | 5 |
| 2 | Explain the architecture of a stream computing system. | 5 |
| 3 | Implement matrix multiplication with Hadoop Map Reduce. | 5 |
| 4 | Illustrate the use of HiveQL queries with an example. | 5 |
| 5 | List and explain three complex data types in Pig. | 5 |

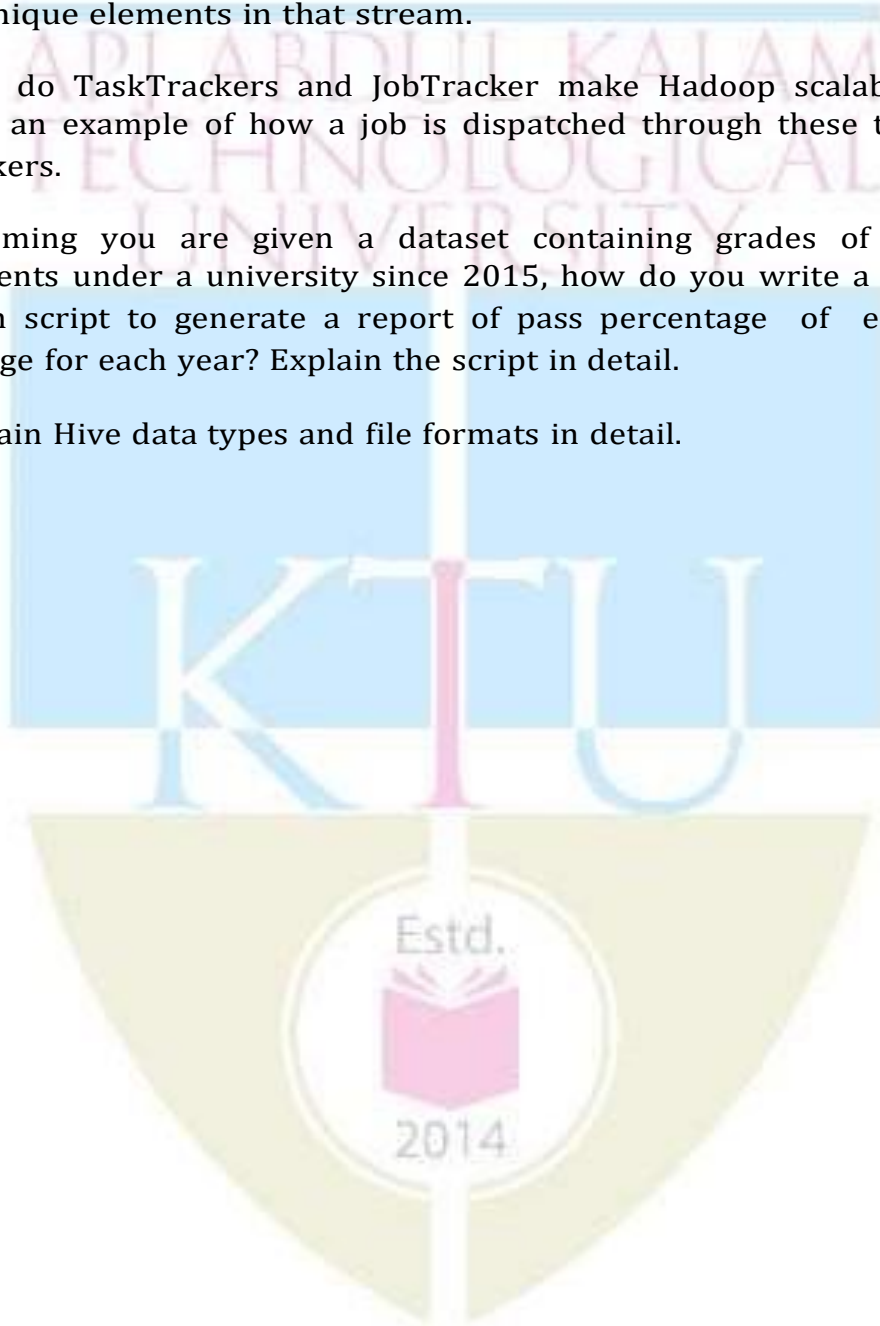
Part B

(Answer any five questions. Each question carries 7 marks)

- | | | |
|-----------|---|------------|
| 6. | Identify challenges associated with Big Data processing and Management. | (7) |
| 7. | For a university management system prepare a rough design for each type of sandbox. | (7) |
| 8. | How are failures handled during Hadoop and MapReduce | (7) |

execution?

9. Compute the second moment and third moment of the stream 3, 1, 4, 1, 3, 4, 2, 1, 2. Use as many estimation variables as number of unique elements in that stream. (7)
10. How do TaskTrackers and JobTracker make Hadoop scalable? Give an example of how a job is dispatched through these two trackers. (7)
11. Assuming you are given a dataset containing grades of all students under a university since 2015, how do you write a Pig Latin script to generate a report of pass percentage of each college for each year? Explain the script in detail. (7)
12. Explain Hive data types and file formats in detail. (7)



SYLLABUS

Module 1

Introduction to big data: Types of big data, 5 v's of big data, Big data architecture, Introduction to big data analytics, Life cycle of Big data analytics, Holistic view of analytics, Big data analytical method, Analytic Processes and Analytic Tools - Analysis vs Reporting.

Introduction to Big Data Platform, Challenges of Conventional Systems - Intelligent data analysis, Nature of Data, data privacy and ethics – privacy landscape, global privacy principles.

Module 2:

Mining data streams: Introduction to Data Streams, Big data streaming, Streaming data architecture, Stream Data Model - Stream Computing – Sampling Techniques, Filtering Streams –Bloom filter, Counting Distinct Elements in a Stream – Flajolet Martin Algorithm

Estimating Moments, Counting Oneness in a Window – DGIM algorithm, Decaying Window – Realtime Analytics Platform (RTAP) Applications.

Module 3

Components of Hadoop – HDFS architecture, Analysing the Data with Hadoop- Scaling Out- HadoopStreaming- Design of HDFS-Java interfaces to HDFS Basics, Hadoop ecosystem

Developing a Map Reduce Application-Anatomy of a Map Reduce Job, Scheduling-Shuffle and Sort -Task execution. Case Study: IBM Infosphere Big Insights and Streams.

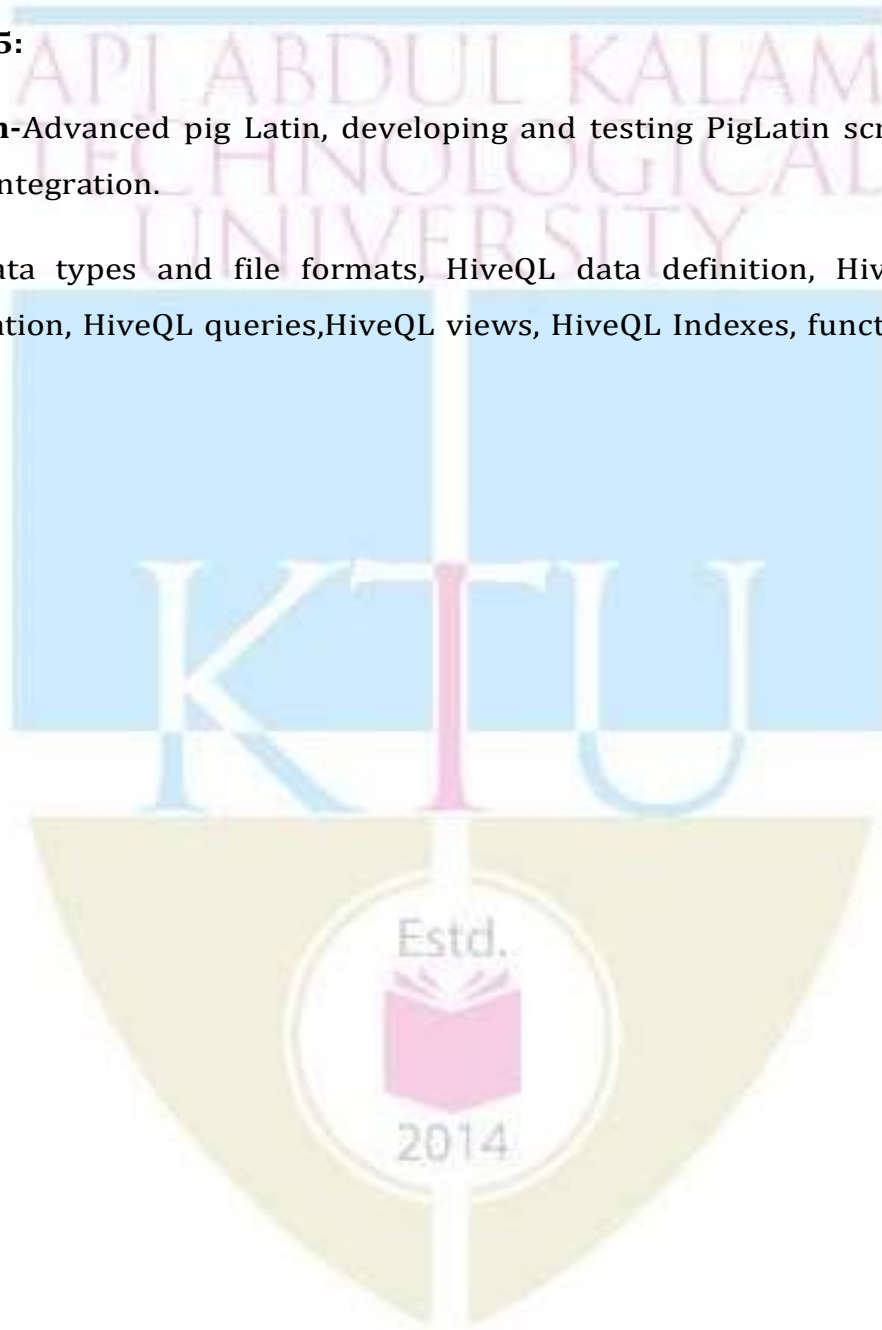
Module 4

Introduction to HBase, Filesystems for HBase, Client API - The Basics, HBase clients – REST, ShellCommands, Map Reduce IntegrationIntroduction to Pig, Grunt, pig data model.

Module 5:

Pig Latin-Advanced pig Latin, developing and testing PigLatin scripts, Map Reduce Integration.

Hive: Data types and file formats, HiveQL data definition, HiveQL data manipulation, HiveQL queries,HiveQL views, HiveQL Indexes, functions.



COURSE PLAN

No	Topic	No. of Lectures (40)
1	Introduction to big data (9 hours)	
1.1	Types of big data, 5 v's of big data	1 hour
1.2	Big data architecture	1 hour
1.3	Introduction to big data analytics, Life cycle of Big data analytics	1 hour
1.4	Holistic view of analytics	1 hour
1.5	Big data analytical method	1 hour
1.6	Analytic Processes and Analytic Tools - Analysis vs Reporting	1 hour
1.7	Introduction to Big Data Platform, Challenges of Conventional Systems	1 hour
1.8	Intelligent data analysis, Nature of Data	1 hour
1.9	Data privacy and ethics – privacy landscape, global privacy principles	1 hour
2	Mining data streams (8 hours)	
2.1	Introduction to Data Streams, Big data streaming	1 hour

INFORMATION TECHNOLOGY

2.2	Streaming data architecture, Stream Data Model - Stream Computing	1 hour
2.3	Sampling Techniques	1 hour
2.4	Filtering Streams - Bloom filter	1 hour
2.5	Estimating Moments ,Counting Distinct Elements in a Stream - Flajolet Martin Algorithm	1 hour
2.6	Counting Oneness in a Window - DGIM algorithm	1 hour
2.7	Decaying Window	1 hour
2.8	Real time Analytics Platform (RTAP) Applications, Case Studies - Real Time Sentiment Analysis- Stock Market Predictions	1 hour
3	Hadoop,HDFS, MapReduce (8 hours)	
3.1	HDFS architecture	1 hour
3.2	Analyzing the Data with Hadoop	1 hour
3.3	Scaling Out- Hadoop, Streaming	1 hour
3.4	Design of HDFS-Java interfaces to HDFS Basics	1 hour
3.5	Hadoop ecosystem	1 hour
3.6	Developing a Map Reduce Application-Anatomy of a Map Reduce Job	1 hour

INFORMATION TECHNOLOGY

3.7	Scheduling-Shuffle and Sort-Task execution	1 hour
3.8	Case Study: IBM Infosphere Big Insights and Streams	1 hour
4	Introduction to HBase and Pig (7 hours)	
4.1	Filesystems for Hbase	1 hour
4.2	Client API - The Basics	1 hour
4.3	HBase clients	1 hour
4.4	REST Shell Commands	1 hour
4.5	Map Reduce Integration	1 hour
4.6	Introduction to Pig, Grunt	1 hour
4.7	Pig data model	1 hour
5	Pig Latin, Hive (8 hours)	
5.1	Pig Latin , Developing and testing Pig Latin scripts	1 hour
5.2	Map Reduce Integration	1 hour
5.3	Hive, data types and file formats	1 hour
5.4	HiveQL data definition	1 hour
5.5	HiveQL data manipulation	1 hour

5.6	HiveQL queries	1 hour
5.7	HiveQL views	1 hour
5.8	HiveQL Indexes, functions.	1 hour

References

1. Michael Minelli, Michelle Chambers, and AmbigaDhiraj, "Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends for Today's Businesses", Wiley,2013.
2. P. J. Sadalage and M. Fowler, "NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence", Addison-Wesley Professional,2012.
3. Tom White, "Hadoop: The Definitive Guide", Third Edition, O'Reilley,2012. Analytics for Enterprise Class Hadoop and Streaming Data", McGrawHill Publishing, 2012.
4. Eric Sammer, "Hadoop Operations", O'Reilley,2012.
5. E. Capriolo, D. Wampler, and J. Rutherglen, "Programming Hive", 'Reilley,2012.
6. Lars George, "HBase: The Definitive Guide", O'Reilley,2011.
7. Alan Gates, "Programming Pig", O'Reilley,2011.
8. Chris Eaton, Dirk De Roos, Tom Deutsch, George Lapis, Paul Zikopoulos,"Understanding Big Data: Analytics for Enterprise Class Hadoop and Streaming Data", McGrawHill Publishing, 2012.
9. Anand Rajaraman and Jeffrey D. Ullman,"Mining of Massive Datasets", Cambridge University Press, 2012.
- 10.Arshdeep Bahga, Vijay Madisetti, "Big Data Science & Analytics:A Hands – On Approach",VPT, 2016

222EIT002	LAWS AND ETHICS IN COMPUTING	CATEGORY	L	T	P	CREDIT
		PROGRAMME ELECTIVE 3	3	0	0	3

Preamble: This course will enable the students to understand the ethical issues in computing, code of conduct in the computing profession, intellectual property rights issues, cybercrimes and cyber laws in India.

Course Outcomes:

After the completion of the course, the student will be able to

CO 1	Identify the ethical issues and understand the characteristics of professional ethics and the code of ethics for professional conduct. (Level: Understand)
CO 2	Explain the ways to protect intellectual property rights in computing and understand the terms and conditions in computing contracts. (Level: Understand)
CO 3	Differentiate between the different computer crimes (Level: Understand)
CO 4	Prepare and conduct a computer investigation with computer forensics (Level: Apply)
CO 5	Explain the cyber laws in India (Level: Understand)

Program Outcomes:

PO 1	An ability to independently carry out research/ investigation and development work in engineering and allied streams.
PO 2	An ability to communicate effectively, and write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.
PO 3	An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor's program.

INFORMATION TECHNOLOGY

PO 4	An ability to apply stream knowledge to design or develop solutions for real-world problems by following the standards.
PO 5	An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tools to model, analyze and solve practical engineering problems.
PO 6	An ability to engage in lifelong learning for the design and development of stream-related problems taking into consideration sustainability, societal, ethical, and environmental aspects.
PO 7	An ability to develop cognitive load management skills related to project management and finance which focus on entrepreneurship and industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1				√	√	√	
CO 2		√		√			
CO 3				√			
CO 4	√		√	√	√		
CO 5				√			

Assessment Pattern

Bloom's Category	End Semester Examination
Understand	80%
Apply	20%

INFORMATION TECHNOLOGY

Analyze	
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

The evaluation shall only be by employing application-based, analysis-based, or design-based questions (for both internal and end-semester examinations)

Continuous Internal Evaluation (CIE) Pattern:

Preparing a review article based on peer-reviewed original publications (minimum 10 Publications shall be referred)/ Micro Project/ Assignment: 10 Marks

Course-based task/Seminar/Data Collection and interpretation: 20 Marks

Test paper (one only, and shall include a minimum of 80% of the syllabus): 10 Marks

End Semester Examination (ESE) Pattern:

The end-semester examination shall be conducted by the respective College. There shall be two parts; Part A and Part B. Part A shall contain five numerical/short answer questions with one question from each module, having five marks for each question (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students). Students shall answer all questions. Part B shall contain seven questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations,

problem-solving and quantitative evaluation), with a minimum one question from each module of which student shall answer any five. Each question shall carry seven marks.

MODEL QUESTIONS:

CO 1:

1. What are the issues related to ethics?
2. Explain the characteristics of professional ethics

CO 2:

1. How can you protect the intellectual property rights of a computer software?
2. What are the minimum components required to be specified in a computing contract?

CO 3:

1. How do you classify computer crimes?
2. Differentiate between different types of computer crimes?

CO 4:

1. How to prepare for a computer investigation?
2. Explain the steps involved in conducting a computer investigation?

CO 5:

1. What is the purpose of cyber laws?
2. Explain the scope of cyber laws?

MODEL QUESTION PAPER FOR END-SEMESTER EXAMINATION

Time: 2½ Hours, Maximum marks: 60

PART A (*Answer all questions. Five marks each*)

1. Differentiate between descriptive and normative claims. [CO1]
2. Explain the law of confidence in Intellectual property rights. [CO2]
3. Explain the term computer fraud by giving any one example of such activity. [CO3]
4. What do you mean by case law? How does it help in computer investigation? [CO4]
5. List any five amendments in Indian Penal Code 1860 for cyber laws. [CO5]

PART B (*Answer any five questions. Seven marks each*)

1. With an example, explain the ethical relativism theory used in ethics. [CO1]
2. How to ensure copyright in computer-generated documents? [CO2]
3. List and explain any seven cybercrimes that are happening in India. What can you do to prevent such crimes? [CO3]
4. What is the aim of computer forensics? Explain the main steps performed in a computer forensics operation. [CO4]
5. Explain the cyber jurisprudence law of digital contracts? [CO5]
6. If you are asked to set up a computer forensic lab, explain the different steps taken by you for setting up the lab. You have to give the specifications for physical, hardware, and software requirements. [CO4]
7. Describe any seven hacking techniques used in cybercrimes. [CO3]

SYLLABUS

Module I: Ethics and Professional Ethics (8 Hours)

Ethics – Philosophy, and issues - Vacuum of policies - conceptual muddles – social context - moral and legal issues - uniqueness of ethical issues - role of analogy - descriptive and normative claims - ethical theories- relativism - utilitarianism- other theories.

Professional Ethics - Characteristics – the system of professions – computing as a profession – professional relationship – responsibilities – code of ethics and professional conduct.

Module II: IPR and Computing Contracts: (8 Hours)

Computing and Intellectual Property Rights - Overview of intellectual property rights - Copyright basics - Computer software and copyright – Copyright in Computer-generated works - The law of confidence - Copyright and electronic publishing - Copyright in the information society - Patent law - Trademarks and passing-off – product designs

Introduction to Computing Contracts - Fundamentals of computer contracts – Liability for defective hardware and software - Contracts for writing software - License agreements for "off-the-shelf" software - Contract between software author and publisher – WEB and Hardware contracts.

Module III: Computer Crime: (8 Hours)

Computer Crime: Computer Fraud – Hacking – Unauthorized modification of information – Piracy – Pornography and harassment.

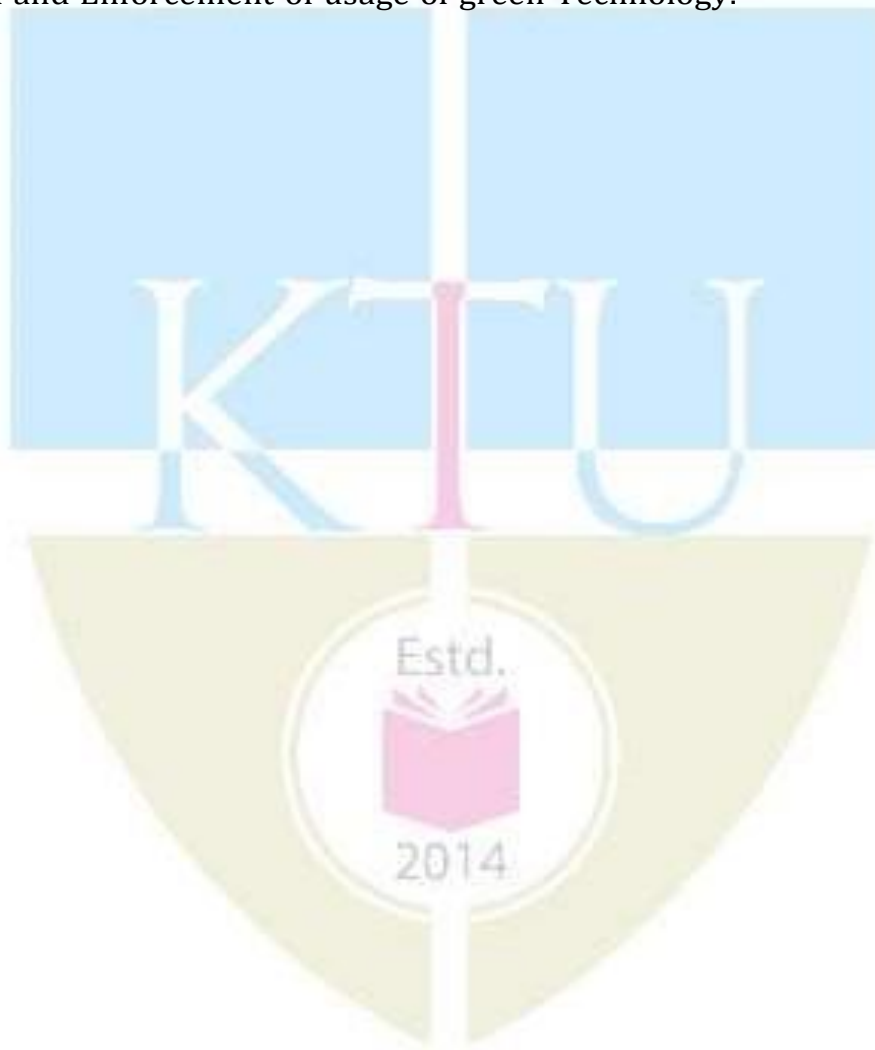
Module IV: Computer Forensics: (8 Hours)

Computer Forensics: Introduction to Computer Forensics- history of computer forensics- understanding case law- developing Computer forensics resources- understanding law enforcement agency investigations- understanding corporate investigations- maintaining professional conduct- Understanding Computer Investigations - Preparing a computer investigation- taking a systematic approach- procedures for corporate high tech investigations- understanding data recovery workstations and software- conducting an investigation- completing the case - Requirements for forensic lab certification-

determining the physical requirements for a computer forensics lab- selecting a basic forensic workstation- building a business case for developing a forensic lab

Module V: Cyber laws in India: (8 Hours)

Cyber laws in India: Basic Concepts of Technology and Law- Understanding the Technology of Internet- Scope of Cyber Laws- Cyber Jurisprudence Law of Digital Contracts - Intellectual Property Issues in CyberSpace - Rights of Netizens and E-Governance - Information Technology Act 2000for data protection - Legislation on waste management and green technology - Legislation and Enforcement of usage of green Technology.



COURSE PLAN

No	Topic	No. of Lectures (40)
1	Ethics and Professional Ethics (8 hours)	
1.1	Ethics – philosophy and issues - Vacuum of policies - conceptual muddles – social context	1
1.2	Moral and legal issues - uniqueness of ethical issues - role of analogy - descriptive and normative claims	1
1.3	Ethical theories- relativism - utilitarianism- other theories	2
1.4	Professional Ethics - Characteristics – the system of professions	1
1.5	Computing as a profession – professional relationship	1
1.6	Responsibilities – code of ethics and professional conduct.	2
2	IPR and Computing Contracts (8 hours)	
2.1	Computing and Intellectual Property Rights - Overview of intellectual property rights	1
2.2	Copyright basics - Computer software and copyright – Copyright in Computer-generated works	1
2.3	The law of confidence - Copyright and electronic publishing - Copyright in the information society	1
2.4	Patent law - Trademarks and passing-off – product designs	1
2.5	Introduction to Computing Contracts - Fundamentals of computer contracts- Liability for defective hardware and software	1
2.6	Contracts for writing software	1
2.7	License agreements for "off-the-shelf" software - Contract between software author and publisher	1

INFORMATION TECHNOLOGY

2.8	WEB and Hardware contracts.	1
3	Computer Crime (8 hours)	
3.1	Computer Crime: Computer Fraud	1
3.2	Hacking	3
3.3	Unauthorized modification of information – Piracy	2
3.4	Pornography and harassment.	2
4	Computer Forensics(8 hours)	
4.1	Introduction to Computer Forensics- history of computer forensics- understanding case law-developing Computer forensics resources	1
4.2	Understanding law enforcement agency investigations- understanding corporate investigations- maintaining professional conduct	1
4.3	Understanding Computer Investigations - Preparing a computer investigation- taking a systematic approach	1
4.4	Procedures for corporate high-tech investigations- understanding data recovery workstations and software	1
4.5	Conducting an investigation- completing the case	1
4.6	Requirements for forensic lab certification- determining the physical requirements for a computer forensics lab-	1
4.7	Selecting a basic forensic workstation	1
4.8	Building a business case for developing a forensic lab	1
5	Cyber laws in India(8 hours)	
5.1	Cyber laws in India: Basic Concepts of Technology and Law	1
5.2	Understanding the Technology of the Internet- Scope of Cyber Laws	1
5.3	Cyber Jurisprudence Law of Digital Contracts	1

INFORMATION TECHNOLOGY

5.4	Intellectual Property Issues in CyberSpace	1
5.5	Rights of Netizens and E-Governance	1
5.6	Information Technology Act 2000 for data protection	1
5.7	Legislation on waste management and green technology	1
5.8	Legislation and Enforcement of usage of green Technology.	1

Reference Books

1. Deborah G Johnson, Computer Ethics (Paperback), 3rd edition, Pearson education.
2. D. Bainbridge, Introduction to Computer Law, 5/e, Pearson Education, 2004.
3. P. Duggal, Cyber Law – The Indian Perspective, Saakshar Law Publications, New Delhi, 2009.
4. Bill Nelson, Amelia Phillips, Frank Enfinger, Christopher Steuart Computer Forensics and Investigations, second Indian Reprint 2009, Cengage Learning India.

INFORMATION TECHNOLOGY

222EIT003	CYBER LEGISLATION AND SECURITY POLICIES	CATEGORY	L	T	P	CREDIT
		PROGRAMME ELECTIVE 3	3	0	0	3

Preamble: This course will enable the students to have an understanding of Cyber rules, regulations, and security policies.

Course Outcomes:

After the completion of the course, the student will be able to

CO 1	Explain the social and intellectual property issues emerging from cyberspace. (Level: Understand)
CO 2	Explore the legal and policy developments in various countries to regulate Cyberspace. (Level: Understand)
CO 3	Describe the relationship between commerce and cyberspace (Level: Understand)
CO 4	Explain the information technology act and legal frame work for the right to privacy, data security, and data protection (Level: Understand)
CO 5	Compare the various security policies and real-time crimes (Level: Understand)

Program Outcomes:

PO 1	An ability to independently carry out research/ investigation and development work in engineering and allied streams.
PO 2	An ability to communicate effectively, and write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.
PO 3	An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor's program.
PO 4	An ability to apply stream knowledge to design or develop solutions for real-world problems by following the standards.

INFORMATION TECHNOLOGY

PO 5	An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tools to model, analyze and solve practical engineering problems.
PO 6	An ability to engage in lifelong learning for the design and development of stream-related problems taking into consideration sustainability, societal, ethical, and environmental aspects.
PO 7	An ability to develop cognitive load management skills related to project management and finance which focus on entrepreneurship and industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	√	√				√	√
CO 2	√	√				√	√
CO 3	√					√	√
CO 4	√	√	√			√	√
CO 5	√					√	√

Assessment Pattern

Bloom's Category	End Semester Examination
Understand	100%
Apply	
Analyze	

INFORMATION TECHNOLOGY

Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

The evaluation shall only be by employing application-based, analysis-based, or design-based questions (for both internal and end-semester examinations)

Continuous Internal Evaluation (CIE) Pattern:

Preparing a review article based on peer-reviewed original publications (minimum 10 Publications shall be referred)/

Assignment:

Course-based task/Seminar/Data Collection and interpretation: 20 Marks

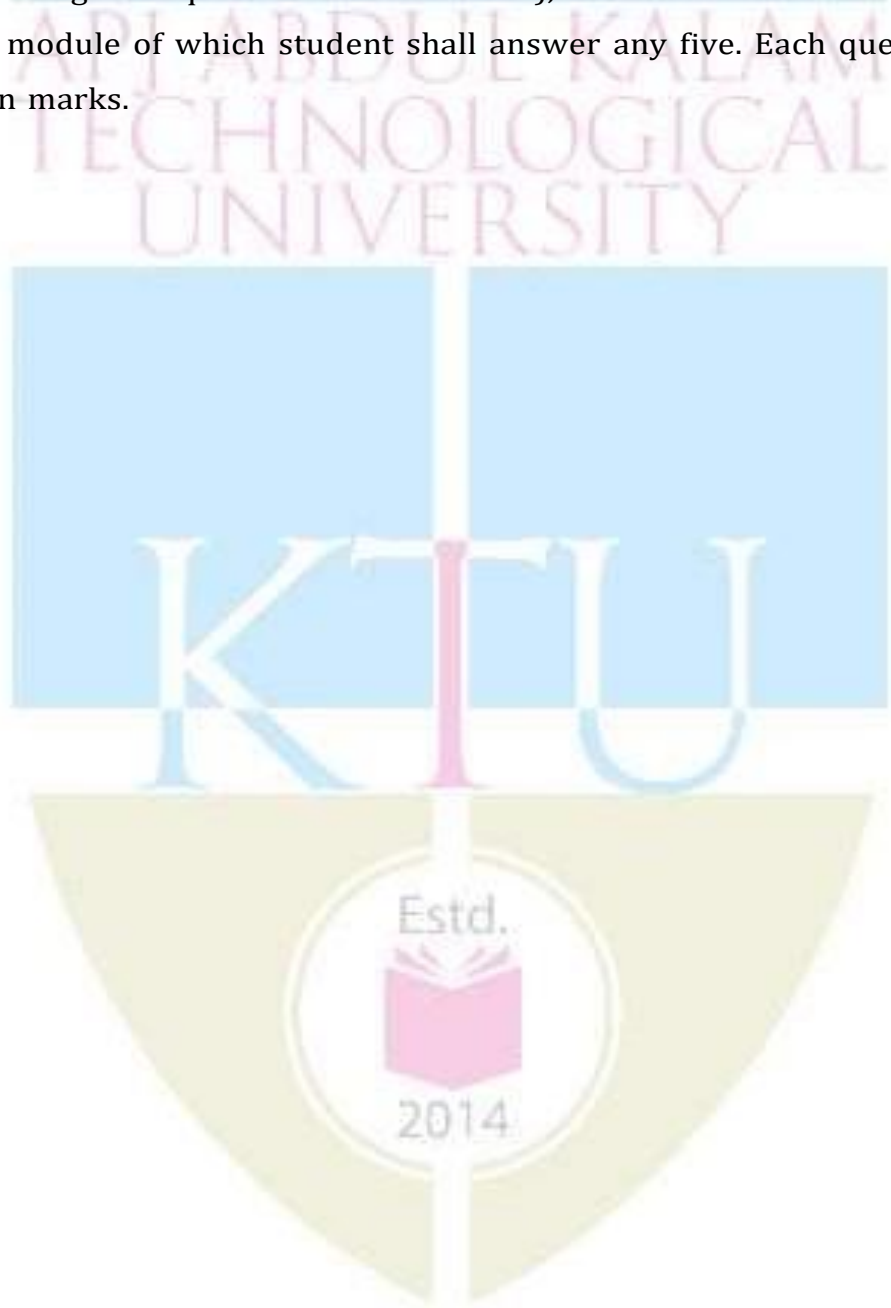
Test paper (one only, and shall include a minimum of 80% of the syllabus): 10 Marks

End Semester Examination (ESE) Pattern:

The end-semester examination shall be conducted by the respective College. There shall be two parts; Part A and Part B. Part A shall contain five numerical/short answer questions with one question from each module, having five marks for each question (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students). Students shall answer all questions. Part

INFORMATION TECHNOLOGY

B shall contain seven questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem-solving and quantitative evaluation), with a minimum one question from each module of which student shall answer any five. Each question shall carry seven marks.



MODEL QUESTION PAPER FOR END-SEMESTER EXAMINATION

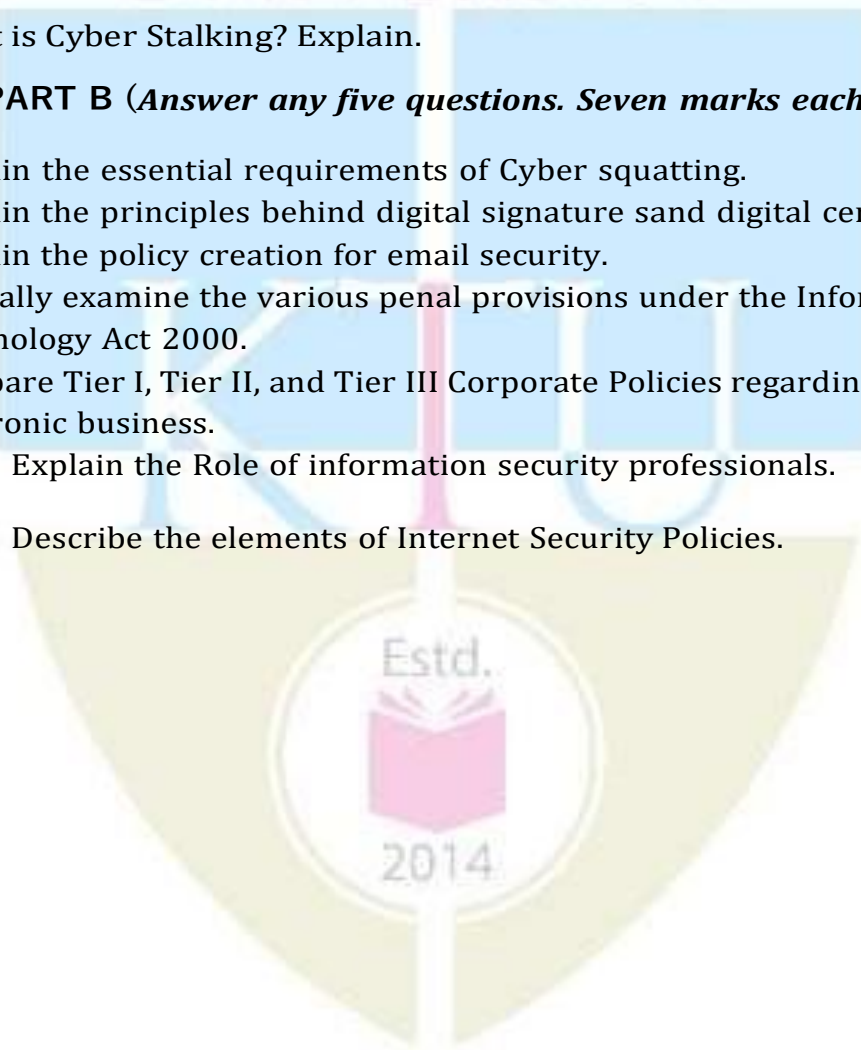
Time: 2½ Hours, Maximum marks: 60

PART A (*Answer all questions. Five marks each*)

1. Explain Computer Security mandates and legislation
2. Explain the trusted computer system evaluation criteria named Orange book
3. What are the government requirements for buying security?
4. What do you mean by Cybercrimes? Discuss the nature and types of Cybercrimes. What are the challenges due to them?
5. What is Cyber Stalking? Explain.

PART B (*Answer any five questions. Seven marks each*)

6. Explain the essential requirements of Cyber squatting.
7. Explain the principles behind digital signature and digital certificates.
8. Explain the policy creation for email security.
9. Critically examine the various penal provisions under the Information Technology Act 2000.
10. Compare Tier I, Tier II, and Tier III Corporate Policies regarding electronic business.
11. Explain the Role of information security professionals.
12. Describe the elements of Internet Security Policies.



SYLLABUS

Module I: Introduction to Cyber Law Evolution of Computer Technology (8 Hours)

The emergence of Cyberspace – Cyber Ethics and Cyber Jurisdiction, Jurisdiction issues under IT Act, 2000. Cyberspace-Web space, Web hosting and web Development agreement, Legal and Technological Significance of Domain Names, Internet as a tool for global access and internet protocols

Module II: Information technology Act (8 Hours)

Overview of IT Act, 2000, Amendments and Limitations of IT Act - Digital Signatures, Cryptographic Algorithm, Public Cryptography, Private Cryptography, Electronic Governance, Legal Recognition of Electronic Records, Legal Recognition of Digital Signature Certifying Authorities, Cyber Crime and Offences, Network Service Providers Liability, Cyber Regulations Appellate Tribunal, Penalties and Adjudication.

Module III: Cyber law and related Legislation (8 Hours)

Patent Law, Trademark Law, Copyright Software – Copyright or Patented, Domain Names and Copyright disputes, Electronic Data Base and its Protection, IT Act and Civil Procedure Code, IT Act and Criminal Procedural Code, Relevant Sections of Indian Evidence Act, Relevant Sections of Bankers Book Evidence Act, Relevant Sections of Indian Penal Code, Computer Crime, Ethical issues in computer security.

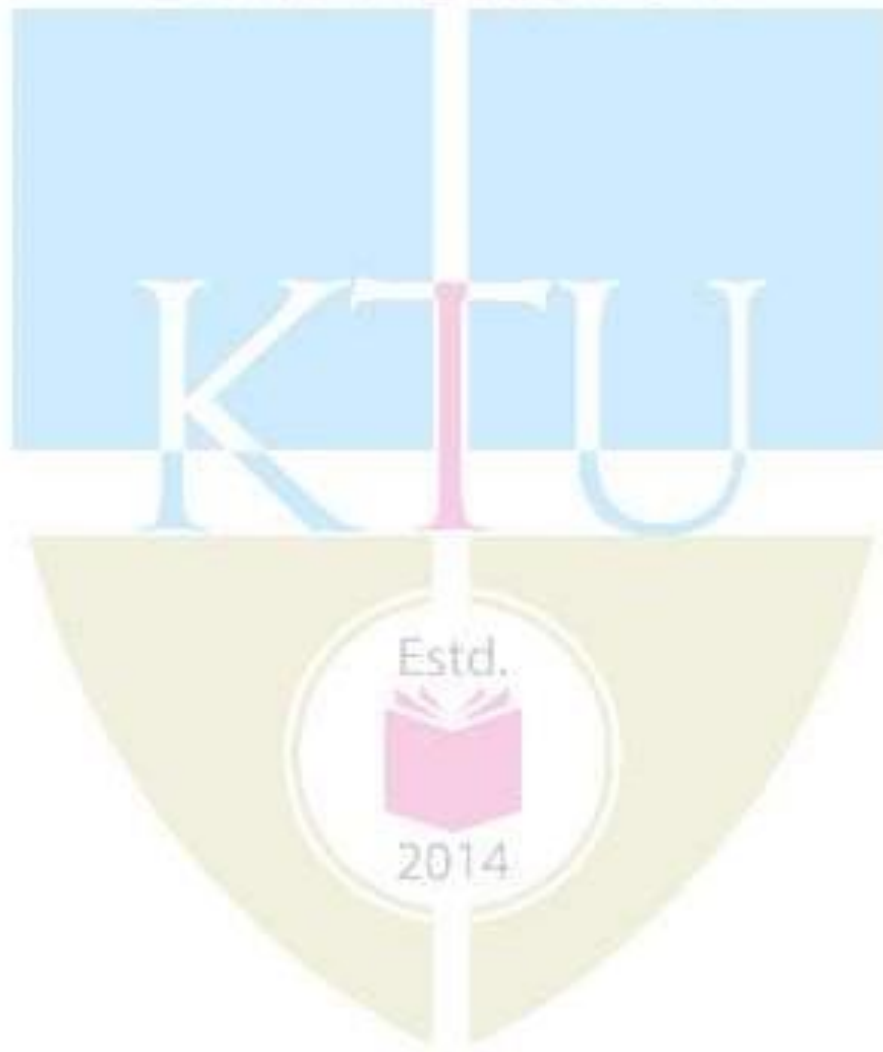
Module IV: Electronic Business and legal issues (8 Hours)

Evolution and development in ecommerce, paper vs paperless contracts, E-Commerce models- B2B,B2C,E-security, Application area: Business, taxation,

electronic payments, supply chain, Corporate policies-Tier1,Tier 2 and Tier 3 policies.

Module V: Writing the Security Policies (8 Hours)

Computer location and facility construction, Contingency planning - Internet security policies, Administrative and User responsibilities, Internet security policies, Human factors in security. Role of an information security professional, Cyber Pornography (Exm.MMS),Cyber-Stalking.



COURSE PLAN

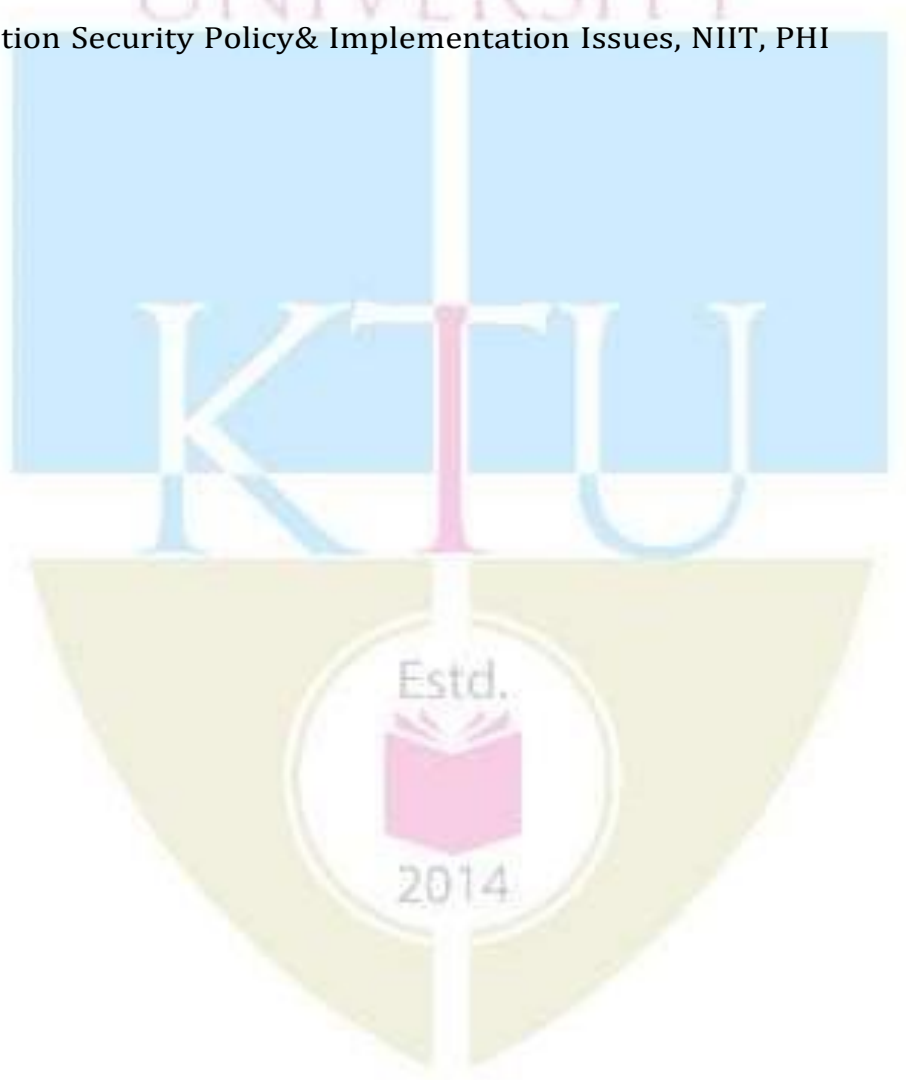
No	Topic	No. of Lectures (40)
1	Introduction to Cyber Law Evolution of Computer Technology(8 hours)	
1.1	Emergence of Cyberspace	2
1.2	Cyber Ethics and Cyber Jurisdiction,	2
1.3	Jurisdiction issues under IT Act, 2000.	1
1.4	Cyberspace-Web space, Web hosting and web Development agreement, Legal and Technological Significance of domain Names.	2
1.5	Internet as a tool for global access and internet protocols	1
2	Information technology Act(8 hours)	
2.1	Overview of IT Act, 2000, Amendments, and Limitations of IT Act	2
2.2	Digital Signatures, Cryptographic Algorithm, Public Cryptography, Private Cryptography	2
2.3	Electronic Governance, Legal Recognition of Electronic Records, Legal Recognition of Digital Signature Certifying Authorities	2
2.4	Cyber Crime and Offences, Network Service Providers Liability, Cyber Regulations Appellate Tribunal, Penalties and Adjudication	2
3	Cyber law and related Legislation(8 hours)	
3.1	Patent Law, Trademark Law, Copyright	1

INFORMATION TECHNOLOGY

3.2	Software – Copyright or Patented, Domain Names and Copyright disputes, Electronic Data Base and its Protection	2
3.3	IT Act and Civil Procedure Code, IT Act and Criminal Procedural Code	1
3.4	Relevant Sections of Indian Evidence Act, Relevant Sections of Bankers Book Evidence Act, Relevant Sections of Indian Penal Code	2
3.5	Computer Crime, Ethical issues in computer security	2
4	Electronic Business and legal issues (8 hours)	
4.1	Evolution and development in Ecommerce	2
4.2	Paper vs paperless contracts, E-Commerce models- B2B, B2C, E security	2
4.3	Application area: Business, taxation, electronic payments, supply chain	2
4.4	Corporate policies-Tier1, Tier 2, and Tier 3 policies	2
5	Writing The Security Policies (8 hours)	
5.1	Computer location and facility construction, Contingency planning	2
5.2	Internet security policies, Administrative and User responsibilities, Internet security policies	1
5.3	Human factors in security.	1
5.4	Role of an information security professional	2
5.5	Cyber Pornography (Exm.MMS), Cyber-Stalking	2

Reference Books

1. Guide to Cyber Laws by Rodney D. Ryder; Wadhwa and Company, Nagpur.
2. Charles P. Pfleeger Security in computing, 4th edition, Prentice Hall, 2006.
3. Thomas R. Peltier, Information Security policies and procedures: A Practitioner's Reference, 2nd Edition Prentice Hall, 2004.
4. K. Kumar," Cyber Laws: Intellectual property &E-Commerce, Security",1st Edition, DominantPublisher,2011.
5. Information Security Policy& Implementation Issues, NIIT, PHI



222EIT004	DESIGN AND ANALYSIS OF NETWORKS	CATEGORY	L	T	P	CREDIT
		PROGRAMME ELECTIVE 3	3	0	0	

Preamble:

This course helps the learner to inspire the students to learn the subject of theoretical foundations in computer networking and understand the fundamental concepts in design of networking also introduce the purpose of analysing different concept of network and routing protocols

Course Outcomes:

After the completion of the course, the student will be able to

CO 1	Analyse user requirement, Application Requirement, device requirement and Network Requirement	Analyse
CO 2	Analyse flows through a given Network Architecture	Analyse
CO 3	Identify the architecture for network management	Apply
CO 4	Choose performance, security and privacy architecture	Apply
CO 5	Applying Interconnection Mechanism to the design.	Apply

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1							
CO 2							
CO 3							
CO 4							
CO 5							

Assessment Pattern

Bloom's Category	End Semester Examination
Understand	20%
Apply	60%
Analyse	20%
Evaluate	0%
Create	0%

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

The evaluation shall only be by employing application-based, analysis-based, or design-based questions (for both internal and end-semester examinations)

Continuous Internal Evaluation (CIE) Pattern:

Preparing a review article based on peer-reviewed original publications (minimum 10 Publications shall be referred)/Assignment: 10 Marks

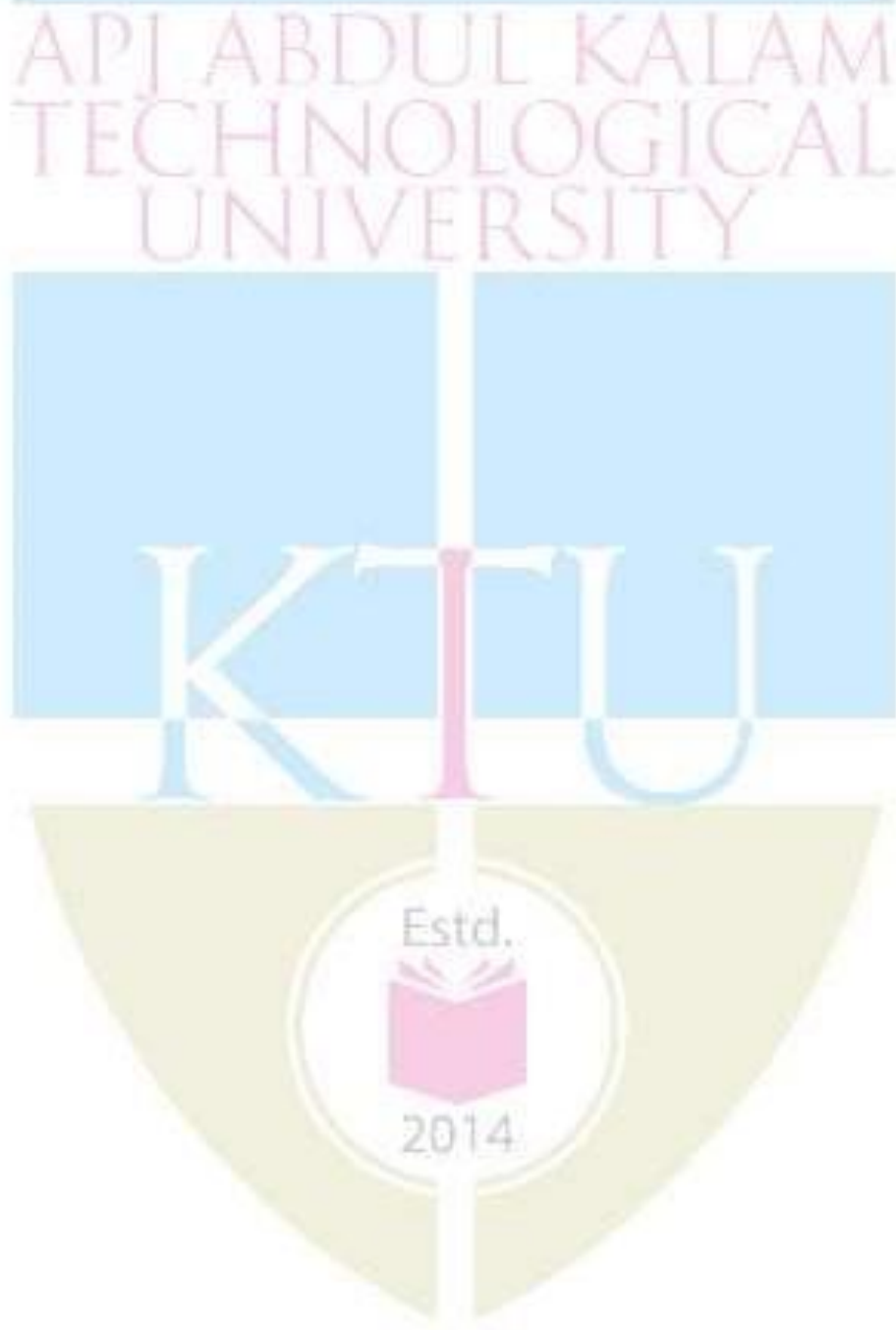
Course-based task/Seminar/Data Collection and interpretation: 20 Marks

Test paper (one only, and shall include a minimum of 80% of the syllabus): 10 Marks

End Semester Examination (ESE) Pattern:

The end-semester examination shall be conducted by the respective College. There shall be two parts; Part A and Part B. Part A shall contain five numerical/short answer questions with one question from each module, having five marks for each question (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students). Students shall answer all questions. Part B shall contain seven questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem-solving and quantitative evaluation), with a minimum one question from each module of which student shall answer any five. Each question shall carry seven marks.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly. For example if the average end semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is $40+20 = 60\%$.



MODEL QUESTION PAPER FOR END-SEMESTER EXAMINATION

Time: 2½ Hours, Maximum marks: 60

PART A (Answer all questions. Five marks each)

1. Assume your institute is selected for a centre of broadcasting lecture videos in realtime by APJ Abdul Kalam Technological University. For that you need to conduct a capacity requirement analysis for the proposed network. How can you perform the analysis? Explain.
2. Which flow models apply to each set of flows described below? Justify.
 - a. Forty workstations processing batch jobs overnight, managed by a central mainframe
 - b. A transaction-processing application, authorizing credit card transactions between a company's retail stores and its headquarters
3. Provide an account on Monitoring mechanisms for managing a network
4. Describe the term Physical Security. With a neat diagram explain the areas of a network where you can implement Physical Security
5. Describe the process of making technology choices for network design

PART B (Answer any five questions. Seven marks each)

6.
 - a. Compare and contrast the Tactical and Strategic significance of networking projects. (3)
 - b. With a diagram mention the components behind Network Analysis. (4)
7.
 - a. Suppose you are a network analyst in an MNC. Your organization has 40 workstations processing batch jobs overnight, managed by a central mainframe. Which flow model you will suggest the organization to follow? Explain why. (4)
 - b. Briefly explain the Monitoring Mechanisms available today for a network (3)
8. Suppose your institution has the below shown departments and labs. It has been given a Class B network of address 142.118.0.0. Perform VLSM on it and provide one subnet to each lab. (7)

Department	Labs	Size
IT	4	900
CCF	1	2000
EE	3	200
ECE	6	90
ME	5	150
CE	10	400

9. a. An IP address of 16.0.0.0/8 is assigned to an organization that has 50 departments. The organization needs to distribute IP addresses among these 50 departments. Identify
- i. The subnet mask used
 - ii. The number of subnets possible
 - iii. The first and last address of the first 10 subnets
- (4)
- b. What is NAT? (3)
10. a. Briefly explain the importance of SLA. (4)
- b. Outline the functionality of Next Hop Resolution Protocol. (3)
11. List different Security and Privacy mechanisms available for a network. (7)
12. Explain how security and privacy administration can be done. (7)

SYLLABUS

MODULE-1 (8 hours)

INTRODUCTION, REQUIREMENT ANALYSIS - Overview of Analysis, Architecture and Design Processes-Performance Characteristics. Requirement analysis - User Requirement- Application Requirement- Device Requirement- Network Requirement- Other Requirement. Gathering and Listing Requirements- Developing RMA, Delay and Capacity Requirements.

MODULE-2 (8 hours)

FLOW ANALYSIS - Flows-Individual and Composite flow -Critical Flow, Flow models, Flow Specification. Network Architecture- Component Architecture, Reference Architecture, Architectural Models.

MODULE-3 (8 hours)

NETWORK MANAGEMENT ARCHITECTURE - Addressing Mechanisms- Classful Addressing - Subnetting - Variable length Subnetting - Supernetting-Private Addressing and NAT, Routing Mechanisms, Addressing Strategies, Routing Strategies, Architectural Consideration. Network management, Network management Mechanisms, Architectural Considerations.

MODULE-4 (8 hours)

PERFORMANCE ARCHITECTURE, SECURITY AND PRIVACY ARCHITECTURE - Performance Mechanisms- Quality Services, Service level Arrangements, Architectural Consideration, Evaluation of Performance Mechanisms- Internal Relationship, External Relationship.

MODULE-5 (8 hours)

Interconnecting Technologies - Developing Criteria for Technology Evaluation, Making Technology Choices for the Network Design. Shared Medium, Switching, hybrid Mechanism-NHRP-MPOA, Applying Interconnection Mechanisms to the Design.

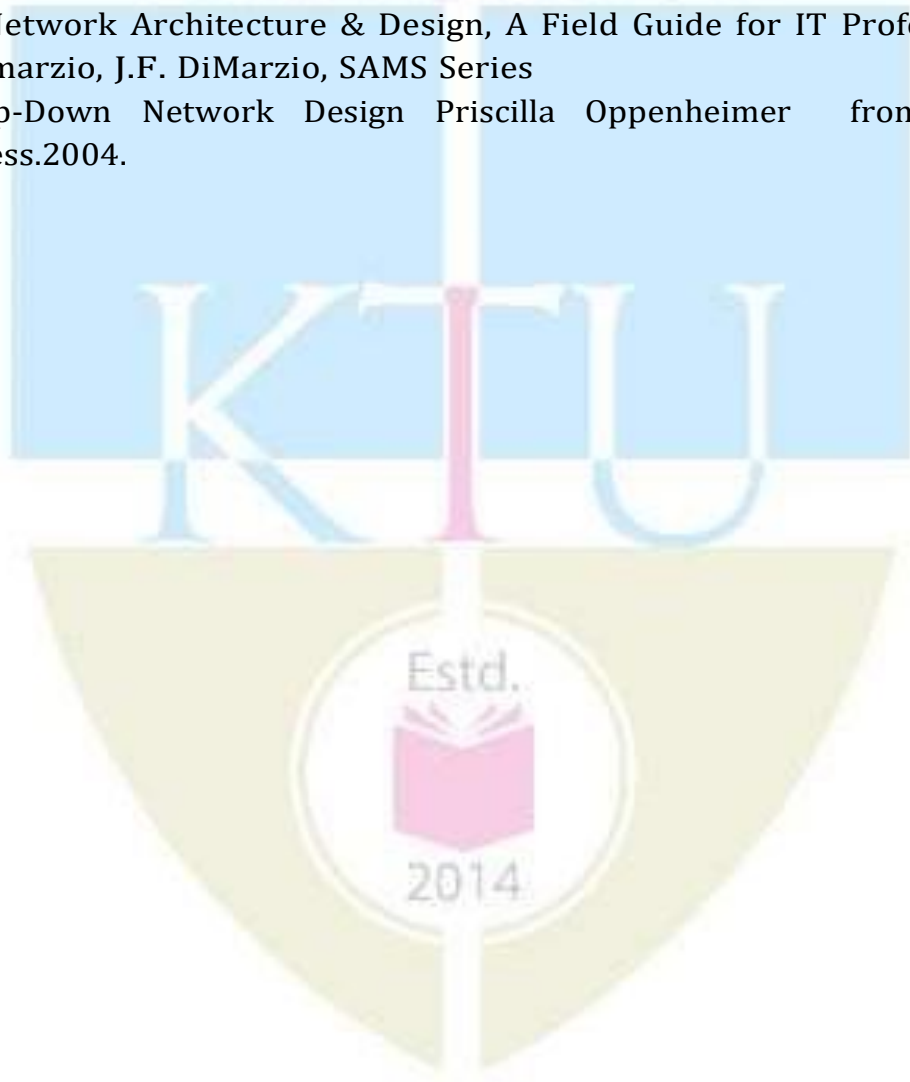
COURSE PLAN

No	Topic	No. of Lectures
1	Introduction, Requirement Analysis	
1.1	Overview of Analysis	1
1.2	Architecture and Design Processes-Performance Characteristics.	1
1.3	Requirement analysis - User Requirement Application Requirement- Device Requirement Network Requirement- Other Requirement.	1 1 1
1.4	Gathering and Listing Requirements Developing RMA, Delay and Capacity Requirements	1 2
2	Flow Analysis	
2.1	Individual and Composite flow Critical Flow	1 1
2.2	Flow models Flow Specification	1 1
2.3	Network Architecture Component Architecture Reference Architecture, Architectural Models.	1 1 2
3	Network Management Architecture	
3.1	Addressing Mechanisms- Classful Addressing – Subnetting Variable length Subnetting – Supernetting Private Addressing and NAT	1 1 1
3.2	Routing Mechanisms, Addressing Strategies Routing Strategies, Architectural Consideration.	1 1
3.3	Network management Network management Mechanisms Architectural Considerations.	1 1 1
4	Performance Architecture, Security and Privacy Architecture	
4.1	Performance Mechanisms Quality Services, Service level Arrangements Architectural Consideration	1 1 1
4.2	Evaluation of Performance Mechanisms Internal Relationship External Relationship.	1 1 1
4.3	Security and privacy Plan- Administration, Security and privacy Mechanisms Architectural considerations- Evaluation of Security Mechanisms	1 1

5	Interconnecting Technologies	
5.1	Developing Criteria for Technology Evaluation	2
	Making Technology Choices for the Network Design.	2
5.2	Shared Medium,	1
	Switching, hybrid Mechanism	1
5.3	NHRP-MPOA	1
	Applying Interconnection Mechanisms to the Design.	1

Reference Books

1. Network Analysis, Architecture and Design, 3rd Edition (The Morgan Kaufmann Series in Networking) James D. McCabe, Elsevier Science (USA), 2007.
2. Network Architecture & Design, A Field Guide for IT Professionals, Dimarzio, J.F. DiMarzio, SAMS Series
3. Top-Down Network Design Priscilla Oppenheimer from Cisco Press.2004.



222EIT005	PERFORMANCE EVALUATION OF COMPUTER SYSTEMS AND NETWORKS	CATEGORY	L	T	P	CREDIT
		PROGRAMME ELECTIVE 3	3	0	0	3

Preamble: The objective of this course is to understand the fundamental concepts of computer system performance evaluation. This will include an introduction to mathematical modeling techniques (Markov Chains, Queuing Theory, and Networks of Queues), discrete event simulation modeling, experiment design, workload characterization, and measurement of performance metrics, analysis, and presentation of results

Course Outcomes: After the completion of the course the student will be able to:

CO 1	Identify the main parameters and metrics related to the performance analysis of systems, particularly networks and computer systems. (Cognitive Knowledge Level: Understand)
CO 2	Compare single and multiple queues systems, with or without priority. (Cognitive Knowledge Level: Understand)
CO 3	Explain different types of queueing networks (Cognitive Knowledge Level: Understand)
CO 4	Use Queueing theory and Markov processes to model and measure the performance of systems (Cognitive Knowledge Level: Apply)
CO 5	Explain the types and uses of Petri nets (Cognitive Knowledge Level: Understand)

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	√	√	√	√	√		√
CO 2			√	√	√		√
CO 3		√	√	√	√		
CO 4	√		√	√	√		
CO 5	√	√	√	√	√		

Assessment Pattern

Bloom's Category	End Semester Examination
Understand	80%
Apply	20%
Analyse	

Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation (CIE) Pattern:

Preparing a review article based on peer-reviewed original publications (minimum 10 Publications shall be referred)/

10 Marks

Assignment:

Course-based task/Seminar/Data Collection and interpretation:

20 Marks

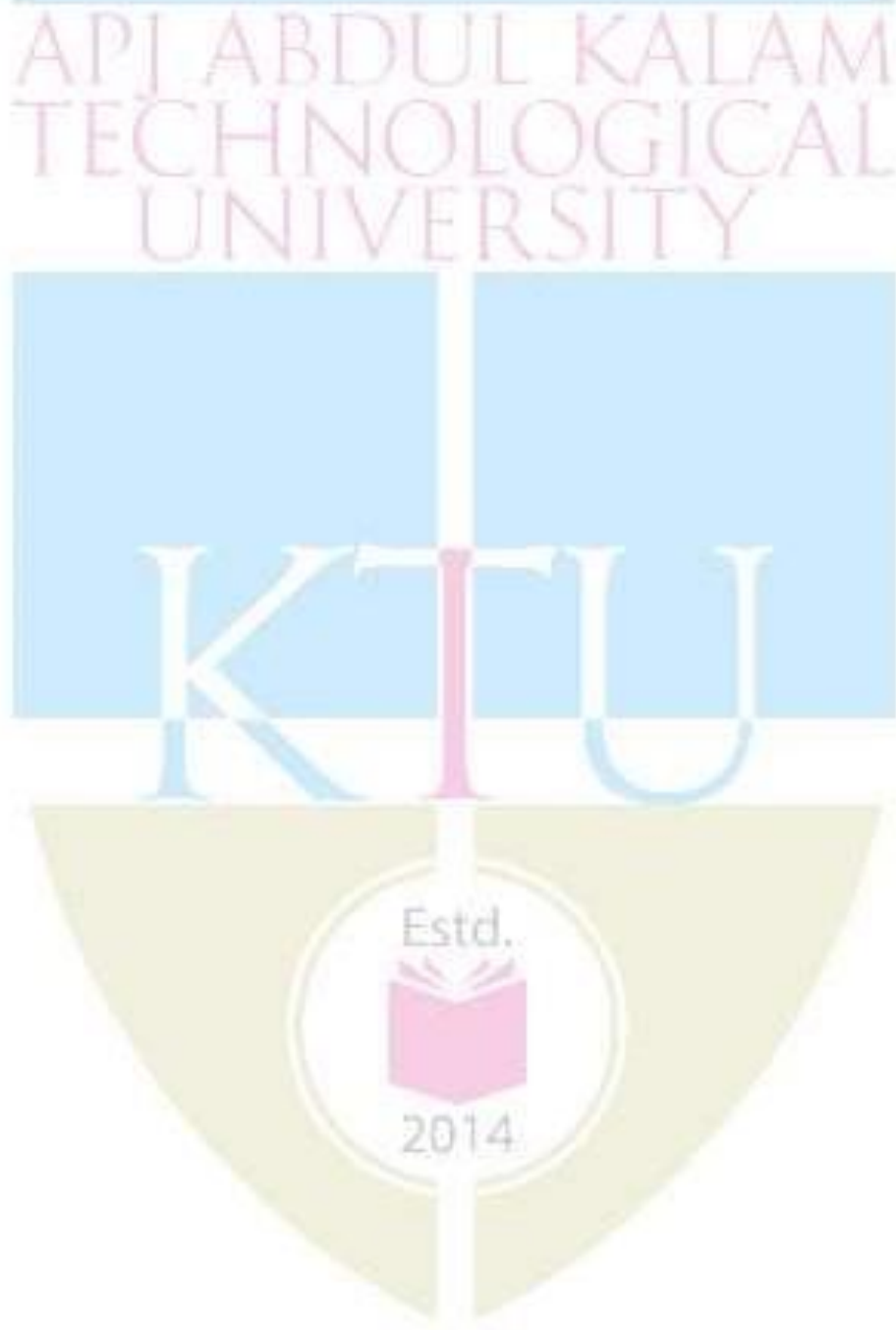
Test paper (one only, and shall include a minimum of 80% of the syllabus):

10 Marks

End Semester Examination (ESE) Pattern:

The end-semester examination shall be conducted by the respective College. There shall be two parts; Part A and Part B. Part A shall contain five numerical/short answer questions with one question from each module, having five marks for each question (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students). Students shall answer all questions. Part B shall contain seven questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem-solving and quantitative evaluation), with a minimum one question from each module of which student shall answer any five. Each question shall carry seven marks.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly. For example, if the average end semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is $40+20 = 60\%$.



MODEL QUESTION PAPER FOR END-SEMESTER EXAMINATION

Time: 2½ Hours, Maximum marks: 60

PART A (Answer all questions. Five marks each)

1. What is meant by Queueing theory?
2. Distinguish between Discrete-Time versus Continuous-Time Markov Chains?
3. What does Load really mean in an M/M/k?
4. What are the basic elements of Petri Net?
5. Compare Computer Time-Sharing and Multi-access Systems

PART B (Answer any five questions. Seven marks each)

6. Explain the Classification of Queueing Networks
7. Describe Little's Law for Open Systems and closed systems.
8. Explain the Umbrella Problem of Finite-State DTMCs
9. Describe Google's DTMC Algorithm.
10. State and prove Burke's Theorem.
11. Explain the procedure to compute the reach ability graph of Petri net.
12. Discuss the firing rules of timed Petri nets.
13. Describe the flow control mechanism of computer networks in a detailed way.

SYLLABUS

Module – 1 (8 hours)

Introduction to Queueing-Motivating Examples of the Power of Analytical Modeling-Queueing Theory Terminology-Probability Review-Generating Random Variables-Sample Paths, Convergence, and Averages-The Predictive Power of Simple Operational Laws- Little's Law and Other Operational Laws-Modification Analysis: What-if for Closed Systems

Module – 2 (8 hours)

From Markov Chains to Simple Queues-Discrete-Time Markov Chains-Ergodicity Theory- Real-World Examples: Google, Aloha, and Harder Chains-Exponential Distribution and the Poisson Process-Transition to Continuous-Time Markov Chains-M/M/1 and PASTA.

MODULE: 3 (7 hours)

Server Farms and Networks: Multi-Server, Multi-Queue Systems-Server Farms: M/M/k and M/M/k/k-Capacity Provisioning for Server Farms-Time-Reversibility and Burke's Theorem-Networks of Queues and Jackson Product Form-Classed Network of Queues-Closed Networks of Queues-Transform Analysis-Power Optimization Application

MODULE: 4 (8 hours)

Computer Time-Sharing and Multi-access Systems- Computer-Communication Networks: Analysis and Design -Computer-Communication Networks: Measurement, Flow Control, and ARPANET Traps

MODULE: 5 (8 hours)

PETRI NETS, classical Petri Nets, Timed Petri Nets Priority-based Petri Nets, Colored Petri Nets. Generalized Petri Nets – Tool Selection – Validation of Results.

COURSE PLAN

No	Topic	No. of Lectures (40)
Module - 1 (8 hours)		
1.1	Introduction to Queueing-Motivating Examples of the Power of Analytical Modeling-Queueing Theory Terminology	2
1.2	Probability Review-Generating Random Variables-Sample Paths, Convergence, and Averages	2
1.3	The Predictive Power of Simple Operational Laws- Little's Law and Other Operational Laws	2
1.4	Modification Analysis: What-if for Closed Systems	2
Module - 2 (8 hours)		
2.1	From Markov Chains to Simple Queues-Discrete-Time Markov Chains-Ergodicity Theory-	2
2.2	Real-World Examples: Google, Aloha, and Harder Chains-.	2
2.3	Exponential Distribution and the Poisson Process-Transition to Continuous-Time Markov Chains	2
2.4	M/M/1 and PASTA.	2
MODULE: 3 (8 hours)		
3.1	Server Farms and Networks: Multi-Server, Multi-Queue Systems-Server Farms: M/M/k and M/M/k/k	2
3.2	Capacity Provisioning for Server Farms-Time-Reversibility and Burke's Theorem	2
3.3	Networks of Queues and Jackson Product Form-Classed Network of Queues-Closed Networks of Queues	2
3.4	Transform Analysis-Power Optimization Application	2
MODULE: 4 (8 hours)		

4.1	Computer Time-Sharing Systems	2
4.2	Multi access systems	1
4.3	Computer-Communication Networks: Analysis and Design.	2
4.4	Computer-Communication Networks: Measurement, Flow Control	2
4.5	ARPANET Traps	1
MODULE: 5 (8 hours)		
5.1	PETRI NETS – Classical Petri Nets	2
5.2	Timed Petri Nets	1
5.3	Priority-based Petri Nets	1
5.4	Colored Petri Nets.	1
5.5	Generalized Petri Nets	1
5.6	Tool Selection – Validation of Results.	2

Reference Books

1. Mor Harchol-Balter, Performance Modeling and Design of Computer Systems: Queueing Theory in Action. Cambridge University Press, 2013
2. L. Kleinrock, Queueing Systems vol I & II , John Wiley and Sons, 1975.
3. Averill, M. "Law. Simulation modeling and analysis." Tata McGraw-Hill (2007).
4. Paul J. Fortier, Howard E. Michael, Computer Systems Performance Evaluation and Prediction, Elsevier Science 2003.
5. Thomas G. Robertazzi, Computer Networks and Systems Queing theory and Performance Evaluation, 3rd Edition, Springer, 2000.
6. JY Le Boudec, Performance Evaluation of Computer and Communication Systems, EPFL
7. D. Menascé et al., Performance by Design, Prentice Hall .
8. Domenico Ferrari, Giuseppe Serazzi ,Alexandro Zeijher, Measurement & Tuning of Computer Systems , Prentice Hall Inc, 1983.
9. Michael F. Mories and Paul F. Roth, Tools and techniques Computer Performance Evaluation, Van Nostrand, 1982.



The logo of API Abdul Kalam Technological University is a shield-shaped emblem. The top section is light blue with the university's name in pink. The middle section is divided into four blue squares. The bottom section is olive green and contains a circular seal with a pink book icon, the text 'Estd. 2014', and the year '2014'.

PROGRAM ELECTIVE IV

222EIT006	VIRTUALIZATION TECHNIQUES	CATEGORY	L	T	P	CREDIT
		PROGRAMME ELECTIVE 4	3	0	0	3

Preamble:

The syllabus is designed with the view of preparing the students capable of understanding the virtualization techniques. The students will learn about network, server, and storage virtualization.

Course Outcomes:

After completion of the course, the student will be able to

CO 1	Explain the fundamentals and current practices in virtualization. (Level: Understand)
CO 2	Discuss and Develop Virtual server platforms (Level: Apply)
CO 3	Apply Network virtualization and Application virtualization concepts in real world scenario. (Level: Apply)
CO 4	Summarise various devices and technologies required for providing virtual storage and discuss the concepts of Virtual Desktop (Level: Understand)
CO 5	Apply the concepts of virtualization in computing, network and storage, and recognize the major providers of such systems in the market today (Level: Apply)

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

PO1: An ability to independently carry out research/investigation and development work in engineering and allied streams

PO2: An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

PO3: An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program

PO4: An ability to apply stream knowledge to design or develop solutions for real-world problems by following the standards

PO5: An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tools to model, analyze and solve practical engineering problems.

PO6: An ability to engage in life-long learning for the design and development related to the stream related problems taking into consideration sustainability, societal, ethical, and environmental aspects

PO7: An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	P 6	PO 7
CO 1	☑	☑					
CO 2	☑			☑	☑		☑
CO 3	☑			☑	☑		
CO 4	☑			☑	☑		
CO 5	☑			☑	☑	☑	☑

Assessment Pattern

Bloom's Category	End Semester Examination
Understand	50%
Apply	50%
Analyse	
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation (CIE) Pattern:

Preparing a review article based on peer-reviewed original publications (minimum 10 Publications shall be referred)/
Assignment: 10 Marks

Course-based task/Seminar/Data Collection and interpretation: 20 Marks

Test paper (one only, and shall include a minimum of 80% of the syllabus): 10 Marks

End Semester Examination (ESE) Pattern:

The end-semester examination shall be conducted by the respective College. There shall be two parts; Part A and Part B. Part A shall contain five numerical/short answer questions with one question from each module, having five marks for each question (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students). Students shall answer all questions. Part B shall contain seven questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem-solving and quantitative evaluation), with a minimum one question from each module of which student shall answer any five. Each question shall carry seven marks.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly.

For example, if the average end-semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is $40+20 = 60$ %.

Course Level Assessment Questions

Course Outcome 1 (CO1):

1. Explain different types of virtualization architectures.
- 2 Give the taxonomy of virtual machines.

Course Outcome 2 (CO2)

1. Discuss the uses of Virtual server consolidation.
2. Describe virtual enterprise transport virtualization
3. What is the need for server virtualization?

Course Outcome 3(CO3):

1. Explain plane virtualization in detail
2. Explain VRF in detail.
3. Discuss how VLAN and VRF are different
4. Give examples for application virtualization.

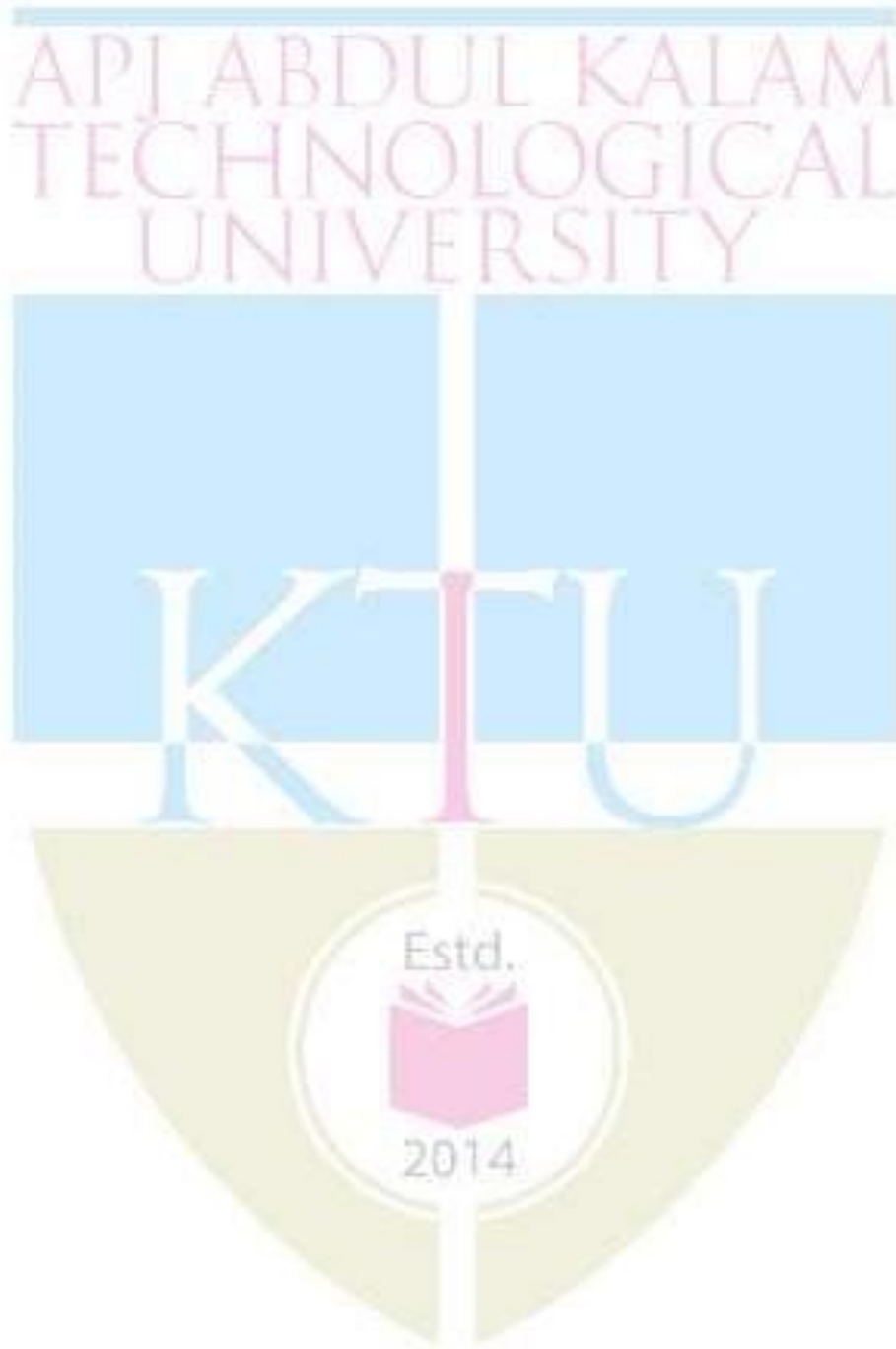
Course Outcome 4 (CO4):

1. Explain the importance of Virtual tape libraries
2. Explain the different common RAID levels, with appropriate diagrams
3. Briefly explain different desktop virtualization scenarios with appropriate diagrams.

Course Outcome 5 (CO5):

1. Apply the various virtualization technologies you studied to design an effective IT infrastructure in your college.
2. What are the differences between SAN and NAS?

3. How virtualization technologies help an IT infrastructure manager in managing resources? Describe with any case study.



MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

SECOND SEMESTER M. TECH DEGREE EXAMINATION, MONTH & YEAR

Course Code: 222EIT006

Course Name: VIRTUALIZATION TECHNIQUES

Max. Marks: 60

Duration: 2.5 Hours

PART A

Answer All Questions. Each Question Carries 5 Marks

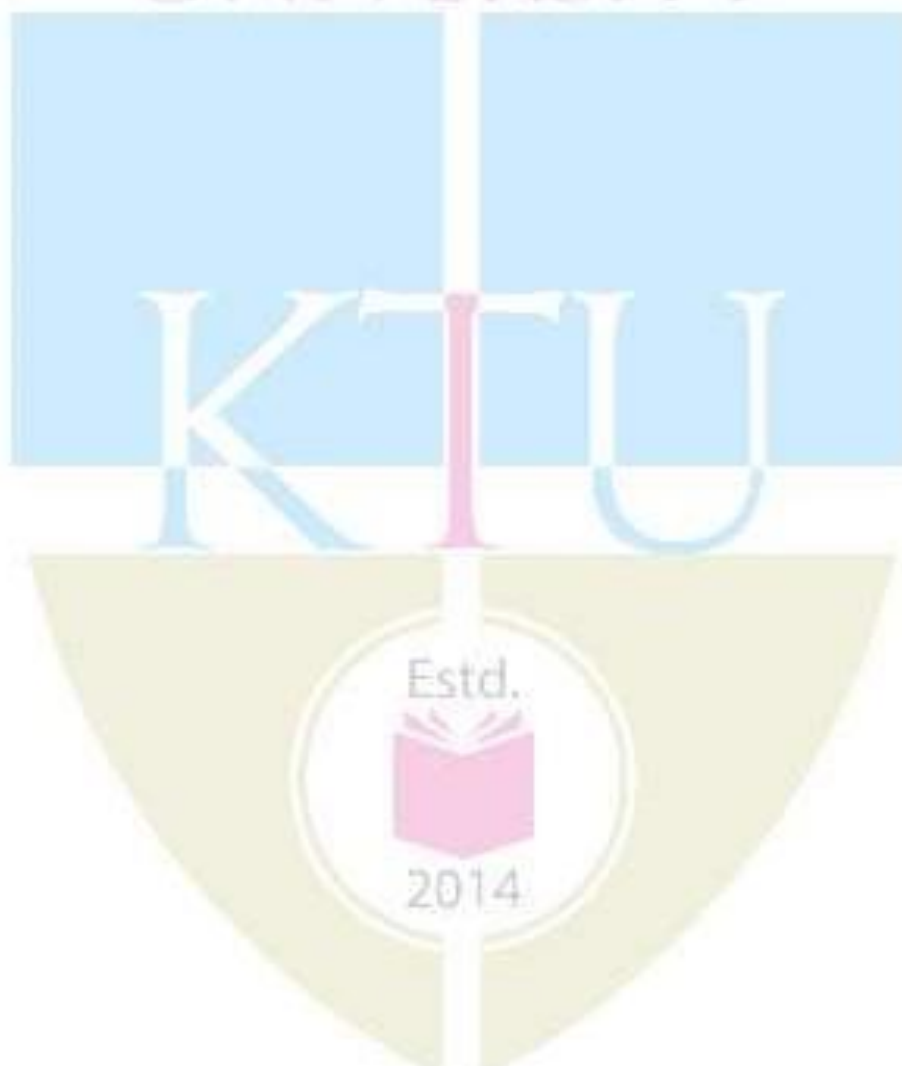
1. Explain hardware partitioning and software partitioning, with their merits and demerits.
2. Draw the figures, with a brief explanation, which show the difference between High-level language virtual machine and conventional systems.
3. Explain Zoning in the context of storage virtualization. How Zoning is done?
4. Compare two VMware products with their characteristic features.
5. Discuss the term DLL Hell. How Application Virtualization helps to overcome such issues.

Part B

(Answer any five questions. Each question carries 7 marks)

6. Briefly explain different desktop virtualization scenarios with appropriate diagrams.
7. Discuss in detail, storage, network, and desktop virtualization.
8. How is Virtualization architecture different from traditional architecture? What are the different virtualization architectures? Explain with diagrams and examples.
9. What are the business cases for server virtualization? Explain the list of factors to be considered in choosing an appropriate server virtualization platform
10. Define the following in the context of network device virtualization:
 - a. Virtual LAN (VLAN)
 - b. Virtual routing and forwarding (VRF)
 - c. Virtual forwarding instance (VFI)
 - d. Virtual firewall context

11. Assume an airport used by several airlines and having several common facilities.
- (i) Briefly describe the scenario.
 - (ii) List different roles that could be played by virtualization in this scenario.
 - (iii) List and describe the different virtualization technologies which will be involved.
12. Explain about SNIA shared storage model.



SYLLABUS

Module 1

INTRODUCTION TO VIRTUALIZATION (8 hours)

System Architectures (Layers-interfaces-API-ABI-ISA) –Virtual Machine

Basics, Types of Virtualization Techniques, Process virtual machines, System virtual Machines, Taxonomy of virtual machines, Emulation: Basic Interpretation – Threaded Interpretation – Pre-coded & Direct Interpretation – Binary translation, Full and Para Virtualization, Types of Hypervisor, Types of Virtualization.

Module 2

HIGH LEVEL LANGUAGE VIRTUAL MACHINES AND SERVER VIRTUALIZATION (8 hours)

HLL Virtual Machines: Object Oriented HLL VMs - Java VM architecture - Java Native Interface - Common Language Infrastructure.

Server virtualization: Partitioning techniques – Physical and Logical Partitioning, Virtual hardware, Virtual server Consolidation, Uses of virtual servers, Server virtualization platforms, Merits and demerits

Module 3

NETWORK AND APPLICATION VIRTUALIZATION(8 hours)

Network Virtualization: Design of Scalable Enterprise Networks, Layer 2 Virtualization – VLAN - VFI, Layer 3 Virtualization – VRF - Virtual Firewall Contexts, Network Device Virtualization, Data Path Virtualization- Generic Routing Encapsulation – IPsec, L2TPv3 Label Switched Paths - Control Plane Virtualization–Routing Protocols-Multi-Topology Routing.

Application Virtualization: Concepts - Application Management Issues - Redesign Application Management – Application Migration

Module 4 STORAGE AND DESKTOP VIRTUALIZATION(9 hours)

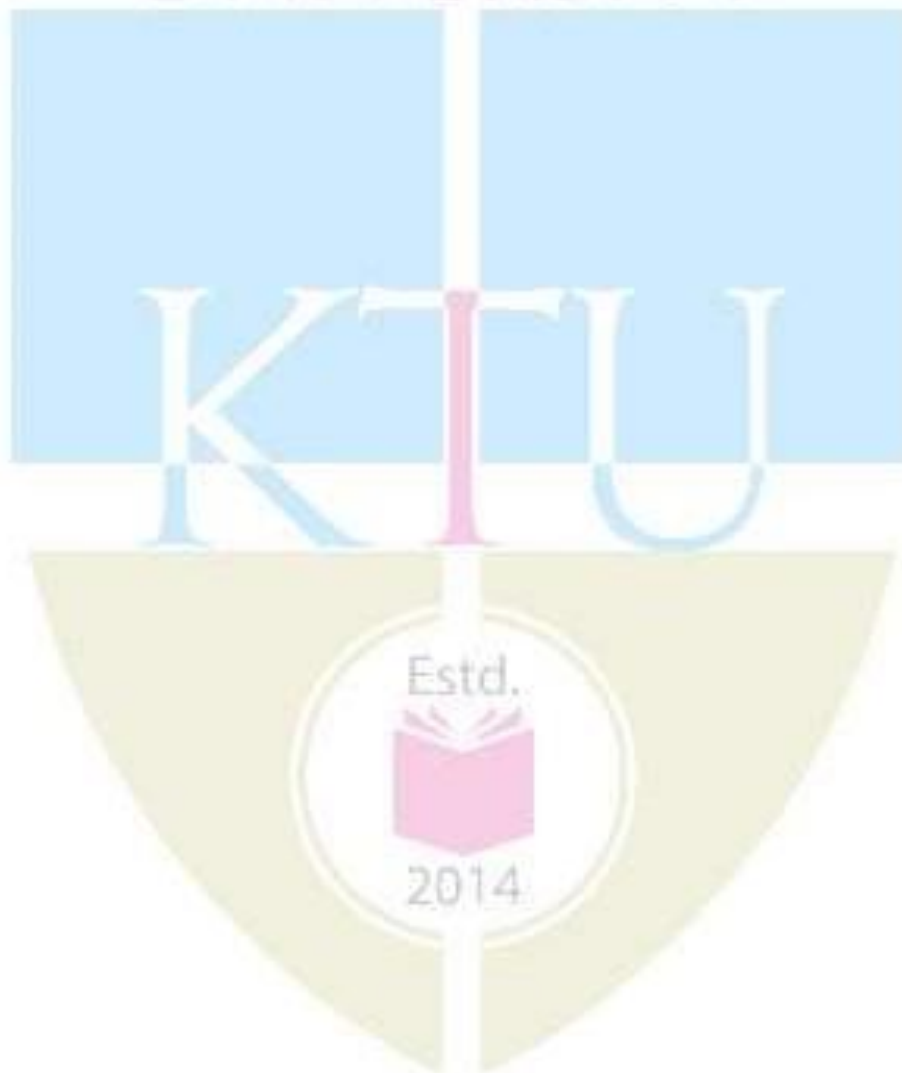
Storage Virtualization: SCSI- SCSI buses, FiberChannel – Fiber Channel Cables – Fiber Channel Hardware Devices, iSCSI Architecture – Securing iSCSI, Backup and Recovery techniques, RAID Levels (RAID 0, RAID 1, RAID 5, RAID 0+1, RAID 1+0, RAID 5+0). SNA Shared Storage Model – Classical Storage Model – SNA Shared Storage Model – Host-based Architecture – Storage based architecture – Network based Architecture – Fault tolerance to SAN – Performing Backups.

Desktop Virtualization: Concepts - Desktop Management Issues - Potential Desktop Virtualization Scenarios - Desktop Virtualization Infrastructures

Module 5

APPLYING VIRTUALIZATION (7 hours)

Comparison of Virtualization Technologies: Guest OS, host OS, hypervisor, emulation, kernel level, shared kernel – Enterprise Solution: VMware Server, VMware Esxi, Citrix XenServer, Microsoft virtual PC, Microsoft Hyper-V, Virtual box – Server Virtualization: configuring Server with server virtualization, Adjusting & tuning virtual servers.VM backup & migration, Desktop Virtualization:terminal services, hosted desktop, web based solutions, localized virtualized desktop, Network & storage virtualization:VPN,VLAN,SAN & VSAN,NAS.



COURSE PLAN

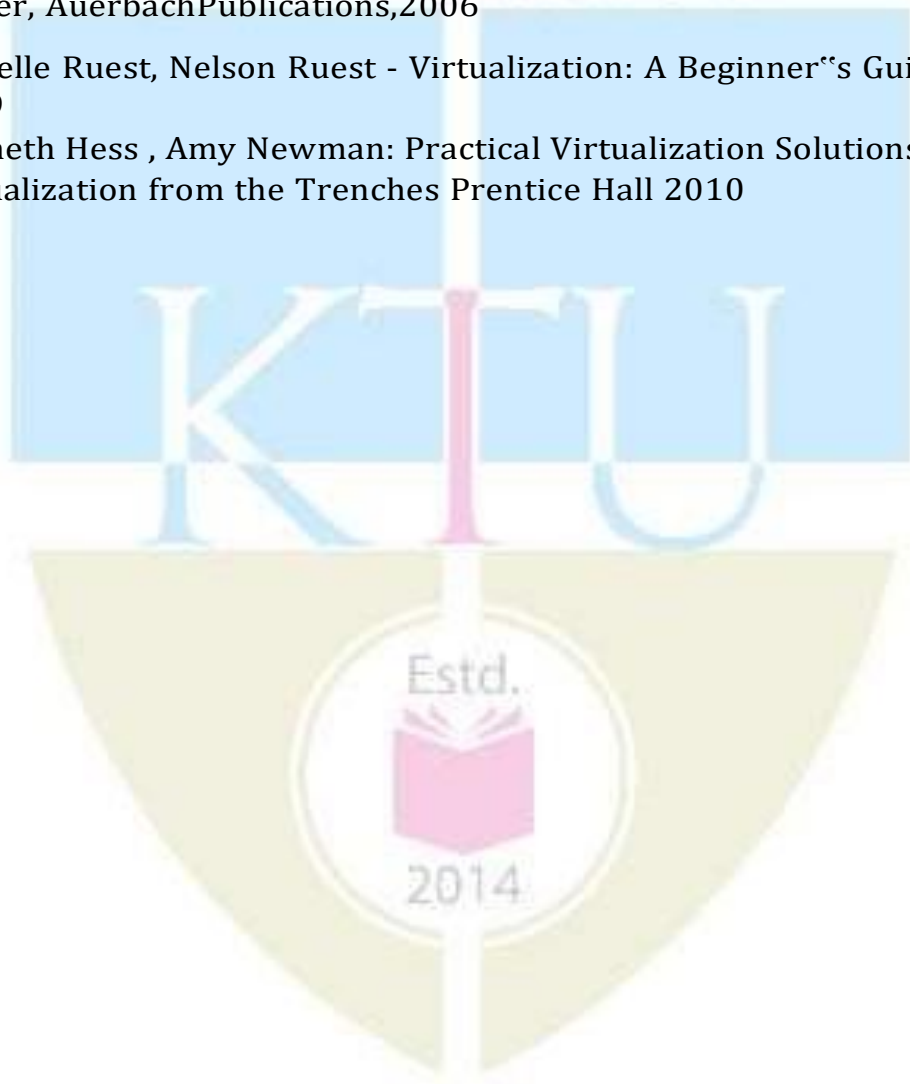
No	Topic	No. of Lectures (40)

1	Module 1 INTRODUCTION TO VIRTUALIZATION (8 hours)	
1.1	System Architectures(Layers-interfaces-API-ABI-ISA) – Virtual Machine Basics	1
1.2	Types of Virtualization Techniques	1
1.3	Process virtual machines – System virtual Machines	1
1.4	Taxonomy of virtual machines	1
1.5	Emulation: Basic Interpretation	1
1.6	Threaded Interpretation	1
1.7	Pre-coded & Direct Interpretation – Binary translation	1
1.8	Full and Para Virtualization, Types of Hypervisor, Types of Virtualization	1
2	Module 2 HIGH LEVEL LANGUAGE VIRTUAL MACHINES AND SERVER VIRTUALIZATION (8 hours)	
2.1	HLL Virtual Machines: Object Oriented HLL VMs - Java VM architecture	1
2.2	Java Native Interface - Common Language Infrastructure.	1
2.3	Server virtualization: Partitioning techniques - Physical and Logical Partitioning	2
2.4	Virtual hardware, Virtual server Consolidation	1
2.5	Uses of virtual servers	1
2.6	Server virtualization platforms	1
2.7	Merits and Demerits (contexts where server virtualization can be and can't be used)	1
3	Module 3 NETWORK AND APPLICATION VIRTUALIZATION(8 hours)	
3.1	Network virtualization : Design of Scalable Enterprise Networks	1
3.2	Layer2 Virtualization – VLAN - VFI	1
3.3	Layer 3 Virtualization – VRF - Virtual Firewall Contexts	1
3.4	Network Device Virtualization, Data Path Virtualization- Generic Routing Encapsulation	1
3.5	IPsec, L2TPv3 Label Switched Paths	1
3.6	Control Plane Virtualization–Routing Protocols - Multi-Topology Routing	1
3.7	Application Virtualization: Concepts – Application Management Issues	1
3.8	Redesign Application Management – Application Migration	1

4	Module 4 STORAGE AND DESKTOP VIRTUALIZATION (9 Hours)	
4.1	SCSI- SCSI buses – FiberChannel – Fiber Channel Cables – Fiber Channel Hardware Devices, iSCSIArchitecture – Securing iSCSI	1
4.2	SAN backup and recovery techniques	1
4.3	RAID – RAID 0, RAID 1, RAID 5, RAID 0+1, RAID 1+0, RAID 5+0	2
4.4	SNIASharedStorageModel - ClassicalStorageModel	1
4.5	SNIASharedStorageModel	1
4.6	HostbasedArchitecture–Storagebasedarchitecture – Network based Architecture, Fault tolerance to SAN – Performing Backups	1
4.7	Desktop Virtualization:Concepts - Desktop Management Issues -	1
	Potential Desktop Virtualization Scenarios - Desktop Virtualization Infrastructures	1
5	Module 5 APPLYING VIRTUALIZATION (7 hours)	
5.1	Comparison of Virtualization Technologies: Guest OS, Host OS, hypervisor, emulation, kernel level, shared kernel	1
5.2	Enterprise Solution: VMware Server, VMware Esxi, Citrix XenServer, Microsoft virtual PC, Microsoft Hyper-V	1
5.3	Virtual box – Server Virtualization: configuring Server with server virtualization	1
5.4	Adjusting & tuning virtual servers	1
5.5	VM backup & migration	1
5.6	Desktop Virtualization:terminal services, hosted desktop, web based solutions, localized virtualized desktop	1
5.7	Network & storage virtualization: VPN,VLAN,SAN &VSAN,NAS	1

Reference Books

1. James E. Smith, Ravi Nair, Virtual Machines: Versatile Platforms for Systems and Processes, Elsevier/Morgan Kaufmann, 2005.
2. William von Hagen, Professional Xen Virtualization, Wrox Publications, January, 2008.
3. Chris Wolf, Erick M. Halter, Virtualization: From the Desktop to the Enterprise, A Press 2005.
4. Kumar Reddy, Victor Moreno, Network virtualization, Cisco Press, July, 2006.
5. David Marshall, Wade A. Reynolds, Advanced Server Virtualization: VMware and Microsoft Platform in the Virtual Data Center, Auerbach Publications, 2006
6. Danielle Ruest, Nelson Ruest - Virtualization: A Beginner's Guide, TMH, 2009
7. Kenneth Hess, Amy Newman: Practical Virtualization Solutions: Virtualization from the Trenches Prentice Hall 2010



INFORMATION TECHNOLOGY

222EIT007	BLOCKCHAIN TECHNOLOGY	CATEGORY	L	T	P	CREDIT
		PROGRAMME ELECTIVE 4	3	0	0	3

Preamble: This course is intended for students to have a deeper knowledge of the Blockchain Framework, Blockchain Architecture, Future systems, Blockchain in business, and its applications.

Course Outcomes:

After the completion of the course, the student will be able to

CO 1	Describe the history, types, and applications of Blockchain. (Level: Understand)
CO 2	Explain Blockchain Architecture and Consensus mechanisms. (Level: Understand)
CO 3	Develop projects using Web3J. (Level: Apply)
CO 4	Build an ICO on Ethereum with Blockchain in business. (Level: Apply)
CO 5	Construct blockchain-based application with Swarm. (Level: Apply)

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

P01:An ability to independently carry out research/investigation and development work in engineering and allied streams

P02:An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

P03:An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program

P04:An ability to apply stream knowledge to design or develop solutions for real world problems by following the standards

P05:An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tool to model, analyse and solve practical engineering problems.

P06:An ability to engage in life-long learning for the design and development related to the stream related problems taking into consideration sustainability, societal, ethical and environmental aspects

P07:An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	☑		☑		☑	☑	
CO 2	☑		☑	☑	☑	☑	☑
CO 3	☑	☑	☑	☑	☑	☑	☑
CO 4	☑	☑	☑	☑	☑	☑	☑
CO 5	☑	☑	☑	☑	☑	☑	☑

Assessment Pattern

Bloom's Category	End Semester Examination
Understand	40%
Apply	60%

INFORMATION TECHNOLOGY

Analyse	
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

The evaluation shall only be by employing application-based, analysis-based, or design-based questions (for both internal and end-semester examinations)

Continuous Internal Evaluation Pattern:

Evaluation shall only be based on application, analysis or design based questions (for both internal and end semester examinations).

Continuous Internal Evaluation (CIE) Pattern:

Preparing a review article based on peer-reviewed original publications (minimum 10 Publications shall be referred)/

10 Marks

Assignment:

Course-based task/Seminar/Data Collection and interpretation:

20 Marks

Test paper (one only, and shall include a minimum of 80% of the syllabus):

10 Marks

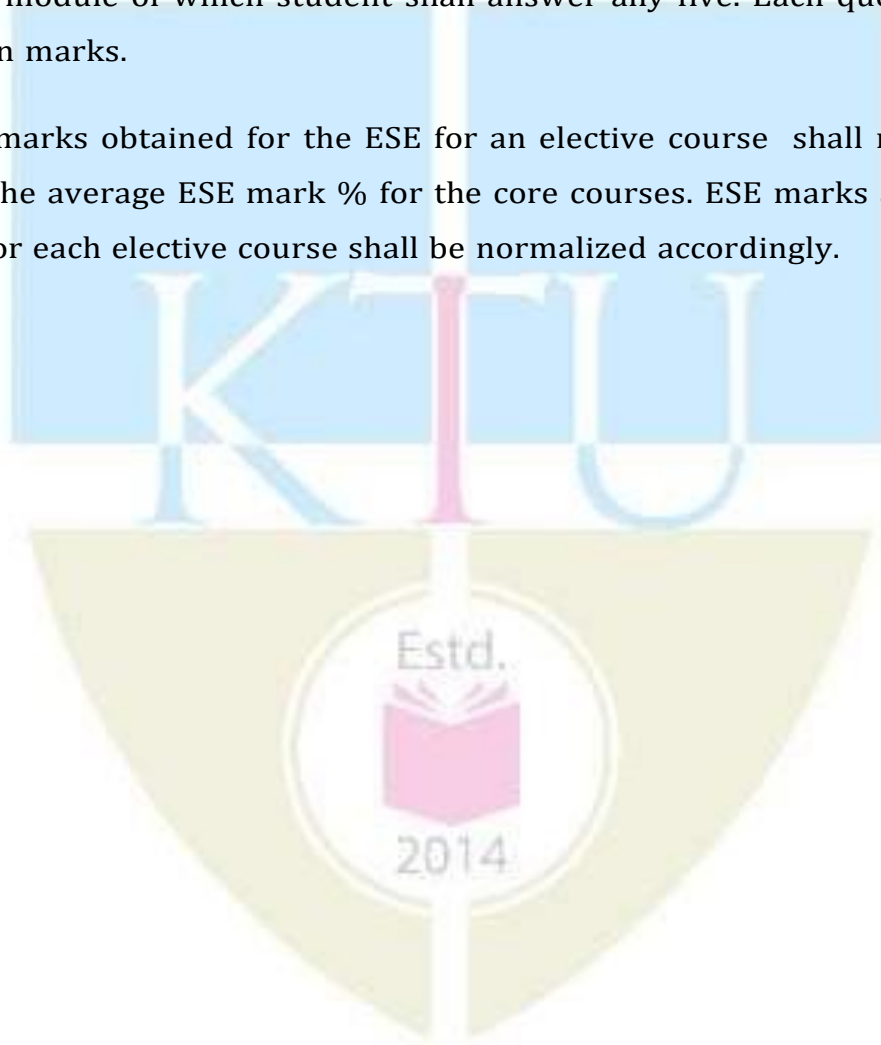
End Semester Examination (ESE) Pattern:

The end-semester examination shall be conducted by the respective College. There shall be two parts; Part A and Part B. Part A shall contain five

INFORMATION TECHNOLOGY

numerical/short answer questions with one question from each module, having five marks for each question (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students). Students shall answer all questions. Part B shall contain seven questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem-solving and quantitative evaluation), with a minimum one question from each module of which student shall answer any five. Each question shall carry seven marks.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly.



Model Question Paper

QP CODE:

Reg No: _____

Name: _____

PAGES :

4

**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
SECOND SEMESTER M.TECH DEGREE EXAMINATION, MONTH & YEAR**

Course Code:

Course Name: BLOCKCHAIN TECHNOLOGY

Max. Marks : 60

Duration: 2.5 Hours

PART A

Answer All Questions. Each Question Carries 5 Marks

1. Discuss the various attacks of vectors and vulnerabilities that may be present in blockchain. **5**
2. Explain the shortcomings of current transaction systems. **5**
3. Describe the concept of a block and its components. **5**
4. Define a Merkle hash tree, describe its role in blockchain, and explain the meaning of a Merkle root in the block header. **5**
5. Justify whether a public blockchain requires issuing its own native cryptocurrency to provide incentives to its validator network? **5**

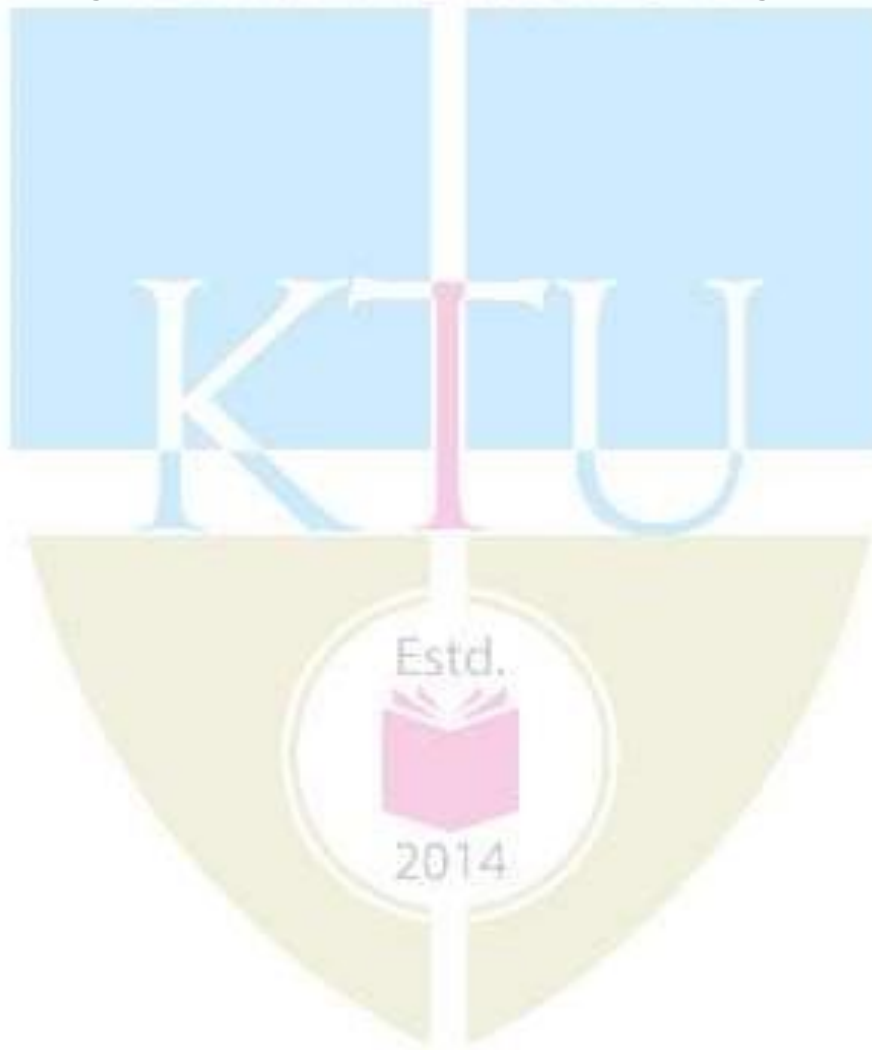
Part B

(Answer any five questions. Each question carries 7 marks)

6. (a) Differentiate between a public/permissionless and a private/permissioned blockchain. **(7)**
7. (a) Explain how consumer privacy is relevant to CBDCs. **(7)**
8. (a) Discuss whether the SHA-256 hash is appropriate for most blockchains **(7)**

INFORMATION TECHNOLOGY

9. (a) Define a utility token and indicate how it differs from a SoV/MoE token (7)
- 10 (a) Provide three reasons why blockchain is an improvement to the current system of data security and data transfer (7)
- 11 (a) Identify the two branches of AI research and the branch more relevant to blockchain (7)
- 12 (a) Discuss how cryptocurrencies are classified by the various government entities described in the reading (7)



SYLLABUS

Module 1: Introduction to blockchain : (8 hours) History, Definition, Distributed Ledger, Blockchain Categories – Public, Private, Consortium, Blockchain Network and Nodes, Peer-to-Peer Network, Mining Mechanism, Generic elements of Blockchain, Features of Blockchain, and Types of Blockchain.

Module 2: Blockchain architecture : (8 hours) Operation of Bitcoin Blockchain, Blockchain Architecture – Block, Hash, Distributer P2P, Structure of Blockchain-Consensus mechanism: Proof of Work (PoW), Proof of Stake (PoS), Byzantine Fault Tolerance (BFT), Proof of Authority (PoA) and Proof of Elapsed Time (PoET)

Module 3: Blockchain-based future system : (8 hours) Project presentation- Futures smart contract: Blockchain oracles-Web3j: Setting up the Web3J- Installing web3J- Wallet creation, Java client: The wrapper generator- Initializing web3j- Setting up Ethereum accounts- Deploying the contract.

Module 4: Blockchains in business and creating ico: (8 hours) Public versus private and permissioned versus permission less blockchains- Privacy and anonymity in Ethereum, Why are privacy and anonymity important? - The Ethereum Enterprise Alliance- Blockchain- as-a-Service, Initial Coin Offering (ICO): Project setup for ICO implementation- Token contracts- Token sale contracts- Contract security and testing the code.

Module 5: Distributed storage ipfs and swarm: (8 hours) Ethereum Virtual Machine- Swarm and IPFS: Installing IPFS, Hosting our frontend: Serving your frontend using IFPS, Serving your frontend using Swarm, IPFS file uploader project: Project setup - the web page.

INFORMATION TECHNOLOGY

COURSE PLAN (For 3-credit courses, the content can be for 40 hours, and for 2-credit courses, the content can be for 26 hours. The audit course in the third semester can have content for 30 hours).

No	Topic	No. of Lectures
1	INTRODUCTION TO BLOCK CHAIN(8 hours)	
1.1	History, Definition, Distributed Ledger	1
1.2	Blockchain Categories – Public, Private	1
1.3	Consortium, Blockchain Network and Nodes	1
1.4	Peer-to-Peer Network	1
1.5	, Mining Mechanism	1
1.6	Generic elements of Blockchain	1
1.7	Features of Blockchain	1
1.8	Types of Blockchain	1
2	BLOCKCHAIN ARCHITECTURE(8 hours)	
2.1	Operation of Bitcoin Blockchain,	1
2.2	Blockchain Architecture – Block hash	1
2.3	Distributer P2P	1
2.4	Structure of Blockchain-Consensus mechanism:	1

INFORMATION TECHNOLOGY

2.5	Proof of Work (PoW)	1
2.6	Proof of Stake (PoS)	1
2.7	Byzantine Fault Tolerance (BFT)	1
2.8	Proof of Authority (PoA) and Proof of Elapsed Time (PoET)	1
3	BLOCKCHAIN-BASED FUTURES SYSTEM(8 hours)	
3.1	Project presentation- Futures smart contract:	1
3.2	Blockchain oracles	1
3.3	Web3j: Setting up the Web3J	1
3.4	Installing web3J- Wallet creation	1
3.5	Java client: The wrapper generator	1
3.6	Initializing web3j	1
3.7	Setting up Ethereum accounts	1
3.8	Deploying the contract	1
4	BLOCKCHAINS IN BUSINESS AND CREATING ICO(8 hours)	
4.1	Public versus private blockchains	1
4.2	Permissioned versus permission less blockchains	1
4.3	Privacy and anonymity in Ethereum	1

INFORMATION TECHNOLOGY

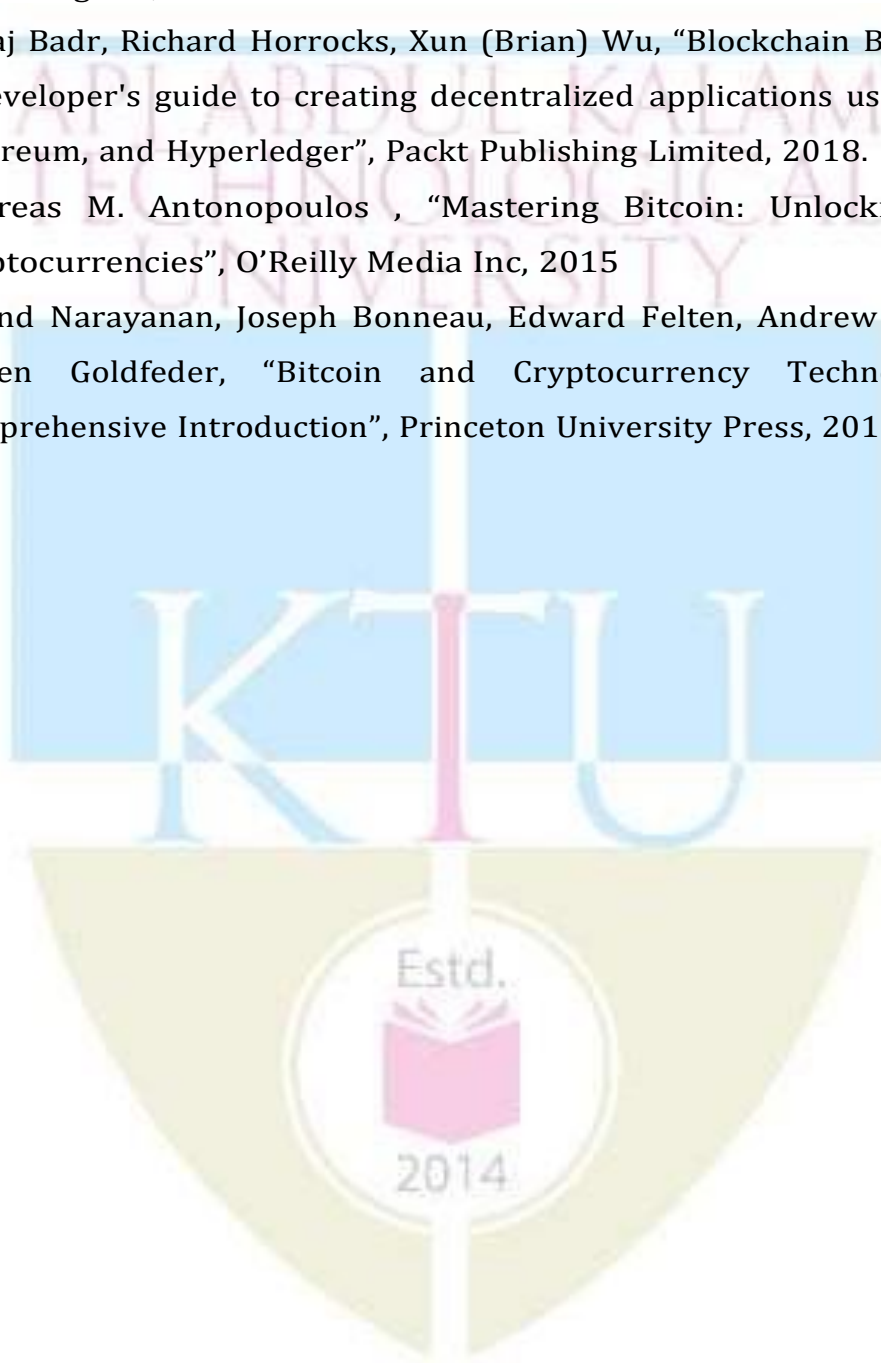
4.4	Why are privacy and anonymity important? The Ethereum Enterprise Alliance	1
4.5	Blockchain- as-a-Service	1
4.6	Initial Coin Offering (ICO): Project setup for ICO implementation	1
4.7	Token contracts- Token sale contracts	1
4.8	Contract security and testing the code	1
5	DISTRIBUTED STORAGE IPFS AND SWARM(8 hours)	
5.1	Ethereum Virtual Machine	1
5.2	Swarm and IPFS:	1
5.3	Installing IPFS	1
5.4	Hosting our frontend	1
5.5	Serving your frontend using IPFS	1
5.6	Serving your frontend using Swarm	1
5.7	IPFS file uploader project	1
5.8	Project setup - the web page	1

Sample Mini Project work questions:

- * Develop a frontend using Swarm, IPFS file uploader
- * Project setup - the web page creation

Reference Books

1. Imran Bashir, "Mastering Blockchain: Distributed Ledger Technology, decentralization, and smart contracts explained", 2nd Edition, Packt Publishing Ltd, March 2018.
2. Bellaj Badr, Richard Horrocks, Xun (Brian) Wu, "Blockchain By Example: A developer's guide to creating decentralized applications using Bitcoin, Ethereum, and Hyperledger", Packt Publishing Limited, 2018.
3. Andreas M. Antonopoulos , "Mastering Bitcoin: Unlocking Digital Cryptocurrencies", O'Reilly Media Inc, 2015
4. Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller and Steven Goldfeder, "Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction", Princeton University Press, 2016.



222EIT008	SOCIAL NETWORK ANALYSIS	CATEGORY	L	T	P	CREDIT
		PROGRAMME ELECTIVE 4	3	0	0	3

Preamble: This course is intended for students to have a deeper knowledge of selected topics in Social Network Analysis including semantic web, modeling, aggregating and knowledge representation, predicting human behavior and privacy issues, and analyzing social networks.

Course Outcomes:

After the completion of the course, the student will be able to

CO 1	Apply network concepts in social network analysis (Level: Apply)
CO 2	Make use of ontological representation of social relationships to reason with social network data. (Level: Apply)
CO 3	Explain how network analysis can contribute to increasing knowledge about diverse aspects of society (Level: Understand)
CO 4	Describe human behavior and privacy issues. (Level: Understand)
CO 5	Experiment with social network data using various techniques. (Level: Apply)

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

P01:An ability to independently carry out research/investigation and development work in engineering and allied streams

P02:An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

INFORMATION TECHNOLOGY

P03:An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program

P04:An ability to apply stream knowledge to design or develop solutions for real world problems by following the standards

P05:An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tool to model, analyse and solve practical engineering problems.

P06:An ability to engage in life-long learning for the design and development related to the stream related problems taking into consideration sustainability, societal, ethical and environmental aspects

P07:An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1							
CO 2							
CO 3							
CO 4							
CO 5							

Assessment Pattern

Bloom's Category	End Semester Examination
-------------------------	---------------------------------

INFORMATION TECHNOLOGY

Understand	40%
Apply	60%
Analyze	
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation Pattern:

Evaluation shall only be based on application, analysis or design based questions (for both internal and end semester examinations).

Continuous Internal Evaluation (CIE) Pattern:

Preparing a review article based on peer-reviewed original publications (minimum 10 Publications shall be referred)/

10 Marks

Assignment:

Course-based task/Seminar/Data Collection and interpretation:

20 Marks

Test paper (one only, and shall include a minimum of 80% of the syllabus):

10 Marks

End Semester Examination (ESE) Pattern:

INFORMATION TECHNOLOGY

The end-semester examination shall be conducted by the respective College. There shall be two parts; Part A and Part B. Part A shall contain five numerical/short answer questions with one question from each module, having five marks for each question (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students). Students shall answer all questions. Part B shall contain seven questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem-solving and quantitative evaluation), with a minimum one question from each module of which student shall answer any five. Each question shall carry seven marks.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly. For example, if the average end-semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is $40+20 = 60$ %.



MODEL QUESTION PAPER

QP CODE:

Reg No: _____

Name: _____

PAGES : 4

**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
SECOND SEMESTER M.TECH DEGREE EXAMINATION, MONTH & YEAR**

Course Code:

Course Name: SOCIAL NETWORK ANALYSIS

Max. Marks : 60

Duration: 2.5 Hours

PART A

Answer All Questions. Each Question Carries 5 Marks (5x5=25)

1. What are the limitations of the current Web? Explain the development of the Semantic Web and the emergence of the Social Web (5)
2. Illustrate Ontological representation of social relationships (5)
3. Enumerate the different dimensions of social capital and their related concepts and measures. (5)
4. Discuss the Global structure of networks with an example. (5)
5. How will you compute the entity similarity matrix? (5)

Part B

(Answer any five questions. Each question carries 7 marks)

6. (a) Briefly explain the following: (7)
 - i) Electronic discussion networks
 - ii) Blogs and online communities
 - iii) Web-based Networks
 - iv) Personal Networks
- (b) How will you extract the evolution of the Web Community from a series of Web Archives?

7. (a) Explain how communities evolve into the learning process as smoothly evolving constellations of interacting entities. (7)
- (b) Discuss the four dimensions that are associated with knowledge discovery in social networks and elaborate on their interplay in the context of evolution.
8. (a) Explain how to visualize social networks with matrix-based representation. Also, discuss the pros and cons of matrix-based representation (7)
9. (a) Discuss the various approaches to scale node-link diagrams to large networks with several thousand or millions of nodes. (7)
- 10 (a) Explain the importance of detecting and evaluating communities in social networks (7)
- 11 (a) Explain how to apply node classification to large social networks. (7)
- 12 (a) Categorize the Twitter messages in TASS Corpus (7)

SYLLABUS

Module 1: Introduction to semantic web: (7 hours)

Limitations of current Web - Development of Semantic Web, Emergence of the Social Web, Social Network Analysis: Development of Social Network Analysis, Key concepts and measures in network analysis, Electronic sources for network analysis, Electronic discussion networks, Blogs and online communities, Web-based networks, Applications of Social Network Analysis.

Module 2: Modelling, aggregating, and knowledge representation(8 hours)

Ontology and their role in the Semantic Web, Ontology-based knowledge Representation, Ontology languages for the Semantic Web: Resource Description Framework, Web Ontology Language - Modelling and aggregating social network data, State-of-the-art in network data representation, Ontological representation of social individuals, Ontological representation of social relationships, Aggregating and reasoning with social network data, Advanced representations.

Module 3: Extraction and mining communities in web social networks(7 hours)

Extracting evolution of Web Community from a Series of Web Archive, Detecting communities in social networks, Definition of community - Evaluating communities, Methods for community detection and mining, Applications of community mining algorithms, Tools for detecting communities social network infrastructures and communities, Decentralized online social networks - Multi-Relational characterization of dynamic social network communities.

Module 4:Predicting human behavior and privacy issues :(9 hours)

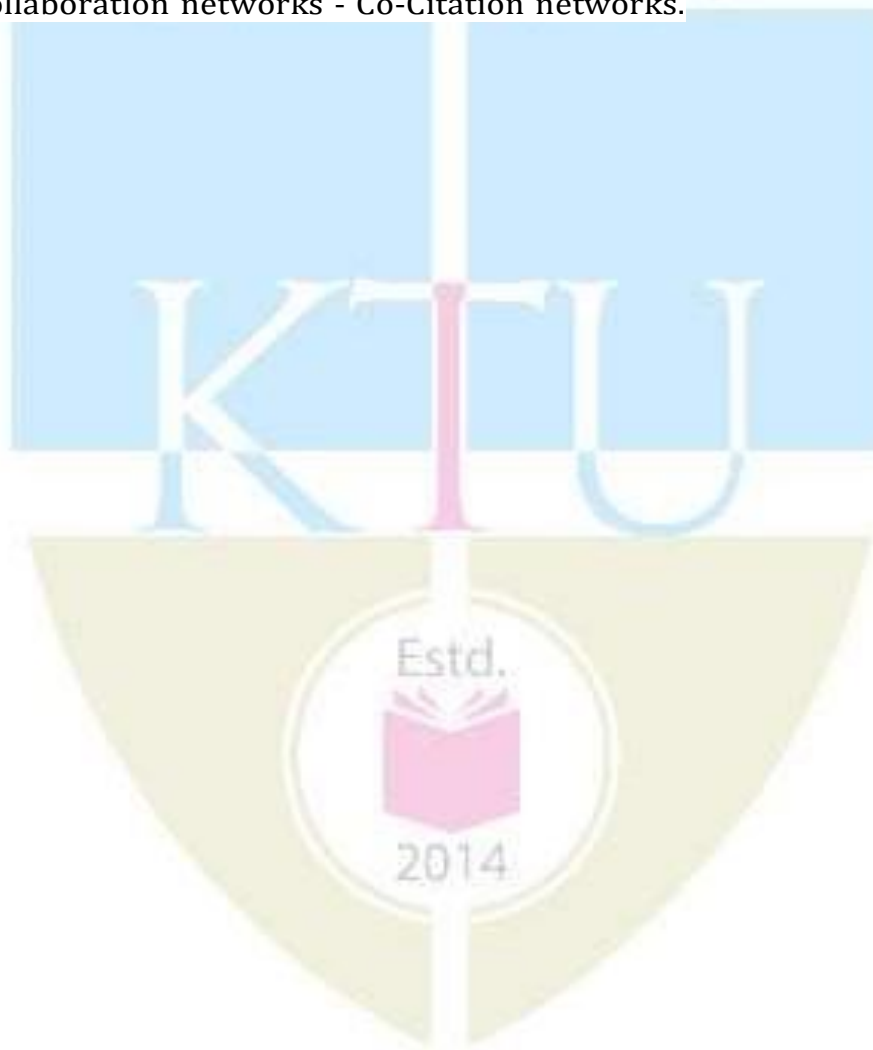
Understanding and predicting human behavior in social communities, User data management - Inference and Distribution, Enabling new human experiences - Reality mining, Context - Awareness - Privacy in online social networks, Trust in the online environment - Trust models based on subjective logic, Trust network analysis - Trust transitivity analysis, Combining trust and

reputation, Trust derivation based on trust comparisons. Attack spectrum and countermeasures.

Module 5: Visualization and applications of social networks:(9 hours)

Graph theory - Centrality - Clustering, Node-Edge Diagrams, Matrix representation, Visualizing online social networks, Visualizing social networks with matrix-based representations, FB/V Kontakte and Twitter analysis:

Natural language processing, sentiment mining, Properties of large social networks: friends, connections, likes, re-tweets, Cover networks - Community welfare, Collaboration networks - Co-Citation networks.



COURSE PLAN

No	Topic	No. of Lectures
1	MODULE 1: INTRODUCTION TO SEMANTIC WEB (7 hours)	
1.1	Limitations of current Web - Development of Semantic Web, Emergence of the Social Web	1
1.2	Social Network analysis: Development of Social Network Analysis	1
1.3	Key concepts and measures in network analysis	1
1.4	Electronic sources for network analysis	1
1.5	Electronic discussion networks, Blogs and online communities	1
1.6	Web-based networks	1
1.7	Applications of Social Network Analysis	1
2	MODULE 2: MODELLING, AGGREGATING AND KNOWLEDGE REPRESENTATION (8 hours)	
2.1	Ontology and their role in the Semantic Web, Ontology-based Knowledge Representation	1
2.2	Ontology languages for the Semantic Web: Resource Description Framework	1
2.3	Web Ontology Language - Modelling and aggregating social network data	1
2.4	State-of-the-art in network data representation	1
2.5	Ontological representation of social individuals	1

INFORMATION TECHNOLOGY

2.6	Ontological representation of social relationships	1
2.7	Aggregating and reasoning with social network data	1
2.8	Advanced representations.	1
3	MODULE 3: EXTRACTION AND MINING COMMUNITIES IN WEB SOCIAL NETWORKS (7 hours)	
3.1	Extracting the evolution of Web Community from a Series of Web Archive	1
3.2	Detecting communities in social networks	1
3.3	Definition of community - Evaluating communities	1
3.4	Methods for community detection and mining	1
3.5	Applications of community mining algorithms	1
3.6	Tools for detecting communities social network infrastructures and communities	1
3.7	Decentralized online social networks -Multi-Relational characterization of dynamic social network communities	1
4	MODULE 4: PREDICTING HUMAN BEHAVIOUR AND PRIVACY ISSUES (9 hours)	
4.1	Understanding and predicting human behavior for social communities	1
4.2	User data management - Inference and Distribution	1
4.3	Enabling new human experiences - Reality mining	1
4.4	Context - Awareness - Privacy in online social networks	1

INFORMATION TECHNOLOGY

4.5	Trust in online environment - Trust models based on subjective logic	1
4.6	Trust network analysis - Trust transitivity analysis	1
4.7	Combining trust and reputation	1
4.8	Trust derivation based on trust comparisons	1
4.9	Attack spectrum and countermeasures	1
5	MODULE 5: VISUALIZATION AND APPLICATIONS OF SOCIAL NETWORKS (9 hours)	
5.1	Graph theory - Centrality – Clustering	1
5.2	Node-Edge Diagrams, Matrix representation	1
5.3	Visualizing online social networks	1
5.4	Visualizing social networks with matrix-based representations	1
5.5	Facebook/Vkontakte and Twitter analysis: Natural language processing	1
5.6	Sentiment mining	1
5.7	Properties of large social networks: friends, connections, likes, re-tweets.	1
5.8	Cover networks - Community Welfare	1
5.9	Collaboration networks - Co-Citation networks.	1

Sample Mini Project work questions:

- How did Covid19 affect SoundCloud users' listening?

- What are the pull/push factors that affects mobility of highly skilled individuals (e.g., professionals on LinkedIn, researchers on MAG)?
- Is it possible to identify bots by the patterns of their online activities?
- What is an echo-chamber and how can we track it on Twitter?

Reference Books

1. Guandong Xu ,Yanchun Zhang and Lin Li, —Web Mining and Social Networking – Techniques and applications, First Edition, Springer, 2011.
2. Dion Goh and Schubert Foo, —Social Information Retrieval Systems: Emerging Technologies and Applications for Searching the Web Effectively, IGI Global Snippet, 2008.
3. Max Chevalier, Christine Julien and Chantal Soulé-Dupuy, —Collaborative and Social Information Retrieval and Access: Techniques for Improved User Modelling, IGI Global Snippet, 2009.
4. John G. Breslin, Alexander Passant and Stefan Decker, —The Social Semantic Web, Springer, 2009.
5. Mohammad Gouse Galety, Chiai Al Atroshi, Buni Balabantaray, Sachi Nandan Mohanty-Social Network Analysis-Theory and Applications-2022 – Scrivener Publishing LLC

222EIT009	NEXT GENERATION NETWORKS	CATEGORY	L	T	P	CREDIT
		PROGRAM ELECTIVE 4	3	0	0	3

Preamble: The next generation network (NGN) is a single converged network that supports all telecom requirements like voice, data, mobility, video etc. and works on packet switching technology and is IP compatible. This course aims to provide an insight into the evolving communication networks in NGN such a 6G network that operates in untapped radio frequencies and uses cognitive technologies like AI to enable high-speed, low-latency communication at a pace multiple times faster than fifth-generation networks.

Course Outcomes: After the completion of the course the student will be able to

CO 1	Illustrate key enabling technologies and also discuss the new emerging use cases of 5G-IoT driven by the advances in technology	Understand
CO 2	Interpret the technologies enabling the NGN	Apply
CO 3	Demonstrate the applicability of machine learning techniques in NGNs	Understand
CO 4	Apply AI in WiFi network to improve the QoS	Apply
CO 5	Examine 6G enabling technology to show the possibility of a future of fully intelligent and autonomous systems	Understand

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

PO1: An ability to independently carry out research/investigation and development work in engineering and allied streams

PO2: An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

PO3: An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program

PO4: An ability to apply stream knowledge to design or develop solutions for real world problems by following the standards

P05: An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tool to model, analyse and solve practical engineering problems.

P06: An ability to engage in life-long learning for the design and development related to the stream related problems taking into consideration sustainability, societal, ethical and environmental aspects

P07: An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	P 6	PO 7
CO 1			✓		✓		
CO 2			✓		✓		
CO 3			✓		✓		
CO 4	✓		✓		✓		
CO 5	✓		✓		✓		

Assessment Pattern

Bloom's Category	End Semester Examination
Understand	60%
Apply	40%
Analyse	
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation Pattern:

Evaluation shall only be based on application, analysis or design-based questions (for both internal and end semester examinations).

Continuous Internal Evaluation (CIE) Pattern:

Preparing a review article based on peer-reviewed original publications (minimum 10 Publications shall be referred)/

10 Marks

Assignment:

Course-based task/Seminar/Data Collection and interpretation:

20 Marks

Test paper (one only, and shall include a minimum of 80% of the syllabus):

10 Marks

End Semester Examination (ESE) Pattern:

The end-semester examination shall be conducted by the respective College. There shall be two parts; Part A and Part B. Part A shall contain five numerical/short answer questions with one question from each module, having five marks for each question (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students). Students shall answer all questions. Part B shall contain seven questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem-solving and quantitative evaluation), with a minimum one question from each module of which student shall answer any five. Each question shall carry seven marks.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly.

For example, if the average end-semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is $40 + 20 = 60\%$.

MODEL QUESTION PAPER

QP Code:

PAGES : 2

Register No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

SECOND SEMESTER M.TECH DEGREE EXAMINATION, MONTH &

YEAR

Course Code: 222EIT009

Course Name: NEXT GENERATION NETWORKS

Max. Marks : 60

Duration: 2.5 Hours

Part A

Answer All Questions. Each Question Carries 5 Marks (5x5=25 Marks)

1. Explain the spectrum requirements of 5G-IoT. Examine the possibility of mmWaves.
2. Describe the different segments of IoT. What are their characteristics?
3. Explain how Swarm Intelligence can be a key for power control and channel allocation?
4. Give any one key problem of 802.11ad/ay networks, which are solved using ML. Explain
5. What are the key requirements of 6G- IoT network?

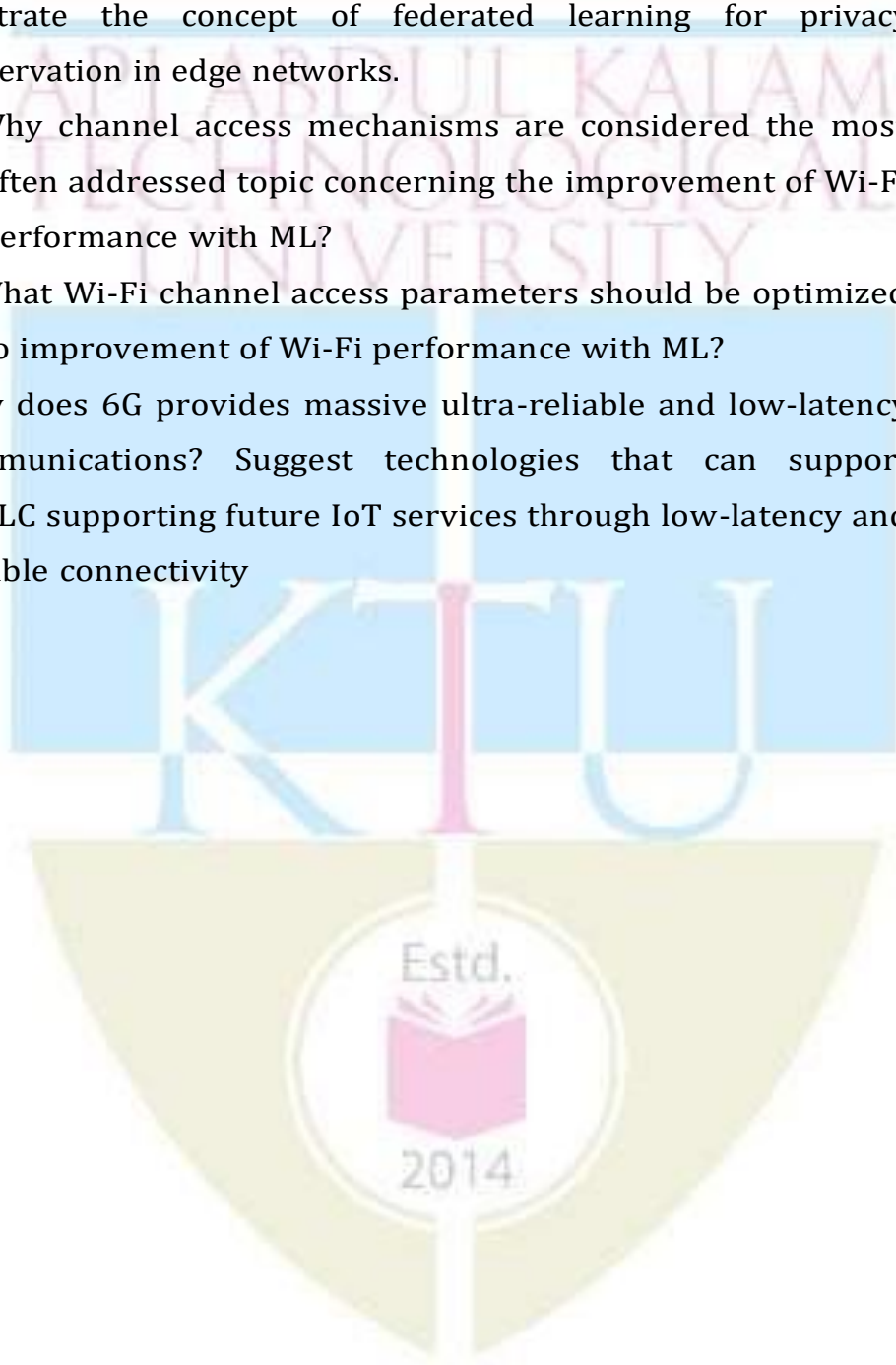
Part B

(Answer any five questions. Each question carries 7 marks)

6. Construct a multi-tier architecture showing the front-haul, back-haul and core network highlighting the peer-to-peer communication, macro- and femto-cells. Explain.
7. Describe an IoT application that can be implemented over an IoT networks over unlicensed spectrum. Also, explain why your application is not best suited for IoT communications over licensed spectrum.
8. Compare the state-of-the-art channel coding techniques for

IoT. Suggests IoT applications that are can make use of these channel coding techniques and why?

9. Using a suitable example, show the use of Reinforced learning technique can be applied in wireless networks.
10. Illustrate the concept of federated learning for privacy preservation in edge networks.
11. a. Why channel access mechanisms are considered the most often addressed topic concerning the improvement of Wi-Fi performance with ML?
b. What Wi-Fi channel access parameters should be optimized to improvement of Wi-Fi performance with ML?
12. How does 6G provides massive ultra-reliable and low-latency communications? Suggest technologies that can support URLLC supporting future IoT services through low-latency and reliable connectivity



SYLLABUS

Module 1: Internet of Things (IoT) for Next-Generation Smart Systems (7 Hours)

Introduction, Background, Motivation and Overview, Vision and Development of 5G-Enabled IoT From 5G Cellular Technologies, Artificial Intelligence Driven Use Cases for 5G-IoT Networks, Challenges and Issues.

Module 2: Cellular, Wide-Area, and Non-Terrestrial IoT (8 Hours)

Introduction, Key Performance Indicators, Enabling Access Solutions, Channel Code Design, Massive Connectivity, IoT Security, Deep Learning for IoT, NTN Integration into IoT Networks, Remaining Challenges and Future Directions

Module 3: Machine Learning Techniques for Next-Generation Networks (9 Hours)

Applying machine learning techniques for caching in next-generation edge networks – Introduction, Fundamentals of edge networks and 5G technologies, Taxonomy of ML based edge caching, State-of-the-art: machine learning based caching, Comparison of ML based edge caching techniques, Research challenges and future directions.

Swarm intelligence for next-generation networks – Introduction, Swarm intelligence: An overview, SI for spectrum management and resource allocation, SI for wireless caching and edge computing, SI for network security, SI for miscellaneous issues, Research challenges and future directions.

Module 4: WiFi Meets AI (7 Hours)

Introduction, Related Surveys, Core Wi-Fi Features, Recent Wi-Fi Features, Connectivity Management, Coexistence Scenarios, Multi-Hop Wi-Fi Networks, Available Tools and Datasets, Future Research Directions.

Module 5: Towards 6G (9 Hours)

Towards 6G - Introduction, Key Driving Trends Towards 6G, 6G Applications, 6G Requirements/Vision, 6G Enabling Technologies, 6G Projects and Research Activities, 6G Standardization Efforts, Future Research Directions and General Challenges in 6G.

6G Internet of Things – Introduction, 6G and Internet of Things: Vision and Requirements, Fundamental Technologies for 6G-IoT, 6G for IoT Applications, Research Challenges and Future Directions.

COURSE PLAN

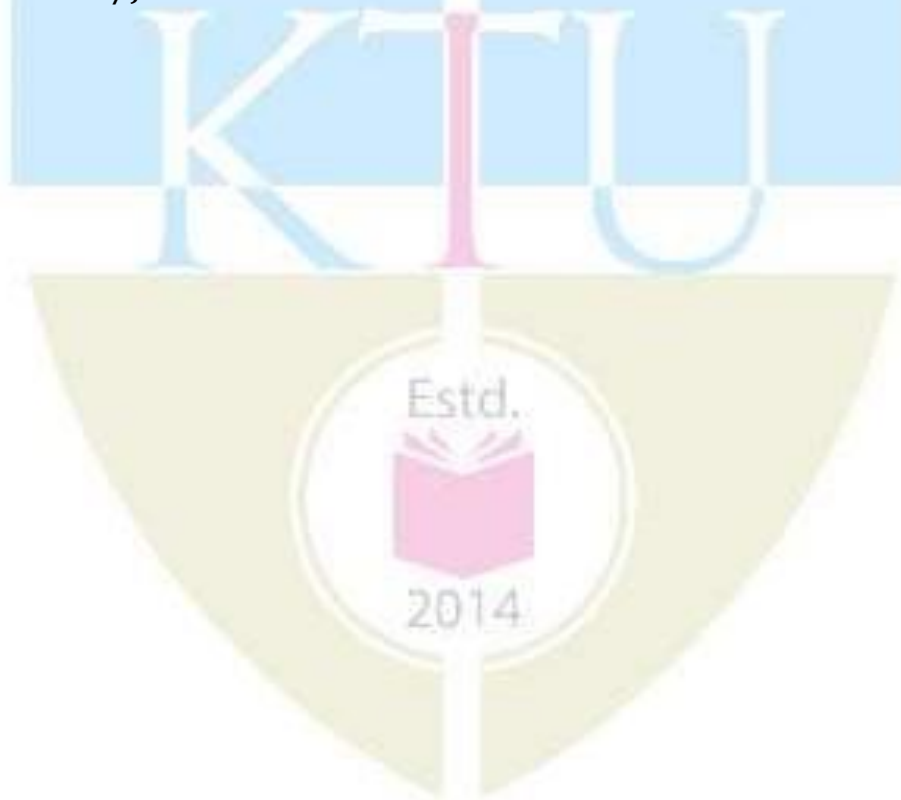
No	Topic	No. of Lectures 40 Hours
1	Internet of Things (IoT) for Next-Generation Smart Systems	
1.1	Introduction, Background	1 Hour
1.2	Motivation and Overview	1 Hour
1.3	Vision and Development of 5G-Enabled IoT From 5G Cellular Technologies	2 Hours
1.4	Artificial Intelligence Driven Use Cases for 5G-IoT Networks	2 Hours
1.5	Challenges and Issues.	1 Hour
2	Cellular, Wide-Area, and Non-Terrestrial IoT	
2.1	Introduction, Key Performance Indicators.	1 Hour
2.2	Enabling Access Solutions, Channel Code Design	2 Hours
2.3	Massive Connectivity, IoT Security, Deep Learning for IoT	2 Hours
2.4	NTN Integration into IoT Networks	2 Hour
2.5	Remaining Challenges and Future Directions	1 Hour
3	Machine Learning Techniques for Next-Generation Networks	
3.1	Applying machine learning techniques for caching in next-generation edge networks – Introduction, Fundamentals of edge networks and 5G technologies	2 Hours
3.2	Taxonomy of ML based edge caching, State-of-the-art: machine learning based caching	2 Hours
3.3	Comparison of ML based edge caching techniques, Research challenges and future directions.	1 Hours
3.4	Swarm intelligence for next-generation networks – Introduction, Swarm intelligence: An overview, SI for spectrum management and resource allocation	2 Hours
3.5	SI for wireless caching and edge computing, SI for network security	1 Hours

3.6	SI for miscellaneous issues, Research challenges and future directions.	1 Hour
4	WiFi Meets AI	
4.1	Introduction, Related Surveys	1 Hour
4.2	Core Wi-Fi Features, Recent Wi-Fi Features,	2 Hours
4.3	Connectivity Management	1 Hours
4.4	Coexistence Scenarios	1 Hour
4.5	Multi-Hop Wi-Fi Networks, Available Tools and Datasets, Future Research Directions	2 Hours
5	Towards 6G	
5.1	Towards 6G - Introduction, Key Driving Trends Towards 6G	1 Hour
5.2	6G Applications	1 Hour
5.3	6G Requirements/Vision, 6G Enabling Technologies, 6G Projects and Research Activities	2 Hours
5.4	6G Standardization Efforts, Future Research Directions and General Challenges in 6G	2 Hours
5.5	6G Internet of Things – Introduction, 6G and Internet of Things: Vision and Requirements.	1 Hours
5.6	Fundamental Technologies for 6G-IoT, 6G for IoT Applications, Research Challenges and Future Directions	2 Hours

References:

1. K. Shafique, B. A. Khawaja, F. Sabir, S. Qazi and M. Mustaqim, "Internet of Things (IoT) for Next-Generation Smart Systems: A Review of Current Challenges, Future Trends and Prospects for Emerging 5G-IoT Scenarios," in *IEEE Access*, vol. 8, pp. 23022-23040, 2020, doi: 10.1109/ACCESS.2020.2970118.
2. M. Vaezi *et al.*, "Cellular, Wide-Area, and Non-Terrestrial IoT: A Survey on 5G Advances and the Road Toward 6G," in *IEEE Communications Surveys & Tutorials*, vol. 24, no. 2, pp. 1117-1174, Secondquarter 2022, doi: 10.1109/COMST.2022.3151028.
3. Junaid Shuja, Kashif Bilal, Waleed Alasmay, Hassan Sinky, Eisa Alanazi, "Applying machine learning techniques for caching in next-generation edge networks: A comprehensive survey", in *Journal of*

- Network and Computer Applications*, Volume 181, 2021, 103005, ISSN 1084-8045, <https://doi.org/10.1016/j.jnca.2021.103005>.
4. Quoc-Viet Pham, Dinh C. Nguyen, Seyedali Mirjalili, Dinh Thai Hoang, Diep N. Nguyen, Pubudu N. Pathirana, Won-Joo Hwang, "Swarm intelligence for next-generation networks: Recent advances and applications", *Journal of Network and Computer Applications*, Volume 191, pp. 103-141, 2021, ISSN 1084-8045, <https://doi.org/10.1016/j.jnca.2021.103005>.
 5. S. Szott et al., "Wi-Fi Meets ML: A Survey on Improving IEEE 802.11 Performance with Machine Learning," in *IEEE Communications Surveys & Tutorials*, vol. 24, no. 3, pp. 1843-1893, thirdquarter 2022, doi: 10.1109/COMST.2022.3179242.
 6. C. D. Alwis et al., "Survey on 6G Frontiers: Trends, Applications, Requirements, Technologies and Future Research," in *IEEE Open Journal of the Communications Society*, vol. 2, pp. 836-886, 2021, doi: 10.1109/OJCOMS.2021.3071496.
 7. D. C. Nguyen et al., "6G Internet of Things: A Comprehensive Survey," in *IEEE Internet of Things Journal*, vol. 9, no. 1, pp. 359-383, 1 Jan.1, 2022, doi: 10.1109/JIOT.2021.3103320.



INFORMATION TECHNOLOGY

222EIT010	SECURE HARDWARE DESIGN	CATEGORY	L	T	P	CREDIT
		PROGRAMME ELECTIVE 4	3	0	0	3

Preamble: This course is intended for students to understand the importance of addressing different security threats in modern hardware design, installation, and operating practices.

Course Outcomes:

After the completion of the course, the student will be able to

CO 1	Explain the mathematical foundation and basics of the techniques to provide hardware security. (Cognitive Level : Understand)
CO 2	Examine the implementation quality of cryptographic hardware. (Cognitive Level : Understand)
CO 3	Identify side-channel attacks and choose appropriate techniques to mitigate them. (Cognitive Level : Understand)
CO 4	Describe the security measures employed in modern IC design and manufacturing. (Cognitive Level : Understand)
CO 5	Employ established and novel techniques to identify hardware trojans. (Cognitive Level : Apply)

Program Outcomes:

PO 1	An ability to independently carry out research/ investigation and development work in engineering and allied streams.
PO 2	An ability to communicate effectively, and write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.
PO 3	An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor's program.

INFORMATION TECHNOLOGY

PO 4	An ability to apply stream knowledge to design or develop solutions for real-world problems by following the standards.
PO 5	An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tools to model, analyze and solve practical engineering problems.
PO 6	An ability to engage in lifelong learning for the design and development of stream-related problems taking into consideration, sustainability societal, ethical, and environmental aspects.
PO 7	An ability to develop cognitive load management skills related to project management and finance which focus on entrepreneurship and industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	√			√			
CO 2	√		√	√	√		
CO 3			√	√	√		
CO 4			√	√	√		
CO 5			√	√	√		

Assessment Pattern

Bloom's Category	End Semester Examination
Understand	80%

INFORMATION TECHNOLOGY

Apply	20%
Analyze	
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

The evaluation shall only be by employing application-based, analysis-based, or design-based questions (for both internal and end-semester examinations)

Continuous Internal Evaluation (CIE) Pattern:

Preparing a review article based on peer-reviewed original publications (minimum 10 Publications shall be referred)/

10 Marks

Assignment:

Course-based task/Seminar/Data Collection and interpretation:

20 Marks

Test paper (one only, and shall include a minimum of 80% of the syllabus):

10 Marks

End Semester Examination (ESE) Pattern:

The end-semester examination shall be conducted by the respective College. There shall be two parts; Part A and Part B. Part A shall contain five numerical/short answer questions with one question from each module, having

five marks for each question (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students). Students shall answer all questions. Part B shall contain seven questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem-solving and quantitative evaluation), with a minimum one question from each module of which student shall answer any five. Each question shall carry seven marks.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly.

MODEL QUESTIONS:

For assessing CO 1:

1. IoT devices are especially risky. Justify/ contradict the statement with respect to hardware-based security.
2. Prove that $2^2 + 1$ is divisible by 641. using the concept of modular reductions.

For assessing CO 2:

1. With a neat diagram explain the Secret Key Cryptosystem Model
2. Compare Montgomery's Algorithm and LSB first algorithm., used for the Scalar Multiplication.

For assessing CO 3:

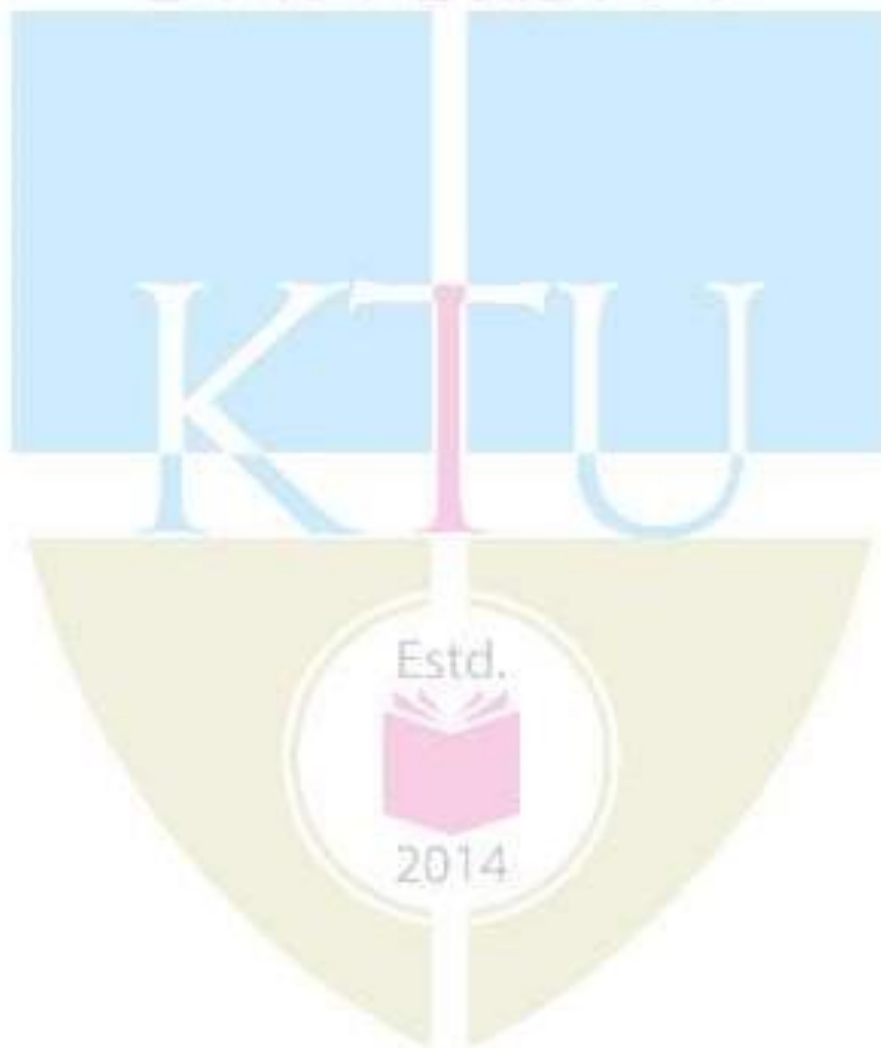
1. State the Difference of Side Channel Analysis and Conventional Cryptanalysis
2. Explain the working principle of time-driven side channel attacks

For assessing CO 4:

1. Explain the hardware obfuscation design flow and the application of iterative ranking algorithm in the design flow.
2. Analyze how hardware obfuscation challenges Structural Analysis-based Attacks

For assessing CO 5:

1. Discuss various triggering mechanisms of hardware Trojans.
2. Analyze the effects of Hardware Trojan on Circuit Reliability



MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
FIRST SEMESTER M. TECH DEGREE EXAMINATION, MONTH & YEAR

Course Code: **222EIT010**

Course Name: Secure Hardware Design

Max. Marks : 60

Duration: 2.5 Hours

PART A (*Answer all questions. Five marks each*)

1. What is the implication of the expression $a \equiv b \pmod{m}$ where a , b and m are integers? What are the properties of this relation?
2. Explain the constraints provided by the designer while transforming HDL to actual hardware through CAD tools and the significance of the constraints.
3. How is a differential timing attack performed?
4. Define obfuscation and compare hardware and software obfuscation techniques.
5. Explain the two most important security parameters of PUF.

PART B (*Answer any five questions. Seven marks each*)

6. Explain the Block cipher technique, clearly indicating the inner structure of a blockcipher with a diagram.
7. Explain FPGA island-style architecture and the logic block with a neat sketch.
8. Show how Theoretical Cache Trace Attack is possible on a DES implementation.
9. Explain the concept of CDFG Obfuscation Methodology
10. Explain about the designer-embedded Trojan in the hardware implementation of the AES cryptographic hardware.
11. Explain how PUFs can be used in Cryptographic Key Generation hardware.
12. Analyse the effect of hardware Trojans on circuit reliability.

SYLLABUS

Module I: Preliminaries: (8 Hours)

Overview of Different Issues of Hardware Security, Algebra of Finite Fields, Mathematical Theory of Public Key Cryptography, Basics of Digital Design on Field-programmable Gate Array (FPGA). Mapping an Algorithm to Hardware: Components of a Hardware Architecture

Module II: Hardware Security Primitives: (8 Hours)

Cryptographic Hardware and their Implementation, Optimization of Cryptographic Hardware on FPGA, Physically Unclonable Functions (PUFs) – PUF implementations and quality evaluation, Design Techniques to Increase PUF Response Quality. Attacks Models against PUFs- Model-building Attack on PUFs

Module III: Side-channel Attacks: (8 Hours)

The basic idea of side-channel attacks, Current-measurement-based side-channel attacks, DES, Case Study: Kocher's Attack on DES, Design Techniques to Prevent Side-channel Attacks, Improved Side-channel Attack Algorithms - Template Attack, Cache Attacks.

Module IV: Modern IC Design and Manufacturing – Security aspects: (8 Hours)

Hardware Intellectual Property (IP) Piracy and IC Piracy, Design Techniques to Prevent IP and IC Piracy, Hardware Intellectual Property Protection through Obfuscation. Measure of Obfuscation Level Obfuscation through Control and Dataflow Graph (CDFG).

Module V: Hardware Trojans: (8 Hours)

Hardware Trojan Nomenclature and Operating Modes, Countermeasures - Design and Manufacturing Techniques to Prevent/Detect Hardware Trojans, Logic Testing and Side-channel Analysis based Techniques for Trojan Detection, Techniques to Increase Testing Sensitivity.

COURSE PLAN

No	Topic	No. of Lectures
1	Preliminaries(8 Hours)	
1.1	Overview of Different Issues of Hardware Security	1
1.2	Algebra of Finite Fields	2
1.3	Mathematical Theory of Public Key Cryptography	2
1.4	FPGA Architecture and Design Flow	1
1.5	Basics of Digital Design on Field-programmable Gate Array (FPGA)	1
1.6	Mapping an Algorithm to Hardware: Components of a Hardware Architecture	1
2	Hardware Security Primitives(8 Hours)	
2.1	Cryptographic Hardware and their Implementation	1
2.2	Optimization of Cryptographic Hardware on FPGA	2
2.3	Physically Unclonable Functions (PUFs) –	1
2.4	PUF implementations and quality evaluation	1
2.5	Design Techniques to Increase PUF Response Quality.	1
2.6	Attack models against PUFs	1
2.7	Model building attack on PUFs	1
3	Side-channel Attacks(8 Hours)	
3.1	The basic idea of side-channel attacks	1
3.2	Current-measurement-based side-channel attacks	1
3.3	DES	1

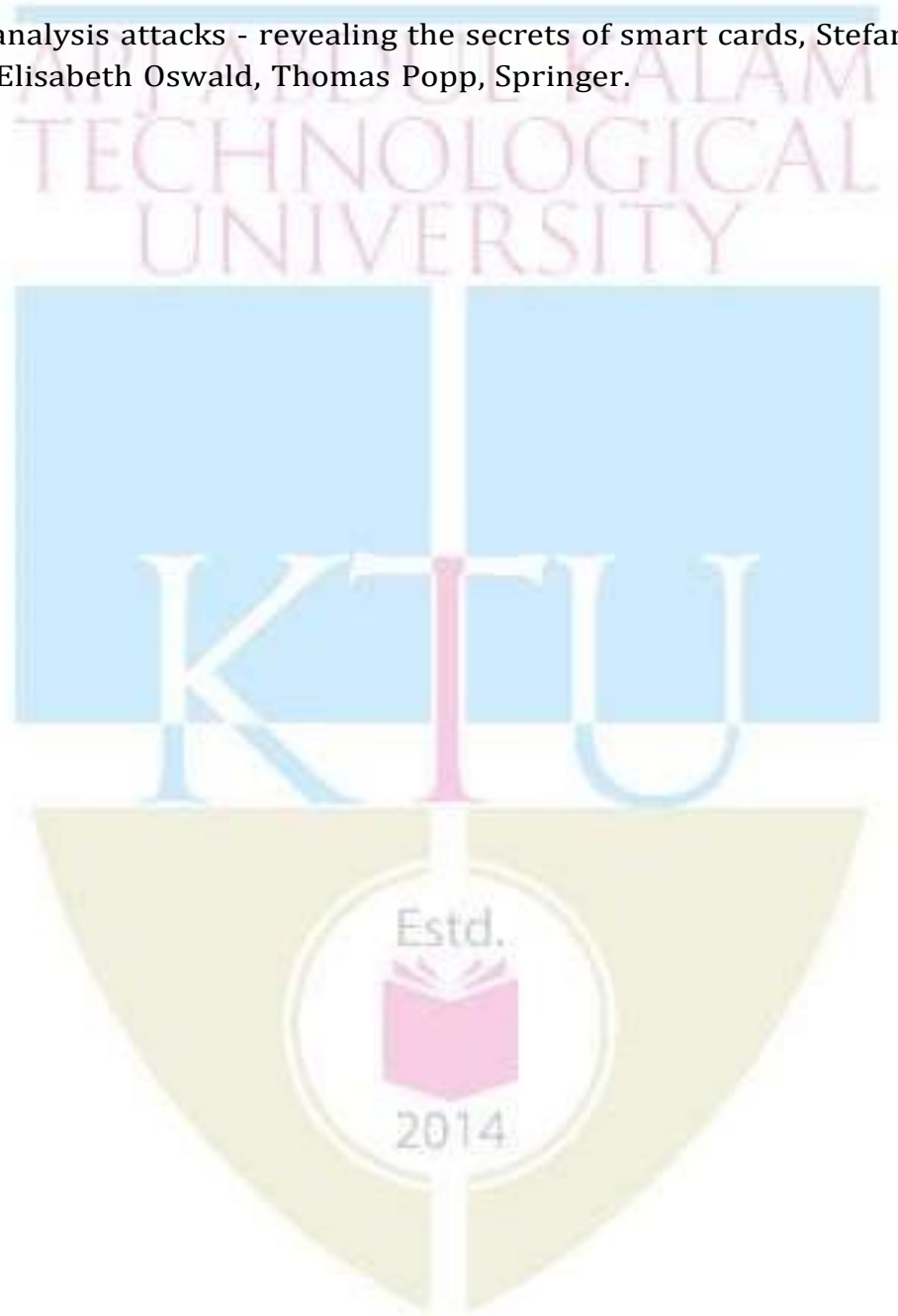
INFORMATION TECHNOLOGY

3.4	Case Study: Kocher's Attack on DES.	1
3.5	Design Techniques to Prevent Side-channel Attacks	1
3.6	Improved Side-channel Attack Algorithms - Template Attack	2
3.7	Cache Attacks	1
4	Modern IC Design and Manufacturing – Security aspects(8 Hours)	
4.1	Hardware Intellectual Property (IP) Piracy and IC Piracy	2
4.2	Design Techniques to Prevent IP and IC Piracy	1
4.3	Hardware Intellectual Property Protection Through Obfuscation	2
4.4	Measure of Obfuscation Level	1
4.4	Obfuscation through Control and Dataflow Graph (CDFG).	2
5	Hardware Trojans(8 Hours)	
5.1	Hardware Trojan Nomenclature and Operating Modes	2
5.2	Countermeasures - Design and Manufacturing Techniques to Prevent/Detect Hardware Trojans.	2
5.3	Logic Testing and Side-channel Analysis based Techniques for Trojan Detection	2
5.4	Techniques to Increase Testing Sensitivity	2

Reference Books

1. Hardware Security: Design, Threats, and Safeguards, Debdeep Mukhopadhyay and Rajat Subhra Chakraborty, CRC Press, 2014.

2. Towards Hardware-intrinsic Security: Theory and Practice, Ahmad-Reza Sadeghi and David Naccache, Springer.
3. Handbook of FPGA Design Security, Ted Huffmire, et al., Springer.
4. Power analysis attacks - revealing the secrets of smart cards, Stefan Mangard, Elisabeth Oswald, Thomas Popp, Springer.



222EIT011	OPTIMIZATION TECHNIQUES	CATEGORY	L	T	P	CREDIT
		PROGRAM ELECTIVE 4	3	0	0	3

Preamble: The area of optimization plays a critical role designing and implementations of modern computer networks and security. Many issues can be formulated into an optimization problem and can be solved using of analytical and computational approach to different optimization techniques.

Course Outcomes: After the completion of the course the student will be able to

CO 1	Identify appropriate optimization method to solve complex problems involved in various industries.(Level: Apply)
CO 2	Demonstrate the optimized material distribution schedule using transportation model to minimize total distribution cost.(Level: Apply)
CO 3	Find the appropriate algorithm to find shortest path for given network.(Level: Apply)
CO 4	Determine the optimal value of the objective function in two ways using the primal and dual problems with realistic bounds on the variables.(Level: Analyse)
CO 5	Identify appropriate equipment replacement technique to be adopted to minimize maintenance cost by eliminating equipment break-down.(Level: Apply)
CO 6	Apply the knowledge of game theory concepts to articulate real-world competitive situations to identify strategic decisions to counter the consequences.(Level: Apply)

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

PO1: An ability to independently carry out research/investigation and development work in engineering and allied streams

PO2: An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

PO3: An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program

PO4: An ability to apply stream knowledge to design or develop solutions for real world problems by following the standards

PO5: An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tool to model, analyse and solve practical engineering problems.

PO6: An ability to engage in life-long learning for the design and development related to the stream related problems taking into consideration sustainability, societal, ethical, and environmental aspects

PO7: An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1							
CO 2							
CO 3							
CO 4							
CO 5							
CO 6							

Assessment Pattern

Bloom's Category	End Semester Examination
Apply	80%
Analyse	20%
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation Pattern:

Evaluation shall only be based on application, analysis or design-based questions (for both internal and end semester examinations).

Continuous Internal Evaluation (CIE) Pattern:

Preparing a review article based on peer-reviewed original publications (minimum 10 Publications shall be referred)/

10 Marks

Assignment:

Course-based task/Seminar/Data Collection and interpretation:

20 Marks

Test paper (one only, and shall include a minimum of 80% of the syllabus):

10 Marks

End Semester Examination (ESE) Pattern:

The end-semester examination shall be conducted by the respective College. There shall be two parts; Part A and Part B. Part A shall contain five numerical/short answer questions with one question from each module, having five marks for each question (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students). Students shall answer all questions. Part B shall contain seven questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem-solving and quantitative evaluation), with a minimum one question from each module of which student shall answer any five. Each question shall carry seven marks.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly.

For example, if the average end-semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is $40 + 20 = 60\%$.

Course Level Assessment Questions**Course Outcome 1 (CO1):**

1. Consider the following LP:

$$\text{Maximize } z = 2x_1 + 3x_2$$

subject to

$$x_1 + 3x_2 \leq 12$$

$$3x_1 + 2x_2 \leq 12$$

$$x_1, x_2 \geq 0$$

- (a) Express the problem in equation form.
 - (b) Determine all the basic solutions of the problem and classify them as feasible and infeasible.
2. Three types of coal, C1, C2, and C3, are pulverized and mixed to produce 50 tons per hour needed to power a plant for generating electricity. The burning of coal emits sulfur oxide (in parts per million) which must meet the EPA specifications of no more than 2000 parts per million. The following table summarizes the data of the situation:

	C1	C2	C3
Sulfur (parts per million)	2500	1500	1600
Pulverizer capacity (ton/hr)	30	30	30
Cost per ton	\$30	\$35	\$33

Determine the optimal mix of the coals.

Course Outcome 2 (CO2)

1. Four factories are engaged in the production of four types of toys. The following table lists the toys that can be produced by each factory.

Factory	Toys productions mix
1	1, 2, 3
2	2, 3
3	1, 3, 4
4	1, 3, 4

All toys require approximately the same per-unit labor and material. The daily capacities of the four factories are 250, 180, 300, and 200 toys, respectively. The daily demands for the four toys are 200, 150, 350, and 100 units, respectively. Determine the factories' production schedules that will most satisfy the demands for the four toys.

2. In intermodal transportation, loaded truck trailers are shipped between railroad terminals on special flatbed carts. Figure below shows the location of the main railroad terminals in the United States and the existing railroad tracks. The objective is to decide which tracks should be "revitalized" to handle the intermodal traffic. In particular, the Los Angeles (LA) terminal must be linked directly to Chicago (CH) to accommodate expected heavy traffic. Other than that, all the remaining terminals can be linked, directly or indirectly, such that the total length (in miles) of the selected tracks is minimized. Determine the segments of the railroad tracks that must be included in the revitalization program.

Course Outcome 3(CO3):

1. Consider the implementation of the feasibility condition of the simplex method. Specify the mathematical conditions for encountering a degenerate solution (at least one basic variable = 0) for the first time, for continuing to obtain a degenerate solution in the next iteration, and for removing degeneracy in the next iteration.

2. Consider the LP

$$\text{Maximize } z = CX \text{ subject to } AX \leq b, X \geq 0, \text{ where } b \geq 0$$

After obtaining the optimum solution, it is suggested that a non-basic variable x_j can be made basic (profitable) by reducing the resource requirements per unit of x_j to $\frac{1}{a}$ of their original values, $a > 1$. Since the requirements per unit are reduced, it is expected that the profit per unit of x_j will also be reduced to $\frac{1}{a}$ of its original value. Will these changes make x_j a profitable variable? Explain mathematically.

Course Outcome 4 (CO4):

1. The height of a cylindrical water tank must be at least twice as much as its base diameter. Neither the diameter nor the height can exceed 10 ft. The volume of the tank must be at least 300 ft³. The cost of the elevated structure on which the tank is installed is proportional to the area of the base. The sheet metal cost is \$8/ft², and the cost of the supporting structure is \$15/ft². Formulate the problem as a mathematical model, and develop a random-walk heuristic to estimate the diameter and height of the tank.
2. A tourist in New York City uses local transportation to visit 8 sites. The start and end and the order in which the sites are visited are unimportant. What is important is to spend the least amount of money on transportation. Matrix $\|C_{ij}\|$ below provides the fares in dollars between the different locations.

$$\|C_{ij}\| = \begin{pmatrix} 0 & 20 & 30 & 25 & 12 & 33 & 44 & 57 \\ 22 & 0 & 19 & 20 & 20 & 29 & 43 & 45 \\ 28 & 19 & 0 & 17 & 38 & 48 & 55 & 60 \\ 25 & 20 & 19 & 0 & 28 & 35 & 40 & 55 \\ 12 & 18 & 34 & 25 & 0 & 21 & 30 & 40 \\ 35 & 25 & 45 & 30 & 20 & 0 & 25 & 39 \\ 47 & 39 & 50 & 35 & 28 & 20 & 0 & 28 \\ 60 & 38 & 54 & 50 & 33 & 40 & 25 & 0 \end{pmatrix}$$

The tourist is budgeting \$120 for cab cost to all eight sites. Is this a realistic expectation?

Course Outcome 5 (CO5):

1. A school district is in dire need to reduce expenses to meet new budgetary restrictions at its elementary schools. Two options are available: delete the physical education program (E), or delete the music program (M). The superintendent has formed a committee with equal-vote representation from the school board (S) and the parent-teacher association (P) to study the situation and make a recommendation. The committee has decided to study the issue from the standpoint of budget restriction (B) and students needs (N). The analysis produced the following comparison matrices:

$$\begin{aligned}
 \mathbf{A}_S &= \begin{matrix} & \begin{matrix} B & N \end{matrix} \\ \begin{matrix} B \\ N \end{matrix} & \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \end{matrix} & \mathbf{A}_P &= \begin{matrix} & \begin{matrix} B & N \end{matrix} \\ \begin{matrix} B \\ N \end{matrix} & \begin{pmatrix} 1 & \frac{1}{2} \\ 2 & 1 \end{pmatrix} \end{matrix} \\
 \mathbf{A}_{SB} &= \begin{matrix} & \begin{matrix} E & M \end{matrix} \\ \begin{matrix} E \\ M \end{matrix} & \begin{pmatrix} 1 & \frac{1}{2} \\ 2 & 1 \end{pmatrix} \end{matrix} & \mathbf{A}_{SN} &= \begin{matrix} & \begin{matrix} E & M \end{matrix} \\ \begin{matrix} E \\ M \end{matrix} & \begin{pmatrix} 1 & \frac{1}{3} \\ 3 & 1 \end{pmatrix} \end{matrix} \\
 \mathbf{A}_{PB} &= \begin{matrix} & \begin{matrix} E & M \end{matrix} \\ \begin{matrix} E \\ M \end{matrix} & \begin{pmatrix} 1 & \frac{1}{3} \\ 3 & 1 \end{pmatrix} \end{matrix} & \mathbf{A}_{PN} &= \begin{matrix} & \begin{matrix} E & M \end{matrix} \\ \begin{matrix} E \\ M \end{matrix} & \begin{pmatrix} 1 & 2 \\ \frac{1}{2} & 1 \end{pmatrix} \end{matrix}
 \end{aligned}$$

Analyze the decision problem, and make a recommendation.

2. You are the author of what promises to be a successful novel. You have the option to either publish the novel yourself or through a publisher. The publisher is offering you \$20,000 for signing the contract. If the novel is successful, it will sell 200,000 copies. Else, it will sell 10,000 copies only. The publisher pays a \$1 royalty per copy. A market survey indicates that there is a 70% chance that the novel will be successful. If you undertake publishing, you will incur an initial cost of \$90,000 for printing and marketing, but each copy sold will net you \$2. Based on the given information, would you accept the publisher's offer or publish the novel yourself?

MODEL QUESTION PAPER																								
QP CODE:																								
Reg No: _____																								
Name: _____			PAGES :																					
3																								
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY																								
SECOND SEMESTER M.TECH DEGREE EXAMINATION, MONTH & YEAR																								
Course Code: 222EIT011																								
Course Name: OPTIMIZATION TECHNIQUES																								
Max. Marks : 60																								
Duration: 2.5 Hours																								
PART A																								
Answer All Questions. Each Question Carries 5 Marks (5x5=25 Marks)																								
1.	A large department store operates 7 days a week. The manager estimates that the minimum number of salespersons required to provide prompt service is 12 for Monday, 18 for Tuesday, 20 for Wednesday, 28 for Thursday, 32 for Friday, and 40 for each of Saturday and Sunday. Each salesperson works 5 days a week, with the two consecutive off-days staggered throughout the week. For example, if 10 salespersons start on Monday, 2 can take their off days on Tuesday and Wednesday, 5 on Wednesday and Thursday, and 3 on Saturday and Sunday. Develop the LP model.																							
2.	Three refineries with daily capacities of 6, 5, and 8 million gallons, respectively, supply three distribution areas with daily demands of 4, 8, and 7 million gallons, respectively. Gasoline is transported to the three distribution areas through a network of pipelines. The transportation cost is 10 cents per 1000 gallons per pipeline mile. Table 1 gives the mileage between the refineries and the distribution areas. Refinery 1 is not connected to distribution area 3. Construct the associated transportation model. <table border="1" style="margin-left: auto; margin-right: auto; border-collapse: collapse; text-align: center;"> <thead> <tr> <th colspan="4">Table 1: Mileage Chart</th> </tr> <tr> <th></th> <th colspan="3">Distribution Area</th> </tr> <tr> <th></th> <th>1</th> <th>2</th> <th>3</th> </tr> </thead> <tbody> <tr> <td>Refinery 1</td> <td>180</td> <td>180</td> <td>--</td> </tr> <tr> <td>Refinery 2</td> <td>300</td> <td>800</td> <td>900</td> </tr> </tbody> </table>			Table 1: Mileage Chart					Distribution Area				1	2	3	Refinery 1	180	180	--	Refinery 2	300	800	900	
Table 1: Mileage Chart																								
	Distribution Area																							
	1	2	3																					
Refinery 1	180	180	--																					
Refinery 2	300	800	900																					

		Refinery 3	220	200	120		
3.	Show that the set $Q = \{x_1, x_2 x_1 + x_2 \leq 3, x_1 \geq 0, x_2 \geq 0\}$ is convex. Is the nonnegativity condition essential for the proof?						
4.	Apply the uniform sampling heuristic to estimate the minimum solution of the following two-variable function: $f(x) = 3x^2 + 2y^2 - 4xy - 2x - 3y, 0 \leq x \leq 5, 0 \leq y \leq 5.$						
5.	Using an example, explain the decision tree analysis.						
Part B (Answer any five questions. Each question carries 7 marks)							
6.	Determine the optimum solution for each of the following LPs by enumerating all the basic solutions. Maximize $z = 2x_1 - 4x_2 + 5x_3 - 6x_4$ Subject to $x_1 + 4x_2 - 2x_3 + 8x_4 \leq 2$ $-x_1 + 2x_2 + 3x_3 + 4x_4 \leq 1$ $x_1, x_2, x_3, x_4 \geq 0$						(7)
7.	For the network in figure below, find the shortest routes between every two nodes. The distances (in miles) are given on the arcs. Arc (3, 5) is directional—no traffic is allowed from node 5 to node 3. All the other arcs allow two-way traffic. Use Floyd's algorithm to determine the shortest route between any two nodes in the network.						(7)
8.	Explain the mathematical model for the travelling salesman problem. Evaluate the model.						(7)
9.	In the following LP, compute the entire simplex tableau associated with $X_B = (x_1, x_2, x_5)^T$.						(7)

	Minimize $z = 2x_1 + x_2$ subject to $3x_1 + x_2 - x_3 = 2$$4x_1 + 3x_2 - x_4 = 4$$x_1 + 2x_2 + x_5 = 2$$x_1, x_2, x_3, x_4, x_5 \geq 0$																															
10	Three orchards supply crates of oranges to four retailers. The daily demand amounts at the four retailers are 150, 150, 400, and 100 crates, respectively. Supplies at the three orchards are dictated by available regular labour and are estimated at 150, 200, and 250 crates daily. The transportation costs per crate from the orchards to the retailers are given in Table 2. Formulate the problem as a transportation model. Solve the problem. <table><tr><th colspan="5">Table 2: Transportation Cost/Crate</th></tr><tr><th></th><th colspan="4">Retailer</th></tr><tr><th></th><th>1</th><th>2</th><th>3</th><th>4</th></tr><tr><td>Orchard 1</td><td>₹100</td><td>₹200</td><td>₹300</td><td>₹200</td></tr><tr><td>Orchard 2</td><td>₹200</td><td>₹400</td><td>₹100</td><td>₹200</td></tr><tr><td>Orchard 3</td><td>₹100</td><td>₹300</td><td>₹500</td><td>₹300</td></tr></table>	Table 2: Transportation Cost/Crate						Retailer					1	2	3	4	Orchard 1	₹100	₹200	₹300	₹200	Orchard 2	₹200	₹400	₹100	₹200	Orchard 3	₹100	₹300	₹500	₹300	(7)
Table 2: Transportation Cost/Crate																																
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Orchard 2	₹200	₹400	₹100	₹200																												
Orchard 3	₹100	₹300	₹500	₹300																												
11	A book salesperson who lives in Basin must call once a month on four customers located in Wald, Bon, Mena, and Kiln before returning home to Basin. The following table gives the distances in miles among the different cities. The objective is to minimize the total distance travelled by the salesperson. Write down the LP for computing a lower-bound estimate on the optimum tour length. <table><tr><th colspan="6">Table 3: Miles between cities</th></tr><tr><th></th><th>Basin</th><th>Wald</th><th>Bon</th><th>Mena</th><th>Kiln</th></tr><tr><td>Basin</td><td>₹0</td><td>₹1250</td><td>₹2250</td><td>₹1550</td><td>₹2150</td></tr></table>	Table 3: Miles between cities							Basin	Wald	Bon	Mena	Kiln	Basin	₹0	₹1250	₹2250	₹1550	₹2150	(7)												
Table 3: Miles between cities																																
	Basin	Wald	Bon	Mena	Kiln																											
Basin	₹0	₹1250	₹2250	₹1550	₹2150																											

		Wald	₹1250	₹0	₹850	₹1150 0	₹1350 0		
		Bon	₹2250	₹850	₹0	₹1650 0	₹1900 0		
		Mena	₹1550	₹1150	₹1650 0	₹0	₹1950 0		
		Kiln	₹2150	₹1350	₹1900 0	₹1950 0	₹0		
12	Consider the following two-person, zero-sum game:								(7)
			B1	B2	B3				
	A1		5	50	50				
	A2		1	1	1				
	A3		10	1	10				
Verify that the strategies $(\frac{1}{6}, 0, \frac{5}{6})$ for A and $(\frac{49}{54}, \frac{5}{54}, 0)$ for B are optimal and determine the value of the game.									



SYLLABUS**Module 1: Modelling with Linear Programming**

Two-Variable LP Model, Graphical LP Solution, LP Model in Equation Form, Transition from Graphical to Algebraic Solution. The Simplex Method - Iterative Nature of the Simplex Method, Computational Details of the Simplex Algorithm. Artificial Starting Solution - M-Method, Two-Phase Method. Special Cases in the Simplex Method – Degeneracy, Alternative Optima, Unbounded Solution, Infeasible Solution. Sensitivity Analysis - Algebraic Sensitivity Analysis—Changes in the Right-Hand Side, Objective Function. Definition of the Dual Problem. Primal–Dual Relationships - Simplex Tableau Layout, Optimal Dual Solution.

Module 2: Transportation and Network Model

Definition of the Transportation Model, Non-traditional Transportation Models. The Transportation Algorithm - Determination of the Starting Solution, Iterative Computations of the Transportation Algorithm, Simplex Method Explanation of the Method of Multipliers. The Assignment Model - The Hungarian Method. Scope and Definition of Network Models, Minimal Spanning Tree Algorithm, Shortest-Route Problem - Examples of the Shortest-Route Applications, Shortest-Route Algorithms, Linear Programming Formulation of the Shortest-Route Problem. Maximal Flow Model - Enumeration of Cuts, Maximal Flow Algorithm, Linear Programming Formulation of Maximal Flow Mode.

Module 3: Advanced Linear Programming

Simplex Method Fundamentals - From Extreme Points to Basic Solutions, Generalized Simplex Tableau in Matrix Form. Revised Simplex Method - Development of the Optimality and Feasibility Conditions, Revised Simplex Algorithm, Computational Issues in the Revised Simplex Method. Bounded-Variables Algorithm. Duality - Matrix Definition of the Dual Problem, Optimal Dual Solution. Parametric Linear Programming - Parametric Changes in C , Parametric Changes in b .

Module 4: Heuristic Programming

Heuristic Programming – Introduction, Greedy (Local Search) Heuristics - Discrete Variable Heuristic, Continuous Variable Heuristic. Metaheuristic - Tabu Search Algorithm, Simulated Annealing Algorithm, Genetic Algorithm. Scope of the TSP, TSP Mathematical Model. Exact TSP Algorithms - B&B Algorithm, Cutting-Plane Algorithm. Local Search Heuristics - Nearest-Neighbor Heuristic, Reversal Heuristic.

Module 5: Decision Analysis and Games

Decision Making Under Certainty—Analytic Hierarchy Process (AHP). Decision Making Under Risk - Decision Tree–Based Expected Value Criterion, Variants of the Expected Value Criterion. Decision Under Uncertainty. Game Theory - Optimal Solution of Two-Person Zero-Sum Games, Solution of Mixed Strategy Games



COURSE PLAN

No	Topic	No. of Lectures 40 Hours
1	Modelling with Linear Programming	
1.1	Two-Variable LP Model, Graphical LP Solution, LP Model in Equation Form, Transition from Graphical to Algebraic Solution. The Simplex Method - Iterative Nature of the Simplex Method, Computational Details of the Simplex Algorithm.	2 Hours
1.2	Artificial Starting Solution - M-Method, Two-Phase Method. Special Cases in the Simplex Method - Degeneracy, Alternative Optima, Unbounded Solution, Infeasible Solution.	2 Hours
1.3	Sensitivity Analysis - Algebraic Sensitivity Analysis—Changes in the Right-Hand Side, Objective Function.	1 Hours
1.4	Definition of the Dual Problem. Primal–Dual Relationships - Simplex Tableau Layout, Optimal Dual Solution.	2 Hours
2	Transportation and Network Model	
2.1	Definition of the Transportation Model, Non-traditional Transportation Models.	1Hour
2.2	The Transportation Algorithm - Determination of the Starting Solution, Iterative Computations of the Transportation Algorithm,	2Hours
2.3	Simplex Method Explanation of the Method of Multipliers.	1Hour
2.4	The Assignment Model - The Hungarian Method.	1Hour
2.5	Scope and Definition of Network Models, Minimal Spanning Tree Algorithm, Shortest-Route Problem - Examples of the Shortest-Route Applications.	2 Hours

2.6	Shortest-Route Algorithms, Linear Programming Formulation of the Shortest-Route Problem.	1Hour
2.7	Maximal Flow Model - Enumeration of Cuts, Maximal Flow Algorithm, Linear Programming Formulation of Maximal Flow Mode.	2 Hours
3	Advanced Linear Programming	
3.1	Simplex Method Fundamentals - From Extreme Points to Basic Solutions, Generalized Simplex Tableau in Matrix Form.	2 Hours
3.2	Revised Simplex Method - Development of the Optimality and Feasibility Conditions, Revised Simplex Algorithm, Computational Issues in the Revised Simplex Method.	2 Hours
3.3	Bounded-Variables Algorithm. Duality - Matrix Definition of the Dual Problem, Optimal Dual Solution.	2 Hours
3.4	Parametric Linear Programming - Parametric Changes in C, Parametric Changes in b.	1 Hours
4	Heuristic Programming	
4.1	Heuristic Programming – Introduction, Greedy (Local Search) Heuristics - Discrete Variable Heuristic, Continuous Variable Heuristic.	2 Hours
4.2	Metaheuristic - Tabu Search Algorithm, Simulated Annealing Algorithm.	2 Hours
4.3	Genetic Algorithm.	1 Hour
4.4	Scope of the TSP, TSP Mathematical Model. Exact TSP Algorithms - B&B Algorithm, Cutting-Plane Algorithm.	2 Hours
4.5	Local Search Heuristics - Nearest-Neighbor Heuristic, Reversal Heuristic.	2 Hours
5	Decision Analysis and Games	
5.1	Decision Making Under Certainty—Analytic	2 Hours

	Hierarchy Process (AHP).	
5.2	Decision Making Under Risk - Decision Tree-Based Expected Value Criterion, Variants of the Expected Value Criterion.	2 Hours
5.3	Decision Under Uncertainty	1 Hours
5.4	Game Theory - Optimal Solution of Two-Person Zero-Sum Games, Solution of Mixed Strategy Games	2 Hours

Reference Books

1. H. A. Taha, "Operations Research: An Introduction" 10th Edition, Pearson Education Limited 2017.
2. S.S.Rao, "Engineering optimization: Theory and practice", New Age International (P) Limited.
3. Bertsimas and J. Tsitsiklis, Introduction to Linear Optimization, Athena Scientific, 1997.
4. Nocedal J and Wright S, Numerical Optimization, 2nd Edition, Springer (2006).
5. Winston W L, Operations Research: Applications and Algorithms, 4th Edition, Thomson Learning (2004).

APJ ABDUL KALAM
TECHNOLOGICAL
UNIVERSITY

INTERDISCIPLINARY ELECTIVE



222EIT012	IOT BASED SMART SYSTEMS	CATEGORY	L	T	P	CREDIT
		INTERDISCIPLINARY ELECTIVE	3	0	0	3

reamble:

This course is aimed at providing an introduction to the field of IoT and IoT-based smart systems. The course provides a detailed discussion on the types and uses of sensors used in IoT. The protocols and standards used in IoT is also discussed along with potential IoT applications.

CourseOutcomes:

After the completion of the course, the student will be able to

CO 1	Explain IoT architecture and functional blocks of IoT ecosystem. (Knowledge Level: Understand)
CO 2	Discuss advanced techniques and tools of sensing and computation to solve multi- disciplinary challenges in industry and society. (Knowledge Level: Understand)
CO 3	Explain the protocols used in various layers to support IoT (Knowledge Level: Understand)
CO 4	Apply IoT in real-world applications (Knowledge Level: Apply)

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	✓		✓	✓	✓		
CO 2	✓	✓	✓	✓	✓	✓	
CO 3	✓		✓	✓	✓		
CO 4	✓	✓	✓	✓	✓	✓	✓

Assessment Pattern

Bloom's Category	End Semester Examination
-------------------------	---------------------------------

INFORMATION TECHNOLOGY

Understand	80%
Apply	20%
Analyse	
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation Pattern:

Evaluation shall only be based on application, analysis or design-based questions (for both internal and end semester examinations).

Continuous Internal Evaluation (CIE) Pattern:

Preparing a review article based on peer-reviewed original publications (minimum 10 Publications shall be referred)/ Assignment: 10 Marks

Course-based task/Seminar/Data Collection and interpretation: 20 Marks

Test paper (one only, and shall include a minimum of 80% of the syllabus): 10 Marks

End Semester Examination (ESE) Pattern:

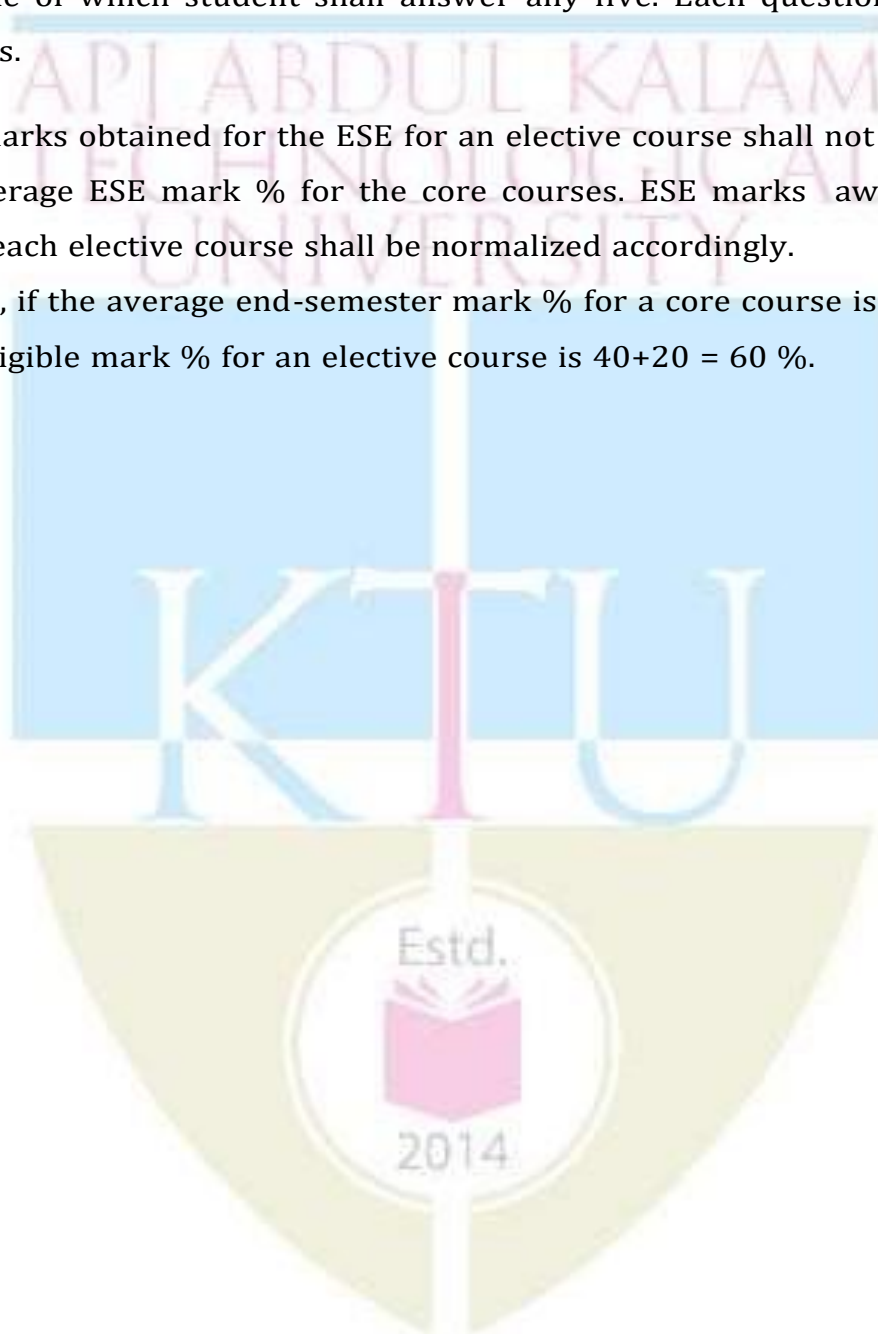
The end-semester examination shall be conducted by the respective College. There shall be two parts; Part A and Part B. Part A shall contain five numerical/short answer questions with one question from each module, having five marks for each question (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students). Students shall answer all questions. Part B

INFORMATION TECHNOLOGY

shall contain seven questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem-solving and quantitative evaluation), with a minimum one question from each module of which student shall answer any five. Each question shall carry seven marks.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly.

For example, if the average end-semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is $40+20 = 60\%$.



MODEL QUESTION PAPER

QPCODE:

RegNo: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

SECOND SEMESTER M.TECH DEGREE EXAMINATION, MONTH & YEAR

222EIT012

IoT BASED SMART SYSTEMS

Max.Marks:60 Duration:2.5hrs

PART A

(Answer all questions. Each Question Carries 5 Marks)

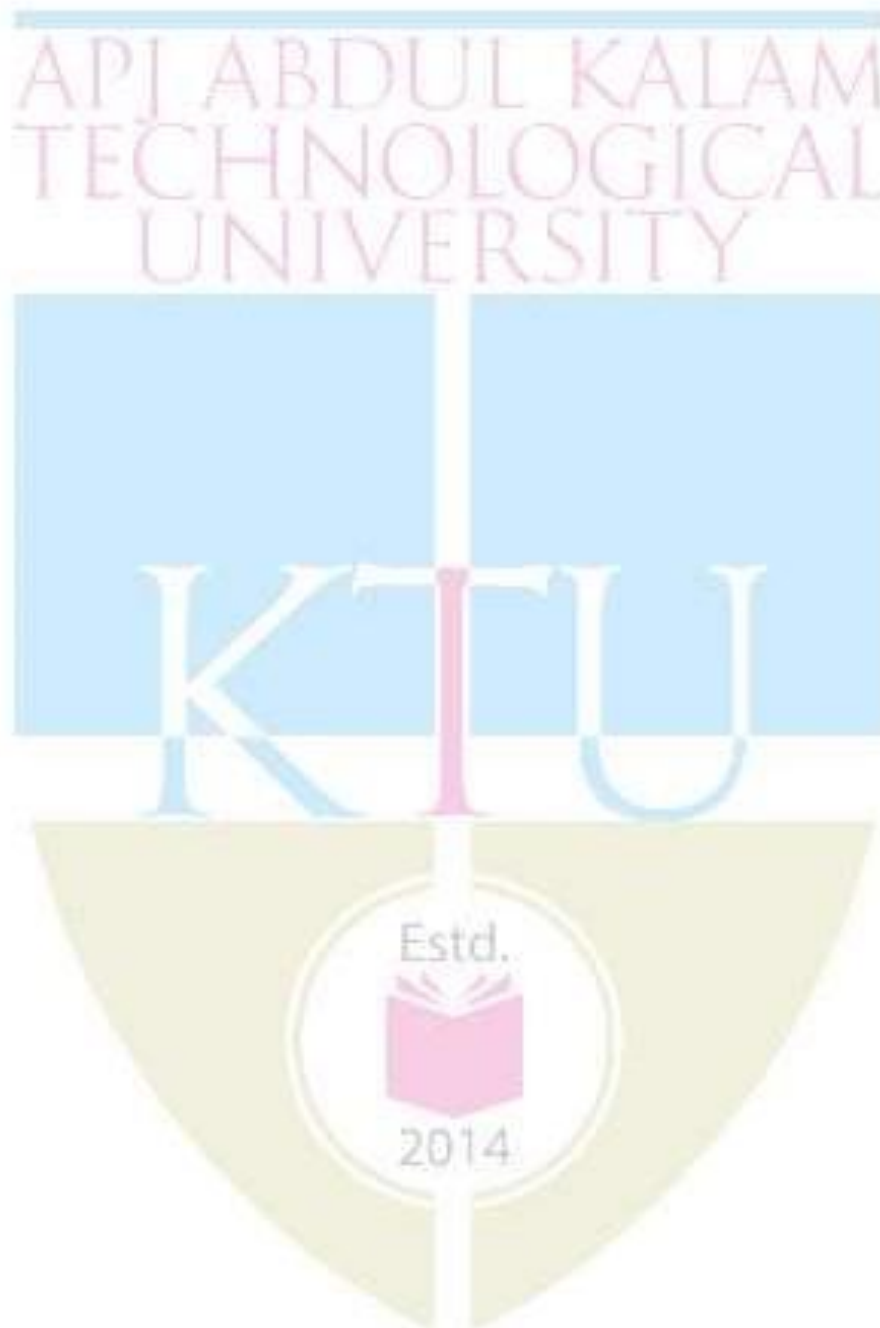
1. What is M2Mcommunication?
2. List the different types of data which are generated at IoT devices.
3. Briefly mention the various design steps in 'Tracking of Customers Carrying Internet Connected Digital Devices' (TCCICDD) for IoT applications.
4. Compare Internet, Intranet and Extranet.
5. Consider a scenario where you're working on a smart home project. Which OS will you prefer to use on your project and why?

PART B

(Answer any FIVE Questions. Each Question Carries 7 Marks)

6. With the help of a neat diagram explain IoT reference model.
7. What is network topology? Which topology will you prefer to develop a smart city? Justify your answer.
8. Briefly explain the various sensor data communication protocols.
9. Briefly discuss the various tasks of a smart irrigation monitoring service and list out the tasks of an air-pollution monitoring service.

10. Explain the CoAP and MQTT.
11. Explain the cloud service model in detail.
12. Using an example application, compare and contrast Web of Things and Internet of Things.



SYLLABUS

Module I (7 Hours)

Fundamentals of IoT : Evaluation of Internet of Things – Enabling Technologies, IoT Architecture: oneM2M, IoT World Forum (IoTWF), Alternative IoT Models - Simplified IoT Architecture and Core IoT Functional Stack – Fog, Edge and Cloud in IoT, Functional Block of IoT Ecosystem –Sensors, Actuators, and Smart Objectives – Open Hardware Platforms for IoT.

Module II (8 Hours)

Sensors Classification, Performance and Types, Error Analysis: Characteristics Optical Sources and Detectors, Electronic and Optical properties of semiconductor as sensors, LED, Semiconductor lasers, Fiber optic sensors, Thermal detectors, Photo multipliers, Photoconductive detectors, Photodiodes, Avalanche photodiodes, CCDs, Acoustic sensors – Microphones – Resistive, Capacitive, Piezoelectric, Fiber optic, solid state – Electret microphone.

Module III (9 Hours)

Electromagnetic velocity sensor, Doppler with sound, light Accelerometer characteristics, capacitive, piezo –resistive, piezoelectric accelerometer, thermal accelerometer, rotor, monolithic and optical gyroscope. Flow sensors: Pressure gradient technique, Thermal transport, Ultrasonic, Electromagnetic and Laser anemometer. Microflow sensor, Coriolis mass flow and Drag flow sensor. Temperature sensors –Thermoresistive, Thermoelectric, Semiconductor and Optical. Piezoelectric temperature sensor.

Module IV (8 Hours)

IoT Protocols and Standards : Application Layer Protocols: CoAP, MQTT, Service discovery – mDNS. Service layer and Security protocols : M2M, OMA, IPSec, Secure DNS Routing over Low Power and Lossy Network (RPL). Data Link Layer Protocols - WiFi, Bluetooth Low Energy, Zigbee Smart Energy.

Module V (8 Hours)

IoT Applications : Smart and Connected Cities: Street Layer, City Layer, Data Center Layer and Services Layer, Street Lighting, Smart Parking Architecture and Smart Traffic Control – Smart Transportation – connected car. Case studies: IoT for health care, agriculture, smart meters, M2M, Web of things, Cellular IoT, Industrial IoT, Industry 4.0

COURSE PLAN

No	Topic	No. of Lectures
1	Fundamentals of IoT (7 Hours)	
1	Evaluation of Internet of Things – Enabling Technologies.	1
1.1	IoT Architecture: oneM2M, IoT World Forum (IoTWF)	2
1.2	Alternative IoT Models -Simplified IoT Architecture and Core IoT Functional Stack – Fog, Edge and Cloud in IoT	2
1.3	Functional Block of IoT Ecosystem –Sensors, Actuators, and SmartObjectives – Open Hardware Platforms for IoT.	2
2	Introduction to sensors (8 Hours)	
2.1	Sensors Classification, Performance and Types, Error Analysis	2
2.2	Characteristics Optical Sources and Detectors.	1
2.3	Electronic and Optical properties of semiconductor as sensors, LED, Semiconductor lasers, Fiber optic sensors	2
2.4	Thermal detectors, Photomultipliers, Photoconductive detectors, Photodiodes, Avalanche photodiodes, CCDs.	1
2.5	Acoustic sensors – Microphones – Resistive, Capacitive, Piezoelectric, Fiber optic, solid state – Electret microphone.	2
3	Introduction to sensors (9 Hours)	
3.1	Electromagnetic velocity sensor, Doppler with sound, light Accelerometer characteristics, capacitive, piezo –resistive, piezoelectric accelerometer, thermal accelerometer, rotor, monolithic and optical gyroscope.	2
		1
3.2	Flow sensors: Pressure gradient technique, Thermal transport, Ultrasonic, Electromagnetic and Laser anemometer.	2
3.3	Microflow sensor, Coriolis mass flow, and Drag	1

	flow sensor.	
3.4	Temperature sensors – Thermo-resistive, Thermoelectric, Semiconductor and Optical.	2
3.5	Piezoelectric temperature sensor.	1
4	IoT Protocols and Standards (8 Hours)	
4.1	Application Layer Protocols: CoAP, MQTT, Service discovery –mDNS.	2
4.2	Service layer and Security protocols: M2M, OMA, IPSec, Secure DNS	2
4.3	Routing over Low Power and Lossy Network (RPL).	2
4.4	Data Link Layer Protocols - WiFi, Bluetooth Low Energy, Zigbee Smart Energy	2
5	IoT Applications (8 Hours)	
5.1	Smart and Connected Cities: Street Layer, City Layer, Data Center Layer and Services Layer, Street Lighting	2
5.2	Smart Parking Architecture and Smart Traffic Control – Smart Transportation – connected car	2
5.3	Smart Parking Architecture and Smart Traffic Control – Smart Transportation – connected car	2
5.4	M2M, Web of things, Cellular IoT,	1
5.5	Industrial IoT, Industry4.0	2

Reference Books

1. David Hanes, GonzolaSalgueiro, Patrick Grossetete, Rob Barton, Jerome Henry, “IoT Fundamentals: Network Technologies, Protocols and Use Cases for Internet of Things”, CISCO Press,2017.
2. Perry Lea, “Internet of things for architects”, Packt,2018.
3. Jan Ho”ller, VlasiosTsiatsis, Catherin Mulligan, Stamatic, Karnouskos, Stefan Avesand, David Boyle, “From Machine-to-Machine to the Internet of Things – Introduction to a New Age of Intelligence”, Elsevier,2014.

222EIT013	SECURITY IN COMPUTING	CATEGORY	L	T	P	CREDIT
		INTERDISCIPLINARY ELECTIVE	3	0	0	3

Preamble: The syllabus aims to make the students capable of understanding the basic principles and concepts of computer and network security. At the end of the course, the students should be able to explain what it means for a system to be secure. Furthermore, the students will get to know about computing systems vulnerabilities, threats, and security controls.

Course Outcomes:

After the completion of the course, the student will be able to

CO 1	Outline the basic concepts and techniques of computer security (Level: Understand)
CO 2	Summarize the methods for achieving security at computer program level. (Level: Understand)
CO 3	Model security schemes for secure and trusted operating systems (Level: Apply)
CO 4	Summarize the requirements and features of schemes for database security. (Level: Understand)
CO 5	Identify the security issues in a network and suggest appropriate security measures (Level: Apply)

Program Outcomes:

PO 1	An ability to independently carry out research/ investigation and development work in engineering and allied streams.
PO 2	An ability to communicate effectively, and write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.
PO 3	An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level

INFORMATION TECHNOLOGY

	higher than the requirements in the appropriate bachelor's program.
PO 4	An ability to apply stream knowledge to design or develop solutions for real-world problems by following the standards.
PO 5	An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tools to model, analyze and solve practical engineering problems.
PO 6	An ability to engage in lifelong learning for the design and development of stream-related problems taking into consideration, sustainability societal, ethical, and environmental aspects.
PO 7	An ability to develop cognitive load management skills related to project management and finance which focus on entrepreneurship and industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	√	√	√		√	√	
CO 2	√	√	√		√	√	
CO 3	√	√	√		√	√	
CO 4	√	√	√		√	√	
CO 5	√	√	√		√	√	

Assessment Pattern

Bloom's Category	End Semester Examination

INFORMATION TECHNOLOGY

Understand	60%
Apply	40%
Analyze	
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

The evaluation shall only be by employing application-based, analysis-based, or design-based questions (for both internal and end-semester examinations)

Continuous Internal Evaluation (CIE) Pattern:

Preparing a review article based on peer-reviewed original publications (minimum 10 Publications shall be referred)/

10 Marks

Assignment:

Course-based task/Seminar/Data Collection and interpretation:

20 Marks

Test paper (one only, and shall include a minimum of 80% of the syllabus):

10 Marks

End Semester Examination (ESE) Pattern:

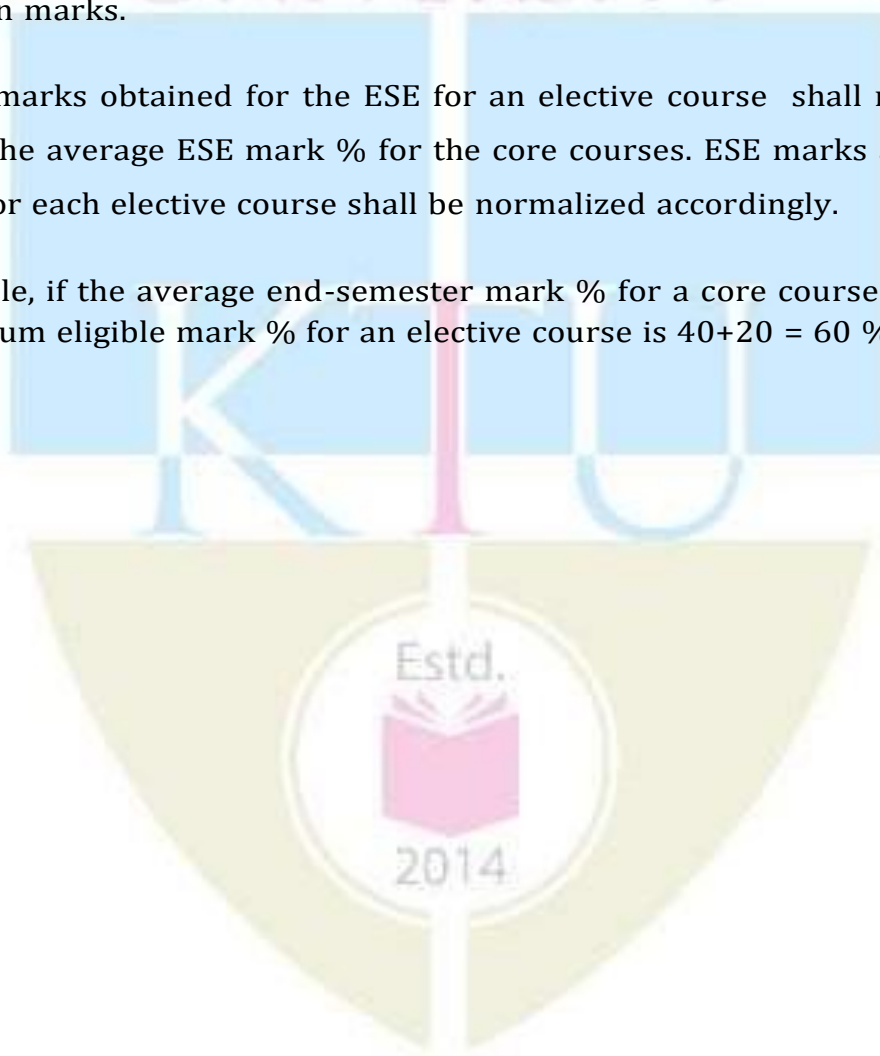
The end-semester examination shall be conducted by the respective College. There shall be two parts; Part A and Part B. Part A shall contain five numerical/short answer questions with one question from each module, having

INFORMATION TECHNOLOGY

five marks for each question (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students). Students shall answer all questions. Part B shall contain seven questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem-solving and quantitative evaluation), with a minimum one question from each module of which student shall answer any five. Each question shall carry seven marks.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly.

For example, if the average end-semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is $40 + 20 = 60$ %.



MODEL QUESTION PAPER FOR END-SEMESTER EXAMINATION

Time: 2½ Hours, Maximum marks: 60

PART A (*Answer all questions. Five marks each*)

1. What do you mean when you say that a system is secure? Explain.
2. Explain with an example, how non-malicious program errors become a threat to security.
3. Differentiate between the types of program security flaws.
4. Explain the two-phase database update with an example.
5. Explain the different types of Intrusion Detection Systems.

PART B (*Answer any five questions. Seven marks each*)

6. Explain the internals of the AES algorithm.
7. What do you mean by non-malicious program errors? Outline the various methods to control different program threats.
8. Give an outline of the Bell La-Padula model. Compare the Simple Security Condition Preliminary version and Star Property Preliminary version in the Bell-La-Padula model.
9. Explain the models for designing a multilevel secure database. Summarize the disadvantages of partitioning as means of implementing multilevel security for databases.
10. What are firewalls for? Explain the working of any three kinds of firewalls in detail.
11. Give an outline of the techniques behind Secure Electronic Transactions. What is the significance of dual signature in secure electronic transactions?
12. According to Bell-La Padula Model, identify the restrictions placed on two active subjects that need to send and receive signals to and from each other. Justify your identification.

SYLLABUS

Module I: Introduction to Security in Computing (8 Hours)

Introduction: Security Problem in Computing, Elementary Cryptography- Terminology and Background, Substitution Ciphers, Transposition Ciphers, Encryption Algorithms, AES, Introduction to Public Key Encryption, Uses of Encryption.

Module II: Program Security (8 Hours)

Secure Programs, Non-malicious Program Errors, Viruses and other Malicious Code, Targeted Malicious Code, Controls against Program Threats.

Module III: Operating System Security (8 Hours)

Protected Objects and Methods of Protection, Memory Address Protection, Control of Access to General Objects, File Protection Mechanisms, User Authentication, Designing Trusted Operating Systems- Security Policies, Models of Security, Trusted Operating System Design.

Module IV: Database Security (8 Hours)

Introduction to Databases, Security Requirements, Reliability and Integrity, Sensitive Data, Inference, Multilevel Databases, Proposals for Multilevel Security.

Module V: Network Security (8 Hours)

Security in Networks- Threats in Networks, Network Security Controls, Secure Electronic Transactions, Firewalls, Intrusion Detection Systems

COURSE PLAN

No	Topic	No. of Lectures
1	Introduction to Security in Computing (8 Hours)	
1.1	Introduction: Security Problem in Computing	1
1.2	Elementary Cryptography- Terminology and Background	1
1.3	Substitution Ciphers, Transposition Ciphers	2
1.4	Encryption Algorithms, AES	2
1.5	Introduction to Public Key Encryption, Uses of Encryption.	2
2	Program Security (8 Hours)	
2.1	Secure Programs	1
2.2	Non-malicious Program Errors	1
2.3	Viruses and other Malicious Code	2
2.4	Targeted Malicious Code	2
2.5	Controls against Program Threats	2
3	Operating System Security (8 Hours)	
3.1	Protected Objects and Methods of Protection	1
3.2	Memory Address Protection	1
3.3	Control of Access to General Objects	1
3.4	File Protection Mechanisms	1
3.5	User Authentication	1
3.6	Designing Trusted Operating Systems- Security Policies, Models of Security	2

3.7	Trusted Operating System Design	1
4	Database Security (8 Hours)	
4.1	Introduction to Databases	1
4.2	Security Requirements	1
4.3	Reliability and Integrity	1
4.4	Sensitive Data, Inference	2
4.5	Multilevel Databases	1
4.6	Proposals for Multilevel Security	2
5	Network Security (8 Hours)	
5.1	Introduction to Computer Networks and the Internet	1
5.2	Threats in Networks	1
5.3	Network Security Controls	2
5.4	Secure Electronic Transactions	1
5.5	Firewalls	1
5.6	Intrusion Detection Systems	2

Reference Books:

1. Charles P. Pfleeger, Shari Lawrence Pfleeger and Deven N. Shah, Security in Computing, 4th Edition
2. William Stallings, Cryptography and Network Security: Principles and Practice, Pearson Education, 7th Edition.
3. William Stallings, Network Security Essentials, Applications and Standards, Pearson Education.
4. Michael E. Whitman and Herbert J Mattord, Principles of Information Security, 4th Edition
5. Kurose and Ross, Computer Networking: A top-down approach, 8th Edition, Pearson

222EIT014	CLOUD COMPUTING	CATEGORY	L	T	P	CREDIT
		INTERDISCIPLINARY ELECTIVE	3	0	0	3

Preamble: This course helps learners to understand cloud computing concepts. This course includes an analysis of cloud computing models and services, cloud security, cloud computing enabling technologies, and popular industry cloud computing platforms. This course enables the learners to suggest cloud-based solutions to real-world problems.

Course Outcomes:

After the completion of the course, the student will be able to

CO 1	Explain the various cloud computing models and services. (Cognitive Knowledge Level: Understand)
CO 2	Illustrate the different cloud-enabling technologies(Cognitive Knowledge Level: Understand)
CO 3	Discuss the need for security mechanisms in the cloud(Cognitive Knowledge Level: Understand)
CO 4	Illustrate various load-balancing methods in the cloud. (Cognitive Knowledge Level: Apply)
CO 5	Contrast the different popular cloud computing platforms(Cognitive Knowledge Level: Understand)

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

PO1: An ability to independently carry out research/investigation and development work in engineering and allied streams.

P02: An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

P03: An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.

P04: An ability to apply stream knowledge to design or develop solutions for real world problems by following the standards.

P05: An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tool to model, analyze and solve practical engineering problems.

P06: An ability to engage in life-long learning for the design and development related to the stream related problems taking into consideration sustainability, societal, ethical and environmental aspects.

P07: An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	√	√	√		√	√	
CO 2	√	√	√		√	√	
CO 3	√	√	√		√	√	
CO 4	√	√	√		√	√	
CO 5	√	√	√		√	√	

Assessment Pattern

Bloom's Category	End Semester Examination
Understand	80%
Apply	20%
Analyze	
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

The evaluation shall only be by employing application-based, analysis-based, or design-based questions (for both internal and end-semester examinations)

Continuous Internal Evaluation (CIE) Pattern:

Preparing a review article based on peer-reviewed original publications (minimum 10 Publications shall be referred)/

10 Marks

Assignment:

Course-based task/Seminar/Data Collection and interpretation:

20 Marks

Test paper (one only, and shall include a minimum of 80% of the syllabus): 10 Marks

End Semester Examination (ESE) Pattern:

The end-semester examination shall be conducted by the respective College. There shall be two parts; Part A and Part B. Part A shall contain five numerical/short answer questions with one question from each module, having five marks for each question (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students). Students shall answer all questions. Part B shall contain seven questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem-solving and quantitative evaluation), with a minimum one question from each module of which student shall answer any five. Each question shall carry seven marks.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly.

For example, if the average end-semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is $40 + 20 = 60\%$.

MODEL QUESTION PAPER FOR END-SEMESTER EXAMINATION

Time: 2½ Hours, Maximum marks: 60

PART A (*Answer all questions. Five marks each*)

1. "A hybrid cloud is a combination of two or more other cloud deployment models". Justify the statement with an example.
2. What are the characteristics of multi-tenant technology?
3. Why is it harder to establish security in the cloud?
4. What is an auto-scaling feature in cloud computing? How has it been achieved based on three pillars?
5. Differentiate between amazon SimpleDB and Amazon RDS.

PART B (*Answer any five questions. Seven marks each*)

6. What are the basic components of an IaaS-based solution for cloud computing? Also provide some examples of IaaS implementations.
7. Explain the broadband networks and internet architecture.
8. Explain common threats and vulnerabilities in cloud-based environments with suitable examples.
9. Auto-scaling implementation requires a mixture of both reactive and proactive scaling approaches. Justify this statement.
10. Explain the architecture of Windows Azure.
11. Differentiate between vertical and horizontal scaling
12. "Integration of cloud computing in mobile computing provides advantages by overcoming the limitations of mobile computing". Justify this statement.

SYLLABUS

Module I: Introduction to Cloud Computing (7 Hours)

Traditional computing: Limitations, Overview of Computing Paradigms: Grid Computing, Cluster Computing, Distributed Computing, Utility Computing, Cloud Computing, NIST reference Model, Basic terminology and concepts, Cloud characteristics, benefits and challenges, Cloud delivery (service) models: Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS), Software-as-a-Service (SaaS), XaaS (Anything-as-a-service), Cloud deployment models: Public cloud, Community cloud, Private cloud, Hybrid cloud, Open Cloud Services.

Module II: Cloud-Enabling Technology (9 Hours)

Broadband Networks and Internet Architecture: Internet Service Providers (ISPs), Router-Based Interconnectivity, Technical and Business Considerations, Data Center Technology, Virtualization Technology, Web Technology, Multitenant Technology, Service Technology.

Module III: Fundamental Cloud Security (8 Hours)

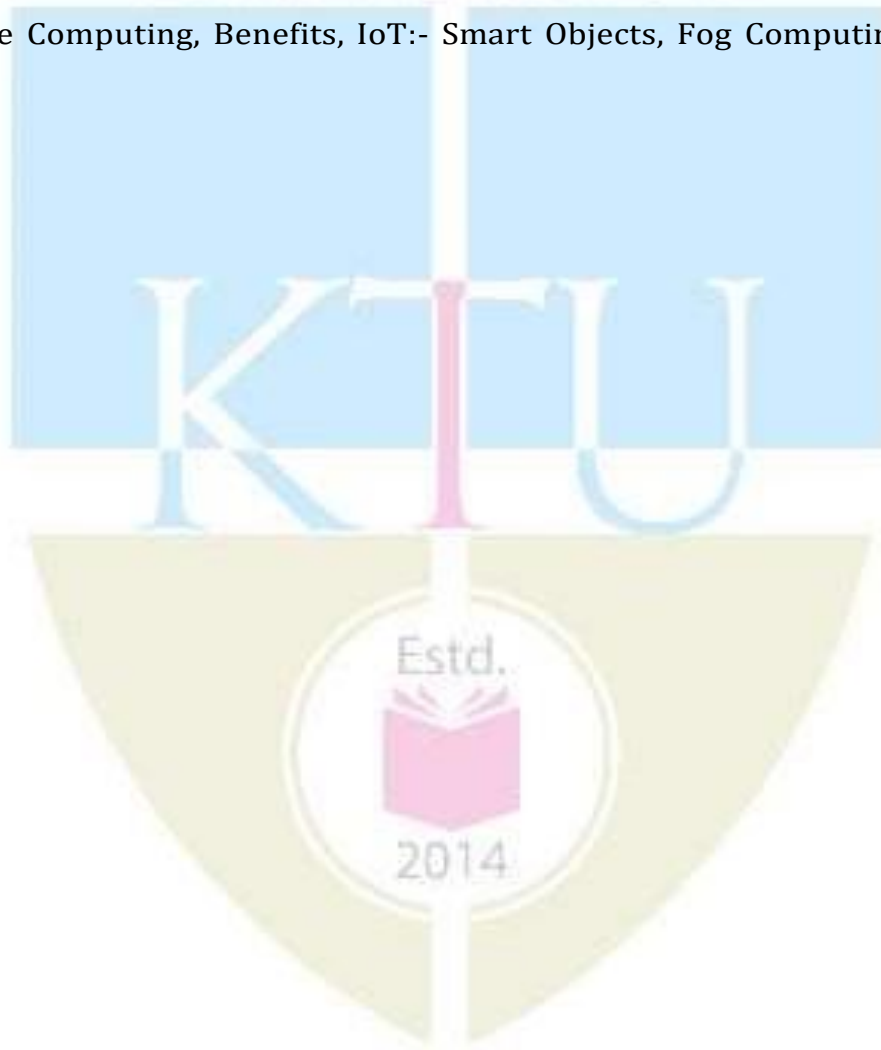
Basic Terms and Concepts in Security, Threat Agents, Cloud Security Threats, Identity Management and access Control, Cloud Security Working Groups, Elements of Cloud Security Model, Cloud Security Reference Model, Examining Cloud Security against Traditional Computing.

Module IV: Cloud Mechanisms (7 Hours)

Scaling, Foundation of Cloud Scaling, Scaling Strategies in Cloud, Auto Scaling in Cloud, Cloud Bursting, Types of Scaling, Capacity Planning, Capacity Planning at Different Service Levels, Load Balancer, Two Levels of Balancing, Goals of Load Balancing, Categories of Load Balancing, Exploring Dynamic Load Balancing.

Module V: Popular Cloud Platforms, Mobile Cloud Computing and IoT (9 Hours)

Amazon web services (AWS):- Amazon Elastic Compute Cloud, Amazon Storage Systems, Database Services of AWS, Amazon CDN Service: CloudFront, Google Cloud:- Google's IaaS Offerings, Google's PaaS Offering, Google's SaaS Offerings, Microsoft Azure:- Azure Virtual Machine, Azure Storage, Azure Database Services, Azure Content Delivery Network, Mobile Cloud Computing:- Introduction, Limitations, Offloading of Tasks, Integration of Cloud Computing into Mobile Computing, Benefits, IoT:- Smart Objects, Fog Computing, Benefits of IoT.



COURSE PLAN

No	Topic	No. of Lectures
1	Introduction to Cloud Computing	7 Hours
1.1	Traditional computing: Limitations, Overview of Computing Paradigms: Grid Computing, Cluster Computing.	1
1.2	Distributed Computing, Utility Computing, Cloud Computing.	1
1.3	NIST reference Model, Basic terminology and concepts,	1
1.4	Cloud characteristics, benefits and challenges	1
1.5	Cloud delivery (service) models: Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS), Software-as-a-Service (SaaS), XaaS (Anything-as-a-service),	1
1.6	Cloud deployment models: Public cloud, Community cloud, Private cloud, Hybrid cloud.	1
1.7	Open Cloud Services.	1
2	Cloud-Enabling Technology	9 Hours
2.1	Broadband Networks and Internet Architecture: Internet Service Providers (ISPs), Router-Based Interconnectivity,	1
2.2	Technical and Business Considerations.	1
2.3	Data Center Technology.	2
2.4	Virtualization Technology.	2

INFORMATION TECHNOLOGY

2.5	Web Technology	1
2.6	Multitenant Technology	1
2.7	Service Technology	1
3	Fundamental Cloud Security	8 Hours
3.1	Basic Terms and Concepts in Security.	1
3.2	Threat Agents	1
3.3	Cloud Security Threats.	1
3.4	Identity Management and access Control	1
3.5	Cloud Security Working Groups, Elements of Cloud Security Model.	1
3.6	Cloud Security Reference Model.	2
3.7	Examining Cloud Security against Traditional Computing.	1
4	Cloud Mechanisms	7 Hours
4.1	Scaling, Foundation of Cloud Scaling, Scaling Strategies in Cloud.	1
4.2	Auto Scaling in Cloud, Cloud Bursting.	1
4.3	Types of Scaling.	1
4.4	Capacity Planning, Capacity Planning at Different Service Levels.	1
4.5	Load Balancer, Two Levels of Balancing, Goals of Load Balancing.	1
4.6	Categories of Load Balancing, Exploring Dynamic Load Balancing	2

5	Popular Cloud Platforms, Mobile Cloud Computing and IoT	9 Hours
5.1	Amazon web services (AWS):- Amazon Elastic Compute Cloud, Amazon Storage Systems.	1
5.2	Database Services of AWS, Amazon CDN Service: CloudFront.	2
5.3	Google Cloud:- Google's IaaS Offerings, Google's PaaS Offering, Google's SaaS Offerings.	1
5.4	Microsoft Azure:- Azure Virtual Machine, Azure Storage, Azure Database Services, Azure Content Delivery Network.	2
5.5	Mobile Cloud Computing:- Introduction, Limitations.	1
5.6	Offloading of Tasks, Integration of Cloud Computing into Mobile Computing, Benefits.	1
5.7	IoT:- Smart Objects, Fog Computing, Benefits of IoT.	1

Reference Books

1. *Cloud computing*. Bhowmik, S., 2017. Cambridge University Press.
2. *Cloud Computing Concepts, Technology & Architecture*. Thomas, E., Zaigham, M., & Ricardo, P. (2013). Prentice Hall.
3. *"Mastering cloud computing: foundations and applications programming"*, Buyya, R., Vecchiola, C., & Selvi, S. T. (2017 Edition)., Morgan Kaufmann.
4. *"Distributed and Cloud Computing from Parallel Processing to the Internet of Things"* K.Hwang, G.C Fox, J.J Dongarra., (2012 Edition), Morgan Kaufmann.

MINIPROJECT

API ABDUL KALAM
TECHNOLOGICAL
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KTU

Estd.



2014

INFORMATION TECHNOLOGY

222PIT100	MINI PROJECT	CATEGORY	L	T	P	CREDIT
		PROJECT	0	0	4	2

Mini project can help to strengthen the understanding of student's fundamentals through application of theoretical concepts and to boost their skills and widen the horizon of their thinking. The ultimate aim of an engineering student is to resolve a problem by applying theoretical knowledge. Exposure to more projects enhances problem solving skills.

The introduction of mini projects ensures preparedness of students to undertake dissertation. Students should identify a topic of interest in consultation with PG Programme Coordinator that should lead to their dissertation/research project. Demonstrate the novelty of the project through the results and outputs.

The progress of the mini project is evaluated based on three reviews, two interim reviews and a final review. A report is required at the end of the semester.

Evaluation Committee - Programme Coordinator, One Senior Professor and Guide.

Sl. No	Type of evaluations	Mark	Evaluation criteria
1	Interim evaluation 1	20	
2	Interim evaluation 2	20	

INFORMATION TECHNOLOGY

3	Final evaluation by a Committee	35	Will be evaluating the level of completion and demonstration of functionality/ specifications, clarity of presentation, oral examination, work knowledge and involvement
4	Report	15	the committee will be evaluating for the technical content, adequacy of references, templates Followed and permitted plagiarism level(not more than 25%)
5	Supervisor/Guide	10	
Total Marks		100	



LABORATORY

API ABDUL KALAM
TECHNOLOGICAL
UNIVERSITY



INFORMATION TECHNOLOGY

222LIT100	SECURE COMPUTING LAB	CATEGORY	L	T	P	CREDIT
		LABORATORY	0	0	2	1

Preamble: This course enables the learners to perform experiments in computer security and experience various tools for network monitoring and security.

Course Outcomes:

After the completion of the course, the student will be able to

CO 1	Implement various cryptographic algorithms (Level: Apply)
CO 2	Evaluate the performance of various cryptographic algorithms (Level: Analyze)
CO 3	Employ and experiment with tools for monitoring network traffic (Level: Analyze)
CO 4	Demonstrate the use of various tools for network security (Level: Apply)

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1		✓		✓	✓		
CO 2		✓		✓	✓		
CO 3		✓		✓	✓		

INFORMATION TECHNOLOGY

CO 4		✓		✓	✓		
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Continuous Internal Evaluation Pattern:

CIE: 100%

Attendance: 10%

Preparation and completion of experiments by the student: 60%

Final Assessment: 30%

SYLLABUS AND COURSE PLAN

No	Topic	No. of Hrs
1	Implement a program to calculate MD5 and SHA hash values 2 Hrs	
2	Using a given private key, generate a public key in an elliptic curve algorithm.	2 Hrs
3	Implement digital signature using RSA algorithm	2 Hrs
4	Implement a protocol for authenticated delivery of data by considering encryption and decryption using AES 128-bit key.	2 Hrs
5	Implementation of web-browser communication using SSL by ensuring strong encryption, authentication and data integrity. 2 Hrs	

INFORMATION TECHNOLOGY

6	Program to sign messages which conform to the S/MIME standard and verify its digital signatures based on certificates containing RSA keys.	2 Hrs
7	Implementation of a sniffer for monitoring network communication.	2 Hrs
8	Program to send an encrypted string via Bluetooth from a PC as a client to a mobile as a server.	2 Hrs
9	Implement a program for secure password storage 2 Hrs	
10	Implement a program for distributed Denial of service	2 Hrs
11	Implement a technique to extract the database information through a web application(SQL Injection)	2 Hrs
12	Performance Evaluation of Cryptographic Algorithms: AES and DES	2 Hrs
13	Use NMAP for port monitoring 1 Hr	
14	Use IPTABLES/NETFILTER on Linux for Firewalls to configure the IP packet filter rules	1 Hr

References

1. https://www.suse.com/documentation/sles11/book_sle_admin/data/sec_a_pache2_ssl.html
2. <http://linuxconfig.org/apache-web-server-ssl-authentication>

3. <http://www.tomcatexpert.com/knowledge-base/using-openssl-onfigure-ssl-certificates- tomcat>
4. http://linuxcommand.org/man_pages/openssl1.html
5. https://www.owasp.org/index.php/Digital_Signature_Implementation_in_Java
6. <https://www.openssl.org/docs/apps/smime.html>
7. http://fedoraproject.org/wiki/How_to_edit_iptables_rules
8. <https://www.frozentux.net/iptables-tutorial/iptables-tutorial.html>
9. <https://www.frozentux.net/iptables-tutorial/iptables-tutorial.html>
10. <http://nmap.org/bennioston-tutorial/>



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SEMESTER III

KTU



SEMESTER III

Slot	Course code	Courses	Marks		L-T-P	Hours	Credit
			CIE	ESE			
TRACK 1							
A*	223MITXXX	MOOC	To be completed successfully		--	--	2
B	223AGEXXX	Audit Course	40	60	3-0-0	3	-
C	223IIT100	Internship	50	50	--	--	3
D	223PIT100	Dissertation Phase 1	100	--	0-0-17	17	11
TRACK 2							
A*	223MITXXX	MOOC	To be completed successfully		--	--	2
B	223AGEXXX	Audit Course	40	60	3-0-0	3	-
C	223IIT100	Internship	50	50	---	--	3
D	223PIT001	Research Project Phase1	100	--	0-0-17	17	11
TOTAL			190	110		20	16

Teaching Assistance: 6 hours

*MOOC Course to be successfully completed before the commencement of fourth semester (starting from semester 1).

AUDIT COURSE

AUDIT COURSE						
SLOT	SL NO	COURSE CODE	COURSE NAME	L-T-P	HOURS	CREDIT
B	1	223AGE100	ACADEMIC WRITING	3-0-0	3	-
	2	223AGE001	ADVANCED ENGINEERING MATERIALS	3-0-0	3	-
	3	223AGE002	FORENSIC ENGINEERING	3-0-0	3	-
	4	223AGE003	DATA SCIENCE FOR ENGINEERS	3-0-0	3	-
	5	223AGE004	DESIGN THINKING	3-0-0	3	-
	6	223AGE005	FUNCTIONAL PROGRAMMING IN HASKELL	3-0-0	3	-
	7	223AGE006	FRENCH LANGUAGE (A1 LEVEL)	3-0-0	3	-
	8	223AGE007	GERMAN LANGUAGE (A1 LEVEL)	3-0-0	3	-
	9	223AGE008	JAPANESE LANGUAGE (N5 LEVEL)	3-0-0	3	-
	10	223AGE009	PRINCIPLES OF AUTOMATION	3-0-0	3	-
	11	223AGE010	REUSE AND RECYCLE TECHNOLOGY	3-0-0	3	-
	12	223AGE011	SYSTEM MODELING	3-0-0	3	-
	13	223AGE012	EXPERT SYSTEMS	3-0-0	3	-

MOOC COURSES

The MOOC course shall be considered only if it is conducted by the agencies namely AICTE/NPTEL/SWAYAM or NITTTR. The MOOC course should have a minimum duration of 8 weeks and the content of the syllabus shall be enough for at least 40 hours of teaching. The course should have a proctored/offline end semester examination. The students can do the MOOC according to their convenience, but shall complete it by third semester. The list of MOOC courses will be provided by the concerned BoS if at least 70% of the course content match with the area/stream of study. The course shall not be considered if its content has more than 50% of overlap with a core/elective course in the concerned discipline or with an open elective.

MOOC Course to be successfully completed before the commencement of fourth semester (starting from semester 1). A credit of 2 will be awarded to all students whoever successfully completes the MOOC course as per the evaluation pattern of the respective agency conducting the MOOC.

TEMPLATE FOR SYLLABUS

CODE		CATEGORY	L	T	P	CREDIT
223AGE100	ACADEMIC WRITING	AUDIT COURSE	3	0	0	NIL

Preamble: Learning academic writing sharpens minds, teaches students how to communicate, and develops their thinking capacities and ability to understand others. Writing is thinking, and every student deserves to be a strong thinker. It can also make them think more carefully about what they write. Showing work to others can help to foster a better culture of learning and sharing among students. It also gives students a sense of how they are contributing to the body of work that makes up an academic subject.

Course Outcomes: The COs shown are only indicative. For each course, there can be 4 to 6 COs.

After the completion of the course the student will be able to

CO 1	Understand the principles of scientific/ academic writing
CO 2	Analyse the technique of scientific writing from the reader's perspective
CO 3	Apply the concepts of setting expectations and laying the progression tracks
CO 4	Evaluate the merits of a title, abstract, introduction, conclusion and structuring of a research paper
CO 5	Justify the need using a project proposal or a technical report
CO 6	Prepare a review paper, an extended abstract and a project proposal

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1		3	1				
CO 2		3	1				
CO 3		3	1			2	
CO 4		3	1				
CO 5		3	2	2		2	
CO 6	1	3	3	2		2	

Assessment Pattern

Bloom's Category	End Semester Examination
Apply	40%
Analyse	30%
Evaluate	30%



Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation Pattern: 40 marks

Course based task : 15 marks

Seminar/Quiz : 15 marks

Test paper, 1 no. : 10 marks

Test paper shall include minimum 80% of the syllabus.

End Semester Examination Pattern: 60 marks

The examination will be conducted by the respective College. The examination will be for 150 minutes and will contain 7 questions, with minimum one question from each module of which student should answer any five. Each question can carry 12 marks.

Model Question paper

		SET1	Total Pages:
Reg No.:			Name:
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY			
THIRD SEMESTER M.TECH DEGREE EXAMINATION, MARCH 2024			
Course Code: 223AGE100			
Course Name: Academic Writing			
Max. Marks: 60			Duration: 2.5 Hours
Answer any five full questions, each carries 12 marks.			
1 a)	Make clear-cut distinctions between 6 factors that take their toll on readers' memory.		6
1 b)	How can you sustain the attention of the reader to ensure continuous reading?		6
2 a)	What are the different methods by which you can create expectations in the reader?		6
2 b)	Give an account of the topic and non-topic based progression schemes.		6
3 a)	Bring out the differences between an abstract and the introduction of a research paper.		8
3 b)	How are the title of the research paper and its structure related?		4
4	What are 7 principles for including visuals in your research paper. What are the recommended constituents of a conclusion segment of a research paper?		12



5	Give a detailed description of the process and contents of a project proposal for funding.	12
6 a)	What are the contexts recommended for choosing between active and passive voices in technical writing?	8
6 b)	What are the different visual forms that are relevant in a research paper and how do you choose them?	4
7	Give the design of a research paper with the purposes each part serves.	12

Syllabus and Course Plan (For 3 credit courses, the content can be for 40 hrs and for 2 credit courses, the content can be for 26 hrs. The audit course in third semester can have content for 30 hours).

Syllabus:

CODE 223AG E100	ACADEMIC WRITING	Audit
Module No.	Topics in a module	Hours
1	Fundamentals of Academic writing from a reader's perspective: acronyms, synonyms, pronouns, disconnected phrases, background ghetos, abusive detailing, cryptic captions, long sentences : all that take their toll on readers' memory.	6
2	Fluid reading & reading energy consumption: setting expectations and laying Progression tracks; Reading energy consumption	6
3	How to write the Title, abstract, introduction ; Structure the writing with headings & subheadings	6
4	Visuals: Resources, Skills, and Methods; Conclusion; References; Bibliography; Grammar in technical writing	6
5	Techniques of writing: An extended abstract, a project proposal, a research paper, a technical report.	6

Course Plan:

No	Topic	No. of Lectures
1	Fundamentals of Academic writing from a reader's perspective: acronyms, synonyms, pronouns, disconnected phrases, background ghetos, abusive detailing, cryptic captions, long sentences all take their toll on readers' memory.	
1.1	The Reading tool-kit to reduce memory required; reduce reading time	1
1.2	Acronyms, Pronouns, Synonyms; Background, broken couple, words overflow	1
1.3	Sustain attention: Keep the story moving forward; Twists, shouts, Pause to clarify, recreate suspense	2



1.4	Keep the reader motivated: Fuel and meet Expectations; Bridge knowledge gap: ground level; Title words; Just In Time to local background	2
2	Fluid reading & reading energy consumption: setting expectations and laying Progression tracks; Reading energy consumption	
2.1	Setting expectations of the reader from Grammar, from theme	1
2.2	Progression tracks for fluid reading: Topic & stress; topic and non topic based progression tracks; pause in progression	2
2.3	Detection of sentence fluidity problems: No expectations/ Betrayed expectations	2
2.4	Controlling reading energy consumption: the energy bill; Energy fuelling stations: Pause	1
3	How to write the Title, abstract, introduction ; Structure the writing with headings & subheadings	
3.1	Title: Face of the paper: Techniques, Qualities & Purpose of title; Metrics	1
3.2	Abstract: Heart of the paper: 4 parts; coherence; tense of verbs, precision; purpose & qualities of the abstract; Metrics	2
3.3	Structure: Headings & sub-headings: Skeleton of the paper: principles for a good structure; Syntactic rules; Quality & Purpose of structures; Metrics	1
3.4	Introduction: Hands of the paper: Start, finish; scope, definitions; answers key reader questions; As a personal active story; Traps, qualities; Metrics	2
4	Visuals: Resources, Skills, and Methods; Conclusion; References; Bibliography; Grammar in technical writing	
4.1	Visuals as the voice of your paper: principles; purpose & qualities of visuals; metrics	2
4.2	Conclusion: contents; purpose, quality; metrics; Abstracts Vs. Conclusion; examples, counter-examples	1
4.3	References, Bibliography: Styles, punctuation marks, quotes, citations	1
4.4	Grammar in Technical writing: Articles, Syntax, Main and subordinate clauses; Active & passive voices; some commonly made mistakes in technical writing.	2
5	Techniques of writing: An extended abstract, a project proposal, a research paper, a technical report.	
5.1	Extended abstract: abstract and keywords, introduction and objective, method, findings and argument, conclusion and suggestions and references.	1
5.2	Project Proposal:Types, executive summary, background including status, objectives, solution, milestones, deliverables, timelines, resources, budgeting, conclusion	2
5.3	Research paper: writing an overview article: provide a comprehensive foundation on a topic; explain the current state of knowledge; identify gaps in existing studies for potential future research; highlight the main methodologies and research techniques	2



5.4	Writing Technical Reports: Title page; Summary; Table of contents; Introduction; Body; Figures, tables, equations and formulae; Conclusion; Recommendations.	1
		30

Reference Books

1. SCIENTIFIC WRITING 2.0 A Reader and Writer's Guide: Jean-Luc Lebrun, World Scientific Publishing Co. Pte. Ltd., 2011
2. How to Write and Publish a Scientific Paper: Barbara Gastel and Robert A. Day, Greenwood publishers, 2016
3. Grammar, Punctuation, and Capitalisation; a handbook for technical writers and editors.
www.sti.nasa.gov/publish/sp7084.pdf www.sti.nasa.gov/sp7084/contents.html
4. Everything You Wanted to Know About Making Tables and Figures. <http://abacus.bates.edu/%7Eganderso/biology/resources/writing/HTWtableVigs.html>



223AGE001	ADVANCED ENGINEERING MATERIALS	CATEGORY	L	T	P	CREDIT
		AUDIT COURSE	3	0	0	-

Preamble: This course is designed in a way to provide a general view on typically used advanced classes of engineering materials including metals, polymers, ceramics, and composites.

Course Outcomes: After the completion of the course the student will be able to

CO 1	Analyse the requirement and find appropriate solution for use of materials.
CO 2	Differentiate the properties of polymers, ceramics and composite materials.
CO 3	Recognize basic concepts and properties of functional materials.
CO 4	Comprehend smart and shape memory materials for various applications.
CO 5	Appraise materials used for high temperature, energy production and storage applications.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	☐				☐	☐	
CO 2	☐				☐	☐	
CO 3	☐				☐	☐	
CO 4	☐				☐	☐	
CO 5	☐				☐	☐	

Assessment Pattern

Bloom's Category	End Semester Examination
Understand	60%
Apply	20%
Analyse	20%

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours



Continuous Internal Evaluation Pattern: 40 marks

Course based task : 15 marks

Seminar/Quiz : 15 marks

Test paper, 1 no. : 10 marks

Test paper shall include minimum 80% of the syllabus.

End Semester Examination Pattern: 60 marks

The examination will be conducted by the respective College. The examination will be for 150 minutes and will contain 7 questions, with minimum one question from each module of which student should answer any five. Each question can carry 12 marks.

Model Question paper

AUDIT COURSE

223AGE001 - ADVANCED ENGINEERING MATERIALS

(Answer any five questions. Each question carries 12 Marks)

1. a) State the relationship between material selection and processing. 5
b) Write about the criteria for selection of materials with respect to the cost and service requirements for engineering applications. 7
2. a) Differentiate thermosetting and thermoplastics with suitable examples. 5
b) Briefly discuss about the properties and applications of polymer nano composite materials. 7
3. a) Write about the potential application areas of functionally graded materials. 5
b) With a neat sketch describe any one processing technique of functionally graded materials. 7
4. a) “Smart materials are functional”? Justify the statement. 5
b) Explain the terms electrostriction and magnetostriction with its application. 7



5. a) What are the factors influencing functional life of components at elevated temperature? 5
- b) What are super alloys and what are their advantages? 7
- 6 a) What is a shape memory alloy? What metals exhibit shape memory characteristics? 4
- b) Explain about the detection capabilities and uses of pyroelectric sensors. 8
- 7 a) Differentiate between conventional batteries and fuel cells. 4
- b) Explain the construction and working of a Li-ion battery. 8

Syllabus

Module	Content	Hours	Semester Exam Marks (%)
I	Requirements / needs of advanced materials. Classification of materials, Importance of materials selection, Criteria for selection of materials; motivation for selection, cost basis and service requirements. Relationship between materials selection and processing.	5	20
II	Classification of non-metallic materials. Polymer, Ceramics: Properties, processing and applications. Nano Composites - Polymer nanocomposites (PNCs), Processing and characterisation techniques – properties and potential applications.	7	20
III	Functionally graded materials (FGMs), Potential Applications of FGMs, classification of FGMs, processing techniques. limitations of FGMs.	6	20
IV	Smart Materials: Introduction, smart material types - pyroelectric sensors, piezoelectric materials, electrostrictors and magnetostrictors, shape memory alloys – associated energy stimulus and response forms, applications.	5	20
V	High Temperature Materials: super alloys – main classes, high temperature properties of superalloys, applications. Energy Materials: materials for batteries.	7	20



Course Plan

No	Topic	No. of Lectures
1	Selection of materials for engineering applications	
1.1	Benefits of advanced materials, classification of materials, importance of materials selection	2
1.2	Selection of materials for different properties, strength, toughness, fatigue and creep	1
1.3	Selection for surface durability, corrosion and wear resistance	1
1.4	Relationship between materials selection and processing	1
2	Classification of non-metallic materials & nano composites	
2.1	Rubber: properties, processing and applications.	1
2.2	Plastics: thermosetting and thermoplastics, applications and properties.	2
2.3	Ceramics: properties and applications.	1
2.4	Introduction to nano composites, classification	1
2.5	Processing and characterisation techniques applicable to polymer nanocomposites.	2
3	Functionally graded materials	
3.1	General concept, Potential Applications of FGMs	2
3.2	Classification of FGMs	1
3.3	FGMs processing techniques: powder metallurgy route, melt-processing route	2
3.4	Limitations of FGMs	1
4	Smart materials	
4.1	Introduction to smart materials, types	1
4.2	Pyroelectric sensors-material class, stimulus, detection capabilities and uses	1
4.3	Piezoelectric materials- material class, stimulus, sensing and actuating applications	1
4.4	Electrostrictors and magnetostrictors - material class, stimulus, micro positioning capabilities and applications	1
4.5	Shape memory alloys (SMAs) - material class, stimulus, temperature sensing and high strain responses, applications.	1
5	High Temperature Materials and Energy Materials	
5.1	Characteristics of high-temperature materials, superalloys as high-temperature materials	1
	superalloys - properties and applications	2
5.2	Introduction to lithium-ion battery (LIBs), operating mechanisms and applications	2
5.3	Introduction to Zn-based battery system, types and existing challenges	2



Reference Books

1. DeGarmo et al, “Materials and Processes in Manufacturing”, 10th Edition, Wiley, 2008.
2. R.E. Smallman and A.H.W. Ngan, Physical Metallurgy and Advanced Materials, Seventh Edition, Butterworth-Heinemann, 2007
3. Vijayamohanan K. Pillai and Meera Parthasarathy, “Functional Materials: A chemist’s perspective”, Universities Press Hyderabad (2012).
4. M.V. Gandhi, B.S. Thompson: Smart Materials and Structures, Chapman & Hall, 1992.
5. G. W. Meetham and M. H. Van de Voorde, Materials for High Temperature Engineering Applications (Engineering Materials) Springer; 1 edition (May 19, 2000)
6. Inderjit Chopra, Jayant Sirohi, “Smart Structures Theory”, Cambridge University Press, 2013



223AGE003	DATA SCIENCE FOR ENGINEERS	CATEGORY	L	T	P	CREDIT
		AUDIT COURSE	3	0	0	0

Preamble: This course covers essentials of statistics and Linear Algebra and how to prepare the data before processing in real time applications. The students will be able to handle missing data and detection of any outliers available in the dataset. This course explores data science, Python libraries and it also covers the introduction to machine learning for engineers.

Course Outcomes: After the completion of the course the student will be able to

CO 1	Study Data Science Concepts and statistics
CO 2	Demonstrate Understanding of Mathematical Foundations needed for Data Science
CO 3	Understand Exploratory analysis and Data Visualization and Preprocessing on given dataset
CO 4	Implement Models such as Naive Bayes, K-Nearest Neighbors, Linear and Logistic Regression
CO 5	Build real time data science applications and test use cases

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	2		2			2	
CO 2	2		2	1		2	
CO 3	2		2	2	2	2	
CO 4	2		2	2	3	2	
CO 5	2		2	3	3	3	2

Assessment Pattern

Bloom's Category	End Semester Examination
Understand	50%
Apply	30%
Analyse	20%

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours



Continuous Internal Evaluation Pattern: 40 marks

Course based task (Project/Assignments/Simulations/Case studies): 15 marks

Seminar/Quiz : 15 marks

Test paper, 1 no. : 10 marks

Test paper shall include minimum 80% of the syllabus.

End Semester Examination Pattern:60 marks

The examination will be conducted by the respective College. The examination will be for 150 minutes and will contain 7 questions, with minimum one question from each module of which student should answer any five. Each question can carry 12 mark.

Syllabus

Module	Content	Hours	Semester Exam Marks (%)
I	Statistics for Data science Probability: Basic concepts of probability, conditional probability, total probability, independent events, Bayes' theorem, random variable, Population, Sample, Population Mean, Sample Mean, Population Distribution, Sample Distribution and sampling Distribution, Mean, Mode, Median, Range, Measure of Dispersion, Variance, Standard Deviation, Gaussian/Normal Distribution, covariance, correlation.	6	20
II	Linear Algebra Vectors and their properties, Sum and difference of Vectors, distance between Vectors, Matrices, Inverse of Matrix, Determinant of Matrix, Trace of a Matrix, Dot Product, Eigen Values, Eigen Vectors, Single Value Decomposition	6	20
III	Hypothesis Testing Understanding Hypothesis Testing, Null and Alternate Hypothesis, Non-directional Hypothesis, Directional Hypothesis Critical Value Method, P-Value Method, Types of Errors-Type1 Error, Type2 Error, Types of Hypothesis Test Z Test, Chi-Square	6	20



IV	Exploratory Data Analysis Data Collection –Public and Private Data, Data Cleaning-Fixing Rows and Columns, Missing Values, Standardizing values, invalid values, filtering data, Data-Integration,Data-Reduction,Data Transformation	6	20
V	Machine Learning and Python for Data Science Python Data structures-List, Tuple, Set, Dictionary, Pandas, Numpy, Scipy, Matplotlib, Machine Learning-Supervised Machine Learning, Unsupervised Machine Learning,Regression, Classification, Naïve-Bayes	6	20

Course Plan

No	Topic	No. of Lectures
1	Statistics for Data science	
1.1	Probability: Basic concepts of probability, conditional probability, total probability	1
1.2	independent events, Bayes' theorem, random variable, Population	1
1.3	Sample, Population Mean, Sample Mean, Population Distribution	1
1.4	Sample Distribution and sampling Distribution, Mean, Mode, Median, Range, Propositional logic and predicate logic	1
1.5	Measure of Dispersion, Variance, Standard Deviation	1
1.6	Gaussian/Normal Distribution, covariance, correlation.	1
2	Linear Algebra	
2.1	Vectors and their properties,	1
2.2	Sum and difference of Vectors, distance between Vectors	1
2.3	Matrices,Inverse of Matrix,	2
2.4	Determinant of Matrix, Trace of a Matrix, Dot Product, Eigen Values, Eigen Vectors, Single Value Decomposition	2
3	Hypothesis Testing	
3.1	Understanding Hypothesis Testing, Null and Alternate Hypothesis	1
3.2	Non-directional Hypothesis, Directional Hypothesis Critical Value Method, P-Value Method,	2
3.3	Types of Errors-Type1 Error,Type2 Error,	1
3.4	Types of Hypothesis Test Z Test, Chi-Square,	2
4	Exploratory Data Analysis	
4.1	Data Collection –Public and Private Data	1
4.2	Data Cleaning-Fixing Rows and Columns	1
4.3	Missing Values	1
4.4	Standardizing values	1
4.5	Invalid values, filtering data	1
4.6	Data Integration, Data Reduction, Data Transformation	1



5	Machine Learning and Python for Data Science	
5.1	Python Data structures-List, Tuple, Set,	1
5.2	Dictionary, Pandas, Numpy, Matplotlib	2
5.3	Machine Learning-Supervised Machine Learning, Unsupervised Machine Learning	1
5.4	Regression, Classification	1
5.5	Naïve-Bayes	1

Reference Books

1. Python Data Science Handbook. Essential Tools for Working with Data, Author(s): Jake VanderPlas, Publisher: O'Reilly Media, Year: 2016
2. Practical Statistics for Data Scientists: 50 Essential Concepts, Author(s): Peter Bruce, Andrew Bruce, Publisher: O'Reilly Media, Year: 2017
3. Practical Linear Algebra for Data Science, by Mike X Cohen, Released September 2022, Publisher(s): O'Reilly Media, Inc.
4. Data Science from Scratch 'by Joel Grus, Released, April 2015, Publisher(s): O'Reilly Media, Inc.
5. Hands-On Exploratory Data Analysis with Python, by Suresh Kumar Mukhiya, Usman Ahmed, Released March 2020, Publisher(s): Packt Publishing



Reg
No.: _

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
THIRD SEMESTER M.TECH DEGREE EXAMINATION, MARCH 2024

Course Code: 223AGE003

Course Name: DATA SCIENCE FOR ENGINEERS

Max. Marks: 60

Duration: 2.5 Hours

Answer any five full questions, each carries 12 marks.

1. a) It is observed that 50% of mails are spam. There is software that filters spam mail before reaching the inbox. Its accuracy for detecting a spam mail is 99% and chances of tagging a non-spam mail as spam mail is 5%. If a certain mail is tagged as spam, find the probability that it is not a spam mail. 5
 b) Depict the relevance of measures of central tendency in data wrangling with a suitable example. 7
2. a) Calculate the inverse of the Matrix 4

2	4	-6
7	3	5
1	-2	4

 b) Find all Eigenvalues and Corresponding Eigenvectors for the matrix if 8

2	-3	0
2	-5	0
0	0	3
3. a) A statistician wants to test the hypothesis $H_0: \mu = 120$ using the alternative hypothesis $H_a: \mu > 120$ and assuming that $\alpha = 0.05$. For that, he took the sample values as $n=40$, $\sigma = 32.17$ and $\bar{x} = 105.37$. Determine the conclusion for this hypothesis? 5
 b) Hypothesis testing is an integral part of statistical inference, list out the various types of hypothesis testing and also mention their significances in data science. 7
4. a) Brief in detail directional and non-directional hypothesis 6
 b) Differentiate null and alternate hypothesis and also elaborate on type 1 and type 2 errors 6
5. a) Explain the concepts of Tuple, List and Directory in python with example 6
 b) Elucidate reinforcement learning and application in real world. 6



6.
 - a) What is Feature Engineering , demonstrate with an example 6
 - b) Describe in detail different steps involved in data preprocessing. 6
7.
 - a) Illustrate supervised learning model with linear regression model 5
 - b) Predict the probability for the given feature vector if an accident will happen or not? 7

Weather condition: rain, Road condition: good, Traffic condition: normal, Engine problem: no, the task is to predict using Naïve Bayes classification.

SNo.	Weather condition	Road condition	Traffic condition	Engine problem	Accident
1	Rain	bad	high	no	yes
2	snow	average	normal	yes	yes
3	clear	bad	light	no	no
4	clear	good	light	yes	yes
5	snow	good	normal	no	no
6	rain	average	light	no	no
7	rain	good	normal	no	no
8	snow	bad	high	no	yes
9	clear	good	high	yes	no
10	clear	bad	high	yes	yes



223AGE004	DESIGN THINKING	CATEGORY	L	T	P	CREDIT
		AUDIT COURSE	3	0	0	-

Preamble:

This course offers an introductory exploration of fundamental engineering concepts and techniques, the design process, analytical thinking and creativity, as well as the fundamentals and development of engineering drawings, along with their application in engineering problems.

Course Outcomes:

After the completion of the course the student will be able to

CO 1	Identify and frame design challenges effectively.
CO 2	Generate creative ideas through brainstorming and ideation
CO 3	Iterate on designs based on user insights
CO 4	Apply Design Thinking to real-world problems and projects.

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1				2		2	2
CO 2	2		2	2			2
CO 3		2		2		2	2
CO 4	2		2	3	2		2

Assessment Pattern

Bloom's Category	End Semester Examination
Apply	40
Analyse	30
Evaluate	30
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation Pattern:

AUDIT COURSES



Continuous Internal Evaluation Pattern: 40 marks

Course based task : 15 marks

Seminar/Quiz : 15 marks

Test paper, 1 no. : 10 marks

Test paper shall include minimum 80% of the syllabus.

End Semester Examination Pattern: 60 marks

The examination will be conducted by the respective College. The examination will be for 150 minutes and will contain 7 questions, with minimum one question from each module of which student should answer any five. Each question can carry 12 marks.

Model Question paper

		SET1		Total Pages:	
Reg No.: _____				Name: _____	
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY THIRD SEMESTER M.TECH DEGREE EXAMINATION, MARCH 2024					
Course Code: 223AGE004					
Course Name: DESIGN THINKING					
Max. Marks: 60				Duration: 2.5 Hours	
<i>Answer any five full questions, each carries 12 marks.</i>					
1 a)	How can a multidisciplinary team collaborate effectively to implement design principles?				7
1 b)	What are the key differences between human-centred design and other design methodologies?				5
2 a)	How do you measure the success of a design project in terms of user satisfaction and impact?				7
2 b)	How does the iterative nature of the design process contribute to better outcomes				5



3 a)	What are the fundamental principles of effective brainstorming, and how do they differ from traditional problem-solving approaches?	7
3 b)	What are some key principles of ergonomic design, and how do they contribute to the usability and comfort of products?	5
4 a)	Enumerate some examples of successful and unsuccessful market testing scenarios, and what lessons can be learned from these experiences to improve future product or service launches?	7
4b)	What is the primary purpose of creating prototypes in the design and development process?	5
5	What strategies and methodologies can designers use to embrace agility and respond quickly to changing user needs and market dynamics?	12
6	Illustrate any four examples of successful bio-mimicry applications in various industries.	12
7	What ethical considerations should designers keep in mind when designing for diverse user groups?	12



Syllabus:

Module 1

Design process: Traditional design, Design Thinking Approach, Introduction to Design Thinking, History and evolution of Design Thinking, Role of design thinking in the human-centred design process. Design space, Design Thinking in a Team Environment, Team formation.

Module 2

Design Thinking Stages: Empathize, Define, Ideate, Prototype and Test. The importance of empathy, Building a user-centred mindset. Problem statement formulation, User needs and pain points, establishing target specifications, Setting the final specifications.

Module 3

Generating Ideas, Brainstorming techniques, Application of Aesthetics and Ergonomics in Design. Bio-mimicry, Conceptualization, Visual thinking, Drawing/Sketching, Presenting ideas.

Module 4

Use of prototyping, Types of prototypes, Rapid prototyping techniques, User testing and feedback collection, Iterative prototyping, testing to gauge risk and market interest

Module 5

Entrepreneurship/business ideas, Patents and Intellectual Property, Agility in design, Ethical considerations in design. Overcoming common implementation challenges

Corse Plan Syllabus and Corse Plan (For 3credit courses, the content can be for 40 hrs. and for 2credit courses, the content can be for 26 hrs. The audit course in third semester can have content for 30hours).

No	Topic	No. of lectures
1	Design process:	
1.1	Design process: Traditional design, Design Thinking Approach, Introduction to Design Thinking, History and evolution of Design Thinking.	3
1.2	Role of design thinking in the human-centred design process. Design space,	2
1.3	Design Thinking in a Team Environment, Team formation.	2



2	Design Thinking Stages:	
2.1	Design Thinking Stages: Empathize, Define, Ideate, Prototype and Test.	2
2.2	The importance of empathy, Building a user-centred mindset.	2
2.3	Problem statement formulation, User needs and pain points, establishing target specifications, Setting the final specifications.	3
3	Ideation	
3.1	Generating Ideas, Brainstorming techniques.	2
3.2	Application of Aesthetics and Ergonomics in Design. Bio-mimicry.	3
3.3	Conceptualization, Visual thinking, Drawing/Sketching, Presenting ideas.	2
4	Prototyping and testing	
4.1	Use of prototyping, Types of prototypes, Rapid prototyping techniques.	3
4.2	User testing and feedback collection, Iterative prototyping, testing to gauge risk and market interest	2
5	IPR in design	
5.1	Entrepreneurship/business ideas, Patents and Intellectual Property.	2
5.2	Agility in design, Ethical considerations in design. Overcoming common implementation challenges	2

Reference Books

1. Christoph Meinel, Larry Leifer and Hasso Plattner- "Design Thinking: Understand – Improve – Apply", Springer Berlin, Heidelberg, 2011.
2. Thomas Lockwood and Edgar Papke – "Design Thinking: Integrating Innovation, Customer Experience, and Brand Value", Allworth Press, 2009.
3. Pavan Soni – "Design Your Thinking", Penguin Random House India Private Limited, 2020.
4. Andrew Pressman- "Design Thinking : A Guide to Creative Problem Solving for Everyone", Taylor & Francis, 2018.
5. N Siva Prasad, "Design Thinking Techniques an Approaches" Ane Books Pvt. Ltd., 2023



SYLLABUS

CODE	COURSE NAME	CATEGORY	L	T	P	CREDIT
223AGE005	FUNCTIONAL PROGRAMMING IN HASKELL	AUDIT COURSE	3	0	0	-

Preamble: This course introduces a functional programming approach in problem solving. Salient features of functional programming like recursion, pattern matching, higher order functions etc. and the implementation in Haskell are discussed.

Course Outcomes:

After the completion of the course the student will be able to

CO 1	Understand the functional programming paradigm which is based on the mathematics of lambda calculus.
CO 2	Develop Haskell programs using functions, guards and recursive functions
CO 3	Apply the concept of tuples, lists and strings in Haskell programming
CO 4	Apply the concept of algebraic data types, abstract data types, modules, recursive data types and user defined data types in Haskell programming
CO 5	Develop Haskell programs with files for reading input and storing output

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1					3		
CO 2	2			2	3		
CO 3	2			2	3		
CO 4	2			2	3		
CO 5	2			2	3		

Assessment Pattern

Bloom's Category	End Semester Examination
Apply	40%
Analyse	40%
Evaluate	20%
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours



Continuous Internal Evaluation: 40 marks

Course based task : 15 marks

Seminar/Quiz : 15 marks

Test paper, 1 no. : 10 marks

Test paper shall include minimum 80% of the syllabus.

End Semester Examination: 60 marks

The examination will be conducted by the respective College. The examination will be for 150 minutes and will contain 7 questions, with minimum one question from each module of which student should answer any five. Each question can carry 12 marks.

Model Question paper

		Total Pages:	
Reg No.: _____		Name: _____	
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY THIRD SEMESTER M.TECH DEGREE EXAMINATION, DECEMBER 2023			
Course Code: 223AGE005			
Course Name: Functional Programming in Haskell			
Max. Marks: 60		Duration: 2.5 Hours	
Answer any five full questions, each carries 12 marks.			
1 a.	Explain the basic differences between imperative style programming and functional style programming.	3	
1 b.	Analyse each of the following lambda expressions to clarify its structure. If the expression is a function, identify the bound variable and the body expression, and then analyse the body expression. If the expression is an application, identify the function and argument expressions, and then analyse the function and argument expressions: i) $\lambda a.(a \lambda b.(b a))$ ii) $\lambda x.\lambda y.\lambda z.((z x) (z y))$ iii) $(\lambda f.\lambda g.(\lambda h.(g h) f) \lambda p.\lambda q.p)$	9	
2 a.	Design a recursive function to find 2^n where n is a natural number.	4	



2 b.	Explain various forms of function definitions in Haskell with the help of examples.	8
3 a.	Explain any three list operations along with function definitions and examples.	6
3 b.	Write a program to duplicate only even numbers among the elements of a list using a Haskell function by (i) Recursion (ii) List Comprehension and explain. Example : $\lambda > \text{dupli } [1, 2, 3]$ ANS: [2,2]	6
4	Write Recursive definitions along with an explanation for the below arithmetic operations. Illustrate the recursive flow with the help of a diagram. i. add x y ii. mult x y iii. div x y	12
5	Write the Haskell code to split a list into two lists such that the elements with odd index are in one list while the elements with even index are in the other list.	12
6 a	Give the type definition of a binary tree along with explanation of two functions on binary trees.	6
6 b	Define a queue data type in Haskell along with any two operations on it with examples.	6
7 a.	Explain the basic steps of reading from files and writing to files in Haskell.	4
7 b.	Write a Haskell program to read from the file "input.txt", display the contents on the screen and write the contents to another file "output.txt".	8

Syllabus and Corse Plan (For 3 credit courses, the content can be for 40 hrs and for 2 credit courses, the content can be for 26 hrs. The audit course in third semester can have content for 30 hours).

Module 1 (5 Hrs)

Introduction to Functional Programming: Programming language paradigms, imperative style programming, comparison of programming paradigms.

Functional programming, Functions - Mathematical concepts and terminology, Lambda calculus, Function definitions, programs as functions, Functional programming Languages. Haskell basics, GHCi interpreter.



Module 2 (6 Hrs)

Programming in Haskell: Expressions and evaluation, Lazy evaluation, let expressions, scopes.

Basic data types in Haskell, operators, infix operators, associativity and precedence, Arithmetic functions.

types, definitions, currying and uncurrying, type abstraction.

Function definitions, pattern matching, guards, anonymous functions, higher order functions.

Recursion, Programming exercises.

Module 3 (7 Hrs)

Data types: tuples and lists: Tuples , Lists: building lists, decomposing lists, functions on lists, built-in functions on lists, primitive and general recursion over lists, infinite lists.

Strings: functions on strings.

Polymorphism and overloading, conditional polymorphism

Module 4 (6 Hrs)

Type classes, Algebraic data types, Modules, Recursive data types.

User defined data types, Records, Stacks, Queues, Binary trees, Constructors, Destructors.

Module 5 (6 Hrs)

Functor, Applicative functor, Monad

Programming with actions: Functions vs actions, Basics of input / output, the do notation, interacting with the command line and lazy I/O, File I/O.

No	Topic	No. of Lectures
1	Introduction to Functional Programming	
1.1	Programming language paradigms, imperative style programming, comparison of programming paradigms	1
1.2	Functional programming, Functions - Mathematical concepts and terminology	1
1.3	Lambda calculus	1
1.4	Function definitions, programs as functions, Functional programming Languages	1
1.5	Haskell basics, GHCi interpreter	1
2	Haskell basics	
2.1	Expressions and evaluation, Lazy evaluation	1
2.2	let expressions, scopes, Basic data types in Haskell	1
2.3	operators, infix operators, associativity and precedence, Arithmetic	1



	functions	
2.4	types, definitions, currying and uncurrying, type abstraction.	1
2.5	Function definitions, pattern matching, Guards	1
2.6	anonymous functions, higher order functions, Recursion	1
3	Data types: tuples and lists	
3.1	Tuples , Lists: building lists, decomposing lists	1
3.2	functions on lists, built-in functions on lists	1
3.3	primitive and general recursion over lists	1
3.4	infinite lists	1
3.5	Strings: functions on strings	1
3.6	Polymorphism and overloading	1
3.7	conditional polymorphism	1
4	User defined data types	
4.1	Type classes, Algebraic data types, Modules	1
4.2	Recursive data types	1
4.3	User defined data types, Records	1
4.4	Stacks, Queues	1
4.5	Binary trees	1
4.6	Constructors, Destructors	1
5	Programming with actions	
5.1	Functor, Applicative functor,	1
5.2	Monad	1
5.3	Programming with actions: Functions vs actions, Basics of input / output, the do notation	1
5.4	interacting with the command line and lazy I/O	1
5.5	File I/O	2

Reference Books

[1] Richard Bird, "Introduction to functional programming using Haskell", second edition, Prentice hall series in computer science

[2] Bryan O'Sullivan, Don Stewart, and John Goerzen, "Real World Haskell"



- [3] Richard Bird, “Thinking Functionally with Haskell”, Cambridge University Press, 2014
- [4] Simon Thompson, “Haskell: The Craft of Functional Programming”, Addison-Wesley, 3rd Edition, 2011
- [5] H. Conrad Cunningham, “Notes on Functional Programming with Haskell”, 2014
- [6] Graham Hutton, “Programming in Haskell”, Cambridge University Press, 2nd Edition, 2016
- [7] Alejandro Serrano Mena, “Practical Haskell: A Real-World Guide to Functional Programming”, 3rd Edition, Apress, 2022
- [8] Miran Lipovaca, “Learn You a Haskell for Great Good!: A Beginner's Guide”, No Starch Press, 2011



223AGE010	REUSE AND RECYCLE TECHNOLOGY	CATEGORY	L	T	P	CREDIT
		AUDIT COURSE	3	0	0	-

Preamble: "Reuse and Recycle Technology" typically focuses on sustainable practices and technologies aimed at reducing waste, conserving resources, and promoting environmental responsibility.

Course Outcomes: After the completion of the course the student will be able to

CO 1	Explain the principles and technologies behind waste reduction, resource conservation, and sustainable practices
CO 2	Describe and Analyze waste generation and management.
CO 3	Apply the knowledge of various reuse strategies and their application in different industries and Analyze various recycling technologies
CO 4	Appraise the methods of E-waste management and Eco friendly packaging
CO 5	Comprehend Environmental Regulations and Policies, Understand the importance of environmental regulations and policies in addressing environmental challenges

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6
CO 1			3			
CO 2				3		
CO 3				3		
CO 4					3	
CO 5			3			

Assessment Pattern

Bloom's Category	End Semester Examination
Understand	60%
Apply	20%
Analyse	20%

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours



Continuous Internal Evaluation Pattern: 40 marks

Course based task : 15 marks

Seminar/Quiz : 15 marks

Test paper, 1 no. : 10 marks

Test paper shall include minimum 80% of the syllabus.

End Semester Examination Pattern: 60 marks

The examination will be conducted by the respective College. The examination will be for 150 minutes and will contain 7 questions, with minimum one question from each module of which student should answer any five. Each question can carry 12 marks.

Model Question paper**AUDIT COURSE****223AGE010 - REUSE AND RECYCLE TECHNOLOGY***Answer any five full questions, each carries 12 marks.*

1.	(a) What are the 3 pillars of sustainability?	5
	(b) What is sustainable waste management? What makes sustainable waste management so important?	7
2.	(a) How do the three categories of municipal solid waste differ?	5
	(b) Discuss the municipal waste collection and management?	7
3.	(a) Explain the major differences between Reuse and Recycle?	5
	(b) Give an overview of recycling technologies used for any two materials. Discuss the Process involved.	7
4.	(a) What are the common source of E-waste	5
	(b) What are the challenges and opportunities in E-waste management	7
5.	(a) What is the case law for waste recycling in India	5
	(b) Discuss sustainable packaging and its environmental impacts	7
6.	Explain the various environmental regulations in India for addressing Environmental challenges	12
7.	a) Give examples of water reuse technologies in circular economy	5
	b) How can we reduce e-waste with sustainable solutions	7



Syllabus

Module	Content	Hours	Semester Exam Marks (%)
I	Introduction to Sustainability , Understanding sustainability and its importance, The three pillars of sustainability: Environmental, Social, and Economic. Biodiversity conservation, Climate change and mitigation Sustainable resource management.	6	20
II	Waste Management , Definition and classification of waste, Waste Generation and Composition, Waste Collection and Transportation, Waste Segregation and Sorting. Waste Disposal Methods Historical perspectives on waste management, The three Rs: Reduce, Reuse, and Recycle.	6	20
III	Recycling and Reuse : Importance of reuse, Application of reuse in various industries, Challenges and opportunities in reuse, Overview of recycling technologies, Circular economy, Sorting and processing of recyclable materials, Advanced recycling methods. Emerging technologies in recycling.	6	20
IV	E-waste Recycling , Challenges and environmental impact of electronic waste, E-waste recycling methods and regulations, Sustainable electronics design, Sustainable Packaging , Packaging materials and their environmental impact, Eco-friendly packaging alternatives, Packaging design for sustainability	6	20
V	Environmental Regulations and Policies , Understand the importance of environmental regulations and policies in addressing environmental challenges, National and international waste and recycling regulations, Compliance and enforcement, Industry standards and certifications	6	20

Course Plan



No	Topic	No. of Lectures
1	Introduction to Sustainability (6)	
1.1	Understanding sustainability and its importance	1
1.2	The three pillars of sustainability: Environmental, Social, and Economic.	3
1.3	Biodiversity conservation, Climate change and mitigation	1
1.4	Sustainable resource management	1
2	Waste Management (6)	
2.1	Definition and classification of waste	1
2.2	Waste Generation and Composition	1
2.3	Waste Collection and Transportation.	1
2.4	Waste Segregation and Sorting.	1
2.5	Waste Disposal Methods	1
2.6	Historical perspectives on waste management, The three Rs: Reduce, Reuse, and Recycle.	1
3	Recycling and Reuse (6)	
3.1	Importance of reuse, Examples of reuse in various industries.	1
3.2	Challenges and opportunities in reuse	1
3.3	Overview of recycling technologies, Sorting and processing of recyclable materials	2
3.4	Advanced recycling methods	1
3.5	Emerging technologies in recycling.	1
4	E-waste Recycling (6)	
4.1	Challenges and environmental impact of electronic waste	1
4.2	E-waste recycling methods and regulations	1
4.3	Sustainable electronics design	1
4.4	Packaging materials and their environmental impact	1
4.5	Eco-friendly packaging alternatives	1
4.6	Packaging design for sustainability	1
5	Environmental Regulations and Policies (6)	
5.1	Importance of environmental regulations and policies in addressing environmental challenges	2
5.2	National and international waste and recycling regulations	2
5.3	Industry standards and certifications, Compliance and enforcement	2



Reference Books

1. Sustainable Engineering: Concepts, Design and Case Studies, David T. Allen, Pearson Publication.
2. A Comprehensive Book on Solid Waste Management with Application, Dr. H.S. Bhatia , Misha Books, 2019
3. "Cradle to Cradle: Remaking the Way We Make Things" by William McDonough and Michael Braungart.
4. "Recycling of Plastic Materials" edited by Vijay Kumar Thakur
5. E-waste: Implications, Regulations and Management in India and Current Global Best Practices, Rakesh Johri, TERI
6. "Sustainable Packaging”, Subramanian Senthilkannan Muthu , Springer Nature.
7. Indian Environmental Law: Key Concepts and Principles " Orient Black swan Private Limited, New Delhi.



223AGE012	EXPERT SYSTEMS	CATEGORY	L	T	P	CREDIT
		AUDIT COURSE	3	0	0	-

Preamble: The course aims to provide an understanding of the basic concepts of Artificial Intelligence (AI) and Expert Systems. The course also covers the knowledge representation in expert systems, classes of expert systems, applications of expert systems.

Course Outcomes: After the completion of the course the student will be able to:

CO 1	Explain the concepts of Artificial Intelligence and different ways of knowledge representations.
CO 2	Explain the components of expert systems, development stages of expert systems and tools available for expert system design.
CO 3	Apply the concept of knowledge representation in expert systems
CO 4	Differentiate the classes of expert systems and examine properties of existing systems

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO7
CO 1	1		2	1	2	2	
CO 2	1		1	3	2	2	
CO 3	1		1	2	2	2	
CO 4	2		2	2	3	2	

Assessment Pattern

Bloom's Category	End Semester Examination
Understand	60%
Apply	20%
Analyse	20%

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation Pattern: 40 marks

Course based task (Project/Assignments/Simulations/Case studies): 15 marks

Seminar/Quiz : 15 marks

Test paper, 1 no. : 10 marks

Test paper shall include minimum 80% of the syllabus.



End Semester Examination Pattern:60 marks

The examination will be conducted by the respective College. The examination will be for 150 minutes and will contain 7 questions, with minimum one question from each module of which student should answer any five. Each question can carry 12 mark.

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY THIRD SEMESTER M.TECH DEGREE EXAMINATION, MARCH 2024		
Course Code: 223AGE012		
Course Name: EXPERT SYSTEMS		
Max. Marks: 60		Duration: 2.5 Hours
Answer any five full questions, each carries 12 marks.		
1.	a) What are the types of AI? Explain with examples .	6
	b) What do you mean by knowledge in AI and explain the different ways of knowledge representation used in AI?	6
2.	a) Write note on semantic network.	6
	b) What are Predicates? Explain its syntax and semantics.	6
3.	a) Write notes on different tools available for expert system design.	6
	b). What are the different stages in the development of an expert system?	6
4.	a) Illustrate Conceptual Dependencies with an example.	6
	b) Illustrate with an example the Structured Knowledge representation of an Expert System.	6
5.	a) What do you mean by Frame based Expert System? Explain	6
	b) Explain the architecture of MYCIN	6
6.	a) Explain Fuzzy based expert systems	6
	b) Explain the neural network based expert systems	6
7.	a) Explain any two applications of expert systems?	6
	b) What are the limitations of expert system ? Explain	6



Syllabus

Module	Content	Hours	Semester Exam Marks (%)
I	Overview of Artificial Intelligence (AI): Definition & Importance of AI. Knowledge general concepts: Definition and Importance of knowledge, Knowledge-Based Systems, Knowledge organization, Knowledge Manipulation and acquisition. Knowledge Representation: Introduction, Syntax and Semantics- Propositional logic and predicate logic.	6	20
II	Basic concepts of expert systems-Introduction to expert systems, Components of expert systems. Features of Expert System, Stages in the development of expert system, Types of tools available for expert system design	6	20
III	Knowledge representation in expert systems: Structured Knowledge representation: Graphs, Frames and related structures, Associative networks, Conceptual dependencies, Examples of structured knowledge representation.	6	20
IV	Classes of expert systems: Rule-based expert systems, Example- MYCIN, Frame-based expert system, terminologies, IF-THEN structure. Fuzzy and Neural network based expert systems(basic concepts)	7	20
V	Currents trends in expert systems, Advantages and limitations of expert systems, Applications of expert systems.	5	20



Course Plan

No	Topics	No. of Lectures
1	Overview of Artificial Intelligence& Knowledge general concepts	
1.1	Definition & Importance of AI	1
1.2	Definition and Importance of Knowledge,	1
1.3	Knowledge-Based Systems, Knowledge Organization	1
1.4	Knowledge Manipulation and acquisition	1
1.5	Knowledge Representation: Introduction, Syntax and Semantics	1
1.6	Propositional logic and predicate logic	1
2	Basic concepts of expert systems	
2.1	Introduction to Expert System, Components of expert systems	2
2.2	Features of Expert System, Stages in the development of expert system	2
2.3	Types of tools available for expert system design	2
3	Knowledge representation in expert systems	
3.1	Structured Knowledge representation	1
3.2	Graphs, Frames and Related Structures	2
3.3	Associative Networks, Conceptual Dependencies	2
3.4	Examples of structured knowledge representation	1
4	Classes of expert systems	
4.1	A rule-based expert system -Introduction	1
4.2	MYCIN	1
4.3	IF-THEN structure	1
4.4	Frame-based expert system	2
4.5	Fuzzy based expert systems	1
4.6	Neural network based expert systems	1
5	Currents trends and applications of expert systems	
5.1	Currents trends of expert systems	2
5.2	Advantages and limitations of expert systems	1
5.3	Applications of expert systems	2

Reference Books

1. E. Rich & K. Knight - Artificial Intelligence, 2/e, TMH, New Delhi, 2005.
2. P.H. Winston - Artificial Intelligence, 3/e, Pearson Edition, New Delhi, 2006.
3. D.W. Rolston - Principles of AI & Expert System Development, TMH, New Delhi
4. Kevin Night and Elaine Rich, Nair B., "Artificial Intelligence (SIE) ", McGraw Hill – 2010
5. Dan W Patterson, 'Introduction to Artificial intelligence and Expert systems', Prentice Hall of India Pvt. Ltd, 2007
6. Russel (Stuart), 'Artificial Intelligence- Modern approach, Pearson Education series in AI', 3rd Edition, 2009.
7. I. Gupta, G. Nagpal · Artificial Intelligence and Expert Systems, Mercury Learning and Information -2020



223AGE011	SYSTEM MODELLING	CATEGORY	L	T	P	CREDIT
		AUDIT COURSE	3	0	0	-

Preamble: Study of this course provides the learners a clear understanding of fundamental concepts in simulation and modelling. This course covers the different statistical models, importance of data collection and various types of simulations. The course helps the learners to find varied applications in engineering, medicine and bio-technology.

Course Outcomes: After the completion of the course the student will be able to

CO 1	Analyse the requirement and find appropriate tool for simulation.
CO 2	Differentiate the different statistical models.
CO 3	Discuss the different techniques for generating random numbers.
CO 4	Analyse the different methods for selecting the different input models..
CO 5	Discuss the different measures of performance and their estimation

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6
CO 1	2		1	1	2	
CO 2	2		1	1	1	
CO 3	1					
CO 4	1		1	1		
CO 5	2		1	1	1	

Assessment Pattern

Bloom's Category	End Semester Examination
Understand	60%
Apply	20%
Analyse	20%

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation Pattern:

Course based task (Project/Assignments/Simulations/Case studies): 15 marks

Seminar/Quiz: 15 marks

Test paper, 1 no.: 10 marks

Test paper shall include minimum 80% of the syllabus.



End Semester Examination Pattern:

The examination will be conducted by the respective College. The examination will be for 150 minutes and will contain 7 questions, with minimum one question from each module of which student should answer any five. Each question can carry 12 marks.

Model Question paper

AUDIT COURSE

223AGE001 – SYSTEM MODELLING

Answer any five questions Each carries 12 marks

PART A

1. a. Discuss the advantages and disadvantages of simulation. (5marks)
b. What are the areas of applications of simulation (7 marks)
2. a. A bus arrives every 20 minutes at a specified stop beginning at 6:40 A.M. and continuing until 8:40 A.M. A certain passenger does not know the schedule, but arrives randomly (uniformly distributed) between 7:00 A.M. and 7:30 A.M. every morning. What is the probability that the passenger waits more than 5 minutes for a bus? (5 marks)
b. A production process manufactures computer chips on the average at 2% nonconforming. Every day, a random sample of size 50 is taken from the process. If the sample contains more than two nonconforming chips, the process will be stopped. Compute the probability that the process is stopped by the sampling scheme. (7 marks)
3. a. Discuss the different types of tests for random numbers. (5 marks)
b. Generate random numbers using multiplicative congruential method with $X_0 = 5$, $a = 11$, and $m = 64$. (7 marks)
4. a. What are the different methods of data collection. (4marks)
b. Records pertaining to the monthly number of job-related injuries at an underground coalmine were being studied by a federal agency. The values for the past 100 months were as follows:

Injuries per Month	Frequency of Occurrence
0	35
1	40
2	13
3	6
4	4
5	1
6	1



- (a) Apply the chi-square test to these data to test the hypothesis that the underlying distribution is Poisson. Use the level of significance $\alpha = 0.05$.
- (b) Apply the chi-square test to these data to test the hypothesis that the distribution is Poisson with mean 1.0. Again let $\alpha = 0.05$.
- c) What are the differences between parts (a) and (b), and when might each case arise? (8 marks)

5. a. What is the difference between validation and verification. (5 marks)
b. Discuss the different measures of performance and their estimation. (7 marks)
6. a. Discuss the different methods of parameter estimation. (5 marks)
b. With an example, describe the Poisson process. (7 marks)
7. a. Distinguish between discrete and continuous systems. (5 marks)
b. What are the different components of a simulation system. (7 marks)

Syllabus

Module	Content	Hours	Semester Exam Marks (%)
I	When simulation is the appropriate tool. Advantages and disadvantages of Simulation; Areas of application, Systems and system environment; Components of a system; Discrete and continuous systems, Model of a system; Types of Models, Discrete-Event System Simulation, Steps of a simulation study.	6	20
II	Review of terminology and concepts, Useful statistical models, Discrete distributions. Continuous distributions, Poisson process, Empirical distributions. (basic idea only)	6	20
III	Properties of random numbers; Generation of pseudo-random numbers, Techniques for generating random numbers, Tests for Random Numbers	6	20
IV	Data Collection; Identifying the distribution with data, Parameter estimation, Goodness of Fit Tests, Fitting a non-stationary Poisson process, Selecting input models without data, Multivariate and Time-Series input models.	6	20
V	Measures of performance and their estimation, Output analysis for terminating simulations, Output analysis for steady-state simulations, Verification, calibration and validation	6	20



Course Plan

No	Topic	No. of Lectures
1	Introduction	
1.1	When simulation is the appropriate tool	1
1.2	Advantages and disadvantages of Simulation;	1
1.3	Areas of application, Systems and system environment;	1
1.4	Components of a system; Discrete and continuous systems,	1
1.5	Model of a system; Types of Models,	1
1.6	Discrete-Event System Simulation ,Steps of a simulation study	1
2	Statistical Models in Simulation	
2.1	Review of terminology and concepts, Empirical distributions. (basic idea only)	1
2.2	Useful statistical models,	1
2.3	Discrete distributions.	1
2.4	Continuous distributions,.	1
2.5	Poisson process	1
2.6	Empirical distributions	1
3	Random Number Generation	
3.1	Properties of random numbers;	1
3.2	Generation of pseudo-random numbers,	
3.3	Techniques for generating random numbers	1
3.4	Techniques for generating random numbers(cont)	1
3.5	Tests for Random Numbers	1
3.6	Tests for Random Numbers(cont)	1
4	Input Modelling	
4.1	Data Collection;	1
4.2	Identifying the distribution with data.	1
4.3	Parameter estimation, Goodness of Fit Tests	1
4.4	Fitting a non-stationary Poisson process	1
4.5	Selecting input models without data,	1
4.6	Multivariate and Time-Series input models	1
5	Measures of Performance and their Estimation	
5.1	Measures of performance and their estimation	1
5.2	Measures of performance and their estimation(cont)	1
5.3	Output analysis for terminating simulations	1
5.4	Output analysis for steady-state simulations	1
5.5	Verification, calibration and validation	1
5.6	Verification, calibration and validation(cont)	1



Textbooks:

1. Jerry Banks, John S. Carson II, Barry L. Nelson, David M. Nicol: Discrete-Event System Simulation, 5th Edition, Pearson Education, 2010.

Reference Books:

1. Lawrence M. Leemis, Stephen K. Park: Discrete – Event Simulation: A First Course, Pearson Education, 2006.

2. Averill M. Law: Simulation Modeling and Analysis, 4 th Edition, Tata McGraw-Hill, 2007

3. System Modelling and Response by Ernest O. Doebelin

4. Averill M Law, “Simulation Modeling and Analysis”,McGraw-Hill Inc,2007 Geoffrey Gorden, “System Simulation”,Prentice Hall of India,1992.



223AGE009	Principles of Automation	CATEGORY	L	T	P	CREDIT
		AUDIT COURSE	3	0	0	0

Preamble:

This course deals in detail with the various aspects of automation such as sensors, actuators, controllers, mechanical and electrical elements and their integration for automating new and existing manufacturing and process industries and applications. This course will be beneficial to students in designing automation schemes for industries and to design automated systems

Course Outcomes: After the completion of the course the student will be able to

CO 1	Explain the fundamentals of sensor systems and to choose a suitable sensor system for the given application based on the evaluation of the constraints.
CO 2	Explain the fundamentals of signal conditions and to design a suitable signal conditioning scheme for given application.
CO 3	Describe the characteristics of various actuator systems and to decide the right type of actuator for the given application.
CO 4	Describe the importance of an industrial robot and fundamentals of numerical control in automation.
CO 5	Explain the fundamentals of controllers used in industrial automation and to construct simple automation schemes by ladder logic programs.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	2		2	2	2		
CO 2	2		2	2	2		
CO 3	2		2	2	2		
CO 4	2		2	2	2		
CO 5	2		2				

Assessment Pattern

Bloom's Category	End Semester Examination
Understand	70 %
Apply	30 %

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours



Continuous Internal Evaluation Pattern: 40 marks

Course based task (Project/Assignments/Simulations/Case studies): 15 marks

Seminar/Quiz: 15 marks

Test paper, 1 no.: 10 marks

Test paper shall include minimum 80% of the syllabus.

End Semester Examination Pattern: 60 marks

The examination will be conducted by the respective College. The examination will be for 150 minutes and will contain 7 questions, with minimum one question from each module of which student should answer any five. Each question can carry 12 marks.

Model Question Paper
223AGE009 Principles of Automation

Time 2.5 Hrs

Marks 60

Answer any five questions Each carries 12 marks

1. (a) Differentiate the static and dynamic characteristics of a temperature sensor and explain how it affects the selection of a suitable temperature sensor. (6 marks)
(b) Explain the working of a strain-gauge. (6 marks)
2. (a) Explain why anti-aliasing filters are used in analog to digital converters. (3 marks)
(b) Design a first order low pass filter with a cutoff frequency of 2 kHz. (9 marks)
3. (a) What are the factors to consider while deciding choosing between hydraulic, pneumatic or electrical actuation systems for an automation scheme? (4 marks)
(b) Explain the working of a three-way pressure reducing valve. (4 marks)
(c) Explain the working of solenoids. In what applications would you use a Solenoid valve. (4 marks)
4. (a) Explain the principle of the Touch sensor and also mention how they are used in robots. (5 marks)
(b) Explain the basic terminologies in robotic system and also explain the components of robotic system. (7 marks)
5. (a) With neat schematic explain the architecture of the PLC. (6 marks)
(b) Explain the use of an up-down counter in PLC with a suitable example. (6 marks)
6. (a) Write short note on SCADA. What is difference PLC and SCADA? (3 marks)
(b) Construct a ladder logic for controlling a process tank as per the logic given below;
i. The tank should be filled by a valve V1 when low level float switch L1 is ON and an external input S1 is received.



- ii. V1 should be closed when the liquid level reaches a high-level float switch L2.
 - iii. An agitator motor should be turned on after a delay of 5sec after L2 is triggered.
 - iv. After agitating for 30mins, contents of the tank should be emptied by opening another valve V2.
 - v. The temperature should be maintained at 70°C using a thermostat T1 and Heater H
(9 marks)
7. (a) Explain the levels of Automation. (6 marks)
- (b) Explain the working of Flow sensor (6 marks)

Syllabus and Course Plan

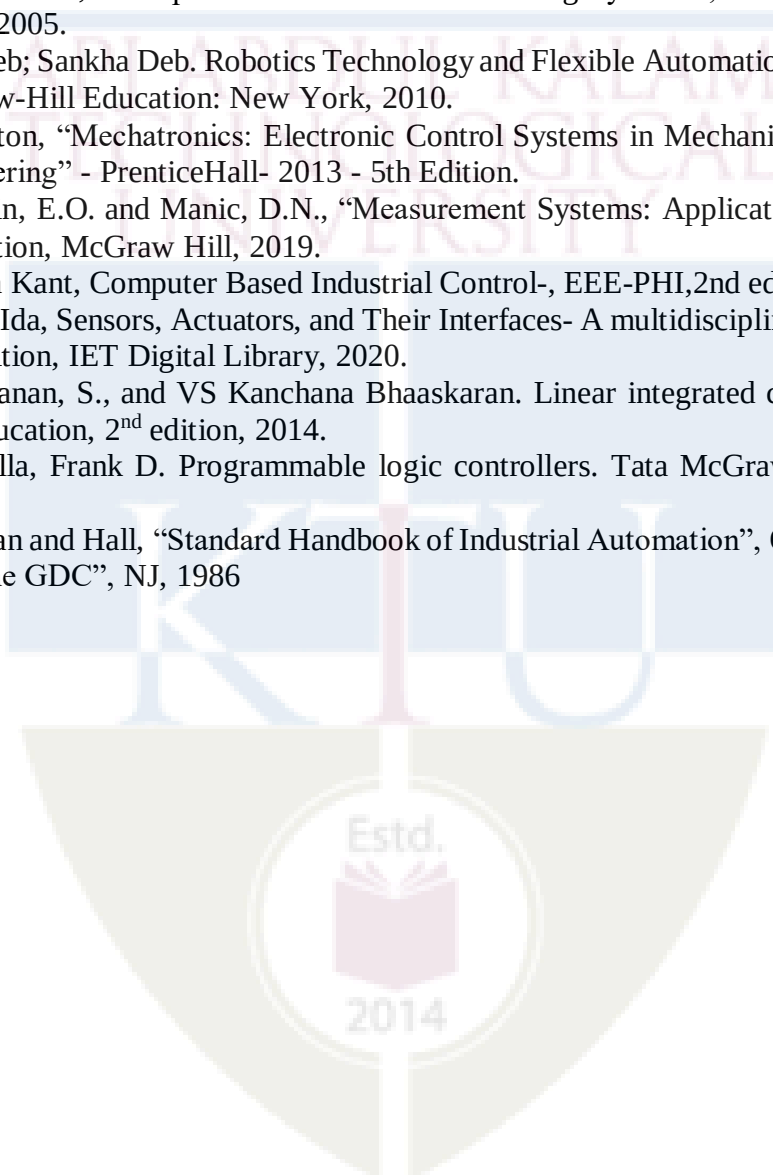
No	Topics	No. of Lectures
1	Introduction to Industrial Automation	
1.1	Basic Elements of an Automated System, Levels of Automation	2
1.2	Hardware components for Automation: Sensors, classification, Static and dynamic behaviour of sensors.	2
1.3	Basic working principle of different sensors: Proximity sensors, Temperature sensors, flow sensors, Pressure sensors, Force sensors. Position sensors	4
2	Signal conditioning	
2.1	Need for signal conditioning, Types of signal conditioning.	2
2.2	Signal conditioning using operational amplifier-Amplifier (Inverting and Non-inverting) and Filter circuits (Basic concepts). Design of first order low pass filter.	2
2.3	Signal conditioning for data acquisition systems, anti-aliasing filters, Analog-Digital Conversions, Analog-to-Digital Converters (ADC)- Steps in analog-to-digital conversion, Successive Approximation Method, Digital-to-Analog Converters (DAC)- Steps in digital to analog conversion, Zero-order and first order data hold circuits	4
3	Actuators	
3.1	Types of actuators- mechanical, electrical, pneumatic and hydraulic actuators. (Basic working principle)	2
3.2	Mechanical systems for motion conversion, transmission systems	3
3.3	Solenoids, Electric and stepper motors control.	3
4	Robotics and Automated Manufacturing Systems	
4.1	Robot Anatomy and Related Attributes: Joints and Links, Common Robot Configurations, Joint Drive Systems, Sensors in Robotics (Basic concepts)	3
4.2	Robot Control Systems, Applications of Industrial Robots- Material handling	4
4.3	Fundamentals of Numerical control (NC) Technology	1
5	Discrete Control and Programmable Logic Controllers	



5.1	Discrete Process Control: Logic and Sequence control	2
5.2	Ladder Logic Diagrams, Programmable Logic Controllers: Components of the PLC, PLC Operating Cycle, Programming the PLC (Basic concepts only)	4
5.3	Introduction to Distributed control system (DCS) and Supervisory Control and Data Acquisition Systems (SCADA)	2

Reference Books

1. Mikell Groover, Automation, Production Systems, and Computer-Integrated Manufacturing, 5th Edition, Pearson, 2019.
2. Yoram Koren, "Computer Control of Manufacturing Systems", Tata McGraw Hill Edition 2005.
3. S. R. Deb; Sankha Deb. Robotics Technology and Flexible Automation, Second Edition McGraw-Hill Education: New York, 2010.
4. W. Bolton, "Mechatronics: Electronic Control Systems in Mechanical and Electrical Engineering" - Prentice Hall- 2013 - 5th Edition.
5. Doebelin, E.O. and Manic, D.N., "Measurement Systems: Applications and Design", 7th Edition, McGraw Hill, 2019.
6. Krishna Kant, Computer Based Industrial Control-, EEE-PHI, 2nd edition, 2010.
7. Nathan Ida, Sensors, Actuators, and Their Interfaces- A multidisciplinary introduction, 2nd Edition, IET Digital Library, 2020.
8. Salivahanan, S., and VS Kanchana Bhaaskaran. Linear integrated circuits. McGraw-Hill Education, 2nd edition, 2014.
9. Petruzella, Frank D. Programmable logic controllers. Tata McGraw-Hill Education, 2005
10. Chapman and Hall, "Standard Handbook of Industrial Automation", Onsidine DM C & Onsidine GDC", NJ, 1986



223AGE002	FORENSIC ENGINEERING	CATEGORY	L	T	P	CREDIT
		Audit Course	3	0	0	-

Preamble: This course explores various aspects of Forensic Engineering and different methods ,tools and procedures used by Engineers to investigate and analyze . The students will learn to develop their awareness in Forensic Engineering .

Pre-requisite: Nil

Course Outcomes:

After the completion of the course the student will be able to

CO 1	Identify the fundamental aspects of forensic Engineering
CO 2	Apply forensic Engineering in Practical work flow and Investigation
CO 3	Apply methods and analysis in Forensic Investigation
CO 4	Develop practical strategies and standards of Investigation
CO 5	Create an awareness in criminal cases and create Engineering expertise in court room on forensic Engineering

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO7
CO 1	2	2	3	3	3	3	
CO 2	2	2	3	3	3	3	1
CO 3	3	3	3	3	3	3	1
CO 4	3	3	3	3	3	3	1
CO 5	3	3	3	3	3	3	

Assessment Pattern

Bloom's Category	Continuous Internal Evaluation	End Semester Examination
Apply	40 %	60 %
Analyse	40 %	40 %
Evaluate	20 %	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation: 40 marks

Course based task	:15marks
Seminar/Quizz	:15marks
Test paper	:10 marks
Test paper shall include minimum 80% of the syllabus.	



End Semester Examination: 60 marks

The examination will be conducted by the respective College. The examination will be for 150 minutes and will contain 7 questions, with minimum one question from each module of which student should answer any five. Each question can carry 12 marks.

Model Question paper

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

THIRD SEMESTER M. TECH DEGREE EXAMINATION

Course Code: 223AG002

Course Name: FORENSIC ENGINEERING

Max. Marks: 60

Duration: 2.5 Hours

PART A

Answer any 5 questions, each question carries 12 marks.

Marks

- | | | |
|----|---|------|
| 1. | (a) What are the uses of forensic engineering in legal laws ? | (7) |
| | (b) Discuss the professional responsibility of a forensic Engineer . | (5) |
| 2. | (a) What are the steps in preliminary on site Investigation ? | (7) |
| | (b) With suitable examples, explain photo cataloguing? | (5) |
| 3. | (a) Discuss STEP method . | (7) |
| | (b) Explain root cause Analysis | (5) |
| 4. | (a) Detail about EDAX Method. | (7) |
| | (b) Enlist the uses of NDT in forensic Analysis with example | (5) |
| 5. | (a) Differentiate NFPA & FMV Standards | (7) |
| | (b) Briefly discuss the term Email Phishing ? | (5) |
| 6. | Define the responsibility and duty of a forensic expert in the court. | (12) |
| 7. | Explain Forensic Engineering workflow with examples | (12) |



Syllabus and Course Plan

Module No	Topic	No. of Lectures (Hours)
1	Module 01: Introduction to Forensic Engineering (6 Hours)	
1.1	Forensic Engineering-Definition, Investigation Pyramid, Eyewitness Information, Role in Legal System	2
1.2	Scientific Method-Appling scientific methods in Forensic Engineering-Engineer as expert Witness-Scientific methods and legal system	2
1.3	Qualification of Forensic Engineer-Technical- Knowledge- Oral-written-Communication- other skills-Personality Characteristics	1
1.4	Ethics and professional responsibilities.	1
2	Module 02: Forensic Engineering Workflow and Investigation Methods (6 Hours)	
2.1	Forensic Engineering Workflow-Team & planning-preliminary onsite investigation. Sampling-selection of sample-collection- packing-sealing of samples.	2
2.2	Source and type of evidence - Paper documentation- digital documentation-electronic data. Physical Evidence-Collection of photograph-cataloguing -Recognizing the Evidence-organizing- Evidence Analysis -Reporting	2
2.3	Investigation Methods- Cause and Causal mechanism analysis-Time and event sequence-STEP method. Human Factors, Human errors - Analysis of Operative Instruction and working Procedures	2
3	Module 03: Physical Product Failure & Analytical Methods (6 Hours)	
3.1	Introduction to typical Forensic Engineering Tool box-NDT, Crack detection and human eye -Hardness testing- and Destructive testing Methods with case studies	2
3.2	Indirect stress strain Analysis-Brittle lacquer technique, Contact Radiography-Metallography-EDAX method	1
3.3	Forensic Optical Microscopy-Examination- Magnification-USB Microscopy -Wifi Enabled microscopy -Reflected microscopy	2
3.4	Novel Tools and System -Contour Method-Flash Thermography- Thermographic signal reconstruction (TSR)-Electromagnetically induced acoustic Emission (EMAE)-Pulsed Eddy Current (PEA)-Theory only	1
4	Module 04: Cyber Forensic , Civil ,Electrical Accidents & Standards (6 Hours)	
4.1	Basics of Digital & Cyber forensics: Technical concepts; labs and tools; collecting evidence Operating System Forensic basics with - Windows, Linux -Mobile Forensic-Anti forensics-Malware- Web attack forensics with Email Crimes-Cyber Laws	3
4.2	Different types of Forensic accident investigations- Civil Engineering-Structural- Road accidents -Fire accidents - Water related accidents- Electrical accidents and Investigation methods	2
4.3	Protocol for forensic Investigations-Standard guides-scope significance - use -procedures- reports. Standards – ASTM standards -FMV Standards - SAE Standards -Relevant Standards -NFPA Standards -International Standards	1



5	Module 05: Engineer in the Court room& Criminal Cases (6 Hours)	
5.1	Role of an Engineering Expert-Report-pre trial meetings-Alternative dispute resolution-Single joint expert. Engineer in the court room	2
5.2	Criminal Cases-Introduction-Counterfeit coins-fraudulent road accidents-Fraudulent Insurance claims.	2
5.3	Cyber Crimes and Cases- SIM Swapping -ATM Cloning-Microsoft Internal Spam- Intellectual property cases.	2

Reference Books

1. Colin R Gagg, *Forensic Engineering The Art & Craft of a failure detective* , Taylor & Francis Publishing, 2020
2. Luca Fiorentini ,Luca Marmo *Principles of Forensic Engineering Applied to Industrial Accidents* , Wiley, 2019
3. Harold Franck, Darren Franck , *Forensic Engineering Fundamentals* ,Taylor & Francis publishing 2013
4. Randall K Noon , *Forensic Engineering Investigation*, CRC press limited , 2001
5. Stephen E Petty , *Forensic Engineering: Damage assessment for residential and commercial structures* CRC press 2nd edition , 2017
6. Joshua B Kardon , *Guideliness for forensic Engineering practice* , ASCE, 2012
7. Richard W. Mclay and Robert N. Anderson, *Engineering standards for forensic Applications* , Academic Press; 1st edition 2018
8. Max M Houck ,*Forensic Engineering (Advanced forensic Science)*, Academic press 1st edition 2017
9. Niranjana Reddy - Practical Cyber Forensics. *An Incident-based Approach to Forensic Investigations-Apress (2019)*
10. Peter Rhys Lewis, Ken Reynolds, Colin Gagg - *Forensic Materials Engineering Case Studies- CRC Press (2003) (1)*



INTERNSHIP

A student shall opt for carrying out the Internship at an Industry/Research Organization or at another institute of higher learning and repute (Academia). The organization for Internship shall be selected/decided by the students on their own with prior approval from the faculty advisor/respective PG Programme Coordinator/Guide/Supervisor. Every student shall be assigned an internship Supervisor/Guide at the beginning of the Internship. The training shall be related to their specialisation after the second semester for a minimum duration of six to eight weeks. On completion of the course, the student is expected to be able to develop skills in facing and solving the problems experiencing in the related field.

Objectives

- Exposure to the industrial environment, which cannot be simulated in the classroom and hence creating competent professionals for the industry.
- Provide possible opportunities to learn understand and sharpen the real time technical / managerial skills required at the job.
- Exposure to the current technological developments relevant to the subject area of training.
- Create conducive conditions with quest for knowledge and its applicability on the job.
- Understand the social, environmental, economic and administrative considerations that influence the working environment.
- Expose students to the engineer's responsibilities and ethics.

Benefits of Internship

Benefits to Students

- An opportunity to get hired by the Industry/ organization.
- Practical experience in an organizational setting & Industry environment.
- Excellent opportunity to see how the theoretical aspects learned in classes are integrated into the practical world. On-floor experience provides much more professional experience which is often worth more than classroom

teaching.

- Helps them decide if the industry and the profession is the best career option to pursue.
- Opportunity to learn new skills and supplement knowledge.
- Opportunity to practice communication and teamwork skills.
- Opportunity to learn strategies like time management, multi-tasking etc in an industrial setup.
- Makes a valuable addition to their resume.
- Enhances their candidacy for higher education/placement.
- Creating network and social circle and developing relationships with industry people.
- Provides opportunity to evaluate the organization before committing to a full time position.

Benefits to the Institute

- Build industry academia relations.
- Makes the placement process easier.
- Improve institutional credibility & branding.
- Helps in retention of the students.
- Curriculum revision can be made based on feedback from Industry/ students.
- Improvement in teaching learning process.

Benefits to the Industry

- Availability of ready to contribute candidates for employment.
- Year round source of highly motivated pre-professionals.
- Students bring new perspectives to problem solving.
- Visibility of the organization is increased on campus.

- Quality candidate's availability for temporary or seasonal positions and projects.
- Freedom for industrial staff to pursue more creative projects.
- Availability of flexible, cost-effective workforce not requiring a long-term employer commitment.
- Proven, cost-effective way to recruit and evaluate potential employees.
- Enhancement of employer's image in the community by contributing to the educational enterprise.

Types of Internships

- Industry Internship with/without Stipend
- Govt / PSU Internship (BARC/Railway/ISRO etc)
- Internship with prominent education/research Institutes
- Internship with Incubation centres /Start-ups

Guidelines

- All the students need to go for internship for minimum duration of 6 to 8 weeks.
- Students can take mini projects, assignments, case studies by discussing it with concerned authority from industry and can work on it during internship.
- All students should compulsorily follow the rules and regulations as laid by industry.
- Every student should take prior permissions from concerned industrial authority if they want to use any drawings, photographs or any other document from industry.
- Student should follow all ethical practices and SOP of industry.
- Students have to take necessary health and safety precautions as laid by the industry.
- Student should contact his /her Guide/Supervisor from college on weekly basis to communicate the progress.
- Each student has to maintain a diary/log book
- After completion of internship, students are required to submit
 - Report of work done
 - Internship certificate copy
 - Feedback from employer / internship mentor
 - Stipend proof (in case of paid internship).

Total Marks 100: The marks awarded for the Internship will be on the basis of (i) Evaluation done by the Industry (ii) Students diary (iii) Internship Report and (iv) Comprehensive Viva Voce.

Continuous Internal Evaluation: 50 marks

Student's diary - 25 Marks

Evaluation done by the industry - 25 Marks

Student's Diary/ Daily Log: The main purpose of writing daily diary is to cultivate the habit of documenting and to encourage the students to search for details. It develops the students' thought process and reasoning abilities. The students should record in the daily training diary the day to day account of the observations,

impressions, information gathered and suggestions given, if any. It should contain the sketches & drawings related to the observations made by the students. The daily training diary should be signed after every day by the supervisor/ in charge of the section where the student has been working. The diary should also be shown to the Faculty Mentor visiting the industry from time to time and got ratified on the day of his visit. Student's diary will be evaluated on the basis of the following criteria:

- Regularity in maintenance of the diary
- Adequacy & quality of information recorded
- Drawings, design, sketches and data recorded
- Thought process and recording techniques used
- Organization of the information.

The format of student's diary

Name of the Organization/Section:

Name and Address of the Section Head:

Name and Address of the Supervisor:

Name and address of the student:

Internship Duration: From To

Brief description about the nature of internship:

Day	Brief write up about the Activities carried out: Such as design, sketches, result observed, issues identified, data recorded, etc.
1	
2	
3	

Signature of Industry Supervisor

Signature of Section Head/HR Manager

Office Seal

Attendance Sheet

Name of the Organization/Section:

Name and Address of the Section Head:

Name and Address of the Supervisor:

Name and address of the student:

Internship Duration: From To

Month & Year	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	...
Month & Year																					
Month & Year																					

Signature of Industry Supervisor

Signature of Section Head/HR Manager

Office Seal

Note:

- Student's Diary shall be submitted by the students along with attendance record and an evaluation sheet duly signed and stamped by the industry to the Institute immediately after the completion of the training.
- Attendance Sheet should remain affixed in daily training diary. Do not remove or tear it off.
- Student shall sign in the attendance column. Do not mark 'P'.
- Holidays should be marked in red ink in the attendance column. Absent should be marked as 'A' in red ink.

Evaluation done by the Industry (Marks 25)

Format for Supervisor Evaluation of Intern

Student Name : _____ Date: _____

Supervisor Name : _____ Designation: _____

Company/Organization : _____

Internship Address: _____

Dates of Internship: From _____ To _____

Please evaluate intern by indicating the frequency with which you observed the following parameters:

Parameters	Marks	Needs improvement (0 – 0.25 mark)	Satisfactory (0.25 – 0.50 mark)	Good (0.75 mark)	Excellent (1 mark)
Behavior					
Performs in a dependable Manner					
Cooperates with coworkers and supervisor					
Shows interest in work					
Learns quickly					
Shows initiative					
Produces high quality work					
Accepts responsibility					
Accepts criticism					
Demonstrates organizational skills					
Uses technical knowledge and expertise					
Shows good judgment					
Demonstrates creativity/originality					
Analyzes problems effectively					
Is self-reliant					
Communicates well					
Writes effectively					
Has a professional attitude					
Gives a professional appearance					
Is punctual					
Uses time effectively					

Overall performance of student

Intern (Tick one) : Needs improvement (0 - 0.50 mark) / Satisfactory (0.50 – 1.0 mark) / Good (1.5 mark) / Excellent (2.0 mark)

Additional comments, if any (2 marks) :

Signature of Industry Supervisor

Signature of Section Head/HR Manager

Office Seal

End Semester Evaluation (External Evaluation): 50 Marks

Internship Report	- 25 Marks
Viva Voce	- 25 Marks

Internship Report: After completion of the internship, the student should prepare a comprehensive report to indicate what he has observed and learnt in the training period and should be submitted to the faculty Supervisor. The student may contact Industrial Supervisor/ Faculty Mentor for assigning special topics and problems and should prepare the final report on the assigned topics. Daily diary will also help to a great extent in writing the industrial report since much of the information has already been incorporated by the student into the daily diary. The training report should be signed by the Internship Supervisor, Programme Coordinator and Faculty Mentor.

The Internship report (25 Marks) will be evaluated on the basis of following criteria:

- Originality
- Adequacy and purposeful write-up
- Organization, format, drawings, sketches, style, language etc.
- Variety and relevance of learning experience
- Practical applications, relationships with basic theory and concepts taught in the course

Viva Voce (25 Marks) will be done by a committee comprising Faculty Supervisor, PG Programme Coordinator and an external expert (from Industry or research/academic Institute). This committee will be evaluating the internship report also.

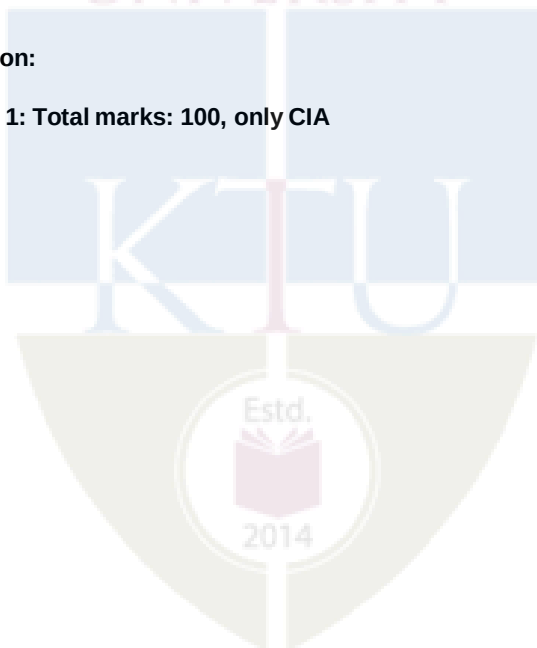
RESEARCH PROJECT/DISSERTATION

Research Project: Students choosing track 2 shall carry out the research project in their parent Institution only under the guidance of a supervisor assigned by the DLAC.

Dissertation: All categories of students in track 1 are to carry out the dissertation in the Institute they are studying or can work either in any CSIR/Industrial R&D organization/any other reputed Institute which have facilities for dissertation work in the area proposed.

Mark Distribution:

Phase 1: Total marks: 100, only CIA



CODE	COURSE NAME	CATEGORY	L	T	P	CREDIT
223PIT100	DISSERTATION PHASE I	Project Work	0	0	17	11

COURSE OBJECTIVES:

Dissertation is aimed to bridge the gap between theoretical knowledge and practical application, fostering a well-rounded skill set that prepares students for success in their future engineering careers. Engineering projects often simulate real-world engineering scenarios. This exposure allows students to become familiar with industry practices, standards, and expectations and preparing them for the challenges they might face in their future careers. Depending on the nature of the project, students may acquire practical skills related to specific tools, software, or equipment. This hands-on experience can be highly beneficial when transitioning to a professional engineering role.

Dissertation Phase I can help to identify the problem based on the area of interest through proper literature survey and to foster innovation in design of products, processes or systems based on the identified problem. perform feasibility study by creative thinking and requirement analysis in finding viable solutions to engineering problems

All categories of students in track 1 are to carry out the dissertation in the Institute they are studying or in any CSIR/Industrial/ R&D organization/any other reputed institute which have facilities for dissertation work in the area proposed.

Course Outcomes:

After the completion of the course the student will be able to

CO 1	Identify and define a relevant and significant problem or challenge in the relevant field
CO2	Formulate research methodologies for the innovative and creative solutions
CO 3	Plan and execute tasks utilizing available resources within timelines, following ethical professional and financial norms
CO 4	Organize and communicate technical and scientific findings effectively in written reports, oral presentation, and visual aids

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO7
CO 1	3		3	2	2	3	2
CO 2	3		3	3	3	2	
CO 3	3		2		3	3	2
CO 4		3	3	2			2

Continuous Internal Assessment (CIA) Total Marks: 100

The evaluation committee comprises

- 1- Project Coordinator(s)
- 2- A Senior faculty member
- 3- Supervisor of the student

Pattern:

Zeroth evaluation by the Evaluation Committee	-
Interim evaluation by the Evaluation Committee	20 marks
Final evaluation by the Evaluation Committee	40 marks
Project Phase - I Report (By Evaluation Committee)	20 marks
Project progress evaluation by supervisor	20 marks

The Plagiarism level in the project report shall be less than 25%.

Interim Review

Literature Survey (CO1- 5 marks)

Comprehension and Problem Identification (CO2-5 marks)

Objective Identification (CO2-5 marks)

Document Preparation and Presentation (CO4-5 marks)

Final Review

Literature Survey (CO1-10 marks)

Project Design (CO2-10 marks)

Execution of tasks by utilizing available resources within timelines (CO3 – 10 marks)

Presentation and document preparation (CO4-10 marks)

Evaluation by the supervisor

The guide/supervisor shall monitor the progress being carried out by the student on a regular basis. In case it is found that progress is unsatisfactory it shall be reported to the Department Evaluation Committee for necessary action.

Student's Diary/ Log book: The main purpose of writing diary/log book is to cultivate the habit of documenting and to encourage the students to search for details. The activity diary shall be signed after every week by the supervisor.

The minimum attendance for completing the course is 75%. The pass minimum for the course is 50% for CIA.

SYLLABUS:

DETAILS	HOURS
1. Literature study/survey of published literature on the assigned topic 2. Formulation of objectives 3. Formulation of hypothesis/ design/ methodology 4. Formulation of work plan and task allocation. 5. Design documentation 6. Preliminary analysis/Modelling/Simulation/Experiment/Design/Feasibility study 7. Preparation of Phase 1 report	150

Dissertation outside the Institute: For doing dissertation outside the Institution, the following conditions are to be met:

- i. They have completed successfully the course work prescribed in the approved curriculum up to the second semester.
- ii. The student has to get prior approval from the DLAC and CLAC.
- iii. Facilities required for doing the dissertation shall be available in the Organization/Industry (A certificate stating the facilities available in the proposed organization and the time period for which the facilities shall be made available to the student, issued by a competent authority from the Organization/Industry shall be submitted by the student along with the application).
- iv. They should have an external as well as an internal supervisor. The internal supervisor should belong to the parent institution and the external supervisor should be Scientists or Engineers from the Institution/Industry/ R&D organization with which the student is associated for doing the dissertation work. The external supervisor shall be with a minimum post graduate degree in the related area.
- v. The student has to furnish his /her monthly progress as well as attendance report signed by the external supervisor and submit the same to the concerned Internal supervisor.
- vi. The external supervisor is to be preferably present during all the stages of evaluation of the dissertation.

Internship leading to Dissertation: The M. Tech students who after completion of 6 to 8 weeks internship at some reputed organizations are allowed to continue their work as dissertation for the third and fourth semester after getting approval from the CLAC. Such students shall make a brief presentation regarding the work they propose to carry out before the DLAC for a detailed scrutiny and to resolve its suitability for accepting it as an M.Tech dissertation. These students will be continuing as regular students of the Institute in third semester for carrying out all academic requirements as per the curriculum/regulation. However, they will be permitted to complete their dissertation in the Industry/Organization (where they have successfully completed their internship) during fourth semester. They should have an external as well as an internal supervisor. The internal supervisor should belong to the parent institution and the external supervisor should be Scientists or Engineers from the external organization with which the student is associated for doing the dissertation work. The external supervisor shall be with a minimum post graduate degree in the related area. The student has to furnish his /her monthly progress as well as attendance report signed by the external guide and submit the same to the concerned internal guide. The external guide is to be preferably present during all the stages of evaluation of the dissertation.

Dissertation as part of Employment: Students may be permitted to discontinue the programme and take up a job provided they have completed all the courses till second semester (FE status students are not permitted) prescribed in the approved curriculum. The dissertation work can be done during a later period either in the organization where they work if it has R&D facility, or in the Institute. Such students should submit application with details (copy of

employment offer, plan of completion of their project etc.) to the Dean (PG) through HoD. The application shall be vetted by CLAC before granting the approval. When the students are planning to do the dissertation work in the organization with R&D facility where they are employed, they shall submit a separate application having following details:

- i. Name of R&D Organization/Industry
- ii. Name and designation of an external supervisor from the proposed Organization/Industry (Scientists or Engineers with a minimum post graduate degree in the related area) and his/her profile with consent
- iii. Name and designation of a faculty member of the Institute as internal supervisor with his/her consent
- iv. Letter from the competent authority from the Organization/Industry granting permission to do the dissertation
- v. Details of the proposed work
- vi. Work plan of completion of project

DLAC will scrutinize the proposal and forward to CLAC for approval. When students are doing dissertation work along with the job in the organization (with R & D facility) where they are employed, the dissertation work shall be completed in four semesters normally (two semesters of dissertation work along with the job may be considered as equivalent to one semester of dissertation work at the Institute). Extensions may be granted based on requests from the student and recommendation of the supervisors such that he/she will complete the M. Tech programme within four years from the date of admission as per the regulation. Method of assessment and grading of the dissertation will be the same as in the case of regular students. The course work in the 3rd semester for such students are to be completed as per the curriculum requirements (i) MOOC can be completed as per the norms mentioned earlier (ii) Audit course are to be carried out either in their parent Institution or by self-learning. However, for self-learning students, all assessments shall be carried out in their parent institution as in the case of regular students.



APJ ABDUL KALAM
TECHNOLOGICAL
UNIVERSITY



SEMESTER - IV



CODE	COURSE NAME	CATEGORY	L	T	P	CREDIT
224PIT100	DISSERTATION PHASE II	Project Work	0	0	24	16

All categories of students in track 1 are to carry out the DISSERTATION PHASE II in the institute they are studying or in any Industrial/ R&D organization/any other reputed institute which have facilities for dissertation work in the area proposed. DISSERTATION PHASE II shall not compulsorily continuation of DISSERTATION PHASE I. The student has to publish a research article in a conference or a reputed journal before appearing for the end-semester examination. The eligibility criteria for registering to the end semester examination are attendance in the course and no pending disciplinary action. The minimum attendance for appearing for the end semester examination is 75%. Students who do not meet these eligibility criteria are ineligible (identified by FE grade) to appear for the ESE. Students, who have completed a course but could not appear for the end semester examination, shall be awarded 'AB' Grade, provided they meet other eligibility criteria. The pass minimum for the course is 45% for ESE and 50% for (CIA and ESE) put together.

Continuous Internal Assessment (CIA) Total Marks: 100

The evaluation committee comprises

- 1- Project Coordinator(s)
- 2- A Senior faculty member
- 3- Supervisor of the student

Pattern (CIA)

Zeroth evaluation by the Evaluation Committee	-
Interim evaluation by the Evaluation Committee	30 marks
Final evaluation by the Evaluation Committee	50 marks
Project progress evaluation by supervisor	20 marks

Evaluation by the supervisor

The guide/supervisor shall monitor the progress being carried out by the student on a regular basis. In case it is found that progress is unsatisfactory it shall be reported to the Department Evaluation Committee for necessary action.

Student's Diary/ Log book: The main purpose of writing diary/log book is to cultivate the habit of documenting and to encourage the students to search for details. The activity diary shall be signed after every week by the supervisor.

End Semester Evaluation (ESE) Total Marks: 100

The evaluation committee comprises

- 1- Project Coordinator(s)
- 2- An external expert (from Industry or research/academic institute)
- 3- Supervisor of the student

Pattern (ESE)

1. Innovation and Originality (10 marks):

Assessment of the uniqueness and innovation demonstrated in the project work.
Original contributions, if any, to the field or problem area.

2. Implementation and Execution (20 marks):

Evaluation of the actual implementation or execution of the project, including:

- Quality of work done
- Demonstrated skills and techniques applied
- Adherence to project timelines and milestones

3. Project Documentation (25 marks):

Comprehensive project report evaluation including:

- Introduction and problem statement
- Literature review
- Methodology and approach
- Results and analysis
- Conclusion and recommendations
- References and citations
- Details of the publications
- Plagiarism certificate

The Plagiarism level in the project report shall be less than 25%.

4. Presentation and Defence (40 marks):

Oral presentation of the project to a panel of examiners, including:

- Clarity and effectiveness of the presentation
- Ability to explain the project objectives, methodologies, and findings
- Handling questions and providing satisfactory answers during the defence

5. Publication of the work either in a conference or in a journal (5 marks)

SYLLABUS:

DETAILS	HOURS
1. Literature study/survey of published literature on the assigned topic 2. Topic Selection and Proposal 3. Formulation of objectives 4. Research and Planning 5. Formulation of work plan and task allocation. 6. Execution 7. Documentation and Reporting 8. Project Showcase reflecting on the project experience and lessons learned	200

Dissertation outside the Institute: For doing dissertation outside the Institution, the following conditions are to be met:

- i. They have completed successfully the course work prescribed in the approved curriculum up to the second semester.
- ii. The student has to get prior approval from the DLAC and CLAC.
- iii. Facilities required for doing the dissertation shall be available in the Organization/Industry (A certificate stating the facilities available in the proposed organization and the time period for which the facilities shall be made available to the student, issued by a competent authority from the Organization/Industry shall be submitted by the student along with the application).
- iv. They should have an external as well as an internal supervisor. The internal supervisor should belong to the parent institution and the external supervisor should be Scientists or Engineers from the Institution/Industry/ R&D organization with which the student is associated for doing the dissertation work. The external supervisor shall be with a minimum post graduate degree in the related area.
- v. The student has to furnish his /her monthly progress as well as attendance report signed by the external supervisor and submit the same to the concerned internal supervisor.
- vi. The external supervisor is to be preferably present during all the stages of evaluation of the dissertation.

